

Microeconomics II

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Disclaimer

- **Incomplete Document:** This file constitutes supporting material under active development. Several sections and asymptotic derivations are still being revised, expanded, and supplemented.
 - **AI-Assisted Construction:** This material was structured, reviewed, and expanded with the assistance of Artificial Intelligence models for didactic organization and Markdown/LaTeX formatting rigor.
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Chapter 1

Choice under Uncertainty

1.1 Introduction

The theory of choice under uncertainty extends the models of choice under certainty to environments in which each action can generate multiple possible outcomes, each with a probability. The central object of the theory is the **lottery**, and the main tool for representation is the von Neumann–Morgenstern **expected utility**.

The objective of this chapter is to:

1. present the formal foundation of choice under risk: lotteries, axioms, the expected utility theorem, and its properties;
2. develop the notions of risk aversion, certainty equivalent, risk premium, Arrow–Pratt measures, and stochastic dominance;
3. discuss descriptive limitations of the theory, especially the Allais and Machina paradoxes.

1.2 Lotteries and Risk Representation

1.2.1 Simple Lotteries and the Lottery Space

In choice under uncertainty, the agent does not directly choose a certain outcome, but rather a probability distribution over possible outcomes. A classic example is an insurance policy: by purchasing it, the agent alters the wealth distribution across states of nature, reducing the loss in the bad state at the cost of lower wealth in the good state.

Definition 1.2.1 (Simple lottery). Let $\mathcal{Z} = \{z_1, \dots, z_N\}$ be a finite set of outcomes, or prizes. A **simple lottery** L is a list

$$L = (p_1, \dots, p_N),$$

with $p_n \geq 0$ for all n and $\sum_{n=1}^N p_n = 1$, where p_n is the probability of outcome z_n occurring.

Definition 1.2.2 (Lottery space). The space of all simple lotteries over $\mathcal{Z}$, denoted by

Δ^{N-1} , is the $(N - 1)$ -dimensional simplex:

$$\Delta^{N-1} = \left\{ p \in \mathbb{R}_+^N : \sum_{n=1}^N p_n = 1 \right\}.$$

The essential assumptions are:

1. **Non-negativity:** $p_n \geq 0$;
2. **Unit sum:** $\sum_{n=1}^N p_n = 1$.

Remark 1.2.1. If the probabilities do not sum to 1, the object is not a valid probability distribution. If there are negative probabilities, the probabilistic interpretation and the convex structure of the lottery space are lost.

1.2.2 Degenerate Lotteries

A degenerate lottery represents formal certainty within the uncertainty environment.

Definition 1.2.3 (Degenerate lottery). A lottery $L \in \Delta^{N-1}$ is **degenerate** if there exists $z_n \in \mathcal{Z}$ such that $p_n = 1$ and $p_m = 0$ for all $m \neq n$. We denote this lottery by δ_n or $[z_n]$.

Properties

1. **Vertices of the simplex:** degenerate lotteries are the extreme points of Δ^{N-1} . Furthermore,

$$\Delta^{N-1} = \text{conv}\{\delta_1, \dots, \delta_N\}.$$

2. **Certain outcomes:** when expected utility is defined, the utility of a degenerate lottery coincides with the Bernoulli utility of the certain outcome:

$$U([z_n]) = u(z_n).$$

3. **Pure strategies:** in game theory, a pure strategy can be viewed as a degenerate distribution over actions.

Example 1.2.1. The choice between receiving \$1,000 for certain and a gamble that pays \$2,000 with probability 1/2 and \$0 with probability 1/2 is, formally, the choice between a degenerate lottery and a non-degenerate lottery.

1.2.3 Compound Lotteries and Reduction

A compound lottery is one whose prizes are other lotteries.

Definition 1.2.4 (Two-stage compound lottery). Let $L_1, \dots, L_K \in \Delta^{N-1}$ be simple lotteries and $\alpha = (\alpha_1, \dots, \alpha_K)$ a probability vector with $\alpha_k \geq 0$ and $\sum_{k=1}^K \alpha_k = 1$. The compound lottery

$$(L_1, \dots, L_K; \alpha_1, \dots, \alpha_K)$$

is the draw that generates lottery L_k with probability α_k .

Definition 1.2.5 (Generalized compound lottery). A generalized compound lottery extends the idea above to any finite number of stages. The set $\mathcal{L}$ is defined recursively by:

1. every simple lottery $L \in \Delta^{N-1}$ belongs to $\mathcal{L}$;
2. if $L_1, \dots, L_K \in \mathcal{L}$ and $\alpha \in \Delta_K$, then $(L_1, \dots, L_K; \alpha) \in \mathcal{L}$.

Remark 1.2.2. This recursive definition represents a decision tree: each intermediate node is a draw, and the terminal nodes are prizes in $\mathcal{Z}$.

Axiom 1.2.1 (Reduction of compound lotteries). The agent is indifferent between a compound lottery and its corresponding reduced lottery; that is, only the final probability distribution over the prizes matters.

For a two-stage compound lottery, the final probability of outcome z_n is

$$p_n = \sum_{k=1}^K \alpha_k p_n^k,$$

where p_n^k is the probability of z_n in lottery L_k .

For a generalized compound lottery with T stages,

$$p_n = \sum_{\text{paths } c \rightarrow z_n} \prod_{t=1}^T \alpha_{c_t}.$$

Example 1.2.2. Suppose that, with probability $1/2$, the agent enters a lottery that pays \$100 with probability 0.8 and \$0 with probability 0.2; with probability $1/2$, they enter another lottery that pays \$100 with probability 0.5 and \$0 with probability 0.5. The reduced lottery is

$$\begin{aligned} p_{\$100} &= \frac{1}{2} \cdot 0.8 + \frac{1}{2} \cdot 0.5 = 0.65, \\ p_{\$0} &= \frac{1}{2} \cdot 0.2 + \frac{1}{2} \cdot 0.5 = 0.35. \end{aligned}$$

1.2.4 Matrix Representation and Comparison

If the K simple lotteries are organized as columns of an $N \times K$ matrix P ,

$$P = \begin{bmatrix} p_1^1 & p_1^2 & \cdots & p_1^K \\ p_2^1 & p_2^2 & \cdots & p_2^K \\ \vdots & \vdots & \ddots & \vdots \\ p_N^1 & p_N^2 & \cdots & p_N^K \end{bmatrix},$$

then the reduced lottery is

$$L = P\alpha = \sum_{k=1}^K \alpha_k L_k.$$

For a compound lottery with T stages, the reduction can be seen as successive applications of transition matrices:

$$p = P_T P_{T-1} \cdots P_1 \alpha.$$

Feature	Degenerate lottery	Generalized compound lottery
Nature	Certainty	Multi-stage uncertainty
Location in the simplex	Vertex	Convex combination of lotteries
Components	A single prize $z \in \mathcal{Z}$	Lotteries $L_k \in \mathcal{L}$
Reduction	Already simple	Reducible via the reduction axiom
Representation	Canonical vector e_n	$L = P\alpha$

Table 1.1: Comparison between degenerate lottery and generalized compound lottery.

Remark 1.2.3 (Theoretical implication). The reduction of compound lotteries is fundamental to the linearity of expected utility in probabilities. If the agent violates this principle, the traditional expected utility representation may fail.

1.3 Preferences over Lotteries and von Neumann–Morgenstern Axioms

1.3.1 Economic Intuition

The central problem is comparing objects that involve risk. For example, how to compare a fixed salary with a startup that might pay a lot or nothing? The idea of the theory is that, if preferences are sufficiently consistent, there is a scale of desirability for the prizes, the Bernoulli utility, such that the value of a lottery is the weighted average of the utilities of the prizes.

Example 1.3.1 (Risk thermometer). The von Neumann–Morgenstern theorem works like a thermometer: it anchors the extremes, the best and worst outcomes, and positions any lottery between them through the probabilities of indifference.

1.3.2 von Neumann–Morgenstern Axioms

Axiom 1.3.1 (Rationality). The preference relation $\succsim$ over $\mathcal{L}$ is complete and transitive.

Axiom 1.3.2 (Continuity). For any $L, L', L'' \in \mathcal{L}$, the sets

$$\{\alpha \in [0, 1] : \alpha L + (1 - \alpha)L' \succsim L''\}$$

and

$$\{\alpha \in [0, 1] : L'' \succsim \alpha L + (1 - \alpha)L'\}$$

are closed.

Axiom 1.3.3 (Independence). For any $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0, 1)$,

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''.$$

1.4 Expected Utility Theorem

1.4.1 Statement

Theorem 1.4.1 (von Neumann–Morgenstern). *Let $\mathcal{L}$ be the space of simple lotteries over a finite set of prizes $\mathcal{Z}$. If $\succsim$ satisfies rationality, continuity, and independence, then there exists a function $u : \mathcal{Z} \rightarrow \mathbb{R}$ such that, for any lotteries $L = (p_1, \dots, p_N)$ and $L' = (p'_1, \dots, p'_N)$,*

$$L \succsim L' \iff \sum_{n=1}^N p_n u(z_n) \geq \sum_{n=1}^N p'_n u(z_n).$$

Furthermore, u is unique up to positive affine transformations.

1.4.2 Proof

The proof constructs the utility function in six linked steps: Claims 1–2 create the anchors and the “no-jump” behavior of mixtures; Claim 3 transforms the order of weights into the order of preference; Claim 4 uses topology to calibrate each lottery by a single number; Claim 5 validates this number as an ordinal representation; Claim 6 converts the ordinal representation into the linear form of expected utility.

Claim 1: Global Bounds

Lemma 1.4.1 (Global bounding elements). *Since $\mathcal{Z}$ is finite and $\succsim$ is complete and transitive, we can order the outcomes such that*

$$L^* \equiv \delta_{z_{\sigma(1)}} \succsim \delta_{z_{\sigma(2)}} \succsim \dots \succsim \delta_{z_{\sigma(N)}} \equiv L_*.$$

Furthermore, for all $L \in \mathcal{L}$,

$$L^* \succsim L \succsim L_*.$$

Proof. For degenerate lotteries, the result follows from the ordering. For an arbitrary L , induction on the size of the support. **Base (two prizes):** if $L = p \delta_a + (1 - p) \delta_b$ with $\delta_a \succsim \delta_b$, independence mixed with δ_a gives $\delta_a \succsim L$, and mixed with δ_b gives $L \succsim \delta_b$; by transitivity, $L^* \succsim L \succsim L_*$. **Inductive step:** write

$$L = p_{\bar{n}} \delta_{z_{\bar{n}}} + (1 - p_{\bar{n}}) L',$$

with L' having a smaller support. By hypothesis, $L^* \succsim L' \succsim L_*$; since $L^* \succsim \delta_{z_{\bar{n}}} \succsim L_*$, independence and transitivity give $L^* \succsim L \succsim L_*$. $\square$

Structural role It provides the “ruler” for measurement: no lottery falls outside the interval $[L_*, L^*]$, without which it would not be possible to calibrate generic lotteries by mixing the extremes.

Claim 2: Strict Betweenness

Lemma 1.4.2 (Strict betweenness). *For all $L, L' \in \mathcal{L}$ and $\alpha \in (0, 1)$,*

$$L \succ L' \implies L \succ \alpha L + (1 - \alpha) L' \succ L'.$$

Proof. If $L \succ L'$, independence implies

$$\alpha L + (1 - \alpha)L \succ \alpha L' + (1 - \alpha)L,$$

so $L \succ \alpha L' + (1 - \alpha)L = (1 - \alpha)L + \alpha L'$. Applying independence again,

$$\alpha L + (1 - \alpha)L' \succ \alpha L' + (1 - \alpha)L' = L'.$$

By transitivity, $L \succ \alpha L + (1 - \alpha)L' \succ L'$. □

Structural role It prevents illogical preference reversals when adding uncertainty; it is the technical engine of Claim 3 and the topological validity of Claim 4.

Claim 3: Strict Monotonicity

Lemma 1.4.3 (Strict monotonicity). *Assume $L^* \succ L_*$. For $\alpha, \beta \in [0, 1]$,*

$$\beta > \alpha \iff \beta L^* + (1 - \beta)L_* \succ \alpha L^* + (1 - \alpha)L_*.$$

Proof. If $\beta > \alpha$, define $\psi = \frac{\beta - \alpha}{1 - \alpha} \in (0, 1]$. Then

$$\beta L^* + (1 - \beta)L_* = \psi L^* + (1 - \psi)[\alpha L^* + (1 - \alpha)L_*].$$

Since $L^* \succ \alpha L^* + (1 - \alpha)L_*$ (Claim 2), Claim 2 itself implies $\beta L^* + (1 - \beta)L_* \succ \alpha L^* + (1 - \alpha)L_*$. The converse follows by exclusion: if $\beta = \alpha$ the lotteries are identical; if $\beta < \alpha$ the strict preference would be the opposite. □

Structural role It guarantees that the numerical order of the weights reflects the order of preferences and is the ingredient for uniqueness in Claim 4.

Claim 4: Existence and Uniqueness of the Indifference Weight

Lemma 1.4.4 (Calibration weight). *For every lottery $L \in \mathcal{L}$, there exists a unique $\alpha_L \in [0, 1]$ such that*

$$L \sim \alpha_L L^* + (1 - \alpha_L)L_*.$$

Proof. Define $A = \{\alpha : \alpha L^* + (1 - \alpha)L_* \precsim L\}$ and $B = \{\alpha : L \precsim \alpha L^* + (1 - \alpha)L_*\}$. By continuity, A and B are closed; by Claim 1, $1 \in A$ and $0 \in B$ (non-empty); by completeness, $A \cup B = [0, 1]$. Since $[0, 1]$ is connected, $A \cap B \neq \emptyset$: there exists α_L with $L \sim \alpha_L L^* + (1 - \alpha_L)L_*$. Uniqueness: if $\alpha_L > \alpha'_L$, Claim 3 would give $\alpha_L L^* + (1 - \alpha_L)L_* \succ \alpha'_L L^* + (1 - \alpha'_L)L_*$, contradicting both being indifferent to L . □

Structural role This is the moment of creation of the utility function: the rule $L \mapsto \alpha_L$ is well-defined (exists and is unique), allowing us to define $U(L) \equiv \alpha_L$.

Claim 5: Ordinal Representation

Lemma 1.4.5 (Ordinal representation). *Define $U : \mathcal{L} \rightarrow \mathbb{R}$ by $U(L) = \alpha_L$. Then*

$$L \succsim L' \iff U(L) \geq U(L').$$

Proof. By Claim 4, $L \sim U(L)L^* + (1 - U(L))L_*$ and $L' \sim U(L')L^* + (1 - U(L'))L_*$. By transitivity,

$$L \succsim L' \iff U(L)L^* + (1 - U(L))L_* \succsim U(L')L^* + (1 - U(L'))L_*,$$

and by Claim 3 this is equivalent to $U(L) \geq U(L')$. $\square$

Structural role It converts the abstract binary relation $\succsim$ into a scalar function that preserves the ordinal order.

Claim 6: Linearity

Lemma 1.4.6 (Linearity of the representation). *For any $L, L' \in \mathcal{L}$ and $\beta \in [0, 1]$,*

$$U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L').$$

Proof. By the calibration indifference and the independence axiom,

$$\beta L + (1 - \beta)L' \sim \beta[U(L)L^* + (1 - U(L))L_*] + (1 - \beta)[U(L')L^* + (1 - U(L'))L_*].$$

Grouping the terms,

$$\beta L + (1 - \beta)L' \sim [\beta U(L) + (1 - \beta)U(L')]L^* + [1 - \beta U(L) - (1 - \beta)U(L')]L_*.$$

By the uniqueness of the calibration weight, $U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L')$. $\square$

Remark 1.4.1 (The key insight: why uniqueness forces linearity). The definition of U and the uniqueness in Claim 4 do all the work. Reducing the compound lottery, the total probability of obtaining L^* is $\gamma = \beta U(L) + (1 - \beta)U(L')$; hence the mixture is indifferent to $\gamma L^* + (1 - \gamma)L_*$. But Claim 4 states that there is a *unique* weight that generates this indifference, and this weight is, by definition, $U(\beta L + (1 - \beta)L')$. Since γ also generates it, necessarily $U(\beta L + (1 - \beta)L') = \gamma$. In short: the “label” of the mixture is the weighted average of the labels, because the label is unique.

Conclusion

For an arbitrary lottery, $L = \sum_n p_n \delta_{z_n}$; by linearity, $U(L) = \sum_n p_n U(\delta_{z_n})$. Defining $u(z_n) := U(\delta_{z_n})$, we obtain $U(L) = \sum_n p_n u(z_n)$ and, by Claim 5,

$$L \succsim L' \iff \sum_n p_n u(z_n) \geq \sum_n p'_n u(z_n).$$

1.4.3 Geometry of the Simplex: Straight and Parallel Indifference Curves

Consider $N = 3$ ordered prizes and a lottery $L = (p_1, p_2, p_3)$ with $p_2 = 1 - p_1 - p_3$, projected onto the (p_1, p_3) plane. By Claim 6, $U(L) = p_1 u_1 + p_2 u_2 + p_3 u_3$, with $u_n = u(z_n)$ being constants.

Proposition 1.4.1 (Indifference curves are straight lines). *If $L \sim L'$, then the entire segment $\alpha L + (1 - \alpha)L'$ belongs to the same indifference curve; furthermore, the level curve is a straight line in the (p_1, p_3) plane.*

Via independence. If $L \sim L'$, independence with $L'' = L'$ gives $\alpha L + (1 - \alpha)L' \sim \alpha L' + (1 - \alpha)L' = L'$; since $L' \sim L$, the entire segment is indifferent to L . $\square$

Via linearity. If $U(L) = U(L') = \bar{U}$, then $U(\alpha L + (1 - \alpha)L') = \alpha \bar{U} + (1 - \alpha)\bar{U} = \bar{U}$. The level curve $p_1 u_1 + (1 - p_1 - p_3)u_2 + p_3 u_3 = \bar{U}$ resolves to

$$p_3 = \left(\frac{u_2 - u_1}{u_3 - u_2} \right) p_1 + \left(\frac{\bar{U} - u_2}{u_3 - u_2} \right),$$

a first-degree equation in (p_1, p_3) : a straight line. $\square$

Proposition 1.4.2 (Indifference curves are parallel). *Two distinct indifference curves have the same slope in the (p_1, p_3) plane.*

Via independence. If $L \sim L'$ and L'' is any lottery, independence gives $\alpha L + (1 - \alpha)L'' \sim \alpha L' + (1 - \alpha)L''$. The direction vectors of the two curves are $L' - L$ and $[\alpha L' + (1 - \alpha)L''] - [\alpha L + (1 - \alpha)L''] = \alpha(L' - L)$, positive scalar multiples: same direction, hence same slope. $\square$

Via linearity. From the equation of the line, the slope is $\frac{dp_3}{dp_1} = \frac{u_2 - u_1}{u_3 - u_2}$, a constant that does not depend on $\bar{U}$ nor on (p_1, p_3) ; changing $\bar{U}$ only alters the intercept, translating the line without rotating it. $\square$

Remark 1.4.2 (Geometric synthesis). Straight lines follow from $L \sim L' \implies \alpha L + (1 - \alpha)L' \sim L$ (independence) $\iff$ linearity (Claim 6); parallelism follows from $L \sim L' \implies \alpha L + (1 - \alpha)L'' \sim \alpha L' + (1 - \alpha)L'' \iff$ slope invariant to $\bar{U}$. It is the geometric signature of the independence axiom on the Marschak–Machina simplex.

1.4.4 Correspondence between Intuition, Proof, and Structural Role

Intuition / What was proven	Formal element	Structural role
Best and worst lotteries as anchors	L^* and L_* ; $L^* \succsim L \succsim L_*$	Claim 1: creates the bounds of the choice space
Mixture lies between extremes	$L \succ \alpha L + (1 - \alpha)L' \succ L'$	Claim 2: local monotonicity without jumps
More weight on the best is preferred	$\beta > \alpha \iff \text{mixture } \beta \text{ preferred}$	Claim 3: numerical order = preference order; uniqueness
Lottery calibration	$\exists! \alpha_L : L \sim \alpha_L L^* + (1 - \alpha_L)L_*$	Claim 4: defines $U(L) = \alpha_L$
Numerical representation	$L \succsim L' \iff U(L) \geq U(L')$	Claim 5: validates U as ordinal representation
Linearity in probabilities	$U(\beta L + (1 - \beta)L') = \beta U(L) + (1 - \beta)U(L')$	Claim 6: expected utility form; parallel lines on the simplex

Table 1.2: Logical structure of the von Neumann–Morgenstern theorem: intuition, formalization, and the role of each step.

1.5 Bernoulli Utility, Cardinality, and Linearity

1.5.1 Bernoulli Utility

Definition 1.5.1 (Bernoulli utility). The **Bernoulli utility** is a function $u : \mathcal{Z} \rightarrow \mathbb{R}$ that assigns a real value to each certain outcome $z \in \mathcal{Z}$.

Given a lottery $L = (p_1, \dots, p_N)$, the expected utility is

$$U(L) = \sum_{n=1}^N p_n u(z_n).$$

Unlike ordinary utility under certainty, which is purely ordinal, Bernoulli utility is **cardinal**: only positive affine transformations preserve the expected utility structure.

1.5.2 Cardinality of Bernoulli Utility

Proposition 1.5.1 (Cardinality). *If $u(\cdot)$ represents preferences in the expected utility form, then $v(\cdot)$ also represents them if and only if*

$$v(z) = \alpha u(z) + \beta, \quad \alpha > 0.$$

Proof. Sufficiency. If $v(z) = \alpha u(z) + \beta$, then, for any lottery L ,

$$V(L) = \sum_{n=1}^N p_n v(z_n) = \alpha \sum_{n=1}^N p_n u(z_n) + \beta \sum_{n=1}^N p_n = \alpha U(L) + \beta.$$

Since $\alpha > 0$, $V(L) \geq V(L')$ if and only if $U(L) \geq U(L')$.

Necessity. Suppose u and v represent the same preferences in the expected utility form. Let $\bar{z}$ be the best outcome and $\underline{z}$ the worst outcome. If $u(\bar{z}) = u(\underline{z})$, the case is trivial. Otherwise, for any $z \in \mathcal{Z}$, continuity implies there exists $\lambda_z \in [0, 1]$ such that

$$\delta_z \sim \lambda_z \delta_{\bar{z}} + (1 - \lambda_z) \delta_{\underline{z}}.$$

Since u represents the preference,

$$u(z) = \lambda_z u(\bar{z}) + (1 - \lambda_z) u(\underline{z}).$$

Hence,

$$\lambda_z = \frac{u(z) - u(\underline{z})}{u(\bar{z}) - u(\underline{z})}.$$

Since v represents the same preference,

$$v(z) = \lambda_z v(\bar{z}) + (1 - \lambda_z) v(\underline{z}).$$

Substituting λ_z ,

$$v(z) = \frac{u(z) - u(\underline{z})}{u(\bar{z}) - u(\underline{z})} v(\bar{z}) + \left[1 - \frac{u(z) - u(\underline{z})}{u(\bar{z}) - u(\underline{z})} \right] v(\underline{z}).$$

Rearranging,

$$v(z) = \left[\frac{v(\bar{z}) - v(\underline{z})}{u(\bar{z}) - u(\underline{z})} \right] u(z) + \left[v(\underline{z}) - \frac{v(\bar{z}) - v(\underline{z})}{u(\bar{z}) - u(\underline{z})} u(\underline{z}) \right].$$

Defining

$$\alpha = \frac{v(\bar{z}) - v(\underline{z})}{u(\bar{z}) - u(\underline{z})} > 0,$$

and

$$\beta = v(\underline{z}) - \alpha u(\underline{z}),$$

we have

$$v(z) = \alpha u(z) + \beta.$$

□

Remark 1.5.1. Cardinality implies that utility differences have economic meaning. In particular, the curvature of u cannot be eliminated by arbitrary ordinal transformations, being central to the analysis of risk aversion.

1.5.3 Linearity in Probabilities

Proposition 1.5.2 (Linearity of expected utility). *A function $U : \mathcal{L} \rightarrow \mathbb{R}$ has the expected utility form if and only if, for any lotteries $L_1, \dots, L_K$ and probabilities $\alpha_k \geq 0$ with $\sum_{k=1}^K \alpha_k = 1$,*

$$U \left(\sum_{k=1}^K \alpha_k L_k \right) = \sum_{k=1}^K \alpha_k U(L_k).$$

Proof. If $U(L) = \sum_{n=1}^N p_n u(z_n)$ and the lottery $\sum_k \alpha_k L_k$ has final probabilities

$$q_n = \sum_{k=1}^K \alpha_k p_n^k,$$

then

$$U\left(\sum_{k=1}^K \alpha_k L_k\right) = \sum_{n=1}^N q_n u(z_n) = \sum_{n=1}^N \sum_{k=1}^K \alpha_k p_n^k u(z_n) = \sum_{k=1}^K \alpha_k \sum_{n=1}^N p_n^k u(z_n) = \sum_{k=1}^K \alpha_k U(L_k).$$

Conversely, if U is linear in mixtures of lotteries, write

$$L = \sum_{n=1}^N p_n \delta_n.$$

Defining $u_n := U(\delta_n)$, we obtain

$$U(L) = U\left(\sum_{n=1}^N p_n \delta_n\right) = \sum_{n=1}^N p_n U(\delta_n) = \sum_{n=1}^N p_n u_n.$$

□

Remark 1.5.2. Degenerate lotteries form the canonical basis of the simplex. Linearity allows decomposing any lottery as a convex combination of these degenerate lotteries.

1.6 Paradoxes and Limits of Expected Utility

1.6.1 Allais Paradox

Intuition

The Allais paradox shows that individuals may violate the independence axiom by assigning excessive weight to certainty. The observed behavior is not consistent with any expected utility.

Example 1.6.1. The difference between 99% and 100% chance may seem larger than the difference between 10% and 11%, even though both are 1 percentage point. This reflects the certainty effect.

Formal Formulation

Consider the prizes:

$$z_1 = 5 \text{ million}, \quad z_2 = 1 \text{ million}, \quad z_3 = 0.$$

Situation 1:

$$L_1 = (0, 1, 0), \\ L'_1 = (0.10, 0.89, 0.01).$$

Many people choose $L_1 \succ L'_1$.

Situation 2:

$$L_2 = (0, 0.11, 0.89), \\ L'_2 = (0.10, 0, 0.90).$$

Many people choose $L'_2 \succ L_2$.

Contradiction

Assume $u(0) = 0$. If $L_1 \succ L'_1$, then

$$u(1) > 0.10u(5) + 0.89u(1).$$

Hence,

$$0.11u(1) > 0.10u(5).$$

If $L'_2 \succ L_2$, then

$$0.10u(5) > 0.11u(1).$$

The two inequalities cannot hold simultaneously.

Corollary 1.6.1. *The typical choices in the Allais paradox cannot be represented by expected utility.*

1.6.2 Machina Paradox

Intuition

The Machina paradox explores feelings such as disappointment. The utility of an outcome may depend on what almost happened, violating independence with respect to the global distribution of the lottery.

Example 1.6.2. Winning a movie about Venice might be disappointing if the agent almost won a trip to Venice.

Formal Formulation

Let the outcomes be:

z_1 = trip to Venice,

z_2 = watching a movie about Venice,

z_3 = staying home.

Assume

$$z_1 \succ z_2 \succ z_3.$$

Consider

$$L = (0.999, 0.001, 0),$$

$$L' = (0.999, 0, 0.001).$$

By expected utility, since $u(z_2) > u(z_3)$,

$$U(L) = 0.999u(z_1) + 0.001u(z_2),$$

$$U(L') = 0.999u(z_1) + 0.001u(z_3).$$

Hence,

$$U(L) > U(L').$$

However, individuals may prefer $L' \succ L$ due to the disappointment associated with z_2 .

Corollary 1.6.2. *The Machina paradox suggests that utility may depend on counterfactuals or the entire distribution of the lottery, not just the final outcome and its probability.*

1.6.3 Implications and Extensions

The paradoxes show that expected utility is strong as a normative model but may fail as a descriptive model. This motivated alternative theories, such as:

- prospect theory;
- non-expected utility;
- models with non-linear probability weighting;
- models with regret or disappointment.

Remark 1.6.1 (Possible extensions of the chapter). For a more complete coverage, the following could be included:

1. first-order stochastic dominance;
2. applications to portfolio, insurance, and principal–agent problems;
3. formal definitions of risk neutrality and risk seeking;
4. extensions to infinite outcome spaces;
5. additional characterizations of risk aversion and behavioral theories.

1.7 Risk Aversion: Formal Characterization and Measures

1.7.1 Economic Intuition

An agent is risk-averse when they prefer to receive the expected value of a lottery with certainty rather than face the lottery itself. In economic terms, the agent accepts giving up part of the expected value to eliminate uncertainty.

There are three objects that need to be distinguished precisely:

1. the utility of the expected value received with certainty;
2. the expected utility of the lottery;
3. the certainty equivalent of the lottery.

For a risk-averse agent, the utility of the certain expected value is greater than or equal to the expected utility of the lottery. Since the certainty equivalent generates exactly the same utility as the lottery, it follows that the expected value is greater than or equal to the certainty equivalent. The difference between them is the risk premium.

1.7.2 Monetary Lotteries and Functional Representation

In monetary applications, the set of outcomes is usually a subset of the real line, for example $C \subseteq \mathbb{R}_+$. A monetary lottery is described by a probability distribution F .

If preferences satisfy the von Neumann–Morgenstern axioms, there exists a Bernoulli utility function $u(\cdot)$ such that the lottery F is evaluated by

$$V(F) = \int u(x) dF(x).$$

In this context:

- $u(x)$ assigns utility to a certain amount of wealth x ;
- $V(F)$ assigns utility to the lottery F as a whole;
- the expected value of the lottery is

$$E_F[x] = \int x dF(x).$$

In the discrete case, the integral is replaced by

$$V(F) = \sum_{n=1}^N p_n u(x_n).$$

1.7.3 Formal Definitions

Strict Monotonicity

Definition 1.7.1 (Strict monotonicity). A preference relation $\succsim$ over monetary lotteries is **strictly monotone** if, for any certain wealth levels x and y ,

$$x > y \iff \delta_x \succ \delta_y,$$

where δ_x and δ_y are degenerate lotteries, or Dirac measures, that pay x and y with probability 1.

Remark 1.7.1. Under expected utility, strict monotonicity is equivalent to $u(\cdot)$ being strictly increasing. This expresses that more wealth is always preferable to less wealth; that is, the marginal utility of wealth is positive.

Risk Aversion

Definition 1.7.2 (Risk aversion). A decision maker is **risk-averse** if, for every non-degenerate monetary lottery F with a well-defined expected value,

$$\delta_{E_F[x]} \succsim F.$$

That is, the agent prefers to receive the expected value of the lottery with certainty rather than face the lottery.

Definition 1.7.3 (Strict risk aversion). The agent is **strictly risk-averse** if, for every non-degenerate lottery F ,

$$\delta_{E_F[x]} \succ F.$$

1.7.4 von Neumann–Morgenstern Structural Equivalence

The mathematical characterization of risk aversion connects behavioral properties of preference to the shape of the Bernoulli function.

Proposition 1.7.1 (vNM structural equivalence). *Let $\succsim$ be a preference relation represented by a vNM functional of the form*

$$V(F) = \int u(x) dF(x).$$

Then:

1. $\succsim$ is strictly monotone if and only if $u(\cdot)$ is strictly increasing.
2. $\succsim$ is risk-averse if and only if $u(\cdot)$ is concave.

Furthermore, if $u(\cdot)$ is strictly concave, the agent is strictly risk-averse for non-degenerate lotteries.

Proof. Proof of item (a): strict monotonicity and increasing utility.

For any certain amount z , the utility of the degenerate lottery δ_z is

$$V(\delta_z) = \int u(x) d\delta_z(x) = u(z).$$

Direction 1: strict monotonicity implies u is increasing.

Suppose $\succsim$ is strictly monotone. If $x > y$, then, by definition,

$$\delta_x \succ \delta_y.$$

Since V represents $\succsim$,

$$V(\delta_x) > V(\delta_y).$$

But $V(\delta_x) = u(x)$ and $V(\delta_y) = u(y)$. Hence,

$$u(x) > u(y).$$

Therefore, $u(\cdot)$ is strictly increasing.

Direction 2: u increasing implies strict monotonicity.

Suppose $u(\cdot)$ is strictly increasing. If $x > y$, then

$$u(x) > u(y).$$

Since $V(\delta_x) = u(x)$ and $V(\delta_y) = u(y)$,

$$V(\delta_x) > V(\delta_y).$$

Hence,

$$\delta_x \succ \delta_y.$$

Thus, $\succsim$ is strictly monotone.

Proof of item (b): risk aversion and concavity.

Direction 1: risk aversion implies concavity.

Suppose the agent is risk-averse. Take $x, y \in \mathbb{R}$ and $\alpha \in [0, 1]$. Consider the simple lottery F that pays x with probability α and y with probability $1 - \alpha$. The expected value of this lottery is

$$\bar{x} = \alpha x + (1 - \alpha)y.$$

By the definition of risk aversion,

$$\delta_{\bar{x}} \succsim F.$$

Therefore,

$$V(\delta_{\bar{x}}) \geq V(F).$$

Using the expected utility form,

$$u(\alpha x + (1 - \alpha)y) \geq \alpha u(x) + (1 - \alpha)u(y).$$

This is exactly the definition of concavity of $u(\cdot)$.

Direction 2: concavity implies risk aversion.

Suppose $u(\cdot)$ is concave. By Jensen's inequality,

$$u\left(\int x dF(x)\right) \geq \int u(x) dF(x).$$

Hence,

$$V(\delta_{E_F[x]}) = u(E_F[x]) \geq V(F).$$

Thus,

$$\delta_{E_F[x]} \succsim F,$$

which characterizes risk aversion.

If $u(\cdot)$ is strictly concave and F is non-degenerate, Jensen's inequality is strict, implying

$$\delta_{E_F[x]} \succ F.$$

□

Remark 1.7.2. This equivalence transforms behavioral properties into analytical properties of the utility function: monotonicity is associated with the positive slope of u , while risk aversion is associated with the concavity of u .

1.7.5 Expected Value, Expected Utility, Certainty Equivalent, and Probability Premium

To avoid terminological confusion, it is useful to formally separate the concepts that appear in the analysis of risk aversion.

The Fundamental Concepts

Definition 1.7.4 (Utility of the expected value). The utility of the expected value of lottery F received with certainty is

$$V(\delta_{E_F[x]}) = u(E_F[x]) = u\left(\int x dF(x)\right).$$

Definition 1.7.5 (Expected utility of the lottery). The expected utility of lottery F is

$$V(F) = \int u(x) dF(x) = E_F[u(x)].$$

Definition 1.7.6 (Certainty equivalent). The **certainty equivalent** of a lottery F for an agent with Bernoulli utility $u(\cdot)$ is the certain value $c(F, u)$ that satisfies

$$u(c(F, u)) = V(F).$$

Equivalently,

$$V(\delta_{c(F, u)}) = V(F).$$

Definition 1.7.7 (Risk premium). The **risk premium** is the difference between the expected value of the lottery and the certainty equivalent:

$$P(F, u) = E_F[x] - c(F, u).$$

Definition 1.7.8 (Probability premium). For a wealth level x and an increment $\epsilon > 0$, the **probability premium** $\pi(x, \epsilon, u)$ is the excess probability over $1/2$ that makes the agent indifferent between receiving x with certainty and facing the lottery that pays $x + \epsilon$ or $x - \epsilon$:

$$u(x) = \left(\frac{1}{2} + \pi(x, \epsilon, u)\right) u(x + \epsilon) + \left(\frac{1}{2} - \pi(x, \epsilon, u)\right) u(x - \epsilon).$$

Concept	Expression	Interpretation
Utility of expected value	$u(E_F[x]) = V(\delta_{E_F[x]})$	Utility of receiving the mean with certainty
Expected utility	$V(F) = \int u(x) dF(x)$	Average utility of outcomes weighted by probabilities
Certainty equivalent	$u(c(F, u)) = V(F)$	Certain value that yields the same utility as the lottery
Risk premium	$P(F, u) = E_F[x] - c(F, u)$	How much the agent gives up to avoid risk
Probability premium	$\pi(x, \epsilon, u)$	Excess probability required to accept the gamble

Table 1.3: Expected value, expected utility, certainty equivalent, risk premium, and probability premium.

The Ordering for Risk-Averse Agents

Proposition 1.7.2 (Ordering for risk-averse agents). *If $u(\cdot)$ is increasing and concave, then, for any lottery F ,*

$$u(E_F[x]) \geq V(F) = u(c(F, u)).$$

Since $u(\cdot)$ is increasing, it follows that

$$E_F[x] \geq c(F, u),$$

and therefore

$$P(F, u) \geq 0.$$

If $u(\cdot)$ is strictly concave and F is non-degenerate, all these inequalities are strict.

Proof. By the concavity of u and Jensen's inequality,

$$u(E_F[x]) \geq E_F[u(x)] = V(F).$$

By the definition of certainty equivalent,

$$u(c(F, u)) = V(F).$$

Hence,

$$u(E_F[x]) \geq u(c(F, u)).$$

Since u is increasing,

$$E_F[x] \geq c(F, u).$$

Therefore,

$$P(F, u) = E_F[x] - c(F, u) \geq 0.$$

If u is strictly concave and F is non-degenerate, Jensen implies strict inequality. $\square$

Remark 1.7.3 (Correct terminology). For a risk-averse agent, one should not say that the utility of the certainty equivalent is greater than the utility of the expected value. The correct statement is:

$$V(\delta_{E_F[x]}) \geq V(F) = V(\delta_{c(F, u)}).$$

Since u is increasing, this implies

$$E_F[x] \geq c(F, u).$$

The certainty equivalent is therefore less than or equal to the expected value. The difference is the risk premium.

Equivalent Characterizations of Risk Aversion

Proposition 1.7.3 (Fundamental equivalence). *For an expected utility maximizer with increasing and continuous Bernoulli utility $u(\cdot)$, the following properties are equivalent:*

1. **Risk aversion:** $\delta_{E_F[x]} \succsim F$ for all F .
2. **Concavity:** $u''(x) \leq 0$, when u is twice differentiable.
3. **Certainty equivalent:** $c(F, u) \leq \int x dF(x)$ for all F .

4. **Probability premium:** $\pi(x, \epsilon, u) \geq 0$ for all x and ϵ .

Sketch of the proof. (i) $\iff$ (ii): This is exactly the vNM structural equivalence proven above: by Jensen's inequality, u is concave if and only if $u(E_F[x]) \geq E_F[u(x)]$ for all F , that is, $\delta_{E_F[x]} \succsim F$.

(ii) $\iff$ (iii): From the definition of certainty equivalent, $u(c(F, u)) = V(F)$. If u is concave, $V(F) \leq u(E_F[x])$; since u is increasing, $c(F, u) \leq E_F[x]$. Conversely, if $c(F, u) \leq E_F[x]$ for all F , then $u(E_F[x]) \geq u(c(F, u)) = V(F)$, so the agent is risk-averse and, by equivalence (i) $\iff$ (ii), u is concave.

(ii) $\iff$ (iv): The second-order Taylor expansion of the indifference condition defining $\pi(x, \epsilon, u)$ shows that, for small ϵ , the sign of π is determined by the sign of $-u''(x)$. Hence, $\pi(x, \epsilon, u) \geq 0$ for all x and ϵ if and only if $u''(x) \leq 0$. $\square$

Corollary 1.7.1 (Jensen's inequality, strict version). *If $u(\cdot)$ is strictly concave and F is non-degenerate, then*

$$c(F, u) < E_F[x],$$

that is, the risk premium is strictly positive.

Proof. By the definition of certainty equivalent,

$$u(c(F, u)) = E_F[u(x)].$$

By the strict Jensen's inequality,

$$E_F[u(x)] < u(E_F[x]).$$

Hence,

$$u(c(F, u)) < u(E_F[x]).$$

Since u is strictly increasing,

$$c(F, u) < E_F[x].$$

$\square$

Concept	Mathematical condition	Economic meaning
Risk aversion	$\delta_{E_F[x]} \succsim F$ for all F	Prefers the certain expected value
Concavity	$u''(x) \leq 0$	Decreasing marginal utility
Certainty equivalent	$c(F, u) \leq E_F[x]$	Accepts less than the expected value
Probability premium	$\pi(x, \epsilon, u) \geq 0$	Requires a favorable gamble

Table 1.4: Equivalent characterizations of risk aversion.

1.7.6 Intensity of Risk Aversion: The Arrow–Pratt Coefficient

Concavity characterizes the presence of risk aversion, but the magnitude of u'' depends on the normalization of utility. To measure the intensity of this aversion in a way that is invariant to positive affine transformations, the Arrow–Pratt coefficient is used.

Definition and Invariance

Motivation

Example 1.7.1. Multiplying u by 2 doubles u'' , but does not alter preferences. Therefore, we need to normalize the curvature by the marginal utility.

Formal definition

Definition 1.7.9 (Coefficient of absolute risk aversion). Let $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ be twice continuously differentiable, with $u'(w) > 0$. The Arrow–Pratt coefficient of absolute risk aversion is

$$r_A(w) = -\frac{u''(w)}{u'(w)}.$$

Remark 1.7.4. If $u'(w) = 0$, the coefficient is not well-defined. If u is not twice differentiable, the local measure may not exist.

Remark 1.7.5. The coefficient $r_A(w)$ measures the local curvature of the Bernoulli utility. The higher $r_A(w)$ is, the more concave the function is locally, and the higher the risk premium the agent tends to demand to accept uncertainty. For small additive zero-mean risks with variance σ^2 , the approximation holds:

$$P \approx \frac{1}{2}\sigma^2 r_A(w).$$

Invariance under affine transformations

Proposition 1.7.4. If $v(w) = \alpha u(w) + \beta$, with $\alpha > 0$, then

$$r_A^v(w) = r_A^u(w).$$

Proof. Since

$$v'(w) = \alpha u'(w), \quad v''(w) = \alpha u''(w),$$

we have

$$r_A^v(w) = -\frac{v''(w)}{v'(w)} = -\frac{\alpha u''(w)}{\alpha u'(w)} = -\frac{u''(w)}{u'(w)} = r_A^u(w).$$

□

Relation to the Risk Premium

Proposition 1.7.5 (Pratt approximation). For small risks with zero mean and variance σ^2 , the risk premium P satisfies

$$P \approx \frac{1}{2}\sigma^2 r_A(w).$$

Proof. Let the risk be $\tilde{\epsilon}$, with $E[\tilde{\epsilon}] = 0$ and $\text{Var}(\tilde{\epsilon}) = \sigma^2$. The risk premium P is defined by

$$u(w - P) = E[u(w + \tilde{\epsilon})].$$

Expanding the left side around w ,

$$u(w - P) \approx u(w) - Pu'(w).$$

Expanding the right side,

$$E[u(w + \tilde{\epsilon})] \approx E \left[u(w) + \tilde{\epsilon}u'(w) + \frac{1}{2}\tilde{\epsilon}^2u''(w) \right].$$

Hence,

$$E[u(w + \tilde{\epsilon})] \approx u(w) + u'(w)E[\tilde{\epsilon}] + \frac{1}{2}u''(w)E[\tilde{\epsilon}^2].$$

Since $E[\tilde{\epsilon}] = 0$ and $E[\tilde{\epsilon}^2] = \sigma^2$,

$$E[u(w + \tilde{\epsilon})] \approx u(w) + \frac{1}{2}\sigma^2u''(w).$$

Equating the two expressions,

$$u(w) - Pu'(w) \approx u(w) + \frac{1}{2}\sigma^2u''(w).$$

Canceling $u(w)$,

$$-Pu'(w) \approx \frac{1}{2}\sigma^2u''(w).$$

Thus,

$$P \approx \frac{1}{2}\sigma^2 \left(-\frac{u''(w)}{u'(w)} \right) = \frac{1}{2}\sigma^2 r_A(w).$$

□

Mechanism	Mathematical element	Part
Preference invariance	Cancellation of α	Invariance proof
Intensity of aversion	$r_A(w)$	Definition
Role of variance	σ^2	Pratt approximation
Normalization	Division by $u'(w)$	Definition

Table 1.5: Connection between intuition and proof of the ARA coefficient.

Connection between intuition and proof

1.7.7 Comparing Risk Aversion across Individuals

Intuition and Equivalent Criteria

To state that one individual is more risk-averse than another, we need criteria invariant to affine transformations. The central idea is that the more risk-averse agent rejects more lotteries, has a lower certainty equivalent, and has a more concave Bernoulli utility.

Example 1.7.2. If two agents evaluate the same gamble, the more risk-averse one demands a higher risk premium to accept it.

Proposition 1.7.6 (Mas-Colell Proposition 6.C.2). *Let u_1 and u_2 be increasing and twice differentiable Bernoulli utility functions. The following are equivalent:*

1. $r_A(x, u_2) \geq r_A(x, u_1)$ for all x .
2. There exists an increasing and concave ψ such that $u_2(x) = \psi(u_1(x))$.
3. $c(F, u_2) \leq c(F, u_1)$ for all lotteries F .
4. $\pi(x, \epsilon, u_2) \geq \pi(x, \epsilon, u_1)$ for all x and ϵ .
5. If u_2 accepts a lottery instead of the certain expected value, then u_1 also accepts.

Remark 1.7.6. If the inequality of the coefficients holds only on part of the domain, the comparison is only local or partial.

Proofs of Two Equivalences

Proof A: concave transformation implies higher coefficient. Assume

$$u_2(x) = \psi(u_1(x)),$$

with $\psi' > 0$ and $\psi'' \leq 0$. Then

$$u_2'(x) = \psi'(u_1(x))u_1'(x),$$

and

$$u_2''(x) = \psi''(u_1(x))[u_1'(x)]^2 + \psi'(u_1(x))u_1''(x).$$

Hence,

$$r_A(x, u_2) = -\frac{u_2''(x)}{u_2'(x)}.$$

Substituting,

$$r_A(x, u_2) = -\frac{\psi''(u_1(x))[u_1'(x)]^2 + \psi'(u_1(x))u_1''(x)}{\psi'(u_1(x))u_1'(x)}.$$

Separating the terms,

$$r_A(x, u_2) = -\frac{\psi''(u_1(x))u_1'(x)}{\psi'(u_1(x))} - \frac{u_1''(x)}{u_1'(x)}.$$

Since

$$r_A(x, u_1) = -\frac{u_1''(x)}{u_1'(x)},$$

it follows that

$$r_A(x, u_2) = r_A(x, u_1) + \left[-\frac{\psi''(u_1(x))}{\psi'(u_1(x))}u_1'(x) \right].$$

Since $\psi'' \leq 0$, $\psi' > 0$, and $u_1' > 0$, the term in brackets is non-negative. Therefore,

$$r_A(x, u_2) \geq r_A(x, u_1).$$

□

Proof B: concave transformation implies lower certainty equivalent. Let c_1 be the certainty equivalent of u_1 and c_2 that of u_2 :

$$u_1(c_1) = \int u_1(x) dF(x),$$

$$u_2(c_2) = \int u_2(x) dF(x).$$

Since $u_2 = \psi(u_1)$,

$$\psi(u_1(c_2)) = \int \psi(u_1(x)) dF(x).$$

By Jensen,

$$\int \psi(u_1(x)) dF(x) \leq \psi\left(\int u_1(x) dF(x)\right).$$

But

$$\int u_1(x) dF(x) = u_1(c_1).$$

Hence,

$$\psi(u_1(c_2)) \leq \psi(u_1(c_1)).$$

Since ψ and u_1 are increasing,

$$c_2 \leq c_1.$$

□

Mechanism	Mathematical element	Part
More concave utility	$\psi'' \leq 0$	Proof A
Higher local aversion	$r_A(u_2) \geq r_A(u_1)$	Proof A
Lower certainty equivalent	$c_2 \leq c_1$	Proof B
Jensen	$\int \psi(u_1) \leq \psi(\int u_1)$	Proof B

Table 1.6: Connection between intuition and proof of the comparison of aversion.

Connection between intuition and proof

1.7.8 Decreasing Absolute Risk Aversion (DARA)

Intuition and Definition

DARA means that the agent becomes less risk-averse, in absolute terms, as their wealth increases. Thus, risky assets behave like normal goods: wealthier agents tend to accept more absolute risk.

Example 1.7.3. A fixed loss of \$1,000 is more painful for someone with little wealth than for someone very rich. Therefore, the wealthier agent tends to accept more gambles of the same absolute value.

Definition 1.7.10 (DARA). A Bernoulli utility u exhibits **decreasing absolute risk aversion**, DARA, if

$$\frac{d}{dx} r_A(x, u) \leq 0.$$

Remark 1.7.7. If $r'_A(x) > 0$, we have IARA, increasing absolute risk aversion. In this case, wealthier agents would be more averse to absolute risks, which is generally implausible in financial applications.

Main Implication

DARA implies lower aversion at higher wealth levels. Let $x_1 > x_2$ be two wealth levels. Define induced functions for an additive risk z :

$$u_1(z) = u(x_1 + z), \quad u_2(z) = u(x_2 + z).$$

Then

$$r_A(z, u_1) = r_A(x_1 + z, u),$$

$$r_A(z, u_2) = r_A(x_2 + z, u).$$

Since $x_1 > x_2$, we have $x_1 + z > x_2 + z$. If r_A is decreasing,

$$r_A(x_2 + z, u) \geq r_A(x_1 + z, u).$$

Hence,

$$r_A(z, u_2) \geq r_A(z, u_1).$$

By Proposition 6.C.2, u_2 is more risk-averse than u_1 . Therefore, the agent with lower wealth demands a higher risk premium and has a lower certainty equivalent for the same absolute risk. $\square$

Mechanism	Mathematical element	Part
Wealth as a buffer	$r_A(x_1 + z) \leq r_A(x_2 + z)$	Proof
Risky asset as a normal good	Higher wealth accepts more risk	Interpretation
Comparison of aversion	Proposition 6.C.2	Proof
Decreasing risk premium	$P(x)$ decreases with x	Consequence

Table 1.7: Connection between intuition and proof of DARA.

Connection between intuition and proof

1.7.9 Relative Risk Aversion (RRA)

Intuition and Definition

Relative risk aversion measures the agent's attitude towards risks proportional to wealth. If the agent keeps the fraction of wealth invested in risky assets constant, their relative aversion is constant.

Example 1.7.4. Losing 10% of wealth is more relevant for a poor agent than for a rich agent. RRA captures this sensitivity to proportional changes.

The property of **decreasing relative risk aversion (DRRA)** answers the question: if you double your wealth, do you become more or less “brave” to risk the same percentage slice of your net worth? Imagine you always bet 10% of your net worth: with \$1,000, you bet \$100; with \$1,000,000, you bet \$100,000. Under DRRA, as the agent gets richer, those same 10% seem proportionally “less painful”, and they demand less compensation to accept the same percentage bet. It is the **scaling of courage**.

Example 1.7.5 (Analogies: road and armor). Think of wealth as the width of a road. If you are poor, the road is narrow and any 10% deviation in direction (the proportional risk) throws you onto the shoulder. If you are rich under DRRA, the road seems proportionally wider and more stable, allowing you to accept the same percentage deviation with much more peace of mind.

Equivalently, think of wealth as the armor of a tank: a projectile that pierces 10% of the armor of a small tank causes much more panic than a projectile that pierces the same 10% of a massive tank, because the remaining structure in the second case is much more robust.

Mechanisms The central mechanism is the **elasticity of marginal utility**. Under DRRA, marginal utility falls in a way that makes the individual “proportionally bolder” at high wealth levels: when we “stretch” the wealth scale, the utility function becomes locally less curved (less concave), decreasing the resistance to risks that scale with net worth. The coefficient $r_R(x)$ measures exactly this proportional sensitivity; if it decreases, the individual becomes proportionally bolder.

Implications This property is the cornerstone of **portfolio theory**. If an individual exhibits DRRA, they will not only invest more money in stocks when they get rich, but they will allocate a **larger percentage** of their total portfolio to risky assets. This explains why the concentration of wealth in an economy tends to flow towards higher-return and higher relative-risk investments. If RRA were increasing, we would see billionaires investing proportionally more in fixed income than the middle class, which contradicts the empirical evidence on portfolio allocation.

Formal definitions Let $u : \mathbb{R}_{++} \rightarrow \mathbb{R}$ be a C^2 Bernoulli utility function, with $u'(x) > 0$ and $u''(x) < 0$.

Definition 1.7.11 (Coefficient of relative risk aversion). The **Arrow–Pratt coefficient of relative risk aversion** at wealth level x is

$$r_R(x, u) = -x \frac{u''(x)}{u'(x)} = x r_A(x, u).$$

Definition 1.7.12 (Proportional certainty equivalent). For a multiplicative risk $t > 0$ with distribution $F(t)$, the **proportional certainty equivalent** $\bar{c}_x$ is the value such that

$$u(\bar{c}_x) = \int u(tx) dF(t).$$

Variables and assumptions:

- x is the base wealth ($x > 0$); t is the scale multiplier (e.g., an asset return).
- **Assumption (i) — DRRA:** $\frac{dr_R(x, u)}{dx} \leq 0$ for all x .
- **Assumption (iii):** $\frac{d}{dx} \left(\frac{x}{\bar{c}_x} \right) \leq 0$, that is, $x/\bar{c}_x$ decreases with x .

Remark 1.7.8 (What formally “breaks”). If $x \rightarrow 0$, the RRA measure may collapse or become undefined due to the multiplicative term x . If u is not C^2 , we cannot measure the elasticity of marginal utility via derivatives.

Main Property

Proposition 1.7.7 (Proposition 6.C.4, equivalences). *Let $u(\cdot)$ be as above. The following properties are equivalent:*

- (i) $u(\cdot)$ exhibits decreasing relative risk aversion: $r_R(x, u)$ is decreasing in x .
- (ii) For $x_2 < x_1$, the function $u_2(t) = u(x_2 t)$ is a concave transformation of $u_1(t) = u(x_1 t)$.
- (iii) The ratio $x/\bar{c}_x$ is decreasing in x .

The pillar of the comparisons below is **Pratt’s Theorem (Proposition 6.C.2)**: $\tilde{u}_2$ is a concave transformation of $\tilde{u}_1$ if and only if $r_A(t, \tilde{u}_2) \geq r_A(t, \tilde{u}_1)$ for all t ; in this case, $\tilde{u}_2$ is more risk-averse and has a lower certainty equivalent for the same lottery.

Proof of (i) $\Leftrightarrow$ (ii): identifying the coefficients. Step 1: scaled functions and their absolute aversion coefficients. Define $\tilde{u}_i(t) = u(tx_i)$, with $x_2 < x_1$. By the chain rule,

$$\tilde{u}'_i(t) = x_i u'(tx_i), \quad \tilde{u}''_i(t) = x_i^2 u''(tx_i).$$

The absolute aversion coefficient of $\tilde{u}_i$ at point t is

$$r_A(t, \tilde{u}_i) = -\frac{\tilde{u}''_i(t)}{\tilde{u}'_i(t)} = -\frac{u''(tx_i) x_i^2}{u'(tx_i) x_i} = -x_i \frac{u''(tx_i)}{u'(tx_i)}.$$

Step 2: relating $r_A(t, \tilde{u}_i)$ to r_R of the original function. Multiplying and dividing by t ,

$$r_A(t, \tilde{u}_i) = \frac{1}{t} \left[-(tx_i) \frac{u''(tx_i)}{u'(tx_i)} \right] = \frac{1}{t} r_R(tx_i, u),$$

recognizing the definition of r_R in the bracketed term, with wealth $w = tx_i$.

Step 3: applying Pratt’s Theorem (both directions). By Pratt’s Theorem, $\tilde{u}_2$ is a concave transformation of $\tilde{u}_1$ if and only if $r_A(t, \tilde{u}_2) \geq r_A(t, \tilde{u}_1)$ for all t . Substituting Step 2,

$$\frac{1}{t} r_R(tx_2, u) \geq \frac{1}{t} r_R(tx_1, u).$$

Since $t > 0$, we can simplify:

$$r_R(tx_2, u) \geq r_R(tx_1, u).$$

Step 4: connecting to the DRRA property. The above condition must hold whenever $x_2 < x_1$, for any $t > 0$. Defining $w_2 = tx_2$ and $w_1 = tx_1$, we have $x_2 < x_1 \iff w_2 < w_1$. The condition becomes

$$w_2 < w_1 \implies r_R(w_2, u) \geq r_R(w_1, u),$$

which is precisely the definition of a non-increasing function. Hence, u exhibits DRRA if and only if the scaled transformations follow the proposed concavity order. $\square$

Proof of (i) $\implies$ (iii), via (ii). **Step 1.** Consider $x_2 < x_1$. By property (ii), already established as equivalent to (i), $u_2(t) = u(x_2 t)$ is a **concave transformation** of $u_1(t) = u(x_1 t)$.

Step 2. Define $\lambda_x = \bar{c}_x/x$. This term represents the **certain scale factor** that makes the agent indifferent to the multiplicative lottery t .

Step 3. Rewrite the definition of $\bar{c}_x$ using λ_x :

$$u(\lambda_x x) = \int u(tx) dF(t).$$

Step 4. With the local notation $u_x(t) = u(xt)$, the equation becomes

$$u_x(\lambda_x) = \int u_x(t) dF(t).$$

Step 5. Note that λ_x is the **certainty equivalent of lottery** $F(t)$ evaluated by the function u_x .

Step 6. By Pratt's Theorem, if u_2 is a concave transformation of u_1 , then u_2 is more risk-averse and will have a **lower** certainty equivalent for the same lottery $F(t)$.

Step 7. Since u_2 (based on x_2) is the concave transformation,

$$\lambda_{x_2} \leq \lambda_{x_1}.$$

Step 8. Substituting $\lambda_x = \bar{c}_x/x$,

$$\frac{\bar{c}_{x_2}}{x_2} \leq \frac{\bar{c}_{x_1}}{x_1}.$$

Step 9. Inverting the terms (which reverses the inequality, since they are positive),

$$\frac{x_2}{\bar{c}_{x_2}} \geq \frac{x_1}{\bar{c}_{x_1}}.$$

Step 10. Since $x_2 < x_1$ implies a higher value of the ratio at the lower point, we conclude that $x/\bar{c}_x$ is decreasing in x (Assumption iii). $\square$

Proof of (iii) $\implies$ (i): Taylor for small risks. **Step 1.** Consider a small proportional risk $t = 1 + \epsilon$, with $E[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$.

Step 2. Second-order Taylor expansion of $u(x(1 + \epsilon))$ around x :

$$E[u(x + x\epsilon)] \approx u(x) + u'(x)E[x\epsilon] + \frac{1}{2}u''(x)E[x^2\epsilon^2] = u(x) + \frac{1}{2}x^2\sigma^2u''(x).$$

Step 3. First-order Taylor expansion of $u(\bar{c}_x)$ around x :

$$u(\bar{c}_x) \approx u(x) + u'(x)(\bar{c}_x - x).$$

Step 4. Equating the two expressions (indifference condition),

$$u'(x)(\bar{c}_x - x) \approx \frac{1}{2}x^2\sigma^2u''(x).$$

Step 5. Dividing both sides by $xu'(x)$,

$$\frac{\bar{c}_x - x}{x} \approx \frac{1}{2}\sigma^2 \left[\frac{xu''(x)}{u'(x)} \right] = -\frac{1}{2}\sigma^2 r_R(x, u).$$

Step 6. Rearranging,

$$\frac{\bar{c}_x}{x} \approx 1 - \frac{1}{2}\sigma^2 r_R(x, u).$$

Step 7. Condition (iii) says that $x/\bar{c}_x$ decreases, which implies that its inverse $\bar{c}_x/x$ increases in x .

Step 8. For $1 - \frac{1}{2}\sigma^2 r_R(x)$ to increase with x , the subtracted term $r_R(x)$ must necessarily decrease.

Step 9. Hence, $r_R(x, u)$ is decreasing in x (Assumption i). $\square$

Remark 1.7.9. The Taylor direction makes it explicit that the proportional “discount” of the certainty equivalent is governed by r_R : $\bar{c}_x/x \approx 1 - \frac{1}{2}\sigma^2 r_R(x)$. Therefore, the monotonicity of the ratio $x/\bar{c}_x$ reflects exactly the monotonicity of r_R , and decreasing relative aversion guarantees that the agent’s **proportional courage** increases with their capital base.

Mechanism (intuition)	Mathematical element	Step
Risk as proportion of wealth	Argument tx_i in function u ; $u_x(t) = u(tx)$	1
RRA as scaled ARA	$r_R = x r_A$; $r_A(t, \tilde{u}_i) = \frac{1}{t} r_R(tx_i, u)$	2
Rich proportionally bolder	$\tilde{u}_2$ more averse than $\tilde{u}_1 \iff r_A(\tilde{u}_2) \geq r_A(\tilde{u}_1)$	3
Relative aversion falls with wealth	$w_2 < w_1 \implies r_R(w_2) \geq r_R(w_1)$	4

Table 1.8: Connection between intuition and proof of equivalence (i) $\Leftrightarrow$ (ii).

Connection between intuition and proof: (i) $\Leftrightarrow$ (ii)

Mechanism (intuition)	Mathematical element	Step
Proportional daring	Ratio $\lambda_x = \bar{c}_x/x$	2
Rich accepts percentage risk better	$\lambda_{x_1} \geq \lambda_{x_2}$ for $x_1 > x_2$	7
Wealth as armor	u_2 concave transformation of u_1 (local function)	1
Elasticity of satisfaction	$\bar{c}_x/x \approx 1 - \frac{1}{2}\sigma^2 r_R(x, u)$	5–8 (converse)
Increasing proportional courage	$x/\bar{c}_x$ decreasing in x	9–10

Table 1.9: Connection between intuition and proof of equivalence (i) $\Leftrightarrow$ (iii).

Connection between intuition and proof: (i) $\Leftrightarrow$ (iii)

1.7.10 Synthesis: Connection between Intuition and Proof

Economic mechanism	Mathematical element	Part
More wealth is better	$x > y \iff \delta_x \succ \delta_y$	Definition of monotonicity
Preference over certain values	$V(\delta_x) = u(x)$	Proof of item (a)
Positive marginal utility	$u'(x) > 0$ or u increasing	Equivalence of item (a)
Risk aversion	$\delta_{E_F[x]} \succsim F$	Formal definition
Concavity	$u(\alpha x + (1-\alpha)y) \geq \alpha u(x) + (1-\alpha)u(y)$	Proof of item (b)
Jensen	$u(E_F[x]) \geq E_F[u(x)]$	Concavity implies aversion
Lower certainty equivalent	$c(F, u) \leq E_F[x]$	Jensen + monotonicity
Non-negative risk premium	$P(F, u) = E_F[x] - c(F, u) \geq 0$	Consequence of aversion
Non-negative probability premium	$\pi(x, \epsilon, u) \geq 0$	Fundamental equivalence
Local intensity	$r_A(w) = -u''(w)/u'(w)$	Arrow–Pratt

Table 1.10: Connection between economic intuition and mathematical structure of risk aversion.

1.8 Stochastic Dominance: Comparing Lotteries

While the Arrow–Pratt coefficients (ARA/RRA) describe the behavior of a *specific* agent, stochastic dominance allows comparing lotteries in a *universal* way: first-order (FOSD) identifies the lottery preferred by any individual who prefers more wealth to less; second-order (SOSD), by any individual who, in addition, is risk-averse.

1.8.1 First-Order Dominance (FOSD)

Economic Intuition

The essence FOSD solves the problem of **universal desirability**: how to assert that lottery F is better than G without knowing the agent’s exact preferences? The only assumption is **monotonicity**: the agent prefers “more to less”.

Example 1.8.1 (Bags of coins). Imagine two bags of coins. In bag F , the probability of finding at least \$10 is always greater (or equal) than in bag G , for any value you choose. If you like money, you will choose bag F , no matter how conservative or bold you are.

Example 1.8.2 (Travel time). Think of $F(x)$ and $G(x)$ as the probability of you arriving “late”, where x is the travel time. If, for any time x , the chance of spending less than that time is always higher in F than in G , then F dominates G : anyone who values their time will prefer route F .

Mechanisms Agents seek to maximize expected utility, and the dominant distribution places **more weight on the upper tails**: for each possible wealth level x , the probability of obtaining an outcome *worse* than x is lower in the dominant lottery. In short, F offers “less risk of being poor” and “more chance of being rich” for any defined threshold.

Implications FOSD is the **unanimity criterion**: if a new investment or public policy stochastically dominates the previous situation, there is no need for debates about the population’s risk aversion — everyone will be (at least weakly) better off. A classic example is “**good news**”: a signal is good news if the posterior belief dominates the prior belief by FOSD.

Remark 1.8.1 (The “defensive” view of dominance). Dominating by first order is **not** having a higher probability of a specific event occurring (in continuous distributions, the probability of an exact point is zero), but rather having a **lower probability of something “worse” happening**:

- “something worse” $\leftrightarrow$ the value x in the CDF $F(x) = \Pr\{z \leq x\}$;
- “lower probability” $\leftrightarrow$ the inequality $F(x) \leq G(x)$;
- “being more prone” $\leftrightarrow$ unanimous preference for all u with $u' \geq 0$.

The dominant distribution “pushes the probability mass to the right”, reducing the chance of failure for any level of demand. That is why the defensive view (looking at what is worse) builds the offensive preference (choosing the lottery with more value): dominating by first order is guaranteeing that, for any level of “failure” you define, the dominant lottery will always have a lower probability of failing to surpass it.

Formal Definition

Let $F(x)$ and $G(x)$ be cumulative distribution functions (CDFs) defined on a common support $[a, b] \subset \mathbb{R}$.

Definition 1.8.1 (FOSD, behavioral definition). We say that F dominates G by first order, denoted $F \succeq_1 G$, if, for every non-decreasing Bernoulli utility function ($u'(x) \geq 0$),

$$E_F[u(x)] \geq E_G[u(x)].$$

Definition 1.8.2 (FOSD, CDF condition). $F \succeq_1 G$ if and only if

$$F(x) \leq G(x) \quad \text{for all } x \in \mathbb{R}.$$

Remark 1.8.2 (What formally “breaks”). If we remove the assumption of u being increasing, an agent with “disutility for wealth” could prefer G , invalidating the dominance. Furthermore, if the CDFs cross at any point, FOSD cannot be applied, requiring second-order criteria (SOSD) that consider risk aversion.

Proposition 6.D.1 and Proof

Proposition 1.8.1 (Proposition 6.D.1). *The distribution $F(\cdot)$ first-order stochastically dominates $G(\cdot)$ (that is, $E_F[u] \geq E_G[u]$ for all non-decreasing u) if and only if $F(x) \leq G(x)$ for all x .*

Direction ($\Leftarrow$): CDF condition implies preference. Define the difference between expected utilities:

$$\Delta = \int_a^b u(x) dF(x) - \int_a^b u(x) dG(x) = \int_a^b u(x) d[F(x) - G(x)].$$

Integration by parts with $U = u(x)$, $dV = d[F(x) - G(x)]$, $V = F(x) - G(x)$:

$$\Delta = [u(x)(F(x) - G(x))]_a^b - \int_a^b u'(x)[F(x) - G(x)] dx.$$

The boundary term is zero: at b , $F(b) = G(b) = 1$; at a , $F(a) = G(a) = 0$. Hence,

$$\Delta = - \int_a^b u'(x)[F(x) - G(x)] dx = \int_a^b u'(x)[G(x) - F(x)] dx.$$

Since $u'(x) \geq 0$ (monotonicity) and, by hypothesis, $G(x) - F(x) \geq 0$, the integrand is non-negative. Therefore $\Delta \geq 0$, that is, $E_F[u] \geq E_G[u]$. $\square$

Direction ($\Rightarrow$): universal preference implies CDF condition. By contrapositive: suppose there exists x_0 such that $F(x_0) > G(x_0)$. Consider the “step” function

$$u(x) = \begin{cases} 0, & x \leq x_0, \\ 1, & x > x_0, \end{cases}$$

which is non-decreasing. Then

$$E_F[u] = \int_{x_0}^b dF(x) = 1 - F(x_0), \quad E_G[u] = 1 - G(x_0).$$

If $F(x_0) > G(x_0)$, then $E_F[u] < E_G[u]$, contradicting dominance for all increasing functions. Hence, $F(x) \leq G(x)$ for all x . $\square$

Remark 1.8.3. If continuity or differentiability of u is required, the step function can be approximated by a sequence of smooth increasing functions; the conclusion is preserved in the limit. The construction reveals that the “advantage” of F over G is accumulated at each point x where the probability of being below x is lower in F , weighted by the marginal importance u' that the agent gives to this increment of wealth.

Mechanism (intuition)	Mathematical element	Part
Preference for more wealth	Assumption $u'(x) \geq 0$	($\Leftarrow$)
Lower chance of bad outcomes	Condition $F(x) \leq G(x)$	($\Leftarrow$)
Mass shifted to the right	$\Delta = \int u'(x)[G(x) - F(x)]dx$	($\Leftarrow$)
Unanimity among agents	Validity for all increasing u	($\Rightarrow$)
“Fear” of failing at threshold x_0	Test with the step function at x_0	($\Rightarrow$)

Table 1.11: Connection between intuition and proof of FOSD (Prop. 6.D.1).

Connection between intuition and proof

1.8.2 Second-Order Dominance (SOSD)

Economic Intuition

The essence: “pure risk” When two lotteries have **exactly the same mean return**, the choice is not about “how much I earn on average”, but about “how uncertain the outcome is”. SOSD formalizes why any risk-averse agent will prefer the stable lottery to the volatile one, even with identical means: it identifies which lottery is “less risky” in an absolute sense.

Example 1.8.3 (Pile of sand). Think of the probability distribution as a “pile of sand”. In FOSD, we push the whole pile to the right (more wealth). In SOSD, we take the sand from the extremes (the extreme risks) and “squeeze” it to the center (the mean). Anyone who dislikes surprises will prefer the more concentrated pile.

Mechanisms: mean-preserving spread A lottery dominates another by second order if the **accumulated area under its CDF** is consistently smaller: the dominant one “accumulates less probability of bad outcomes” in a weighted way. The dominated lottery is a **mean-preserving spread** of the dominant one: it is the dominant one plus a zero-mean noise (random shock). Since the shock has zero mean, the final mean does not change, but the distribution becomes more dispersed — and anyone who hates risk prefers the original lottery.

Implications SOSD is the foundation of **Modern Portfolio Theory** and explains why diversification is a unanimous rational behavior: it reduces variance without necessarily reducing the mean. In public policy, it justifies transfer programs that reduce inequality (dispersion) even if GDP per capita (mean) remains constant. Proposition 6.D.2 is the basis of the **Rothschild–Stiglitz increase in risk**: being “more risky” is exactly the same as being “rejected by all risk-averse agents”.

FOSD versus SOSD: Single and Double Integrals

The structure of the integrals FOSD directly compares $F(x) = \int_a^x f(t) dt$ (single integral of the density). SOSD compares the **accumulated area under the CDF**, $H(x) = \int_a^x F(t) dt$ (double integral of the density). The double integral “smooths” the comparison: it does not just look at the probability at a point x , but at the accumulated weight of the risk up to that point.

The case of crossing curves This is the great advantage of SOSD. If F crosses G , there will be regions where $F(x) > G(x)$ (which would invalidate FOSD); still, F can dominate by second order if the regions where it is “worse” are **compensated** by earlier regions where it was “better”, so that the net sum of the areas remains less than or equal to that of G . This is possible because SOSD requires risk aversion ($u'' \leq 0$): the agent accepts a slightly higher probability of a bad outcome at one point, as long as the total distribution is less dispersed or has a higher mean that compensates for the volatility.

Example 1.8.4 (Mean-preserving spread and crossings). If two distributions have the same mean, their CDFs **necessarily cross** (if one were always below the other, the means would be different). Hence, FOSD is impossible here; SOSD identifies the distribution with “thinner tails” (less dispersion) as preferable for anyone who hates uncertainty.

Criterion	Operation on density f	Sensitivity to crossings	Preference class
FOSD	$F(x) = \int f(t) dt$	Does not tolerate crossings	$u' \geq 0$ (anyone who likes money)
SOSD	$H(x) = \int F(t) dt$	Tolerates crossings, as long as accumulated area is smaller	$u' \geq 0$ and $u'' \leq 0$ (risk-averse)

Table 1.12: Comparison between FOSD and SOSD.

Formal Definition

Definition 1.8.3 (SOSD, behavioral definition). F dominates G by second order, denoted $F \succeq_2 G$, if, for every increasing ($u' \geq 0$) and concave ($u'' \leq 0$) utility function,

$$E_F[u(x)] \geq E_G[u(x)].$$

Definition 1.8.4 (SOSD, integral condition). $F \succeq_2 G$ if and only if, for all $x \in [a, b]$,

$$\int_a^x F(t) dt \leq \int_a^x G(t) dt.$$

Remark 1.8.4 (What formally “breaks”). If we remove risk aversion ($u'' \leq 0$), a risk-seeking agent (convex utility) will prefer the distribution with higher dispersion, invalidating SOSD.

Proposition 6.D.2 and Proofs

Proposition 1.8.2 (Proposition 6.D.2). Consider two distributions $F(\cdot)$ and $G(\cdot)$ with the same mean, $\int_a^b x dF(x) = \int_a^b x dG(x)$. The following statements are equivalent:

- (i) $F(\cdot)$ second-order stochastically dominates $G(\cdot)$;
- (ii) $G(\cdot)$ is a mean-preserving spread of $F(\cdot)$;
- (iii) $\int_a^x F(t) dt \leq \int_a^x G(t) dt$ for all $x \in [a, b]$.

In condition (ii), G is obtained from F by adding a noise ϵ with $E[\epsilon | x] = 0$, that is, $y = x + \epsilon$.

(iii) $\Rightarrow$ (i): integral condition implies preference. Define $\Delta = \int_a^b u(x) dF(x) - \int_a^b u(x) dG(x) = \int_a^b u(x) d[F(x) - G(x)]$.

First integration by parts: with $U = u(x)$, $V = F(x) - G(x)$,

$$\Delta = [u(x)(F(x) - G(x))]_a^b - \int_a^b u'(x)[F(x) - G(x)] dx.$$

The boundary term is zero, since $F(a) = G(a) = 0$ and $F(b) = G(b) = 1$. Hence,

$$\Delta = \int_a^b u'(x)[G(x) - F(x)] dx.$$

Second integration by parts: define $H(x) = \int_a^x [G(t) - F(t)] dt$, so that $H'(x) = G(x) - F(x)$, $H(a) = 0$, and

$$\Delta = [u'(x)H(x)]_a^b - \int_a^b u''(x)H(x) dx.$$

At the upper limit, by integration by parts of the mean,

$$H(b) = \int_a^b [G(t) - F(t)] dt = E_F[x] - E_G[x] = 0,$$

by the equal means. Therefore,

$$\Delta = \int_a^b [-u''(x)] H(x) dx.$$

By hypothesis (iii), $H(x) \geq 0$ for all x ; by risk aversion, $-u''(x) \geq 0$. The product of non-negative terms is non-negative, hence $\Delta \geq 0$, that is, $E_F[u] \geq E_G[u]$. $\square$

Remark 1.8.5 (Not necessarily equal means). Without the equal means hypothesis, the boundary term $u'(b)H(b)$ survives, and the sufficient condition becomes $H(x) \geq 0$ for all x **and** $E_F[x] \geq E_G[x]$: both terms of $\Delta = u'(b)H(b) + \int_a^b [-u''(x)]H(x) dx$ are then non-negative.

(ii) $\Rightarrow$ (i): mean-preserving spread implies SOSD. Let $y = x + \epsilon$, with $E[\epsilon | x] = 0$. For any concave $u(\cdot)$, by the **conditional Jensen's inequality**,

$$E[u(x + \epsilon) | x] \leq u(E[x + \epsilon | x]) = u(x).$$

Taking the total expectation on both sides (law of iterated expectations),

$$E_G[u] = E[u(y)] = E[E[u(x + \epsilon) | x]] \leq E[u(x)] = E_F[u].$$

Hence, the lottery with noise always gives less expected utility than the original, proving the dominance. $\square$

(i) $\Rightarrow$ (iii): universal preference implies integral condition. For any $y \in [a, b]$, the function $u(x) = \min(x, y)$ is increasing and concave. Using the identity (obtained by integration by parts)

$$E_F[\min(x, y)] = y - \int_a^y F(t) dt,$$

and analogously for G , hypothesis (i) implies

$$y - \int_a^y F(t) dt \geq y - \int_a^y G(t) dt \implies \int_a^y F(t) dt \leq \int_a^y G(t) dt.$$

Since y is arbitrary, (iii) holds. $\square$

Remark 1.8.6 (Closing the equivalence). The directions (iii) $\Rightarrow$ (i), (i) $\Rightarrow$ (iii), and (ii) $\Rightarrow$ (i) establish (i) $\iff$ (iii) and show that every mean-preserving spread is riskier. The remaining direction, (i) $\Rightarrow$ (ii) — any G satisfying the integral condition can be constructed as F plus zero-mean noise — follows from the Rothschild–Stiglitz “mass movement” argument, completing the equivalence of the three statements.

Mechanism (intuition)	Mathematical element	Part
Risk aversion	Assumption $u''(x) \leq 0$	(iii) $\Rightarrow$ (i)
Punishment of dispersion	$H(x) = \int_a^x [G - F] dt$	(iii) $\Rightarrow$ (i)
Reduction of the risk area	$H(x) \geq 0$ for all x	(iii) $\Rightarrow$ (i)
Zero-mean noise	$E[\epsilon x] = 0$; conditional Jensen	(ii) $\Rightarrow$ (i)
Unanimous preference of the risk-averse	Validity for all concave u	(i) $\Rightarrow$ (iii)
Welfare difference	$\Delta = \int_a^b [-u''(x)] H(x) dx$	(iii) $\Rightarrow$ (i)

Table 1.13: Connection between intuition and proof of SOSD (Prop. 6.D.2).

Connection between intuition and proof

Remark 1.8.7 (Unification of the three views). Proposition 6.D.2 unifies the **behavioral** view (what people choose: unanimity of the risk-averse), the **statistical** view (how the data spread: G is F plus noise), and the **analytical** view (how the CDFs behave: smaller accumulated area). The advantage of F over G is accumulated by the integral of the difference of the CDFs, weighted by the “pain” of concavity (u''): if G has more “accumulated area” under its CDF, it is intrinsically riskier.

1.9 MGW Exercises

6.B.1: Show that if the preferences $\succsim$ over $\mathcal{L}$ satisfy the independence axiom, then for all $\alpha \in (0, 1)$ and $L, L', L'' \in \mathcal{L}$ we have

$$L \succ L' \quad \text{if and only if} \quad \alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L''$$

and

$$L \sim L' \quad \text{if and only if} \quad \alpha L + (1 - \alpha)L'' \sim \alpha L' + (1 - \alpha)L''.$$

Show also that if $L \succ L'$ and $L'' \succ L'''$, then

$$\alpha L + (1 - \alpha)L'' \succ \alpha L' + (1 - \alpha)L'''.$$

6.B.2: Show that if the preference relation $\succsim$ on $\mathcal{L}$ is represented by a utility function $U(\cdot)$ that has the expected utility form, then $\succsim$ satisfies the independence axiom.

6.B.3^B Show that if the set of outcomes C is finite and the rational preference relation $\succsim$ on the set of lotteries $\mathcal{L}$ satisfies the independence axiom, then there are best and worst lotteries in $\mathcal{L}$. That is, we can find lotteries $\bar{L}$ and $\underline{L}$ such that

$$\bar{L} \succsim L \succsim \underline{L}$$

for all $L \in \mathcal{L}$.

6.B.4^B The purpose of this exercise is to illustrate how expected utility theory allows us to make consistent decisions when dealing with extremely small probabilities by considering relatively large ones. Suppose that a safety agency is thinking of establishing a criterion under which an area prone to flooding should be evacuated. The probability of flooding is 1%. There are four possible outcomes:

- (A) No evacuation is necessary, and none is performed.
- (B) An evacuation is performed that is unnecessary.
- (C) An evacuation is performed that is necessary.
- (D) No evacuation is performed, and a flood causes a disaster.

Suppose that the agency is indifferent between the sure outcome **B** and the lottery of **A** with probability p and **D** with probability $1 - p$, and between the sure outcome **C** and the lottery of **B** with probability q and **D** with probability $1 - q$. Suppose also that it prefers **A** to **D** and that $p \in (0, 1)$ and $q \in (0, 1)$. Assume that the conditions of the expected utility theorem are satisfied.

- (a) Construct a utility function of the expected utility form for the agency.
- (b) Consider two different policy criteria:

Criterion 1: This criterion will result in an evacuation in 90% of the cases in which flooding will occur and an unnecessary evacuation in 10% of the cases in which no flooding occurs.

Criterion 2: This criterion is more conservative. It results in an evacuation in 95% of the cases in which flooding will occur and an unnecessary evacuation in 5% of the cases in which no flooding occurs.

First, derive the probability distributions over the four outcomes under these two criteria. Then, by using the utility function in (a), decide which criterion the agency would prefer.

6.B.5^B The purpose of this exercise is to show that the Allais paradox is compatible with a weaker version of the independence axiom. We consider the following axiom, known as the *betweenness axiom* [see Dekel (1986)]:

For all L, L' and $\lambda \in (0, 1)$, if $L \sim L'$, then

$$\lambda L + (1 - \lambda)L' \sim L.$$

Suppose that there are three possible outcomes.

- (a) Show that a preference relation on lotteries satisfying the independence axiom also satisfies the betweenness axiom.
- (b) Using a simplex representation for lotteries similar to the one in Figure 6.B.1(b), show that if the continuity and betweenness axioms are satisfied, then the indifference curves of a preference relation on lotteries are straight lines. Conversely, show that if the indifference curves are straight lines, then the betweenness axiom is satisfied. Do these straight lines need to be parallel?
- (c) Using (b), show that the betweenness axiom is weaker, i.e., less restrictive, than the independence axiom.
- (d) Using Figure 6.B.7, show that the choices of the Allais paradox are compatible with the betweenness axiom by exhibiting an indifference map satisfying the betweenness axiom that yields the choices of the Allais paradox.

6.B.6^B Prove that the induced utility function $U(\cdot)$ defined in the last paragraph of Section 6.B is convex. Give an example of a set of outcomes and a Bernoulli utility function for which the induced utility function is not linear.

6.B.7^A Consider the following two lotteries:

$$L : \begin{cases} 200 \text{ dollars with probability } 0.7, \\ 0 \text{ dollars with probability } 0.3, \end{cases}$$

$$L' : \begin{cases} 1200 \text{ dollars with probability } 0.1, \\ 0 \text{ dollars with probability } 0.9. \end{cases}$$

Let x_L and $x_{L'}$ be the sure amounts of money that an individual finds indifferent to L and L' . Show that if his preferences are transitive and monotone, the individual must prefer L to L' if and only if

$$x_L > x_{L'}.$$

[*Note:* In actual experiments, however, a preference reversal is often observed in which L is preferred to L' but $x_L < x_{L'}$. See Grether and Plott (1979) for details.]

6.C.1^B Consider the insurance problem studied in Example 6.C.1. Show that if insurance is not actuarially fair (so that $q > \pi$), then the individual will not insure completely.

6.C.2^B

- (a) Show that if an individual has a Bernoulli utility function $u(\cdot)$ with the quadratic form

$$u(x) = \beta x^2 + \gamma x,$$

then his utility from a distribution is determined by the mean and variance of the distribution and, in fact, by these moments alone. [*Note:* The number β should be taken to be negative in order to get the concavity of $u(\cdot)$. Since $u(\cdot)$ is then decreasing at $x > -\gamma/2\beta$, $u(\cdot)$ is useful only when the distribution cannot take values larger than $-\gamma/2\beta$.]

- (b) Suppose that a utility function $U(\cdot)$ over distributions is given by

$$U(F) = (\text{mean of } F) - r (\text{variance of } F),$$

where $r > 0$. Argue that unless the set of possible distributions is further restricted (see, e.g., Exercise 6.C.19), $U(\cdot)$ cannot be compatible with any Bernoulli utility function. Give an example of two lotteries L and L' over the same two amounts of money, say x' and $x'' > x'$, such that L gives a higher probability to x'' than does L' and yet according to $U(\cdot)$, L' is preferred to L .

6.C.3^B Prove that the four conditions of Proposition 6.C.1 are equivalent. [*Hint:* The equivalence of (i), (ii), and (iii) has already been shown. As for (iv), prove that (i) implies (iv) and that (iv) implies $u(\frac{1}{2}x + \frac{1}{2}y) \geq \frac{1}{2}u(x) + \frac{1}{2}u(y)$ for any x and y , which is, in fact, sufficient for (ii).]

6.C.4^B Suppose that there are N risky assets whose returns z_n ($n = 1, \dots, N$) per dollar invested are jointly distributed according to the distribution function $F(z_1, \dots, z_N)$.

Assume also that all the returns are nonnegative with probability one. Consider an individual who has a continuous, increasing, and concave Bernoulli utility function $u(\cdot)$ over $\mathbb{R}_+$. Define the utility function $U(\cdot)$ of this investor over $\mathbb{R}_+^N$, the set of all nonnegative portfolios, by

$$U(\alpha_1, \dots, \alpha_N) = \int u(\alpha_1 z_1 + \dots + \alpha_N z_N) dF(z_1, \dots, z_N).$$

Prove that $U(\cdot)$ is (a) increasing, (b) concave, and (c) continuous (this is harder).

6.C.5^A Consider a decision maker with utility function $u(\cdot)$ defined over $\mathbb{R}_+^L$, just as in Chapter 3.

- (a) Argue that concavity of $u(\cdot)$ can be interpreted as the decision maker exhibiting risk aversion with respect to lotteries whose outcomes are bundles of the L commodities.
- (b) Suppose now that a Bernoulli utility function $u(\cdot)$ for wealth is derived from the maximization of a utility function defined over bundles of commodities for each given wealth level w , while prices for those commodities are fixed. Show that, if the utility function for the commodities exhibits risk aversion, then so does the derived Bernoulli utility function for wealth. Interpret.
- (c) Argue that the converse of part (b) does not need to hold: There are nonconcave functions $u : \mathbb{R}_+^L \rightarrow \mathbb{R}$ such that for any price vector the derived Bernoulli utility function on wealth exhibits risk aversion.

6.C.6^B For Proposition 6.C.2:

- (a) Prove the equivalence of conditions (ii) and (iii).
- (b) Prove the equivalence of conditions (iii) and (iv).

6.C.7^A Prove that, in Proposition 6.C.2, condition (iii) implies condition (iv), and (iv) implies (i).

6.C.8^A In text.

6.C.9^B (M. Kimball) The purpose of this problem is to examine the implications of uncertainty and precaution in a simple consumption–savings decision problem.

In a two-period economy, a consumer has first-period initial wealth w . The consumer's utility level is given by

$$u(c_1, c_2) = u(c_1) + v(c_2),$$

where $u(\cdot)$ and $v(\cdot)$ are concave functions and c_1 and c_2 denote consumption levels in the first and the second period, respectively. Denote by x the amount saved by the consumer in the first period (so that $c_1 = w - x$ and $c_2 = x$), and let x_0 be the optimal value of x in this problem.

We now introduce uncertainty in this economy. If the consumer saves an amount x in the first period, his wealth in the second period is given by $x + y$, where y is distributed according to $F(\cdot)$. In what follows, $E[\cdot]$ always denotes the expectation with respect to

$F(\cdot)$. Assume that the Bernoulli utility function over realized wealth levels in the two periods (w_1, w_2) is $u(w_1) + v(w_2)$. Hence, the consumer now solves

$$\max_x u(w - x) + E[v(x + y)].$$

Denote the solution to this problem by x^* .

- (a) Show that if $E[v'(x_0 + y)] > v'(x_0)$, then $x^* > x_0$.
- (b) Define the *coefficient of absolute prudence* of a utility function $v(\cdot)$ at wealth level x to be $-v'''(x)/v''(x)$. Show that if the coefficient of absolute prudence of a utility function $v_1(\cdot)$ is not larger than the coefficient of absolute prudence of utility function $v_2(\cdot)$ for all levels of wealth, then $E[v'_1(x_0 + y)] > v'_1(x_0)$ implies $E[v'_2(x_0 + y)] > v'_2(x_0)$. What are the implications of this fact in the context of part (a)?
- (c) Show that if $v'''(\cdot) > 0$, and $E[y] = 0$, then $E[v'(x + y)] > v'(x)$ for all values of x .
- (d) Show that if the coefficient of absolute risk aversion of $v(\cdot)$ is decreasing with wealth, then $-v'''(x)/v''(x) > -v''(x)/v'(x)$ for all x , and hence $v'''(\cdot) > 0$.

6.C.10^A Prove the equivalence of conditions (i) through (v) in Proposition 6.C.3. [*Hint:* By letting $u_1(z) = u(w_1 + z)$ and $u_2(z) = u(w_2 + z)$, show that each of the five conditions in Proposition 6.C.3 is equivalent to the counterpart in Proposition 6.C.2.]

6.C.11^B For the model in Example 6.C.2, show that if $r_R(x, u)$ is increasing in x then the proportion of wealth invested in the risky asset $\gamma = \alpha/x$ is decreasing with x . Similarly, if $r_R(x, u)$ is decreasing in x , then $\gamma = \alpha/x$ is increasing in x . [*Hint:* Let $u_1(t) = u(tw_1)$ and $u_2(t) = u(tw_2)$, and use the fact, stated in the analysis of Example 6.C.2, that if one Bernoulli utility function is more risk averse than another, then the optimal level of investment in the risky asset for the first function is smaller than that for the second function. You could also attempt a direct proof using first-order conditions.]

6.C.12^B Let $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a strictly increasing Bernoulli utility function. Show that

- (a) $u(\cdot)$ exhibits constant relative risk aversion equal to $\rho \neq 1$ if and only if $u(x) = \beta x^{1-\rho} + \gamma$, where $\beta > 0$ and $\gamma \in \mathbb{R}$.
- (b) $u(\cdot)$ exhibits constant relative risk aversion equal to 1 if and only if $u(x) = \beta \ln x + \gamma$, where $\beta > 0$ and $\gamma \in \mathbb{R}$.
- (c) $\lim_{\rho \rightarrow 1} (x^{1-\rho}/(1-\rho)) = \ln x$ for all $x > 0$.

6.C.13^B Assume that a firm is risk neutral with respect to profits and that if there is any uncertainty in prices, production decisions are made after the resolution of such uncertainty. Suppose that the firm faces a choice between two alternatives. In the first, prices are uncertain. In the second, prices are nonrandom and equal to the expected price vector in the first alternative. Show that a firm that maximizes expected profits will prefer the first alternative over the second.

6.C.14^B Consider two risk-averse decision makers (i.e., two decision makers with concave Bernoulli utility functions) choosing among monetary lotteries. Define the utility function $u^*(\cdot)$ to be strongly more risk averse than $u(\cdot)$ if and only if there is a positive constant k

and a nonincreasing and concave function $v(\cdot)$ such that $u^*(x) = ku(x) + v(x)$ for all x . The monetary amounts are restricted to lie in the interval $[0, r]$.

- (a) Show that if $u^*(\cdot)$ is strongly more risk averse than $u(\cdot)$, then $u^*(\cdot)$ is more risk averse than $u(\cdot)$ in the usual Arrow–Pratt sense.
- (b) Show that if $u'(\cdot)$ is bounded, then there is no $u^*(\cdot)$ other than $u^*(\cdot) = ku(\cdot) + c$, where c is a constant, that is strongly more risk averse than $u(\cdot)$ on the entire interval $[0, +\infty)$. [*Hint*: In this part, disregard the assumption that the monetary amounts are restricted to lie in the interval $[0, r]$.]
- (c) Using (b), argue that the concept of a strongly more risk-averse utility function is stronger (i.e., more restrictive) than the Arrow–Pratt concept of a more risk-averse utility function.

6.C.15^A Assume that, in a world with uncertainty, there are two assets. The first is a riskless asset that pays 1 dollar. The second pays amounts a and b with probabilities of π and $1 - \pi$, respectively. Denote the demand for the two assets by (x_1, x_2) .

Suppose that a decision maker's preferences satisfy the axioms of expected utility theory and that he is a risk averter. The decision maker's wealth is 1, and so are the prices of the assets. Therefore, the decision maker's budget constraint is given by

$$x_1 + x_2 = 1, \quad x_1, x_2 \in [0, 1].$$

- (a) Give a simple *necessary* condition (involving a and b only) for the demand for the riskless asset to be strictly positive.
- (b) Give a simple *necessary* condition (involving a , b , and π only) for the demand for the risky asset to be strictly positive.

In the next three parts, assume that the conditions obtained in (a) and (b) are satisfied.

- (c) Write down the first-order conditions for utility maximization in this asset demand problem.
- (d) Assume that $a < 1$. Show by analyzing the first-order conditions that $dx_1/da \leq 0$.
- (e) Which sign do you conjecture for $dx_1/d\pi$? Give an economic interpretation.
- (f) Can you prove your conjecture in (e) by analyzing the first-order conditions?

6.C.16^A An individual has Bernoulli utility function $u(\cdot)$ and initial wealth w . Let lottery L offer a payoff of G with probability p and a payoff of B with probability $1 - p$.

- (a) If the individual owns the lottery, what is the minimum price he would sell it for?
- (b) If he does not own it, what is the maximum price he would be willing to pay for it?
- (c) Are buying and selling prices equal? Give an economic interpretation for your answer. Find conditions on the parameters of the problem under which buying and selling prices are equal.
- (d) Let $G = 10$, $B = 5$, $w = 10$, and $u(x) = \sqrt{x}$. Compute the buying and selling prices for this lottery and this utility function.

6.C.17^B Assume that an individual faces a two-period portfolio allocation problem. In period $t = 0, 1$, his wealth w_t is to be divided between a safe asset with return R and a risky asset with return x . The initial wealth at period 0 is w_0 . Wealth at period $t = 1, 2$ depends on the portfolio α_t chosen at period $t - 1$ and on the return x_t realized at period t , according to

$$w_t = ((1 - \alpha_{t-1})R + \alpha_{t-1}x_t) w_{t-1}.$$

The objective of this individual is to maximize the expected utility of terminal wealth w_2 . Assume that x_1 and x_2 are independently and identically distributed. Prove that the individual optimally sets $\alpha_0 = \alpha_1$ if his utility function exhibits constant relative risk aversion. Show also that this fails to hold if his utility function exhibits constant absolute risk aversion.

6.C.18^B Suppose that an individual has a Bernoulli utility function $u(x) = \sqrt{x}$.

- (a) Calculate the Arrow–Pratt coefficients of absolute and relative risk aversion at the level of wealth $w = 5$.
- (b) Calculate the certainty equivalent and the probability premium for a gamble $(16, 4; \frac{1}{2}, \frac{1}{2})$.
- (c) Calculate the certainty equivalent and the probability premium for a gamble $(36, 16; \frac{1}{2}, \frac{1}{2})$. Compare this result with the one in (b) and interpret.

6.C.19^C Suppose that an individual has a Bernoulli utility function $u(x) = -e^{-ax}$ where $a > 0$. His (nonstochastic) initial wealth is given by w . There is one riskless asset and there are N risky assets. The return per unit invested on the riskless asset is r . The returns of the risky assets are jointly normally distributed random variables with means $\mu = (\mu_1, \dots, \mu_N)$ and variance–covariance matrix V . Assume that there is no redundancy in the risky assets, so that V is of full rank. Derive the demand function for these $N + 1$ assets.

6.C.20^B Consider a lottery over monetary outcomes that pays $x + \epsilon$ with probability $\frac{1}{2}$ and $x - \epsilon$ with probability $\frac{1}{2}$. Compute the second derivative of this lottery's certainty equivalent with respect to ϵ . Show that the limit of this derivative as $\epsilon \rightarrow 0$ is exactly $-r_A(x)$.

6.D.1^A The purpose of this exercise is to prove Proposition 6.D.1 in a two-dimensional probability simplex. Suppose that there are three monetary outcomes: 1 dollar, 2 dollars, and 3 dollars. Consider the probability simplex of Figure 6.B.1(b).

- (a) For a given lottery L over these outcomes, determine the region of the probability simplex in which lie the lotteries whose distributions first-order stochastically dominate the distribution of L .
- (b) Given a lottery L , determine the region of the probability simplex in which lie the lotteries L' such that $F(x) \leq G(x)$ for every x , where $F(\cdot)$ is the distribution of L' and $G(\cdot)$ is the distribution of L . [Notice that we get the same region as in (a).]

6.D.2^B Prove that if $F(\cdot)$ first-order stochastically dominates $G(\cdot)$, then the mean of x under $F(\cdot)$, $\int x dF(x)$, exceeds that under $G(\cdot)$, $\int x dG(x)$. Also provide an example where $\int x dF(x) > \int x dG(x)$ but $F(\cdot)$ does not first-order stochastically dominate $G(\cdot)$.

6.D.3^B Verify that if a distribution $G(\cdot)$ is an elementary increase in risk from a distribution $F(\cdot)$, then $F(\cdot)$ second-order stochastically dominates $G(\cdot)$.

6.D.4^B The purpose of this exercise is to verify the equivalence of the three statements of Proposition 6.D.2 in a two-dimensional probability simplex. Suppose that there are three monetary outcomes: 1, 2, and 3 dollars. Consider the probability simplex in Figure 6.B.1(b).

- (a) If two lotteries have the same mean, what are their positions relative to each other in the probability simplex.
- (b) Given a lottery L , determine the region of the simplex in which lie the lotteries L' whose distributions are second-order stochastically dominated by the distribution of L .
- (c) Given a lottery L , determine the region of the simplex in which lie the lotteries L' whose distributions are mean-preserving spreads of L .
- (d) Given a lottery L , determine the region of the simplex in which lie the lotteries L' for which condition (6.D.2) holds, where $F(\cdot)$ and $G(\cdot)$ are, respectively, the distributions of L' and L .

Notice that in (b), (c), and (d), you always have the same region.

Chapter 2

General Equilibrium

2.1 Pure Exchange Economy and the Edgeworth Box

The Edgeworth box is the classical graphical and analytical representation of a pure exchange economy with two goods and two consumers. It allows simultaneous analysis of feasibility, budget constraints, Pareto efficiency, market equilibrium, and the *core* of the economy.

2.1.1 Geometric Construction of the Box

The box is constructed by rotating consumer 2's (Bob's) preference diagram by 180° and overlaying it on consumer 1's (Alice's) diagram, so that Bob's origin coincides with the point representing the total social endowment in Alice's graph. Alice's origin, O_1 , is in the lower left corner (Southwest); Bob's origin, O_2 , is in the upper right corner (Northeast).

The dimensions of the box are given by the aggregate endowments:

$$\bar{\omega}_1 = \omega_{11} + \omega_{12}, \quad \bar{\omega}_2 = \omega_{21} + \omega_{22},$$

where $\bar{\omega}_1$ is the width (good 1) and $\bar{\omega}_2$ is the height (good 2).

Definition 2.1.1 (Edgeworth Box). The set of feasible allocations in the pure exchange economy is

$$X = \left\{ (x_1, x_2) \in \mathbb{R}_+^2 \times \mathbb{R}_+^2 : x_{11} + x_{12} = \bar{\omega}_1, x_{21} + x_{22} = \bar{\omega}_2 \right\},$$

which can be identified with the rectangle $[0, \bar{\omega}_1] \times [0, \bar{\omega}_2]$.

2.1.2 Fundamental Properties

Feasibility and Resource Exhaustion

Any point in the box represents a feasible, waste-free allocation: the quantities consumed by the two agents sum exactly to the aggregate endowments,

$$x_{11} + x_{12} = \omega_{11} + \omega_{12}, \quad x_{21} + x_{22} = \omega_{21} + \omega_{22}.$$

Shared Budget Line

The initial endowment $\omega = (\omega_1, \omega_2)$ is a single common point in the box. Given a price vector $p = (p_1, p_2)$, the line passing through ω with slope $-p_1/p_2$ is the budget constraint for **both** consumers. If Alice exactly exhausts her income, Bob also exhausts his, by force of the aggregate constraint (Walras' Law).

Contract Curve and Pareto Efficiency

A point is Pareto efficient if it is impossible to improve one agent's welfare without worsening the other's. Under convex and strictly increasing preferences, this is equivalent to the tangency of indifference curves, that is, $MRS_1 = MRS_2$. The set of all efficient allocations is the **contract curve**.

The Core

The core is the subset of the contract curve where both allocations are individually rational (each agent obtains at least the utility of their own endowment). In the two-agent box, any allocation outside this segment is blocked by a coalition of size 1.

Walrasian Equilibrium (Geometric Preview)

Equilibrium occurs at a relative price p^* where the optimal choices of both agents coincide at the same point in the box: the indifference curves are tangent to each other and to the common budget line. Equivalently, it is the intersection of the *offer curves* emanating from the initial endowment. By the First Welfare Theorem, this point belongs to the contract curve and the core.

2.1.3 Walrasian Equilibrium: Definition 15.B.1

Formal Definition

Definition 2.1.2 (Budget Set). Given a price vector p^* , consumer i 's budget set is

$$B_i(p^*) = \{x'_i \in \mathbb{R}_+^2 : p^* \cdot x'_i \leq p^* \cdot \omega_i\}.$$

Definition 2.1.3 (Walrasian Equilibrium, Definition 15.B.1). A **Walrasian (or competitive) equilibrium** for the Edgeworth box economy is a pair (p^*, x^*) , with $x^* = (x_1^*, x_2^*)$ feasible, such that, for $i = 1, 2$,

$$x_i^* \succsim_i x'_i \quad \text{for all } x'_i \in B_i(p^*).$$

Remark 2.1.1 (Prices as Parameters). The vector p^* is common and taken as given by both agents (*price-takers*). The allocation x^* is intrinsically feasible, respecting the physical resource constraint of the box.

Local Nonsatiation and Walras' Law

If preferences are locally nonsatiated, each agent's budget constraint holds with equality in equilibrium, $p^* \cdot x_i^* = p^* \cdot \omega_i$. Summing the two constraints yields **Walras' Law**: if one market is in equilibrium (clears), so is the other.

Geometry of Equilibrium

The common budget line passes through ω with slope $-p_1^*/p_2^*$. In equilibrium, the indifference curves of both consumers are tangent to each other and to this line:

$$\text{MRS}_1(x_1^*) = \frac{p_1^*}{p_2^*} = \text{MRS}_2(x_2^*).$$

Alternatively, equilibrium is the intersection of the two agents' offer curves.

Properties and Results

First Welfare Theorem

Proposition 2.1.1 (First Welfare Theorem). *If preferences are locally nonsatiated, every Walrasian equilibrium (p^*, x^*) generates a Pareto efficient allocation x^* .*

Sketch. If there existed a feasible allocation x' that Pareto-dominated x^* , individual optimization and nonsatiation would imply $p^* \cdot x'_i \geq p^* \cdot \omega_i$ for all i , with strict inequality for some i . Summing, $p^* \cdot (x'_1 + x'_2) > p^* \cdot (\omega_1 + \omega_2)$, contradicting feasibility $x'_1 + x'_2 = \omega_1 + \omega_2$. $\square$

Existence

Proposition 2.1.2 (Existence). *If preferences are continuous, strictly convex, and strictly monotone, there exists at least one equilibrium price vector p^* that equates aggregate supply and demand.*

Second Welfare Theorem Any Pareto efficient allocation can be achieved as a Walrasian equilibrium after appropriate prior redistribution of endowments (see Definition 2.1.6). The crucial hypothesis is the convexity of preferences.

Market Clearing Under monotonicity, every desirable good has a strictly positive price in equilibrium, $p_j^* > 0$, and its market clears: aggregate excess demand is zero, $z_j(p^*) = 0$.

Uniqueness If demands are differentiable and goods behave as gross substitutes in excess demand ($\partial z_i / \partial p_j > 0$ for $i \neq j$), the equilibrium price vector is unique up to a positive multiplier (unique relative prices).

Deepening: Hypotheses of the First Theorem and Budget Lemma

Crucial Hypotheses The First Theorem requires **local nonsatiation** (ensures $p^* \cdot x_i^* = p^* \cdot \omega_i$; with satiation, resources could be wasted) and **absence of externalities** (agent i 's preferences depend only on x_i). **Convexity is not necessary:** if an equilibrium exists (even with non-convex preferences), it is Pareto efficient. Convexity only enters the Second Theorem.

Auxiliary Lemma: Strict Preference Breaks the Budget

Lemma 2.1.1. *If x_i^* is optimal in $B_i(p^*)$ and preferences are locally nonsatiated, then*

$$y_i \succ_i x_i^* \implies p^* \cdot y_i > p^* \cdot \omega_i.$$

Proof. If $p^* \cdot y_i \leq p^* \cdot \omega_i$, then $y_i \in B_i(p^*)$; by optimality, $x_i^* \succsim_i y_i$, contradicting $y_i \succ_i x_i^*$ (asymmetry of strict preference). $\square$

Deepening: Why Convexity is Indispensable (Separating Hyperplane)

The proof of the Second Theorem uses the **Separating Hyperplane Theorem**. Let x^* be Pareto optimal. Define the set of strictly preferred allocations,

$$PS(x^*) = \{(x_1, x_2) : x_i \succ_i x_i^* \forall i\},$$

and the set of feasible allocations FA . If x^* is efficient, $PS(x^*) \cap FA = \emptyset$. If preferences are convex, both sets are convex; hence there exists a non-zero vector p^* whose hyperplane separates them — and this p^* is exactly the supporting price that decentralizes x^* . Without convexity, indifference curves can “bend around” the hyperplane: agents would choose corners of the budget constraint and the market would not clear.

2.1.4 Pareto Efficiency: Definition 15.B.2

Definition and Basic Notions

Definition 2.1.4 (Pareto Optimality, Definition 15.B.2). An allocation x in the Edgeworth box is **Pareto optimal (efficient)** if there is no other feasible allocation x' with

$$x'_i \succsim_i x_i \text{ for } i = 1, 2, \quad \text{and} \quad x'_i \succ_i x_i \text{ for some } i.$$

An allocation x' satisfying these conditions is a **Pareto improvement**: it leaves at least one consumer strictly better off without worsening the other. The **Pareto set** is the collection of all Pareto optimal allocations; in the box, it coincides with the **contract curve**.

Example 2.1.1 (Apples and Oranges). If Alice only has apples but loves oranges, and Bob only has oranges but loves apples, the initial endowment is inefficient: both gain from trading. When all mutually beneficial trades are exhausted, any additional trade that benefits Alice will harm Bob — Pareto efficiency has been reached.

If Alice’s and Bob’s indifference curves cross, a “lens”-shaped region of allocations strictly preferred by both is formed. When the lens disappears, the curves become tangent and the MRSs are equalized:

$$MRS_1(x_1) = MRS_2(x_2).$$

Strong versus Weak Efficiency

Definition 2.1.5 (Strong and Weak Efficiency). A feasible allocation x is **strongly efficient** if there is no feasible x' with $x'_i \succsim_i x_i$ for all i and $x'_j \succ_j x_j$ for some j . It is **weakly efficient** if there is no feasible x' with $x'_i \succ_i x_i$ for **all** i .

Proposition 2.1.3 (Strong/Weak Equivalence). *Under continuous and strictly monotone preferences, the strong and weak notions coincide.*

Lemma: Weak improvement implies strong improvement. Let x' be feasible with $x'_i \succsim_i x_i$ for all i and $x'_j \succ_j x_j$ for some j . By continuity, the strict upper contour set $\{z : z \succ_j x_j\}$

is open; hence there exists $\epsilon > 0$ such that $\tilde{x}_j = x'_j - \epsilon \mathbf{1} \succ_j x_j$. Define $\tilde{x}_i = x'_i + \frac{\epsilon}{m-1} \mathbf{1}$ for $i \neq j$. Then $\sum_i \tilde{x}_i = \sum_i x'_i = \sum_i \omega_i$ (the $\epsilon \mathbf{1}$ terms cancel), so $\tilde{x}$ is feasible. For $i \neq j$, $\tilde{x}_i \gg x'_i$ and, by strict monotonicity and transitivity, $\tilde{x}_i \succ_i x_i$. Therefore $\tilde{x}_i \succ_i x_i$ for all i : strong improvement. $\square$

Interior Characterization: Tangency

Proposition 2.1.4 (Sufficient Condition for Interior Optimality). *Suppose preferences are continuous, strictly convex, and differentiable. If, at an interior allocation $x = (x_1, x_2)$,*

$$MRS_1(x_1) = MRS_2(x_2),$$

then x is Pareto optimal (belongs to the contract curve).

Heuristic Argument (Why $MRS_1 = MRS_2$ Implies Efficiency) Consider a feasible marginal reallocation with $dx_1 + dx_2 = 0$. To keep consumer 1 on the same indifference curve ($du_1 = 0$),

$$du_1 = \frac{\partial u_1}{\partial x_{11}} dx_{11} + \frac{\partial u_1}{\partial x_{21}} dx_{21} = 0 \implies dx_{21} = -MRS_1 dx_{11}.$$

The impact on consumer 2, using $dx_{12} = -dx_{11}$ and $dx_{22} = -dx_{21} = MRS_1 dx_{11}$, is

$$du_2 = \frac{\partial u_2}{\partial x_{22}} (MRS_1 - MRS_2) dx_{11}.$$

If $MRS_1 = MRS_2$, then $du_2 = 0$: there is no direction of trade that locally improves both. Under strict convexity, the local argument extends and excludes global Pareto improvements.

The Case $MRS_1 \neq MRS_2$ If, for example, $MRS_1 = 3 > 1 = MRS_2$, there are mutual gains: consumer 1 is willing to give up to 2.9 units of good 2 for 1 unit of good 1, and consumer 2 is willing to give 1 unit of good 1 for 1.1 of good 2; trading 1 unit of good 1 for 2 of good 2, both move to a higher indifference curve (the “lens” exists).

Corner Conditions

The condition $MRS_1 = MRS_2$ is necessary only for **interior** allocations. At the boundary of the box, the Kuhn–Tucker conditions become inequalities: if consumer 2 does not consume good 1 ($x_{12} = 0$), it may occur that $MRS_1(x_1) \geq MRS_2(x_2)$; if consumer 1 does not consume good 2 ($x_{21} = 0$), it may occur that $MRS_1(x_1) \leq MRS_2(x_2)$.

Properties and Results

Tangency and Contract Curve Under differentiable, convex, and strictly monotone preferences, every interior Pareto efficient allocation satisfies $MRS_1 = MRS_2$; at corner solutions, the corresponding inequality holds. The set of efficient allocations forms the contract curve, a continuous path linking O_1 to O_2 .

Characterization by Weighted Sum

Proposition 2.1.5 (Weighted Utility Maximization). *Every Pareto efficient allocation x^* solves, for some pair of weights $(\alpha_1, \alpha_2) \geq 0$ not both zero,*

$$\max \alpha_1 u_1(x_1) + \alpha_2 u_2(x_2) \quad \text{subject to feasibility in the box.}$$

If the weights are strictly positive, every solution to this problem is strictly Pareto efficient.

Welfare Theorems The First Theorem connects Definition 2.1.3 to 2.1.4: Walrasian equilibrium is Pareto efficient. The Second Theorem goes the other way under convexity: any point on the contract curve is decentralizable as an equilibrium after redistribution.

Core The core is the individually rational segment of the contract curve: both agents are at least as well off as consuming only their own endowment (see the next section).

What Breaks

*Remark 2.1.2. **Consumption Externalities:** if Alice cares about Bob's consumption (envy, pollution), simple tangency no longer guarantees efficiency. **Non-convexities:** with non-convex preferences, tangency points may be local minima or fail to identify global optima.*

2.1.5 Equilibrium with Transfers: Definition 15.B.3

Definition and Intuition

Definition 2.1.6 (Equilibrium with Transfers, Definition 15.B.3). An allocation x^* in the box is **supportable as an equilibrium with transfers** if there exist a price system p^* and transfers T_1, T_2 with $T_1 + T_2 = 0$ such that, for each consumer i ,

$$x_i^* \succsim_i x'_i \quad \text{for all } x'_i \in \mathbb{R}_+^2 \text{ with } p^* \cdot x'_i \leq p^* \cdot \omega_i + T_i.$$

The competitive market exhausts beneficial trades (efficiency), but is “blind” to justice: if the initial endowment is very unequal, the equilibrium is efficient but socially unacceptable. Equilibrium with transfers separates efficiency from distribution: the planner makes *lump-sum* transfers before opening the market and lets competitive prices act.

Example 2.1.2 (Board Game). If the rules of the game are efficient but some players start with few chips, instead of intervening mid-game by changing the value of the pieces (which would distort signals), chips are redistributed at the beginning (T_i) and the game is allowed to run under the same competitive rules.

Geometrically, a transfer T_i shifts the shared budget line in parallel until it passes exactly through the desired Pareto optimal allocation x^* .

Properties and Results

Zero Sum The condition $T_1 + T_2 = 0$ ensures that the planner only redistributes purchasing power, without injecting or destroying wealth: if $T_1 > 0$ (subsidy to Alice), then $T_2 < 0$ (tax on Bob) of equal value.

Second Welfare Theorem

Proposition 2.1.6 (Second Theorem). *If preferences are continuous, strictly monotone, and convex, any Pareto optimal allocation x^* can be supported as an equilibrium with transfers.*

There is no inevitable *trade-off* between efficiency and equity: any point on the contract curve is achievable by the price mechanism.

Indispensable Convexity Without convexity, tangency at the desired point does not guarantee a global optimum: faced with the linear constraint, agents would jump to extremes, and the market would not clear. Convexity ensures that first-order conditions characterize the global maximum under modified wealth.

Valuation of Net Demands Summing the constraints with equality and using feasibility $x_1^* + x_2^* = \omega_1 + \omega_2$, we obtain $T_1 + T_2 = 0$ and the explicit rule

$$T_i = p^* \cdot (x_i^* - \omega_i),$$

that is, the transfer is the market value of the agent's net excess demand at supporting prices.

Budget Equivalence An equilibrium with transfers (p^*, x^*, T_1, T_2) becomes a classical Walrasian equilibrium if we redefine endowments to ω'_i with $p^* \cdot \omega'_i = p^* \cdot \omega_i + T_i$ (in particular, $\omega'_i = x_i^*$). At interior solutions, $\text{MRS}_1(x_1^*) = p_1^*/p_2^* = \text{MRS}_2(x_2^*)$.

Quasi-Equilibrium and Failures If the transfer reduces some agent's wealth to zero ($p^* \cdot \omega_i + T_i = 0$), standard maximization may fail at the boundary; one then resorts to *quasi-equilibrium* with transfers. If $p^* \cdot \omega_i + T_i$ exceeds the minimum cost of survival for all, every quasi-equilibrium is a strict equilibrium. Without local nonsatiation ("thick" preferences), a Pareto optimum may not even be supportable as a quasi-equilibrium.

2.1.6 Examples: Cobb–Douglas in the Edgeworth Box

Example 1: Contract Curve and Walrasian Equilibrium

Setup Two consumers with symmetric preferences,

$$u_1(x_1, y_1) = x_1^{1/2} y_1^{1/2}, \quad u_2(x_2, y_2) = x_2^{1/2} y_2^{1/2},$$

and asymmetric endowments $\omega_1 = (20, 10)$, $\omega_2 = (10, 20)$. The box dimensions are $X = 30$ and $Y = 30$.

Contract Curve The marginal rates of substitution are

$$\text{MRS}_1 = \frac{y_1}{x_1}, \quad \text{MRS}_2 = \frac{y_2}{x_2} = \frac{30 - y_1}{30 - x_1}.$$

Equating and using feasibility,

$$y_1(30 - x_1) = x_1(30 - y_1) \implies 30y_1 = 30x_1 \implies y_1 = x_1.$$

The contract curve is the diagonal of the box, $y_1 = x_1$.

Walrasian Equilibrium Fix $p_y = 1$. Incomes are $I_1 = 20p_x + 10$ and $I_2 = 10p_x + 20$. Since Cobb–Douglas $x^{1/2}y^{1/2}$ spends 50% of income on each good,

$$x_1 = \frac{I_1}{2p_x} = 10 + \frac{5}{p_x}, \quad x_2 = \frac{I_2}{2p_x} = 5 + \frac{10}{p_x}.$$

Market clearing for good x :

$$\left(10 + \frac{5}{p_x}\right) + \left(5 + \frac{10}{p_x}\right) = 30 \implies 15 + \frac{15}{p_x} = 30 \implies p_x^* = 1.$$

Hence $(p_x^*, p_y^*) = (1, 1)$ and the equilibrium allocation is

$$x_1^* = 15, \quad y_1^* = 15, \quad x_2^* = 15, \quad y_2^* = 15,$$

that is, $x^* = ((15, 15), (15, 15))$.

Position on the Contract Curve Since $y_1^* = x_1^* = 15$, the equilibrium lies exactly on the diagonal (midpoint of the contract curve), confirming the First Welfare Theorem.

Table 2.1: Summary of Example ??: equilibrium on the contract curve.

Concept	Equation	Value in Example	Meaning in the Box
Efficient frontier	$MRS_1 = MRS_2$	$y_1 = x_1$	Diagonal linking O_1 to O_2
Price vector	$p^* = (p_x^*, p_y^*)$	$(1, 1)$	Line with slope -1 through $(20, 10)$
Equilibrium allocation	$x^* = (x_1^*, y_1^*)$	$(15, 15)$	Center of the box; tangency to price line

Example 2: Planner's Transfers

Setup General Cobb–Douglas utilities,

$$u_1 = x_{11}^\alpha x_{21}^{1-\alpha}, \quad u_2 = x_{12}^\beta x_{22}^{1-\beta},$$

endowments ω_1, ω_2 , totals $X_1 = \omega_{11} + \omega_{12}$, $X_2 = \omega_{21} + \omega_{22}$, and a desired Pareto optimal allocation $x^* = (x_1^*, x_2^*)$, feasible by definition.

Step 1: Supporting Prices Every interior Pareto optimal allocation equalizes MRSs. For each agent to choose x_i^* decentralizably, the price ratio must coincide with the MRS at the target point:

$$MRS_1(x_1^*) = \frac{\alpha}{1-\alpha} \frac{x_{21}^*}{x_{11}^*}.$$

Normalizing $p_2^* = 1$,

$$p_1^* = \frac{\alpha}{1-\alpha} \frac{x_{21}^*}{x_{11}^*}.$$

Since $MRS_1(x_1^*) = MRS_2(x_2^*)$, the same price supports Bob: $p_1^* = \frac{\beta}{1-\beta} \frac{x_{22}^*}{x_{12}^*}$.

Step 2: Required Wealth The wealth that allows agent i to acquire x_i^* is

$$R_i = p^* \cdot x_i^* = p_1^* x_{1i}^* + x_{2i}^*.$$

Step 3: Exact Transfers From the modified constraint $p^* \cdot x_i^* = p^* \cdot \omega_i + T_i$,

$$T_i = p^* \cdot (x_i^* - \omega_i),$$

that is,

$$T_1 = p_1^*(x_{11}^* - \omega_{11}) + (x_{21}^* - \omega_{21}), \quad T_2 = p_1^*(x_{12}^* - \omega_{12}) + (x_{22}^* - \omega_{22}).$$

If $T_i > 0$, the agent receives a subsidy; if $T_i < 0$, they pay a *lump-sum* tax.

Consistency: Zero Sum Summing,

$$T_1 + T_2 = p^* \cdot [(x_1^* + x_2^*) - (\omega_1 + \omega_2)] = p^* \cdot 0 = 0,$$

by feasibility of x^* . The redistribution is self-sufficient.

Consistency: Decentralization Verify that Alice indeed demands x_1^* at prices p^* and wealth R_1 . The Cobb–Douglas demand for good 1 is $x_{11} = \alpha R_1 / p_1^*$:

$$x_{11} = \frac{\alpha(p_1^* x_{11}^* + x_{21}^*)}{p_1^*} = \alpha x_{11}^* + \alpha \frac{x_{21}^*}{p_1^*} = \alpha x_{11}^* + (1 - \alpha)x_{11}^* = x_{11}^*,$$

using $x_{21}^* / p_1^* = \frac{1-\alpha}{\alpha} x_{11}^*$. For good 2, $x_{21} = (1 - \alpha)R_1$:

$$x_{21} = (1 - \alpha) \left(\frac{\alpha}{1 - \alpha} x_{21}^* + x_{21}^* \right) = \alpha x_{21}^* + (1 - \alpha)x_{21}^* = x_{21}^*.$$

Decentralization is exact (same for Bob).

Table 2.2: Planner’s protocol: decentralizing a Pareto optimum.

Planner’s Step	Equation	Meaning
1. Read preferences	α, β	Taste structure
2. Define target	$x^* = (x_1^*, x_2^*)$	Desired point on contract curve
3. Calculate prices	$p_1^* = \frac{\alpha}{1-\alpha} \frac{x_{21}^*}{x_{11}^*}$	Tangency at target point
4. Measure wealth	$R_i = p^* \cdot x_i^*$	Market value of each basket
5. Execute adjustment	$T_i = p^* \cdot (x_i^* - \omega_i)$	Tax excess, subsidize deficit

Remark 2.1.3 (Section Synthesis). The Edgeworth box unifies the three central concepts: Walrasian equilibrium (Def. 2.1.3) is decentralized coordination via prices; Pareto efficiency (Def. 2.1.4) is the exhaustion of mutually beneficial trades (contract curve); and equilibrium with transfers (Def. 2.1.6) is the bridge that allows the planner to reach any efficient point in a decentralized way, separating efficiency from equity.

2.2 The *Core* of the Exchange Economy

2.2.1 Economic Intuition

The Essence While the Walrasian approach assumes indirect coordination via prices, the **core** asks: which allocations are stable under direct, voluntary negotiations, without prices or an auctioneer? An allocation is stable if **no coalition** of agents can withdraw with their own endowments and redistribute them internally, leaving all members strictly better off.

Example 2.2.1 (Community Picnic). Each guest brings their basket (endowment). A redistribution at the collective table is in the core if no subgroup can take their original baskets, sit in a corner, and prepare a snack that leaves everyone in the subgroup more satisfied. If such a “clique” exists, the allocation was **blocked**. The core is the set of tables where no clique has an incentive to separate.

Example 2.2.2 (Sticker Trading Fair). Children trade stickers at recess, each with their duplicates (initial endowment). A general redistribution proposal will be in the core if no “clique” can gather only the stickers their members brought and reorganize them internally so that everyone is strictly better off. If such a group exists, it blocks the general proposal.

Mechanisms Three institutional pillars: (i) **collective veto and free association** — groups fragment the allocation if they can do better on their own; (ii) **absence of transaction costs** — every mutually advantageous reallocation is discovered; (iii) **sub-coalition veto** — the core is much more severe than Pareto: Pareto only requires that the grand coalition does not block, the core requires that **no** coalition S , of size 1 to m , blocks.

The stability of the core depends on the following properties of preferences:

- **Non-blocking and social stability:** the core is the set of allocations against which no coalition can credibly protest; if a group can do better on its own, the general arrangement is unstable.
- **Monotonicity (nonsatiation):** agents always prefer “more” to “less”; without it, core allocations could waste endowments, as agents would want to get rid of resources.
- **Convexity (preference for diversity):** identical agents with identical endowments must be treated identically in any stable allocation (equal treatment); without strict convexity, this symmetry breaks and the core does not collapse to the Walrasian set.
- **Continuity:** small changes in baskets generate small changes in utility; without it, stable “compromise points” are lacking and the core may be empty.

Implications The core underpins the design of stable markets and agreements (consortia, joint ventures, emission quotas): an agreement collapses if some block prefers to manage their endowments in isolation. The central theoretical result is the **Edgeworth–Debreu–Scarf Equivalence Theorem**: as the economy is replicated, the core shrinks to coincide with Walrasian allocations — perfect competition emerges as the only cooperatively stable allocation in large economies.

Example 2.2.3 (Water and Carbon Markets). In river basins where farmers hold historical water rights (endowments), the regulator can design a decentralized multilateral exchange market instead of imposing centralized prices. Knowing that voluntary negotiation results in a core allocation, the policymaker has a guarantee that the final outcome will be Pareto efficient and stable, as no coalition of farmers will feel harmed enough to abandon the agreement.

2.2.2 Formal Definitions

Consider a pure exchange economy $\mathcal{E} = \{I, (X_i, \succsim_i, \omega_i)_{i \in I}\}$, with $I = \{1, \dots, m\}$, $X_i = \mathbb{R}_+^\ell$, preferences complete, transitive, continuous, and strictly monotone, and aggregate endowment $\bar{\omega} = \sum_i \omega_i \gg 0$. The set of feasible allocations is

$$FA = \left\{ x = (x_1, \dots, x_m) \in \mathbb{R}_+^{\ell m} : \sum_{i \in I} x_i = \sum_{i \in I} \omega_i \right\}.$$

Definition 2.2.1 (Coalition). A **coalition** is any non-empty subset of agents:

$$\emptyset \neq S \subseteq I.$$

Definition 2.2.2 (Blocking). A coalition $S \subseteq I$, $S \neq \emptyset$, **blocks** $x \in FA$ if there exists $(y_i)_{i \in S}$ with

$$\sum_{i \in S} y_i = \sum_{i \in S} \omega_i \quad \text{and} \quad y_i \succ_i x_i \quad \forall i \in S.$$

That is, the coalition can reallocate only its own resources and make all its members strictly better off relative to x .

Definition 2.2.3 (Core). The **core** $\mathcal{N}(\mathcal{E})$ is the set of feasible allocations that no coalition blocks:

$$\mathcal{N}(\mathcal{E}) = \{x \in FA : \nexists S \neq \emptyset \text{ that blocks } x\}.$$

Equivalently: for every $S \neq \emptyset$ and every $(y_i)_{i \in S}$ feasible in S , there exists $i_0 \in S$ with $x_{i_0} \succsim_{i_0} y_{i_0}$; no coalition can propose an internal redistribution strictly accepted by all its members.

Remark 2.2.1 (What Breaks). Without **monotonicity**, core allocations may waste resources ($\sum_i x_i < \sum_i \omega_i$). Without **strict convexity**, equal treatment in replicas fails and the core does not collapse to the Walrasian set. Without **continuity**, the strict upper contour set ceases to be open and the core may be empty (stable limit points are lacking).

2.2.3 Individual Rationality and Fundamental Inclusions

Unit Coalitions and Individual Rationality

For $S = \{i\}$, the coalition's resources are $\sum_{j \in S} \omega_j = \omega_i$; hence the only feasible reallocation is $y_i = \omega_i$, and

$$\{i\} \text{ blocks } x \iff \omega_i \succ_i x_i.$$

Therefore, if $x \in \mathcal{N}(\mathcal{E})$, no unit coalition can block it, and every allocation in the core is **individually rational**:

$$x_i \succsim_i \omega_i, \quad \forall i. \tag{IR}$$

In the Edgeworth box with two consumers, this confines the core to the “lens” of voluntary trades delimited by the indifference curves passing through ω .

Lemma: Weak Improvements Can Be Transformed into Strict Improvements

Lemma 2.2.1 (Conversion of Weak to Strong Improvement). *Suppose preferences are continuous and strictly monotone. If there exists a feasible allocation $x' \in FA$ such that*

$$x'_i \succsim_i x_i \quad \forall i \in I, \quad (1)$$

and

$$x'_j \succ_j x_j \quad \text{for some } j \in I, \quad (2)$$

then there exists a feasible allocation $\tilde{x} \in FA$ such that

$$\tilde{x}_i \succ_i x_i \quad \forall i \in I. \quad (3)$$

Proof. Since $x'_j \succ_j x_j$, by continuity of $\succsim_j$ the strict upper contour set

$$U_j^>(x_j) = \{z \in \mathbb{R}_+^\ell : z \succ_j x_j\}$$

is open; hence there is a neighborhood of x'_j entirely contained in $U_j^>(x_j)$. Moreover, $x'_j \neq 0$ (since, under strict monotonicity, 0 cannot be strictly preferred to any basket); therefore there is at least one good h with $x'_{jh} > 0$. Choose $\epsilon > 0$ small enough that

$$x'_j - \epsilon e_h \in \mathbb{R}_+^\ell \quad \text{and} \quad x'_j - \epsilon e_h \succ_j x_j, \quad (4)$$

where e_h is the canonical vector of good h . Define

$$\tilde{x}_j = x'_j - \epsilon e_h, \quad (5)$$

and, for each $i \neq j$,

$$\tilde{x}_i = x'_i + \frac{\epsilon}{m-1} e_h. \quad (6)$$

Let us verify feasibility:

$$\begin{aligned} \sum_{i \in I} \tilde{x}_i &= \tilde{x}_j + \sum_{i \neq j} \tilde{x}_i \\ &= (x'_j - \epsilon e_h) + \sum_{i \neq j} \left(x'_i + \frac{\epsilon}{m-1} e_h \right) \\ &= \sum_{i \in I} x'_i - \epsilon e_h + (m-1) \frac{\epsilon}{m-1} e_h \\ &= \sum_{i \in I} x'_i - \epsilon e_h + \epsilon e_h \\ &= \sum_{i \in I} x'_i. \end{aligned}$$

Since $x' \in FA$, it follows that $\tilde{x} \in FA$. For agent j , $\tilde{x}_j \succ_j x_j$ by (4). For $i \neq j$, $\tilde{x}_i = x'_i + \frac{\epsilon}{m-1} e_h \geq x'_i$ with strict inequality in at least one component; by strict monotonicity,

$$\tilde{x}_i \succ_i x'_i. \quad (8)$$

Since $x'_i \succsim_i x_i$ by (1), from (8) and transitivity, $\tilde{x}_i \succ_i x_i$ for all $i \neq j$. Together with $\tilde{x}_j \succ_j x_j$, we conclude (3). $\square$

Every Allocation in the Core is Pareto Optimal

Proposition 2.2.1 (Core $\subseteq$ Pareto). *Suppose preferences are continuous and strictly monotone. Then*

$$x \in \mathcal{N}(\mathcal{E}) \implies x \in OP(\mathcal{E}).$$

Proof. Suppose, for contradiction, that

$$x \in \mathcal{N}(\mathcal{E}) \quad \text{and} \quad x \notin OP(\mathcal{E}).$$

Then there exists $x' \in FA$ with

$$x'_i \succsim_i x_i \quad \forall i \in I,$$

and

$$x'_j \succ_j x_j \quad \text{for some } j \in I.$$

By Lemma 2.2.1, there exists $\tilde{x} \in FA$ with

$$\tilde{x}_i \succ_i x_i \quad \forall i \in I.$$

Consider the grand coalition $S = I$. Since $\tilde{x} \in FA$,

$$\sum_{i \in S} \tilde{x}_i = \sum_{i \in I} \tilde{x}_i = \sum_{i \in I} \omega_i = \sum_{i \in S} \omega_i.$$

Moreover,

$$\tilde{x}_i \succ_i x_i \quad \forall i \in S.$$

The above conditions say that the grand coalition $S = I$ blocks x , contradicting $x \in \mathcal{N}(\mathcal{E})$. Hence $x \in OP(\mathcal{E})$. $\square$

Walrasian Equilibrium and Core

A **Walrasian equilibrium** for the economy $\mathcal{E}$ is a pair (p^*, x^*) , with $p^* \in \mathbb{R}_{++}^\ell$ and $x^* \in FA$, such that, for all $i \in I$,

$$p^* \cdot x_i^* \leq p^* \cdot \omega_i, \tag{15}$$

and

$$x_i^* \succsim_i x_i \quad \text{for all } x_i \in \mathbb{R}_+^\ell \text{ with } p^* \cdot x_i \leq p^* \cdot \omega_i. \tag{16}$$

Condition (16) is individual optimization; $x^* \in FA$ is market clearing.

Lemma 2.2.2 (Strict Preference Implies Budget Inaccessibility). *If x_i^* is optimal in consumer i 's Walrasian budget set, then*

$$y_i \succ_i x_i^* \implies p^* \cdot y_i > p^* \cdot \omega_i. \tag{17}$$

Proof. Suppose, for contradiction, that

$$y_i \succ_i x_i^* \quad \text{and} \quad p^* \cdot y_i \leq p^* \cdot \omega_i.$$

From the second inequality, y_i belongs to i 's budget set:

$$y_i \in \{z_i \in \mathbb{R}_+^\ell : p^* \cdot z_i \leq p^* \cdot \omega_i\}.$$

But x_i^* is optimal in this set, hence

$$x_i^* \succsim_i y_i.$$

The relations $y_i \succ_i x_i^*$ and $x_i^* \succsim_i y_i$ are incompatible by the definition of strict preference. Therefore (17) holds. $\square$

Proposition 2.2.2 (Walrasian $\subseteq$ Core). *If (p^*, x^*) is a Walrasian equilibrium, then $x^* \in \mathcal{N}(\mathcal{E})$.*

Proof. Suppose, for contradiction, that $x^* \notin \mathcal{N}(\mathcal{E})$. Then there exists a non-empty coalition $S \subseteq I$ and $(y_i)_{i \in S}$ that blocks x^* :

$$\sum_{i \in S} y_i = \sum_{i \in S} \omega_i, \quad (22)$$

$$y_i \succ_i x_i^* \quad \forall i \in S. \quad (23)$$

Multiplying equality (22) by p^* and using the linearity of the inner product,

$$\sum_{i \in S} p^* \cdot y_i = \sum_{i \in S} p^* \cdot \omega_i. \quad (24)$$

On the other hand, from (23) and Lemma 2.2.2,

$$p^* \cdot y_i > p^* \cdot \omega_i \quad \forall i \in S. \quad (25)$$

Summing (25) over $i \in S$,

$$\sum_{i \in S} p^* \cdot y_i > \sum_{i \in S} p^* \cdot \omega_i. \quad (26)$$

Relations (24) and (26) are contradictory. Hence no coalition blocks x^* , and $x^* \in \mathcal{N}(\mathcal{E})$. $\square$

Chain of Inclusions and False Reciprocals

Combining the propositions,

$$WE(\mathcal{E}) \subseteq \mathcal{N}(\mathcal{E}) \subseteq OP(\mathcal{E}) \subseteq FA. \quad (27)$$

Interpretation: every Walrasian allocation cannot be blocked by any coalition; every allocation immune to coalitional blocking must be Pareto optimal; every Pareto optimal allocation is feasible.

However, the reciprocals are false in general: $OP(\mathcal{E}) \not\subseteq \mathcal{N}(\mathcal{E})$, since an allocation can be Pareto efficient but violate the individual rationality of some agent; and $\mathcal{N}(\mathcal{E}) \not\subseteq WE(\mathcal{E})$, since there may be allocations stable against blocking that are not sustainable as a Walrasian equilibrium without transfers. Edgeworth's question is whether this slack disappears when the economy becomes large — the Edgeworth–Debreu–Scarf Theorem answers affirmatively.

2.2.4 Examples: Boundaries Between the Sets

Pareto Optimal Outside the Core

Two consumers with $u_i(x_{1i}, x_{2i}) = \sqrt{x_{1i}x_{2i}}$, endowments $\omega_1 = (1.5, 0.5)$, $\omega_2 = (0.5, 1.5)$, and the feasible allocation $x_1 = (0.75, 0.75)$, $x_2 = (1.25, 1.25)$. Since $MRS_i = x_{2i}/x_{1i}$, we have $MRS_1 = MRS_2 = 1$: the allocation is interior and Pareto optimal. However,

$$u_1(\omega_1) = \sqrt{0.75} \approx 0.866 > 0.75 = u_1(x_1),$$

so the unit coalition $S = \{1\}$ blocks x with $y_1 = \omega_1$: $x \in OP(\mathcal{E})$ but $x \notin \mathcal{N}(\mathcal{E})$.

Core Outside Walrasian

In the same economy, the contract curve is the diagonal $x_1 = (c, c)$, $x_2 = (2 - c, 2 - c)$. Individual rationality imposes $c \geq \sqrt{0.75}$ and $2 - c \geq \sqrt{0.75}$, hence

$$\mathcal{N}(\mathcal{E}) = \{((c, c), (2 - c, 2 - c)) : c \in [\sqrt{0.75}, 2 - \sqrt{0.75}] \approx [0.866, 1.134]\}.$$

The unique Walrasian equilibrium is $p^* = (1, 1)$, $x^* = ((1, 1), (1, 1))$ (that is, $c = 1$). Taking $c = 0.90$, the allocation is in the core but not Walrasian: $\mathcal{N}(\mathcal{E}) \not\subseteq WE(\mathcal{E})$.

2.2.5 Equal Treatment and the Edgeworth–Debreu–Scarf Equivalence

Proposition 2.2.3 (Equal treatment in the core). *Consider a replicated economy with r agents of each type, agents of the same type having the same endowment and the same preferences, strictly convex, continuous, and strictly monotone. In any allocation belonging to the core, all agents of the same type receive the same bundle.*

Proof for two types and two replicas. Consider two types, 1 and 2, with two replicas each: 11, 12, 21, 22. Suppose x belongs to the core of the replicated economy and, for contradiction, that equal treatment fails for type 1:

$$x_{11} \neq x_{12}. \quad (28)$$

By completeness, order type 1's bundles so that

$$x_{11} \succsim_1 x_{12}. \quad (29)$$

For type 2, choose the worst-treated replica, say 22:

$$x_{21} \succsim_2 x_{22}. \quad (30)$$

Define the average bundles by type:

$$\bar{x}_1 = \frac{x_{11} + x_{12}}{2}, \quad \bar{x}_2 = \frac{x_{21} + x_{22}}{2}. \quad (31)$$

Since $x_{11} \neq x_{12}$, $x_{11} \succsim_1 x_{12}$ and preferences are strictly convex,

$$\bar{x}_1 \succ_1 x_{12}. \quad (32)$$

Since $x_{21} \succsim_2 x_{22}$ and preferences are convex,

$$\bar{x}_2 \succsim_2 x_{22}. \quad (33)$$

Consider the coalition formed by the worst representatives of each type:

$$S = \{12, 22\}. \quad (34)$$

The sum of the average bundles is

$$\begin{aligned} \bar{x}_1 + \bar{x}_2 &= \frac{1}{2}(x_{11} + x_{12} + x_{21} + x_{22}) \\ &= \frac{1}{2}(2\omega_1 + 2\omega_2) \\ &= \omega_1 + \omega_2 \\ &= \sum_{iq \in S} \omega_{iq}. \end{aligned}$$

Hence $(\bar{x}_1, \bar{x}_2)$ is feasible for S . By (32) and (33), the coalition obtains an internal allocation that leaves 12 strictly better off and 22 at least as well off; by Lemma 2.2.1 of conversion of weak to strong improvement, this allocation is slightly perturbed within S to leave both strictly better off. Thus S blocks x , contradicting $x \in \mathcal{N}(\mathcal{E})$. Hence $x_{11} = x_{12}$; the same argument holds for type 2. $\square$

Theorem 2.2.1 (Edgeworth–Debreu–Scarf). *As the number of replicas $r \rightarrow \infty$, the core of the replica economy contracts exactly to the set of Walrasian equilibrium allocations.*

Remark 2.2.2 (Intuition). In small economies, the core is wide (the entire individually rational segment of the contract curve). Replicas increase the blocking power of subcoalitions, which come to “isolate” agents with bargaining power; in the competitive limit, only the allocations that the price mechanism would sustain survive.

2.2.6 Connection between Intuition and Proof

Table 2.3: Correspondence between economic intuition and formal elements of the core.

Economic mechanism	Formal element
Social stability by free association	No coalition $S \neq \emptyset$ blocks x
Individual right of exit	$S = \{i\}$ blocks $x \iff \omega_i \succ_i x_i$
Threat of collective secession	$\sum_{i \in S} y_i = \sum_{i \in S} \omega_i, y_i \succ_i x_i \forall i \in S$
Core implies efficiency	The grand coalition $S = I$ would block any non-Pareto optimal a
Preferred bundle breaks the budget	$y_i \succ_i x_i^* \implies p^* \cdot y_i > p^* \cdot \omega_i$
Walrasian is in the core	Blocking would violate the coalitional budget feasibility
Symmetry in large economies	Identical agents receive the same bundle in the core
Competition eliminates bargaining	$r \rightarrow \infty: \mathcal{N} \rightarrow WE$

2.3 The Fundamental Theorems of Welfare with Production

2.3.1 The Basic Model with Production: Definitions

Consider a private property economy

$$\mathcal{E} = \left(\{(X_i, \succsim_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \{(\omega_i, \theta_{i1}, \dots, \theta_{iJ})\}_{i=1}^I \right),$$

where: L goods indexed by $\ell = 1, \dots, L$; I consumers with consumption sets $X_i \subset \mathbb{R}_+^L$ (closed and convex) and preferences $\succsim_i$ complete, transitive, continuous, and locally nonsatiated; J firms with production sets $Y_j \subset \mathbb{R}^L$ (negative inputs, positive outputs); endowments $\omega_i \in \mathbb{R}_+^L$ with aggregate endowment $\bar{\omega} = \sum_i \omega_i \gg 0$; and shareholdings $\theta_{ij} \in [0, 1]$ with $\sum_{i=1}^I \theta_{ij} = 1$ for each firm j .

Definition 2.3.1 (Feasible allocation, Def. 16.B.1). An **allocation** $(x, y) = (x_1, \dots, x_I, y_1, \dots, y_J)$ specifies a consumption vector $x_i \in X_i$ for each consumer and a

production vector $y_j \in Y_j$ for each firm. It is **feasible** if, for every good ℓ ,

$$\sum_{i=1}^I x_{\ell i} = \bar{\omega}_\ell + \sum_{j=1}^J y_{\ell j} \iff \sum_{i=1}^I x_i = \bar{\omega} + \sum_{j=1}^J y_j. \quad (1)$$

The set of feasible allocations is

$$A = \left\{ (x, y) \in \prod_{i=1}^I X_i \times \prod_{j=1}^J Y_j : \sum_{i=1}^I x_i = \bar{\omega} + \sum_{j=1}^J y_j \right\} \subset \mathbb{R}^{L(I+J)}.$$

Definition 2.3.2 (Pareto optimality, Def. 16.B.2). A feasible allocation $(x, y) \in A$ is **Pareto optimal (efficient)** if there is no $(x', y') \in A$ that Pareto-dominates it, that is, such that

$$x'_i \succsim_i x_i \quad \forall i \quad \text{and} \quad x'_k \succ_k x_k \quad \text{for some } k.$$

Definition 2.3.3 (Walrasian equilibrium, Def. 16.B.3). Given the private property economy, an allocation (x^*, y^*) and a price vector $p = (p_1, \dots, p_L)$ constitute a **Walrasian (competitive) equilibrium** if:

- (i) for every firm j , y_j^* maximizes profits in Y_j : $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$;
- (ii) for every consumer i , x_i^* is maximal for $\succsim_i$ in the budget set $\{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\}$;
- (iii) markets clear: $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

Definition 2.3.4 (Price equilibrium with transfers, Def. 16.B.4). Given an economy $(\{(X_i, \succsim_i)\}_i, \{Y_j\}_j, \bar{\omega})$, an allocation (x^*, y^*) and a price vector $p \neq 0$ constitute a **price equilibrium with transfers** if there exists a wealth assignment $(w_1, \dots, w_I)$ with

$$\sum_{i=1}^I w_i = p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^* \quad (2)$$

such that: (i) for every j , y_j^* maximizes profits in Y_j ; (ii) for every i , x_i^* is maximal for $\succsim_i$ in $\{x_i \in X_i : p \cdot x_i \leq w_i\}$; (iii) $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

Remark 2.3.1 (Walrasian is a special case). The Walrasian equilibrium is the special case where $w_i = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$. Indeed, using $\sum_i \theta_{ij} = 1$,

$$\sum_{i=1}^I w_i = p \cdot \bar{\omega} + \sum_{i=1}^I \sum_{j=1}^J \theta_{ij} (p \cdot y_j^*) = p \cdot \bar{\omega} + \sum_{j=1}^J (p \cdot y_j^*) \left(\sum_{i=1}^I \theta_{ij} \right) = p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^*,$$

so consistency (2) holds automatically. The definition with transfers captures price-taking behavior without any assumption about how wealth is determined.

2.3.2 Intuition, Mechanisms, and Motivation

The Three Pillars **Feasibility** defines the physical limits of reality: what society can consume is what it initially owns plus what its firms can produce. **Pareto optimality** establishes a minimum and consensual criterion of efficiency: an arrangement in which it is impossible to improve someone without harming another. **Walrasian equilibrium** provides the decentralized mechanism of resolution: a price vector that harmonizes firms' profit maximization and consumers' welfare maximization, clearing all markets without a central planner.

Remark 2.3.2 (Pareto is not distribution). The Pareto criterion does not concern distributional issues: in a pure exchange economy, an allocation that delivers the entire aggregate endowment to a single consumer with strongly monotone preferences is necessarily Pareto optimal.

Mechanisms **Firms:** under perfect competition, they take p as given and choose $y_j \in Y_j$ to maximize profit $p \cdot y_j$ (the value generated above the opportunity cost of inputs). **Consumers:** hold endowments ω_i and shares θ_{ij} ; their income is endogenous, $p \cdot \omega_i + \sum_j \theta_{ij}(p \cdot y_j^*)$, and they choose the best bundle within this budget. **Prices:** transmit information about relative scarcity — excess demand raises the price and induces production and rationing; excess supply depresses the price — until supply and demand match.

Implications The market works as a distributed computer: instead of a planner gathering private information about utilities and technologies, each agent only needs to know their own preferences/technologies and the price vector. This illuminates the design of wholesale energy markets (lower-cost plants produce, and consumption goes to those who value it most) and direct income transfer policies as opposed to price controls: subsidizing the price of a good distorts relative prices and generates deadweight loss, while *lump-sum* transfers preserve efficiency and correct distribution.

Example 2.3.1 (Space colony). In a self-sustaining biosphere on Mars, there are fixed endowments (water, oxygen, minerals), greenhouses and power plants (firms), and colonists with shareholdings (consumers). A plan is feasible if aggregate consumption does not exceed generation capacity plus initial stock; it is Pareto optimal if there is no technological or exchange rearrangement that gives more food to one colonist without making another cold; and it is a competitive equilibrium when a table of exchange rates (prices) makes it so that, seeking profit and utility individually, the quantities demanded coincide with those produced.

Example 2.3.2 (Water condominium). In an agricultural valley, the Walrasian equilibrium limits each farmer's consumption to the market value of inherited land; whoever inherited arid land is condemned to misery even with an efficient market. In the equilibrium with transfers, the council assigns net financial credits (w_i) to the poorest and taxes the richest; then, the free market allocates water with maximum efficiency, now under a socially acceptable distribution.

Remark 2.3.3 (What breaks). **Without local nonsatiation:** the budget constraint may hold with strict inequality ($p \cdot x_i^* < w_i$), wasting resources and causing the First Theorem to fail. **Without convexity:** with non-convex Y_j (increasing returns), profit maximization at competitive prices is ill-defined (profit explodes or the choice set empties); moreover, hyperplane separation fails, making the Second Theorem infeasible. **Without continuity:** aggregate demand loses upper semicontinuity, Kakutani's fixed point does not apply, and equilibrium may not exist. **Distortionary transfers:** if wealth depends on agents' decisions (quantity taxes), MRS and MRT diverge and efficiency breaks down.

2.3.3 Local Nonsatiation and Budget Lemmas

Definition 2.3.5 (Local nonsatiation, Def. 16.C.1). The relation $\succsim_i$ on X_i is **locally nonsatiated** if, for every $x_i \in X_i$ and every $\varepsilon > 0$, there exists $x'_i \in X_i$ with

$$\|x'_i - x_i\| \leq \varepsilon \quad \text{and} \quad x'_i \succ_i x_i.$$

Remark 2.3.4. If $\succsim_i$ is continuous and locally nonsatiated, every closed consumption set X_i must be unbounded; otherwise there would be a global (hence local) satiation point.

Lemma 2.3.1 (LNS implies budget equality). *If $\succsim_i$ is locally nonsatiated and x_i^* is maximal in $\{x_i \in X_i : p \cdot x_i \leq w_i\}$ with $p \neq 0$, then*

$$p \cdot x_i^* = w_i. \quad (3)$$

Proof. Suppose, for contradiction, $p \cdot x_i^* < w_i$. Define $\varepsilon = (w_i - p \cdot x_i^*)/\|p\| > 0$. By local nonsatiation, there exists $x'_i \in X_i$ with $\|x'_i - x_i^*\| \leq \varepsilon$ and $x'_i \succ_i x_i^*$. By Cauchy–Schwarz,

$$p \cdot (x'_i - x_i^*) \leq \|p\| \|x'_i - x_i^*\| \leq \|p\| \varepsilon = w_i - p \cdot x_i^*.$$

Adding $p \cdot x_i^*$ to both sides, $p \cdot x'_i \leq w_i$: bundle x'_i is accessible and strictly preferred, contradicting the maximality of x_i^* . Hence $p \cdot x_i^* = w_i$. $\square$

Lemma 2.3.2 (Revealed preference under LNS). *Under the conditions of Lemma 2.3.1:*

- (i) *if $x'_i \succ_i x_i^*$, then $p \cdot x'_i > w_i$;*
- (ii) *if $x'_i \succsim_i x_i^*$, then $p \cdot x'_i \geq w_i$.*

Proof. (i) If $p \cdot x'_i \leq w_i$, then x'_i belongs to the budget set; by maximality, $x_i^* \succsim_i x'_i$, contradicting $x'_i \succ_i x_i^*$ (asymmetry of strict preference).

(ii) Suppose $p \cdot x'_i < w_i$ and define $\varepsilon = (w_i - p \cdot x'_i)/\|p\| > 0$. By local nonsatiation, there exists x''_i with $\|x''_i - x'_i\| \leq \varepsilon$ and $x''_i \succ_i x'_i$. As in the previous lemma, Cauchy–Schwarz gives $p \cdot x''_i \leq w_i$, so x''_i is accessible. By transitivity, $x''_i \succ_i x'_i \succsim_i x_i^*$ implies $x''_i \succ_i x_i^*$, contradicting the maximality of x_i^* . Hence $p \cdot x'_i \geq w_i$. $\square$

2.3.4 First Fundamental Theorem of Welfare

Proposition 2.3.1 (First Theorem, Prop. 16.C.1). *If preferences are locally nonsatiated and (x^*, y^*, p) is a price equilibrium with transfers, then the allocation (x^*, y^*) is Pareto optimal. In particular, any Walrasian equilibrium allocation is Pareto optimal.*

Proof. Suppose, for contradiction, that (x^*, y^*) is not Pareto optimal. Then there exists $(x', y') \in A$ with

$$x'_i \succsim_i x_i^* \quad \forall i \quad \text{and} \quad x'_k \succ_k x_k^* \quad \text{for some } k. \quad (4)$$

By feasibility of (x', y') ,

$$\sum_{i=1}^I x'_i = \bar{\omega} + \sum_{j=1}^J y'_j. \quad (5)$$

Multiplying (5) by p and using the linearity of the inner product,

$$\sum_{i=1}^I p \cdot x'_i = p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y'_j. \quad (6)$$

Applying Lemma 2.3.2 of revealed preference to (4): for consumer k , $x'_k \succ_k x_k^*$ implies $p \cdot x'_k > w_k$; for $i \neq k$, $x'_i \sim_i x_i^*$ implies $p \cdot x'_i \geq w_i$. Summing,

$$\sum_{i=1}^I p \cdot x'_i > \sum_{i=1}^I w_i = p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^*, \quad (7)$$

where the equality uses wealth consistency (2).

On the other hand, since y_j^* maximizes profits and $y'_j \in Y_j$,

$$p \cdot y'_j \leq p \cdot y_j^* \quad \forall j. \quad (8)$$

Summing (8) over j and adding $p \cdot \bar{\omega}$ to both sides,

$$p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y'_j \leq p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^*. \quad (9)$$

Substituting (6) into the left side of (9),

$$\sum_{i=1}^I p \cdot x'_i \leq p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^*. \quad (10)$$

Relations (7) and (10) are contradictory (one real number cannot be strictly greater and, at the same time, less than or equal to another). Hence (x^*, y^*) is Pareto optimal. Since every Walrasian equilibrium is an equilibrium with transfers (Observation 2.3.1), the result holds in particular for it. $\square$

Remark 2.3.5 (What breaks in the proof). **Without LNS:** the consumer may leave budget slack ($p \cdot x_i^* < w_i$), the revealed preference lemma fails, and equilibrium allocations may waste resources. **With market power:** if agents influence p , MRS diverge from MRT and deadweight loss arises. **With externalities:** the aggregate feasibility constraint no longer coincides with the sum of budget constraints valued at p , and the central contradiction of the proof does not hold.

Example 2.3.3 (Cap-and-trade). The government sets the aggregate pollution limit and distributes initial rights; firms freely trade the quotas. By the First Theorem, under competitive prices the emission reduction is done by firms with the lowest marginal abatement cost, achieving the environmental goal at the lowest possible cost to society — efficiency guaranteed without central planning.

2.3.5 Second Fundamental Theorem of Welfare

Economic Intuition

The Essence The First Theorem guarantees that competitive markets produce **efficiency**, but are blind to **equity**: a Walrasian equilibrium can coexist with extremely

Table 2.4: First Theorem: correspondence between intuition and proof.

Economic mechanism	Formal element	Reference
Physical limits of reality	Resource conservation: $\sum_i x_i = \bar{\omega} + \sum_j y_j$	Def. 2.3.1, eqs. (5)–(6)
Endogenous income / consistency	$\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$	Def. 2.3.4, eq. (2)
Firm maximizes profit	$p \cdot y'_j \leq p \cdot y_j^*$ for $y'_j \in Y_j$	Def. 2.3.4(i), eqs. (8)–(9)
Consumer maximizes welfare	Strictly better bundle costs strictly more	Lemma 2.3.2, eq. (7)
Total spending of income (LNS)	$p \cdot x_i^* = w_i$	Lemma 2.3.1, eq. (3)
Efficiency without waste	Contradiction between (7) and (10)	Prop. 2.3.1

unequal wealth distributions and still be Pareto efficient. The **Second Fundamental Theorem of Welfare** (Prop. 2.3.3) resolves the impasse: under convexity of preferences and technologies, **any** Pareto optimal allocation desired by society can be achieved as a decentralized market equilibrium — it suffices for the government to redistribute initial wealth in a non-distortionary way (*lump-sum*) and let prices adjust.

Example 2.3.4 (Cooperative picnic). Each friend brings their endowment (breads or olives) and free negotiation generates an efficient but possibly unfair outcome. The organizer (social planner) has two options: set prices and dictate consumptions (inefficient and coercive), or — as the Second Theorem guarantees — redistribute the food (or purchasing power tokens) **before** the picnic and let the friends negotiate freely at the prices that form. The final result is exactly the desired fair division, allocated with maximum efficiency.

Mechanisms **Consumers (expenditure minimization)**: given p , each consumer's behavior can be read through the dual: x_i^* is the lowest-cost bundle that reaches the utility level promised by the target optimum; under convexity, a tangent budget hyperplane maps this choice. **Firms (profit maximization)**: they take p as given and choose y_j^* that maximizes $p \cdot y_j$ in Y_j , producing on the possibility frontier. **Government (fixed-sum transfers)**: assigns wealth w_i ex-ante with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$; since they do not depend on marginal choices, they do not alter MRS/MRT and preserve the price signals that coordinate efficiency.

Convexity and Quasiequilibrium The convexity of Y_j and $\succsim_i$ is what allows applying the **Separating Hyperplane Theorem**: a “sheet of paper” (the vector p) separates what the economy can produce from what consumers would demand to be strictly better off than the target optimum. Since the separation of convex sets provides only **weak** inequalities, the natural result is a **quasiequilibrium**: strictly preferred bundles cost *at least* w_i ($p \cdot x_i \geq w_i$), and not strictly more. Only under additional conditions (strictly positive wealth / cheaper consumption point) does the quasiequilibrium elevate to a full Walrasian equilibrium. Without convexity (increasing returns, non-convex preferences), separation fails and competitive decentralization is unfeasible.

Implications The central implication is the **institutional separation between efficiency and distribution**: direct interventions in the price system (price controls, freezes,

generalized subsidies) are inefficient, as they distort opportunity costs and generate dead-weight loss; the government should let prices float and redistribute purchasing power through direct transfers (basic income, vouchers, ex-ante progressive tax).

Example 2.3.5 (Cooking gas subsidy). Instead of capping the price of gas for everyone (discouraging refining and causing shortages), the Second Theorem recommends maintaining the free price in the competitive market and compensating vulnerable families with a monthly cash transfer: they buy gas at the real cost price or substitute it with electricity efficiently, without allocative distortions.

Quasiequilibrium: Definition and Properties

The Transfers Embedded in w In the equilibrium with transfers (Def. 2.3.4), the planner does not physically redistribute endowments: they directly assign final wealth levels w_i , and the net transfers are **embedded** in w . Indeed, the “natural” market wealth of consumer i would be

$$\bar{w}_i(p) = p \cdot \omega_i + \sum_{j=1}^J \theta_{ij} (p \cdot y_j^*);$$

the implicit transfer is $T_i = w_i - \bar{w}_i(p)$, and consistency $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$ is equivalent to $\sum_i T_i = 0$: the taxes collected exactly finance the subsidies paid. In short: the sum of final wealth (with transfers) equals the market value of the aggregate endowment plus optimal production — the planner’s budget is balanced.

Criticisms of the Definition (and Why Use It Anyway) Three discomforts: (i) **circularity** — to calibrate w_i the planner needs to know p in advance, but it is the transfers that should guide the market to the right prices; (ii) **informational miracle** — calculating w requires knowing the preferences and technologies of all agents, making the market dispensable; (iii) **lump-sum fiction** — fixed-sum taxes do not exist in practice, as the government does not observe agents’ intrinsic productivity. Still, the definition is the ideal “scaffolding” for the Second Theorem: the separating hyperplane delivers p and $w_i = p \cdot x_i^*$ directly and cleanly, and a second stage of the proof converts this abstract vector back into physical endowments ω'_i that consumers sell in the market.

Definition 2.3.6 (Price quasiequilibrium with transfers, Def. 16.D.1). A feasible allocation (x^*, y^*) and a price vector $p \neq 0$ constitute a **price quasiequilibrium with transfers** if there exists $(w_1, \dots, w_I)$ with $\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$ such that:

- (i) for every firm j : $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$;
- (ii) for every consumer i : if $x_i \succ_i x_i^*$, then $p \cdot x_i \geq w_i$;
- (iii) $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^*$.

Remark 2.3.6 (Equilibrium versus quasiequilibrium: the sign of the inequality). In equilibrium (Def. 2.3.4), x_i^* is maximal in $\{x_i : p \cdot x_i \leq w_i\}$; by the revealed preference lemma this is equivalent to $x_i \succ_i x_i^* \implies p \cdot x_i > w_i$ (**strict** inequality). In quasiequilibrium only $p \cdot x_i \geq w_i$ is required (**weak** inequality): the consumer may prefer a bundle that costs exactly w_i . In the dual language, equilibrium is the primal problem (utility maximization), while quasiequilibrium is the dual problem (expenditure minimization):

$$x_i^* \in \arg \min_{x_i \in X_i} \{p \cdot x_i : x_i \succsim_i x_i^*\}.$$

The weakness is not optional: Minkowski's Theorem only guarantees weak separation of convex sets, so “for free” one only obtains a quasiequilibrium.

Remark 2.3.7 (The zero-wealth trap). If $w_i = \min p \cdot X_i$ (for example $w_i = 0$ with $X_i = \mathbb{R}_+^L$ and $p \gg 0$), the budget set collapses to $\{0\}$ and $x_i^* = 0$. Any strictly preferred bundle costs $p \cdot x_i > 0 \geq w_i$, satisfying condition (ii) of quasiequilibrium vacuously — but the agent is not maximizing utility in the conventional sense: with $\epsilon > 0$ of income they would jump to a much better bundle. At this “non-survival” point the equivalence between utility maximization and expenditure minimization breaks.

From Quasiequilibrium to Equilibrium: The Cheaper Point Lemma The condition that rescues the equivalence is the existence of a bundle strictly cheaper than the agent's wealth:

$$w_i > \min p \cdot X_i \quad \forall i.$$

Proposition 2.3.2 (Elevation of quasiequilibrium to equilibrium). *If X_i is convex, $\succsim_i$ is continuous, and $w_i > \min p \cdot X_i$ for all i , then every price quasiequilibrium with transfers is a price equilibrium with transfers.*

Proof. Suppose, for contradiction, that there exists x_i with $x_i \succ_i x_i^*$ and $p \cdot x_i \leq w_i$. By the quasiequilibrium condition, $p \cdot x_i \geq w_i$; hence $p \cdot x_i = w_i$. By the cheaper point hypothesis, there exists $x'_i \in X_i$ with $p \cdot x'_i < w_i$. For $\alpha \in (0, 1]$, define $x_i(\alpha) = (1 - \alpha)x_i + \alpha x'_i$; by convexity of X_i , $x_i(\alpha) \in X_i$, and by linearity of the inner product,

$$p \cdot x_i(\alpha) = (1 - \alpha)(p \cdot x_i) + \alpha(p \cdot x'_i) < (1 - \alpha)w_i + \alpha w_i = w_i.$$

Since $V_i(x_i^*) = \{z \in X_i : z \succ_i x_i^*\}$ is open (continuity) and $x_i(\alpha) \rightarrow x_i \in V_i(x_i^*)$ as $\alpha \rightarrow 0$, there exists $\alpha^* > 0$ with $x_i(\alpha^*) \succ_i x_i^*$. The quasiequilibrium condition then requires $p \cdot x_i(\alpha^*) \geq w_i$, contradicting $p \cdot x_i(\alpha^*) < w_i$. Hence $x_i \succ_i x_i^* \implies p \cdot x_i > w_i$, and x_i^* is maximal in the budget set. $\square$

Remark 2.3.8 (Logical asymmetry). The direction equilibrium $\implies$ quasiequilibrium always holds, without additional hypotheses (the strict inequality implies the weak one). The converse is conditional: without the cheaper point, Equilibrium $\subsetneq$ Quasiequilibrium; with it, the concepts coincide. For $X_i = \mathbb{R}_+^L$ and $p \neq 0$, we have $\min p \cdot X_i = 0$, so the condition reduces to $w_i > 0$ — satisfied in the vast majority of applications, which makes every positive-wealth quasiequilibrium immediately elevate to a full Walrasian equilibrium.

Remark 2.3.9 (What breaks). **Without convexity of Y_j :** $Y = \sum_j Y_j$ ceases to be convex and the separating hyperplane does not apply; with increasing returns, profit maximization at competitive prices is ill-defined. **Without convexity of $\succsim_i$:** the strictly preferred set V ceases to be convex and consumers jump to extremes of the constraint, abandoning x_i^* . **Without LNS:** the quasiequilibrium condition does not imply $p \cdot x_i^* = w_i$, breaking aggregate budget consistency. **Zero-wealth trap:** if $w_i = \min p \cdot X_i$, condition (ii) does not guarantee that x_i^* maximizes utility under $p \cdot x_i \leq w_i$; hence the theorem only delivers a quasiequilibrium.

Separating Hyperplane Theorem

Theorem 2.3.1 (Separation of convex sets (Minkowski)). *Let $A, B \subset \mathbb{R}^L$ be convex, nonempty, and disjoint. Then there exist $p \in \mathbb{R}^L \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that*

$$p \cdot a \geq \alpha \quad \forall a \in A, \quad p \cdot b \leq \alpha \quad \forall b \in B.$$

Proof of the Second Theorem

Proposition 2.3.3 (Second Fundamental Theorem of Welfare). *Let (x^*, y^*) be a Pareto optimal allocation in an economy where all Y_j and all preferences $\succsim_i$ are convex, continuous, and locally nonsatiated. Then there exist $p \neq 0$ and wealth $(w_1, \dots, w_I)$ such that (x^*, y^*, p) is a price quasiequilibrium with transfers.*

Proof. **Step 1: convex sets to separate.** For each consumer, define the strictly preferred set

$$V_i(x_i^*) = \{x_i \in X_i : x_i \succ_i x_i^*\},$$

which is convex (convexity and continuity of $\succsim_i$). The aggregate set $V = \sum_i V_i(x_i^*)$ is convex (Minkowski sum of convex sets). Also define $Y + \{\bar{\omega}\} = \sum_j Y_j + \{\bar{\omega}\}$, convex since each Y_j is convex and translation preserves convexity.

Step 2: disjointness. Suppose $z \in V \cap (Y + \{\bar{\omega}\})$. Then there exist $x'_i \succ_i x_i^*$ and $y'_j \in Y_j$ with

$$\sum_{i=1}^I x'_i = z = \bar{\omega} + \sum_{j=1}^J y'_j, \quad (11)$$

so (x', y') is feasible and Pareto-dominates (x^*, y^*) , contradicting optimality. Therefore

$$V \cap (Y + \{\bar{\omega}\}) = \emptyset.$$

Step 3: separation. By Theorem 2.3.1 of Minkowski, there exist $p \neq 0$ and α with

$$p \cdot v \geq \alpha \quad \forall v \in V, \quad (12)$$

$$p \cdot z \leq \alpha \quad \forall z \in Y + \{\bar{\omega}\}. \quad (13)$$

Step 4: the price at the optimum. By LNS, $x_i^* \in \overline{V_i(x_i^*)}$ for each i ; hence $x^* = \sum_i x_i^* \in \overline{V}$ and, by continuity, $p \cdot x^* \geq \alpha$. Since (x^*, y^*) is feasible, $\sum_i x_i^* = \bar{\omega} + \sum_j y_j^* \in Y + \{\bar{\omega}\}$, and (13) gives $p \cdot x^* \leq \alpha$. Therefore

$$p \cdot x^* = p \cdot \left(\bar{\omega} + \sum_{j=1}^J y_j^* \right) = \alpha. \quad (14)$$

Step 5: profit maximization. For any $y = \sum_j y_j \in Y$, (13) and (14) give $p \cdot (y + \bar{\omega}) \leq p \cdot (\sum_j y_j^* + \bar{\omega})$, hence

$$\sum_{j=1}^J p \cdot y_j \leq \sum_{j=1}^J p \cdot y_j^*. \quad (15)$$

If some firm f had $y'_f \in Y_f$ with $p \cdot y'_f > p \cdot y_f^*$, the aggregate plan $\tilde{y} = y'_f + \sum_{j \neq f} y_j^*$ would violate (15). Hence

$$p \cdot y_j \leq p \cdot y_j^* \quad \forall y_j \in Y_j, \quad \forall j. \quad (16)$$

Step 6: expenditure minimization and wealth. If $x_i \succ_i x_i^*$, then $\tilde{v} = x_i + \sum_{k \neq i} x_k^* \in \bar{V}$; by (12) and (14), $p \cdot \tilde{v} \geq p \cdot x^*$, and canceling the common terms,

$$p \cdot x_i \geq p \cdot x_i^*. \quad (17)$$

Define $w_i = p \cdot x_i^*$; then $x_i \succ_i x_i^* \implies p \cdot x_i \geq w_i$ (condition ii). Summing and using feasibility,

$$\sum_{i=1}^I w_i = p \cdot \sum_{i=1}^I x_i^* = p \cdot \bar{\omega} + \sum_{j=1}^J p \cdot y_j^*, \quad (18)$$

the financial consistency. With (16), (17)–(18) and feasibility (iii), (x^*, y^*, p) and (w_i) form a price quasiequilibrium with transfers. $\square$

From Quasiequilibrium to Walrasian Equilibrium

Remark 2.3.10 (Cheaper point condition). If, in addition to the hypotheses above, each wealth satisfies $w_i > \min p \cdot X_i$ (there is a strictly cheaper bundle), the quasiequilibrium is a full **equilibrium**. Indeed, suppose $x_i \succsim_i x_i^*$ with $p \cdot x_i \leq w_i$; taking x_i^0 with $p \cdot x_i^0 < w_i$, the combinations $x_i(\lambda) = (1 - \lambda)x_i + \lambda x_i^0$ have $p \cdot x_i(\lambda) < w_i$ and, by convexity and continuity, bundles x'_i arbitrarily close to $x_i(\lambda)$ satisfy $x'_i \succ_i x_i^*$ with $p \cdot x'_i < w_i$, contradicting (ii). Hence x_i^* is maximal in the budget set.

Table 2.5: Second Theorem: correspondence between intuition and proof.

Economic mechanism	Formal element	Reference
Target distributive justice	Pareto optimal allocation (x^*, y^*) chosen ex-ante	Prem
Price separability	Hyperplane separating V (preferred) from $Y + \{\bar{\omega}\}$ (feasible)	Eqs.
Productive maximization (firms)	$p \cdot y_j \leq p \cdot y_j^*$ for $y_j \in Y_j$	Eqs.
Dual consumer behavior	$x_i \succ_i x_i^* \implies p \cdot x_i \geq w_i$ (expenditure minimization)	Eq. (
Aggregate accounting consistency	$\sum_i w_i = p \cdot \bar{\omega} + \sum_j p \cdot y_j^*$	Eq. (
Survival layer	$w_i > \min p \cdot X_i$ elevates quasiequilibrium to equilibrium	Obs.

2.4 Existence of Competitive Equilibrium

2.4.1 Price Normalization and the Reduced-Form Economy

Commodity and Price Space Consider an economy with L goods. The commodity space is $X = \mathbb{R}^L$; a bundle is $x = (x_1, \dots, x_L)$ with $x_\ell \geq 0$; prices are $p = (p_1, \dots, p_L) \in \mathbb{R}_+^L$.

Normalization Since competitive demands depend only on relative prices (zero-degree homogeneity), we normalize $\sum_\ell p_\ell = 1$ on the price simplex

$$\Delta_+ = \left\{ p \in \mathbb{R}_+^L : \sum_{\ell=1}^L p_\ell = 1 \right\}. \quad (1)$$

Since $p_L = 1 - \sum_{\ell=1}^{L-1} p_\ell$, only $L - 1$ coordinates are independent.

Why Two Goods? For $L = 2$, $p = (p_1, 1 - p_1)$ with $p_1 \in [0, 1]$: the simplex becomes the interval $[0, 1]$ and the price problem is one-dimensional, which makes the Intermediate Value Theorem (IVT) available as an existence tool. The one-dimensional case also isolates the logic of fixed-point arguments: for $f : [0, 1] \rightarrow [0, 1]$ continuous, $g(x) = f(x) - x$ satisfies $g(0) \geq 0$ and $g(1) \leq 0$; by IVT there exists x^* with $g(x^*) = 0$, i.e., $f(x^*) = x^*$ — Brouwer’s intuition in one dimension.

Excess Demand We describe the economy in reduced form by $Z : \Delta_+ \rightarrow \mathbb{R}^L$, $Z(p) = (z_1(p), \dots, z_L(p))$, with $z_\ell(p) = d_\ell(p) - s_\ell(p)$. The sign of z_ℓ is interpreted as: $z_\ell(p) > 0$ excess demand; $z_\ell(p) = 0$ market clears; $z_\ell(p) < 0$ excess supply. For $L = 2$, each component becomes a function of one variable, $\tilde{z}_\ell : [0, 1] \rightarrow \mathbb{R}$.

2.4.2 Competitive Equilibrium, Free Goods, and Walras’ Law

Definition 2.4.1 (Competitive equilibrium). A normalized price vector $p^* \in \Delta_+$ is an **equilibrium** if, for every good ℓ ,

$$z_\ell(p^*) \leq 0, \quad \text{and} \quad p_\ell^* > 0 \implies z_\ell(p^*) = 0. \quad (2)$$

Remark 2.4.1 (Free goods). If $p_\ell^* = 0$, market ℓ may not clear: $p_\ell^* = 0$ with $z_\ell(p^*) = -5$ is compatible with equilibrium (surplus of a free good). But $p_\ell^* = 0$ with $z_\ell(p^*) = +5$ is not: a zero-price good can be in excess supply, but there can never be unmet excess demand.

Proposition 2.4.1 (Walras’ Law). For every normalized price $p \in \Delta_+$,

$$p \cdot Z(p) = \sum_{\ell=1}^L p_\ell z_\ell(p) = 0. \quad (3)$$

Pure exchange economy. If preferences are monotone or locally nonsatiated, each consumer’s budget constraint holds with equality at the optimum:

$$p \cdot x_i(p, p \cdot \omega_i) = p \cdot \omega_i. \quad (4)$$

Rewriting in terms of individual excess demand $z_i(p) = x_i(p, p \cdot \omega_i) - \omega_i$,

$$\sum_{\ell=1}^L p_\ell (x_{i\ell}(p, p \cdot \omega_i) - \omega_{i\ell}) = 0 \implies p \cdot z_i(p) = 0. \quad (5)$$

Summing over all I consumers and reversing the order of summation (finite sums),

$$\sum_{\ell=1}^L p_\ell \underbrace{\sum_{i=1}^I (x_{i\ell}(p, p \cdot \omega_i) - \omega_{i\ell})}_{z_\ell(p)} = 0. \quad (6)$$

Hence $\sum_\ell p_\ell z_\ell(p) = 0$, i.e., $p \cdot Z(p) = 0$ for all $p \in \Delta_+$. $\square$

Economy with production. With firms and private property, consumer i ’s wealth includes dividends:

$$w_i(p) = p \cdot \omega_i + \sum_{j=1}^J \theta_{ij} (p \cdot y_j^*), \quad (7)$$

and, under nonsatiation, $p \cdot x_i(p, w_i(p)) = w_i(p)$. Summing over i and using $\sum_i \theta_{ij} = 1$,

$$\sum_{i=1}^I p \cdot x_i = \sum_{i=1}^I p \cdot \omega_i + \sum_{j=1}^J (p \cdot y_j^*) \left(\sum_{i=1}^I \theta_{ij} \right) = \sum_{i=1}^I p \cdot \omega_i + \sum_{j=1}^J p \cdot y_j^*. \quad (8)$$

Rearranging,

$$p \cdot \left[\sum_{i=1}^I x_i - \sum_{i=1}^I \omega_i - \sum_{j=1}^J y_j^* \right] = 0 \implies p \cdot Z(p) = 0. \quad (9)$$

□

Proposition 2.4.2 (Consequence of Walras' Law). *If $z_\ell(p) \leq 0$ for all ℓ , then p is an equilibrium price vector.*

Proof. Since $p_\ell \geq 0$ and $z_\ell(p) \leq 0$, each term $p_\ell z_\ell(p) \leq 0$. But Walras' Law requires $\sum_\ell p_\ell z_\ell(p) = 0$; a sum of non-positive terms that results in zero forces each term to be null: $p_\ell z_\ell(p) = 0$ for all ℓ . Hence $p_\ell > 0 \implies z_\ell(p) = 0$. Thus, to prove existence it suffices to show that demand does not exceed supply in any market — Walras' Law takes care of the rest. □

Remark 2.4.2 (Continuity). Each component $z_\ell : \Delta_+ \rightarrow \mathbb{R}$ is continuous (inherited from the continuity of individual demands via the Maximum Theorem). For $L = 2$, $\tilde{z}_\ell : [0, 1] \rightarrow \mathbb{R}$ is a continuous function of one variable — the setting for IVT.

2.4.3 Existence with Two Goods: The Arrow–Hahn Theorem

For $L = 2$, Walras' Law becomes

$$p_1 \tilde{z}_1(p_1) + (1 - p_1) \tilde{z}_2(p_1) = 0. \quad (10)$$

Theorem 2.4.1 (Arrow–Hahn). *Let $Z : \Delta_+ \rightarrow \mathbb{R}^2$ be continuous and satisfy Walras' Law. Then there exists a normalized price vector $p^* = (p_1^*, 1 - p_1^*) \in \Delta_+$ that is a competitive equilibrium.*

Proof. For $L = 2$, we write $p = (p_1, 1 - p_1)$ with $p_1 \in [0, 1]$ and $Z(p) = (\tilde{z}_1(p_1), \tilde{z}_2(p_1))$; Walras' Law becomes

$$p_1 \tilde{z}_1(p_1) + (1 - p_1) \tilde{z}_2(p_1) = 0. \quad (10)$$

By the consequence of Walras' Law, it suffices to find p_1^* with $\tilde{z}_1(p_1^*) \leq 0$ and $\tilde{z}_2(p_1^*) \leq 0$. The proof distinguishes boundary equilibria and an interior zero obtained by IVT.

Step 1 (first boundary case). Suppose first that $\tilde{z}_1(0) \leq 0$. At $p_1 = 0$, i.e., $p = (0, 1)$, Walras' Law (10) gives $0 \cdot \tilde{z}_1(0) + 1 \cdot \tilde{z}_2(0) = 0$, hence

$$\tilde{z}_2(0) = 0. \quad (11)$$

Combined with the hypothesis, $\tilde{z}_1(0) \leq 0$ and $\tilde{z}_2(0) = 0$: good 2, with strictly positive price, clears exactly; good 1, with zero price, satisfies $\tilde{z}_1(0) \leq 0$ (clears if $= 0$, or exhibits excess supply of the free good if < 0 , but never excess demand). Hence $p^* = (0, 1)$ is a competitive equilibrium.

Step 2 (second boundary case). Suppose instead that $\tilde{z}_2(1) \leq 0$. At $p_1 = 1$, $p = (1, 0)$, Walras' Law gives $1 \cdot \tilde{z}_1(1) + 0 \cdot \tilde{z}_2(1) = 0$, hence $\tilde{z}_1(1) = 0$. Combined with $\tilde{z}_2(1) \leq 0$, we obtain $p^* = (1, 0)$ as a competitive equilibrium.

Step 3 (when there is no boundary equilibrium). We have established that $\tilde{z}_1(0) \leq 0 \implies p^* = (0, 1)$ equilibrium, and $\tilde{z}_2(1) \leq 0 \implies p^* = (1, 0)$ equilibrium. Therefore, if neither boundary case applies, the negations necessarily hold:

$$\tilde{z}_1(0) > 0 \quad \text{and} \quad \tilde{z}_2(1) > 0. \quad (12)$$

We will show that these two inequalities force an interior equilibrium.

Step 4 (continuity near $p_1 = 1$). We know that $\tilde{z}_2(1) > 0$. Since $\tilde{z}_2 : [0, 1] \rightarrow \mathbb{R}$ is continuous, positivity persists in a neighborhood of 1: there exists $\delta \in (0, 1)$ such that

$$\tilde{z}_2(p_1) > 0 \quad \forall p_1 \in (1 - \delta, 1). \quad (13)$$

For all p_1 in this interval, $p_1 > 0$ and $1 - p_1 > 0$, so we can use Walras' Law to determine the sign of $\tilde{z}_1(p_1)$.

Step 5 (Walras forces the sign of $\tilde{z}_1$). Take any $p_1 \in (1 - \delta, 1)$. Walras' Law (10) gives $p_1 \tilde{z}_1(p_1) = -(1 - p_1) \tilde{z}_2(p_1)$. Since $1 - p_1 > 0$ and $\tilde{z}_2(p_1) > 0$, the right side is strictly negative; hence $p_1 \tilde{z}_1(p_1) < 0$ and, since $p_1 > 0$,

$$\tilde{z}_1(p_1) < 0 \quad \forall p_1 \in (1 - \delta, 1). \quad (14)$$

Step 6 (sign change). Choose any $\hat{p}_1 \in (1 - \delta, 1)$. From Step 3, $\tilde{z}_1(0) > 0$; from Step 5, $\tilde{z}_1(\hat{p}_1) < 0$. Thus the continuous function $\tilde{z}_1$ takes opposite signs at the endpoints of the interval $[0, \hat{p}_1]$:

$$\tilde{z}_1(0) > 0 > \tilde{z}_1(\hat{p}_1). \quad (15)$$

Step 7 (Intermediate Value Theorem). Since $\tilde{z}_1 : [0, 1] \rightarrow \mathbb{R}$ is continuous, by IVT applied to $[0, \hat{p}_1]$ there exists $p_1^* \in (0, \hat{p}_1)$ such that

$$\tilde{z}_1(p_1^*) = 0. \quad (16)$$

Since $0 < p_1^* < \hat{p}_1 < 1$, both price components are strictly positive: $p_1^* > 0$ and $1 - p_1^* > 0$.

Step 8 (the second market clears). Evaluating Walras' Law (10) at p_1^* : $p_1^* \tilde{z}_1(p_1^*) + (1 - p_1^*) \tilde{z}_2(p_1^*) = 0$. Since $\tilde{z}_1(p_1^*) = 0$ by (16), we have $(1 - p_1^*) \tilde{z}_2(p_1^*) = 0$; and since $1 - p_1^* > 0$,

$$\tilde{z}_2(p_1^*) = 0. \quad (17)$$

Step 9 (conclusion). Define $p^* = (p_1^*, 1 - p_1^*)$. Since $p_1^* \in (0, 1)$, we have $p^* \in \Delta_+$ and $p^* \gg 0$. Moreover, by (16) and (17), $\tilde{z}_1(p_1^*) = \tilde{z}_2(p_1^*) = 0$: both markets clear at strictly positive prices. Hence p^* is a competitive equilibrium. $\square$

Remark 2.4.3 (What the proof did). (1) Checked whether an equilibrium already existed at a boundary price ($p_1 = 0$ or $p_1 = 1$). (2) In the negative, obtained $\tilde{z}_1(0) > 0$ and $\tilde{z}_2(1) > 0$. (3) Continuity preserved the second inequality near $p_1 = 1$; Walras' Law then forced $\tilde{z}_1(p_1) < 0$ in this neighborhood, producing the sign change. (4) IVT guaranteed p_1^* with $\tilde{z}_1(p_1^*) = 0$, and Walras' Law cleared the second market. In short: economic constraints (Walras) create the sign change; continuity guarantees the zero.

2.4.4 Existence for L Goods: Brouwer's Theorem and the Nikaido–Uzawa Construction

The proof of the Arrow–Hahn Theorem for $L = 2$ uses the Intermediate Value Theorem, which is essentially Brouwer's Fixed Point Theorem in one dimension. For L goods, the simplex Δ_+ is a compact convex set of dimension $L - 1$, and we need the multidimensional version of the fixed point theorem. This is the famous **General Equilibrium Existence proof via Brouwer's Fixed Point Theorem** (often associated with the Nikaido–Uzawa construction).

Economic Intuition: Walrasian Tâtonnement

Before entering topology, we need to understand the **physics** behind the price adjustment function $f(p)$.

Imagine there is a **Walrasian Auctioneer** coordinating the market. Their rule of conduct is extremely intuitive:

- If a good ℓ exhibits excess demand ($z_\ell(p) > 0$), society wants to consume more than the existing endowment. The auctioneer receives a signal to **push the price of this good up** ($p_\ell \mapsto p_\ell + z_\ell^+(p)$).
- If a good ℓ is in equilibrium or excess supply ($z_\ell(p) \leq 0$), there is no scarcity pressure. The auctioneer **makes no positive adjustment** ($p_\ell \mapsto p_\ell + 0$).

Since this crude adjustment takes the price vector out of the simplex Δ_+ (the sum of the new prices would no longer be 1), the auctioneer normalizes by dividing everything by the sum of the new prices, represented by the scale variable $\alpha(p)$.

Competitive equilibrium will occur exactly when the auctioneer looks at the market, runs their adjustment rule, and **prices no longer move**. That is, when we find a fixed point $f(p^*) = p^*$.

Structure and Rigor of the Proof

Let there be a pure exchange economy with L goods where the aggregate excess demand function $Z : \Delta_+ \rightarrow \mathbb{R}^L$ satisfies:

1. **Continuity:** $Z(\cdot)$ is continuous on Δ_+ .
2. **Zero-Degree Homogeneity:** $Z(\lambda p) = Z(p)$ for all $\lambda > 0$.
3. **Walras' Law:** $p \cdot Z(p) = 0$ for all $p \in \Delta_+$.

Step 1: Price Normalization (the compact domain) Due to zero-degree homogeneity, only relative prices matter. We restrict the domain to the unit simplex:

$$\Delta_+ = \left\{ p \in \mathbb{R}_+^L : \sum_{\ell=1}^L p_\ell = 1 \right\}.$$

The set Δ_+ is **nonempty**, **convex**, and **compact** (closed and bounded in $\mathbb{R}^L$). This is the standard domain for applying Brouwer's Fixed Point Theorem.

Step 2: Construction of the Continuous Adjustment Function $f(p)$ For each good ℓ , we isolate only the positive part of excess demand:

$$z_\ell^+(p) = \max\{z_\ell(p), 0\}.$$

We define the renormalization factor $\alpha(p)$ as the sum of adjusted prices:

$$\alpha(p) = \sum_{\ell=1}^L (p_\ell + z_\ell^+(p)) = \underbrace{\sum_{\ell=1}^L p_\ell}_{=1} + \sum_{\ell=1}^L z_\ell^+(p) = 1 + \sum_{\ell=1}^L z_\ell^+(p).$$

Since $z_\ell^+(p) \geq 0$, we have $\alpha(p) \geq 1 > 0$ for all $p \in \Delta_+$. The guarantee that $\alpha(p) > 0$ is crucial, as it eliminates any risk of division by zero.

We define the adjustment mapping $f : \Delta_+ \rightarrow \Delta_+$ component by component:

$$f_\ell(p) = \frac{p_\ell + z_\ell^+(p)}{\alpha(p)} = \frac{p_\ell + z_\ell^+(p)}{1 + \sum_{k=1}^L z_k^+(p)}.$$

- **Feasibility:** Since $p_\ell \geq 0$ and $z_\ell^+(p) \geq 0$, we have $f_\ell(p) \geq 0$. Moreover, $\sum_\ell f_\ell(p) = \frac{\sum p_\ell + \sum z_\ell^+(p)}{\alpha(p)} = \frac{\alpha(p)}{\alpha(p)} = 1$. Hence, $f(p) \in \Delta_+$.
- **Continuity:** Since $Z(p)$ is continuous, $z^+(p)$ and $\alpha(p)$ are also continuous. Since $\alpha(p) \geq 1$, the quotient $f(p)$ is a continuous function.

Step 3: Application of Brouwer's Theorem (existence of the fixed point) Since Δ_+ is a nonempty, compact, and convex set, and $f : \Delta_+ \rightarrow \Delta_+$ is a continuous function, by **Brouwer's Fixed Point Theorem** there exists a price vector $p^* \in \Delta_+$ such that:

$$f(p^*) = p^*.$$

Component by component, this translates to the identity:

$$p_\ell^* = \frac{p_\ell^* + z_\ell^+(p^*)}{\alpha(p^*)} \implies \alpha(p^*)p_\ell^* = p_\ell^* + z_\ell^+(p^*).$$

Substituting $\alpha(p^*) = 1 + \sum_{k=1}^L z_k^+(p^*)$, we obtain the **key fixed-point identity**:

$$z_\ell^+(p^*) = p_\ell^* \sum_{k=1}^L z_k^+(p^*). \quad (\text{ID})$$

Step 4: The Closure — Why the Fixed Point is an Equilibrium? Here is the analytical magic that ties the entire demonstration together. We will prove that identity (ID) forces all markets to clear.

1. Multiply both sides of identity (ID) by the real excess demand of good ℓ , i.e., by $z_\ell(p^*)$:

$$z_\ell(p^*)z_\ell^+(p^*) = p_\ell^* z_\ell(p^*) \sum_{k=1}^L z_k^+(p^*).$$

2. Sum this equality for all L goods in the economy:

$$\sum_{\ell=1}^L z_{\ell}(p^*) z_{\ell}^+(p^*) = \sum_{\ell=1}^L \left(p_{\ell}^* z_{\ell}(p^*) \sum_{k=1}^L z_k^+(p^*) \right).$$

3. Factor out the common term from the right sum, $\sum_{k=1}^L z_k^+(p^*)$ (which is a scalar), outside the sum in ℓ :

$$\sum_{\ell=1}^L z_{\ell}(p^*) z_{\ell}^+(p^*) = \left(\sum_{\ell=1}^L p_{\ell}^* z_{\ell}(p^*) \right) \left(\sum_{k=1}^L z_k^+(p^*) \right).$$

4. By **Walras' Law**, the total value of excess demand is zero: $\sum_{\ell=1}^L p_{\ell}^* z_{\ell}(p^*) = 0$. Substituting this into our equation, the right side collapses to zero:

$$\sum_{\ell=1}^L z_{\ell}(p^*) z_{\ell}^+(p^*) = 0. \quad (\text{EQ-FINAL})$$

5. Let us analyze the terms on the left side of (EQ-FINAL) for each good ℓ :

- If $z_{\ell}(p^*) > 0$: then $z_{\ell}^+(p^*) = z_{\ell}(p^*) > 0$. The product will be strictly positive: $z_{\ell}(p^*) z_{\ell}^+(p^*) = [z_{\ell}(p^*)]^2 > 0$.
- If $z_{\ell}(p^*) \leq 0$: then $z_{\ell}^+(p^*) = 0$. The product will be null: $z_{\ell}(p^*) z_{\ell}^+(p^*) = 0$.

6. From this it follows that all terms in the left sum are **non-negative**: $z_{\ell}(p^*) z_{\ell}^+(p^*) \geq 0$ for all ℓ .

7. Since the sum of non-negative terms equals zero (as required by (EQ-FINAL)), **each individual term must be rigorously null**:

$$z_{\ell}(p^*) z_{\ell}^+(p^*) = 0 \quad \forall \ell = 1, \dots, L.$$

8. This directly implies that there cannot be positive excess demand in any market! If there were some market with $z_{\ell}(p^*) > 0$, the product would be strictly positive, violating the equality. Hence, we are forced to conclude that:

$$z_{\ell}(p^*) \leq 0 \quad \forall \ell = 1, \dots, L. \quad \square$$

Alternative: Proof by Direct Contradiction There is a more elegant alternative proof that avoids multiplication by $z_{\ell}(p^*)$. Define $S = \sum_{k=1}^L z_k^+(p^*) \geq 0$. Identity (ID) becomes:

$$z_{\ell}^+(p^*) = S p_{\ell}^*.$$

Suppose $S > 0$: For every good ℓ whose price is strictly positive ($p_{\ell}^* > 0$), the equation forces us to accept that $z_{\ell}^+(p^*) = S p_{\ell}^* > 0$. By definition, if the positive part of excess demand is strictly greater than zero, the real excess demand must also be strictly positive: $z_{\ell}(p^*) = z_{\ell}^+(p^*) > 0$. This means that **all goods with positive price would have positive excess demand**.

But Walras' Law requires $\sum_{\ell=1}^L p_{\ell}^* z_{\ell}(p^*) = 0$. Since $p^* \in \Delta_+$, at least one price is strictly positive. If for every good with $p_{\ell}^* > 0$ we have $z_{\ell}(p^*) > 0$, then all non-zero terms in the

sum are strictly positive (product of two positive numbers). A sum of positive and null terms (where at least one is strictly positive) **must necessarily be greater than zero**, blatantly violating Walras' Law.

Therefore, the assumption that $S > 0$ is false. We must have $S = 0$. Since $S = \sum z_\ell^+(p^*) = 0$ and each term is non-negative, each individual term must be null: $z_\ell^+(p^*) = 0$ for all ℓ . By the definition of the positive part ($z_\ell^+ = \max\{z_\ell, 0\}$), if $z_\ell^+(p^*) = 0$, the real excess demand can only be less than or equal to zero: $z(p^*) \leq 0$.

Equilibrium Conclusion

We found a price vector $p^* \in \Delta_+$ such that $z(p^*) \leq 0$. By the consequence of Walras' Law, this already characterizes a competitive equilibrium.

- If consumers' preferences are strongly monotone, no good with strictly positive price can have excess supply in equilibrium (if $p_\ell^* > 0 \implies z_\ell(p^*) = 0$).
- If some good ends up with excess supply ($z_\ell(p^*) < 0$), by the complementarity arising from Walras it must have zero price ($p_\ell^* = 0$), characterizing it as a free good.

The mathematical gear closes with absolute perfection!

Table 2.6: Correspondence between economic intuition and formal elements of Brouwer's proof for L goods.

Economic mechanism	Formal element	Reference
Walrasian tâtonnement	Adjustment function $f(p)$ raises prices with excess demand	Step 2
Price normalization	Simplex Δ_+ is compact and convex	Step 1
Topological fixed point	Brouwer guarantees p^* with $f(p^*) = p^*$	Step 3
Fixed-point identity	$z_\ell^+(p^*) = p_\ell^* \sum_k z_k^+(p^*)$	Eq. (ID)
Walras' Law	$\sum_\ell p_\ell^* z_\ell(p^*) = 0$ forces $z(p^*) \leq 0$	Step 4
Free goods	$z_\ell(p^*) < 0 \implies p_\ell^* = 0$	Def. (2)

2.4.5 From Abstract to Concrete: Microfoundation of Z

So far Z has been taken as primitive. In a competitive economy, however, excess demand is *generated* by preferences, endowments, and optimization:

$$\text{preferences} + \text{endowments} \longrightarrow \text{individual demands} \longrightarrow Z(p) \longrightarrow \text{equilibrium}.$$

Constructing Z explicitly in a simple pure exchange economy (for example, with Cobb–Douglas utilities) provides a concrete instance of the existence proof logic: continuity and Walras' Law emerge naturally from the mathematics of individual choices, and the p_1^* that zeroes $\tilde{z}_1$ can be calculated in closed form.

Table 2.7: Correspondence between economic intuition and elements of the existence proof with two goods.

Economic mechanism	Formal element	Reference
Only relative prices matter	Normalization $\sum_{\ell} p_{\ell} = 1$; simplex Δ_+	Eq. (1)
$L = 2$: one-dimensional problem	$\Delta_+ \cong [0, 1]$; IVT available	Previous section
Free goods	$p_{\ell}^* = 0 \implies$ only admits $z_{\ell}(p^*) \leq 0$	Def. (2)
Aggregate budget accounting	$p \cdot Z(p) = 0$ for all p	Eqs. (3)–(9)
Suffices that there is no excess demand	$z_{\ell} \leq 0 \forall \ell \implies$ equilibrium	Consequence of Wa
Scarcity at zero price generates demand	$\tilde{z}_1(0) > 0, \tilde{z}_2(1) > 0$ when boundary fails	Eq. (12)
High price of good 1 $\implies$ excess supply	$\tilde{z}_1(p_1) < 0$ near $p_1 = 1$	Eq. (14)
Continuity guarantees the zero	IVT on $[0, \hat{p}_1]$	Step 7
Second market clears “for free”	$(1 - p_1^*)\tilde{z}_2(p_1^*) = 0$ with $1 - p_1^* > 0$	Step 8

2.4.6 Connection between Intuition and Proof

2.5 Definitions and Properties of Equilibrium

2.5.1 The Walrasian Equilibrium in the Private Property Economy

Structure Agents are connected by a web of property rights: consumers own initial endowments ω_i ; firms own technologies Y_j ; and consumers are the owners of firms, each i holding share $\theta_{ij} \geq 0$ in firm j with $\sum_{i=1}^I \theta_{ij} = 1$ for all j .

Definition 2.5.1 (Walrasian equilibrium, Def. 17.B.1). A triple (x^*, y^*, p) is a **Walrasian equilibrium** if:

- (i) for each j , y_j^* maximizes profits in Y_j : $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$;
- (ii) for each i , x_i^* is maximal for $\succsim_i$ in the budget set $\{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij}(p \cdot y_j^*)\}$;
- (iii) markets clear: $\sum_{i=1}^I x_i^* = \sum_{i=1}^I \omega_i + \sum_{j=1}^J y_j^*$.

Remark 2.5.1 (Endogenous wealth and circularity). Wealth $w_i(p) = \underbrace{p \cdot \omega_i}_{\text{endowment}} + \underbrace{\sum_j \theta_{ij}(p \cdot y_j^*)}_{\text{dividends}}$

is not a fixed number: it varies with p . There is mutual determination: income depends on profits; profits depend on y_j^* chosen under p ; demands depend on income and p ; and p is determined by the global adjustment between demands and supplies. No one determines anything alone — the market coordinates simultaneously via p^* (the formalized “invisible hand”).

Remark 2.5.2 (Profits and returns to scale). Under constant returns to scale, maximum profit in equilibrium is necessarily zero; under decreasing returns (implicit fixed factor), profits are strictly positive and distributed to consumers.

2.5.2 Equilibrium as Absence of Scarcity: Proposition 17.B.1

In a pure exchange economy with continuous, strictly convex, and locally nonsatiated preferences, define the aggregate excess demand $z(p) = \sum_i (x_i(p, p \cdot \omega_i) - \omega_i)$.

Proposition 2.5.1 (Proposition 17.B.1). *A vector $p \geq 0$ is an equilibrium price vector if and only if $z(p) \leq 0$.*

Proof. ($\Rightarrow$) If p is an equilibrium price with allocation x^* , then $x_i^* = x_i(p, p \cdot \omega_i)$ and markets clear with equality, $\sum_i x_i^* = \sum_i \omega_i$. Hence

$$z(p) = \sum_i (x_i(p, p \cdot \omega_i) - \omega_i) = 0 \leq 0. \quad (1)$$

($\Leftarrow$) Suppose $z(p) \leq 0$. By local nonsatiation Walras' Law holds, $p \cdot z(p) = 0$. Since $p_\ell \geq 0$ and $z_\ell(p) \leq 0$, each term $p_\ell z_\ell(p) \leq 0$; a sum of non-positive terms that results in zero forces each term to be null:

$$p_\ell z_\ell(p) = 0 \quad \forall \ell. \quad (2)$$

Take $x_i^* = x_i(p, p \cdot \omega_i)$: consumers maximize by construction. For $p_\ell > 0$, (2) gives $z_\ell(p) = 0$ (market clears); for $p_\ell = 0$, $z_\ell(p) \leq 0$ is exactly the free good condition. Hence (x^*, p) satisfies the equilibrium definition. $\square$

Remark 2.5.3 (Complementary slackness and free goods). The condition $p_\ell z_\ell(p) = 0$ is the Karush–Kuhn–Tucker **complementary slackness**: the price is the Lagrange multiplier (shadow price) of the scarcity constraint $z_\ell(p) \leq 0$. If the constraint does not bind ($z_\ell < 0$, surplus), the shadow price is zero (free good); if $p_\ell > 0$ (scarcity), the market clears with equality. Walras' Law is the engine: without LNS, $p \cdot z(p) < 0$ and the argument collapses (there could be a valuable good in permanent excess supply).

Table 2.8: Methodological structure of Prop. 2.5.1: it suffices to prove that no market has scarcity.

Mathematical step	Economic meaning	Hypothesis
$x_i(p, w)$ is a function (unique)	Well-defined decisions, no jumps	Strict convexity
$p \cdot z(p) = 0$	No one tears up money; closed budget	Local nonsatiation
$p_\ell z_\ell(p) = 0$	Surplus good is free (free good)	Walras + $p \geq 0$

2.5.3 The Excess Demand Function: Definition 17.B.2

Definition 2.5.2 (Excess demand, Def. 17.B.2). The **individual excess demand** is $z_i(p) = x_i(p, p \cdot \omega_i) - \omega_i$; the **aggregate excess demand** is

$$z(p) = \sum_{i=1}^I z_i(p) = \sum_{i=1}^I x_i(p, p \cdot \omega_i) - \sum_{i=1}^I \omega_i.$$

Interpretation (net trades) $z_{i\ell}(p) > 0$: the agent wants to consume more than they own (**net buyer**); $z_{i\ell}(p) < 0$: wants to consume less (**net seller**, financing consumption with these sales); $z_{i\ell}(p) = 0$: self-sufficiency. The budget constraint rewritten in net terms is $p \cdot z_i(p) \leq 0$ (equality under LNS): the value of net purchases is exactly financed by the value of net sales.

Remark 2.5.4 (Domain and zero prices). If the price of a desirable good touches zero, the budget loses the capacity to contain demand: under monotonicity the consumer would demand infinite quantity, and $z(p)$ loses continuity/boundedness at the boundary. That is why the domain of z excludes problematic boundary prices but contains all $p \gg 0$.

Table 2.9: From absolute bundle analysis to net trades.

Concept	Approach x_i (gross)	Approach z_i (net)
Individual decision	Choose x_i^* to maximize u_i	Choose net trades z_i^*
Physical constraint	$\sum_i x_i \leq \sum_i \omega_i$	$z(p) \leq 0$
Walras' Law	$\sum_i p \cdot x_i = \sum_i p \cdot \omega_i$	$p \cdot z(p) = 0$

2.5.4 Properties of $z(p)$: Proposition 17.B.2

Proposition 2.5.2 (Proposition 17.B.2). *With $X_i = \mathbb{R}_+^L$, $\sum_i \omega_i \gg 0$, and preferences continuous, strictly convex, and strongly monotone, $z(p)$ satisfies:*

- (i) **Continuity**: $z(\cdot)$ is continuous for $p \gg 0$;
- (ii) **Zero-degree homogeneity**: $z(\lambda p) = z(p)$ for $\lambda > 0$;
- (iii) **Walras' Law**: $p \cdot z(p) = 0$ for $p \gg 0$;
- (iv) **Lower boundedness**: there exists $s > 0$ such that $z_\ell(p) > -s$ for all ℓ and $p \gg 0$;
- (v) **Boundary behavior**: if $p^n \rightarrow p$ with $p \neq 0$ and $p_\ell = 0$ for some ℓ , then $\max_\ell z_\ell(p^n) \rightarrow \infty$.

(i) *Continuity*. Under strict convexity, Walrasian demand $x_i(p, p \cdot \omega_i)$ is single-valued. For $p \gg 0$ the budget set is compact and the budget correspondence is continuous; by Berge's Maximum Theorem, $x_i(\cdot)$ is continuous. Hence $z_i(p) = x_i(p, p \cdot \omega_i) - \omega_i$ is continuous and $z(p) = \sum_i z_i(p)$, a sum of continuous functions, is continuous. $\square$

(ii) *Homogeneity*. For $\lambda > 0$, dividing the inequality by λ preserves the inequality:

$$\{x_i : \lambda p \cdot x_i \leq \lambda p \cdot \omega_i\} = \{x_i : p \cdot x_i \leq p \cdot \omega_i\}. \quad (3)$$

Since the feasible set and preferences do not change, $x_i(\lambda p, \lambda p \cdot \omega_i) = x_i(p, p \cdot \omega_i)$; hence $z_i(\lambda p) = z_i(p)$ and, summing, $z(\lambda p) = z(p)$ (absence of monetary illusion). $\square$

(iii) *Walras' Law*. Strong monotonicity implies budget constraint at equality: $p \cdot x_i(p, p \cdot \omega_i) = p \cdot \omega_i$, i.e. $p \cdot z_i(p) = 0$ for each i . Summing, $p \cdot z(p) = 0$. $\square$

(iv) *Lower boundedness.* Since $x_{i\ell}(p, p \cdot \omega_i) \geq 0$, we have $z_{i\ell}(p) \geq -\omega_{i\ell}$; summing over i ,

$$z_\ell(p) \geq -\sum_{i=1}^I \omega_{i\ell}. \quad (4)$$

Since $\sum_i \omega_i \gg 0$, choose $s > \max_\ell \sum_i \omega_{i\ell}$; then $z_\ell(p) > -s$ for all ℓ and $p \gg 0$ (no one sells more than they own). $\square$

(v) *Boundary behavior, by contradiction.* Suppose $\max_\ell z_\ell(p^n)$ does not diverge. By (iv), $z(p^n) \geq -s$ vectorially; hence the sequence $\{z(p^n)\}$ is bounded and, since each $z_i(p^n) \geq -\omega_i$ and $\sum_i z_i = z$, each $\{z_i(p^n)\}$ is bounded. By Bolzano–Weierstrass, extract a subsequence with $z_i(p^n) \rightarrow z_i^*$, so that $x_i(p^n, p^n \cdot \omega_i) = z_i(p^n) + \omega_i \rightarrow x_i^* = z_i^* + \omega_i$.

Passing to the limit the individual Walras’ Law $p^n \cdot x_i(p^n, p^n \cdot \omega_i) = p^n \cdot \omega_i$,

$$p \cdot x_i^* = p \cdot \omega_i. \quad (5)$$

Since $p \neq 0$ and $\sum_i \omega_i \gg 0$, there exists i with $p \cdot \omega_i > 0$; fix such i . We claim that x_i^* is optimal in $B_i(p, p \cdot \omega_i)$: for any x_i with $p \cdot x_i \leq p \cdot \omega_i$, define $\lambda^n = \min\{1, \frac{p^n \cdot \omega_i}{p^n \cdot x_i}\}$ (with $\lambda^n \rightarrow 1$); then $\lambda^n x_i \in B_i(p^n, p^n \cdot \omega_i)$ and, by optimality, $x_i(p^n) \succeq_i \lambda^n x_i$; by continuity of preferences, in the limit,

$$x_i^* \succeq_i x_i. \quad (6)$$

But, since $p_\ell = 0$ for some ℓ , the bundle $\hat{x}_i = x_i^* + e_\ell$ satisfies, by strong monotonicity, $\hat{x}_i \succ_i x_i^*$, and its cost is $p \cdot \hat{x}_i = p \cdot x_i^* + p_\ell = p \cdot x_i^* \leq p \cdot \omega_i$, i.e., $\hat{x}_i \in B_i(p, p \cdot \omega_i)$ — contradicting optimality (6) of x_i^* . Hence $\max_\ell z_\ell(p^n) \rightarrow \infty$. $\square$

Remark 2.5.5 (Intuition of the properties). (i) small price adjustments do not cause jumps; (ii) doubling all prices does not alter the budget set (no monetary illusion); (iii) the value of net purchases equals that of net sales; (iv) no consumer sells more than they own; (v) if the price of a good goes to zero, the desire for it explodes — the property that “pushes” equilibrium to the interior and guarantees existence via fixed point.

2.6 General Equilibrium under Uncertainty

2.6.1 Spot Market Equilibrium under Uncertainty

Economic Intuition

The Essence Before introducing the sophisticated structure of Arrow–Debreu contingent commodities, it is fundamental to understand the most elementary institutional arrangement of exchange under uncertainty: the **spot market**. In a spot market, trades of physical goods occur **after** the revelation of the state of nature — that is, at date 1, when uncertainty has already been resolved and all agents know exactly which state $s \in S$ the economy is in.

The real economic problem: If agents need to wait until date 1 to trade, they face uncertainty **without any ex ante protection instrument**. Each consumer arrives at the spot market with their contingent endowment ω_{si} and needs to buy and sell physical goods at the spot prices p_s that form in that specific state. There is no way to transfer wealth between states — each state is an isolated “island” of negotiation.

Example 2.6.1 (Concrete World Analogy — The Unprotected Farmer). Consider a farmer who planted corn in spring without knowing whether summer will be drought (s_1) or abundant rain (s_2). In the spot market arrangement, they simply wait for summer to arrive. If it rains, their harvest is abundant and they sell cheaply in the local market; if there is drought, their harvest is scarce and they need to buy expensive corn to survive. There is no contract, insurance, or financial security that allows them to smooth consumption between the two scenarios. The farmer’s wealth fluctuates wildly with the weather, and the spot market only **reacts** to the realization of the state — it offers no mechanism of **anticipation**.

Mechanisms

- **Market Opening at Date 1:** For each state $s \in S$, an independent spot market opens at date 1. Spot prices $p_s = (p_{1s}, \dots, p_{\bar{L}s}) \in \mathbb{R}_{++}^{\bar{L}}$ are determined by the interaction between supply and demand **in that specific state**.
- **Absence of Inter-State Transfer:** Since there are no contingent securities traded ex ante, each consumer i faces a **separate budget constraint for each state**:

$$p_s \cdot x_{si} \leq p_s \cdot \omega_{si}, \quad \forall s \in S.$$

It is not possible to sell wealth in state s_1 (where endowment is abundant) to buy more in state s_2 (where endowment is scarce).

- **State-by-State Equilibrium:** In each state s , spot equilibrium is a pair (p_s^*, x_s^*) such that, for every consumer i , x_{si}^* maximizes $u_i(x_{si})$ subject to $p_s^* \cdot x_{si} \leq p_s^* \cdot \omega_{si}$, and markets clear: $\sum_i x_{si}^* = \sum_i \omega_{si}$.

Implications

1. **Ex Ante Inefficiency:** Spot equilibrium is Pareto efficient **within each state** (ex post efficiency), but generally **is not** Pareto efficient **ex ante** (before the resolution of uncertainty). Risk-averse agents would like to transfer wealth between states, but have no way to do so.
2. **Consumption Volatility:** Without *hedge* instruments, each agent’s consumption fluctuates along with their contingent endowment. In “lean years” states, the agent consumes little; in “fat years” states, they consume a lot. Expected utility is significantly lower than it would be under complete markets.
3. **Benchmark for Comparison:** Spot equilibrium serves as a *benchmark* to quantify the **value of market completeness**. The welfare difference between spot equilibrium and Arrow–Debreu equilibrium measures the efficiency gain from the possibility of risk mutualization.

Formal Definition

Consider a pure exchange economy with two periods $t \in \{0, 1\}$ under uncertainty.

- **States of Nature:** $S = \{s_1, s_2, \dots, s_{\bar{S}}\}$, revealed at date 1.
- **Physical Goods:** $l \in L = \{1, 2, \dots, \bar{L}\}$ in each state s .

- **Consumers:** $i \in I = \{1, 2, \dots, \bar{I}\}$.
- **Contingent Endowments:** $\omega_i = (\omega_{1i}, \dots, \omega_{\bar{S}i}) \in \mathbb{R}_+^{\bar{L}\bar{S}}$, where $\omega_{si} \in \mathbb{R}_+^{\bar{L}}$ is consumer i 's endowment in state s .
- **Preferences:** $U_i(x_i) = \sum_{s=1}^{\bar{S}} \pi_s v_i(x_{si})$, with $v_i : \mathbb{R}_+^{\bar{L}} \rightarrow \mathbb{R}$ continuous, strictly increasing, and strictly quasi-concave.

Definition of Spot Market Equilibrium A **Spot Market Equilibrium** is a collection of price-allocation pairs $\{(p_s^*, x_s^*)\}_{s \in S}$ such that, for each state $s \in S$:

1. For each consumer i , x_{si}^* solves:

$$\max_{x_{si} \in \mathbb{R}_+^{\bar{L}}} v_i(x_{si}) \quad \text{subject to} \quad p_s^* \cdot x_{si} \leq p_s^* \cdot \omega_{si}.$$

2. Markets clear in state s :

$$\sum_{i=1}^{\bar{I}} x_{si}^* = \sum_{i=1}^{\bar{I}} \omega_{si}.$$

What Formally “Breaks”

- **Impossibility of Insurance:** Since each state has its own budget constraint, there is no way for the agent to transfer purchasing power between states. Spot equilibrium is **autarkic** in inter-state terms.
- **Failure of the First Welfare Theorem (Ex Ante):** Spot equilibrium is ex post Pareto efficient (within each state), but **is not** ex ante Pareto efficient. There are feasible reallocations that would improve all agents' expected utility before the resolution of uncertainty.

Proof: Ex Ante Inefficiency of Spot Equilibrium

Statement: Under strictly concave preferences (risk aversion) and non-perfectly-correlated contingent endowments, spot market equilibrium is not ex ante Pareto efficient.

Proof:

1. In spot equilibrium, each consumer i solves a separate problem in each state s :

$$\max_{x_{si}} v_i(x_{si}) \quad \text{s.t.} \quad p_s \cdot x_{si} \leq p_s \cdot \omega_{si}.$$

2. By local nonsatiability, the constraint holds with equality: $p_s \cdot x_{si}^* = p_s \cdot \omega_{si}$.
3. The equilibrium expected utility is:

$$U_i(x_i^*) = \sum_{s=1}^{\bar{S}} \pi_s v_i(x_{si}^*).$$

4. Consider an alternative allocation $\hat{x}_i$ that **smooths consumption** between states: for each i , define $\hat{x}_{si} = \bar{x}_i$ (constant in all states), where $\bar{x}_i$ is the bundle that costs the same as the average endowment:

$$p \cdot \bar{x}_i = \sum_{s=1}^{\bar{S}} \pi_s (p_s \cdot \omega_{si}).$$

5. By strict concavity of v_i (risk aversion) and Jensen's inequality:

$$v_i(\bar{x}_i) > \sum_{s=1}^{\bar{S}} \pi_s v_i(x_{si}^*)$$

whenever x_{si}^* is not constant in s (that is, whenever the endowment is volatile).

6. Therefore, $U_i(\hat{x}_i) = v_i(\bar{x}_i) > U_i(x_i^*)$ for all i , demonstrating that the spot allocation is ex ante Pareto-dominated. ■

Formal Connection Intuition → Proof

Table 2.10: Correspondence between economic intuition and formal elements of spot equilibrium.

Economic mechanism	Formal element
Markets open only after uncertainty is resolved	Separate budget constraints by state: $p_s \cdot x_{si} \leq p_s \cdot \omega_{si}$
Impossibility of inter-state transfer	Absence of contingent securities traded at date 0
Consumption volatility	x_{si}^* varies with s (is not constant)
Ex ante inefficiency	Jensen's inequality: $v_i(\bar{x}_i) > \sum_s \pi_s v_i(x_{si}^*)$
Efficiency gain via market completeness	Allocation $\hat{x}_i$ with constant consumption Pareto-dominant

2.6.2 Arrow–Debreu Model with Contingent Commodities

Economic Intuition

The Essence In the traditional General Equilibrium model, commodities are defined and indexed only by their physical characteristics and location. When we introduce uncertainty about the future, the nature of the good critically depends on the **state of nature** that comes to be realized.

The real economic problem: How to organize exchanges and allocate resources efficiently before knowing which future will materialize? If economic agents are risk-averse and face uncertainties about their endowments, productivity, or future needs, how does the market allow them to protect themselves against such risks ex ante?

The fundamental solution proposed by Kenneth Arrow and Gérard Debreu is to treat **delivery rights conditioned on future states as distinct economic goods** traded today — the so-called **contingent commodities**. A contingent commodity ls represents the right to receive (or the obligation to deliver) one unit of physical good l if and only if the state of nature $s \in S$ is realized. All $L \times S$ contingent commodity markets open and close at a single initial moment ($t = 0$), before the resolution of uncertainty.

Example 2.6.2 (Concrete World Analogy — The Protected Farmer). Imagine a farmer who will plant wheat. They do not know whether the next season will be severe drought (s_1) or abundant rain (s_2). Instead of waiting for the harvest to sell the wheat in the *spot* market, they go to the market at date 0 and trade two distinct contracts: “wheat delivered in the drought state” and “wheat delivered in the rain state”. If they sell the

wheat contract in case of drought to a buyer who needs wheat exactly in that scenario (and protect themselves by buying the opposite contract), both completely eliminate future risk at date 0 before even the first drop of rain falls.

Mechanisms

- **Agent Behavior:** At date 0, before nature chooses the state $s \in S$, each consumer i evaluates contingent consumption bundles $x_i = (x_{i11}, \dots, x_{iLS}) \in \mathbb{R}_+^{LS}$ through their expected utility function. The consumer faces a **single ex ante budget constraint**: the total present value of their contingent consumption plan quoted at market prices $q = (q_{11}, \dots, q_{LS})$ cannot exceed the total present value of their initial contingent endowment $w_i = (w_{i11}, \dots, w_{iLS})$.
- **Risk Pricing:** The price of a contingent commodity q_{ls} simultaneously reflects the relative scarcity of physical good l , the probability of state s occurring (π_s), and the agents' risk assessment (the marginal utility of wealth in that state).
- **Risk Mutualization (*Risk Sharing*):** Under complete markets of contingent commodities, risk-averse individuals transfer purchasing capacity from states where their endowment is abundant (or their marginal utility is low) to states where they suffer negative shocks (high marginal utility), equalizing the marginal rate of substitution between states to the relative price of securities.
- **Passive Ex Post Execution:** Since all trades and contractual obligations are established and paid at date 0, at date 1 — after the revelation of state s — no new trades or price renegotiations occur. Only the physical deliveries contracted ex ante are strictly fulfilled.

Implications

1. **Unified Validity of Welfare Theorems:** The Arrow–Debreu framework proves that, if there exist contingent markets for all goods and states ($L \times S$ markets), the competitive price mechanism guarantees **ex ante Pareto Efficiency** without the need for state intervention.
2. **Analysis of Market Incompleteness:** In practice, the absence of markets for certain states (due to information asymmetries, transaction costs, or non-applicability of contracts) generates *incomplete markets*, breaking the First Welfare Theorem and justifying the creation of public insurance or social security mechanisms.

Example 2.6.3 (Agricultural Insurance Market). The agricultural insurance market with parametric pricing or climate option contracts. An insurer sells today a security that pays a fixed value if the rainfall index of a region falls below a certain threshold. The producer buys the contingent contract to guarantee their income, neutralizing the productive risk ex ante.

Formal Definition

Consider a pure exchange economy in a static environment under decentralized uncertainty at date 0.

- **State Space of Nature:** $S = \{1, 2, \dots, \bar{S}\}$, finite, with mutually exclusive and exhaustive states. The event set is the algebra $\Sigma = 2^S$.
- **Physical Goods:** $l \in L = \{1, 2, \dots, \bar{L}\}$ available in each state.
- **Contingent Commodity Space:** $\mathbb{R}_+^{\bar{L}\bar{S}}$, of dimension $K = \bar{L} \times \bar{S}$.
- **Consumers:** $i \in I = \{1, 2, \dots, \bar{I}\}$.
- **Contingent Consumption Plan:** $x_i = (x_{i11}, \dots, x_{i\bar{L}\bar{S}}) \in \mathbb{R}_+^{\bar{L}\bar{S}}$, where $x_{is} = (x_{1is}, \dots, x_{\bar{L}is}) \in \mathbb{R}_+^{\bar{L}}$ is the bundle consumed if state s occurs.
- **Initial Contingent Endowments:** $w_i = (w_{i11}, \dots, w_{i\bar{L}\bar{S}}) \in \mathbb{R}_+^{\bar{L}\bar{S}}$.
- **Preferences and Expected Utility:** Each consumer i has expected utility of the form $U_i(x_i) = \sum_{s=1}^{\bar{S}} \pi_{is} v_i(x_{is})$, where $\pi_{is} > 0$ is the subjective/objective probability of state s ($\sum_{s=1}^{\bar{S}} \pi_{is} = 1$) and $v_i : \mathbb{R}_+^{\bar{L}} \rightarrow \mathbb{R}$ is continuous, strictly increasing, and strictly quasi-concave.
- **Ex Ante Contingent Price Vector:** $q = (q_{11}, \dots, q_{\bar{L}\bar{S}}) \in \mathbb{R}_{++}^{\bar{L}\bar{S}}$, quoted at date 0.

Mathematical Definition: Arrow–Debreu Equilibrium with Contingent Commodities An **Arrow–Debreu Equilibrium** is a pair (q^*, x^*) , where $q^* \in \mathbb{R}_{++}^{\bar{L}\bar{S}} \setminus \{0\}$ and $x^* = (x_1^*, \dots, x_{\bar{I}}^*) \in \mathbb{R}_+^{\bar{L}\bar{S}\bar{I}}$, such that:

1. For each consumer $i \in I$, the consumption plan x_i^* solves the problem:

$$\max_{x_i \in \mathbb{R}_+^{\bar{L}\bar{S}}} \sum_{s=1}^{\bar{S}} \pi_{is} v_i(x_{is}) \quad \text{subject to} \quad \sum_{s=1}^{\bar{S}} q_s^* \cdot x_{is} \leq \sum_{s=1}^{\bar{S}} q_s^* \cdot w_{is}$$

2. Markets of all contingent commodities clear (*market clearing*):

$$\sum_{i=1}^{\bar{I}} x_{lis}^* \leq \sum_{i=1}^{\bar{I}} w_{lis}^*, \quad \forall l \in \{1, \dots, \bar{L}\}, \forall s \in \{1, \dots, \bar{S}\}$$

Required Structural Hypotheses

1. **Market Completeness:** There are $\bar{L} \times \bar{S}$ independent prices opened ex ante at date 0.
2. **Local Nonsatiability:** Strictly increasing preferences guarantee that the budget constraint is reached with equality.
3. **Full Contract Enforceability (No-Default):** Absence of bankruptcy or non-fulfillment of physical delivery ex post.

What Formally “Breaks” if Any Hypothesis is Removed

- **If markets are missing ($K < \bar{L}\bar{S}$):** The budget set is restricted to a linear subspace, preventing the equalization of marginal rates of substitution between contingent goods and violating the Pareto efficiency of equilibrium (First Welfare Theorem fails).
- **If there is ex post information asymmetry:** Incentives for fraud or default arise, making the settlement of contracts established at date 0 unfeasible.

Step-by-Step Proofs

Auxiliary Theorem: First Welfare Theorem in Contingent Economies *Statement:* Let (q^*, x^*) be an Arrow–Debreu Equilibrium with contingent commodities for an economy with locally nonsatiable preferences. The allocation x^* is ex-ante Pareto efficient.

Step-by-Step Proof:

1. Consider consumer i 's problem in Arrow–Debreu equilibrium: x_i^* maximizes $U_i(x_i)$ in the budget set $B_i(q^*) = \{x_i \in \mathbb{R}_+^{\bar{L}\bar{S}} \mid q^* \cdot x_i \leq q^* \cdot w_i\}$ (by definition of Arrow–Debreu equilibrium).
2. If a bundle $\tilde{x}_i \in \mathbb{R}_+^{\bar{L}\bar{S}}$ satisfies $U_i(\tilde{x}_i) > U_i(x_i^*)$, then necessarily $q^* \cdot \tilde{x}_i > q^* \cdot w_i$ (by the definition of optimal maximizing choice in the budget set).
3. If a bundle $\tilde{x}_i \in \mathbb{R}_+^{\bar{L}\bar{S}}$ satisfies $U_i(\tilde{x}_i) \geq U_i(x_i^*)$, then necessarily $q^* \cdot \tilde{x}_i \geq q^* \cdot w_i$ (by the hypothesis of local nonsatiability of preferences).
4. Suppose, for contradiction, that there exists an alternative feasible allocation $\hat{x} = (\hat{x}_1, \dots, \hat{x}_{\bar{I}})$ that Pareto-dominates x^* ex ante, that is:
 - (a) $U_i(\hat{x}_i) \geq U_i(x_i^*)$ for all $i \in \{1, \dots, \bar{I}\}$ (weak Pareto improvement).
 - (b) $U_k(\hat{x}_k) > U_k(x_k^*)$ for at least one consumer k (strict improvement for consumer k).
 - (c) $\sum_{i=1}^{\bar{I}} \hat{x}_i \leq \sum_{i=1}^{\bar{I}} w_i$ (aggregate feasibility).
5. Applying the result of Step 3 to all consumers $i \neq k$:

$$q^* \cdot \hat{x}_i \geq q^* \cdot w_i, \quad \forall i \neq k$$

(by applying the cost minimization property of the utility level).

6. Applying the result of Step 2 to consumer k :

$$q^* \cdot \hat{x}_k > q^* \cdot w_k$$

(by the need for greater expenditure to obtain strictly higher utility).

7. Summing the inequalities from Steps 5 and 6 over all $\bar{I}$ consumers:

$$\sum_{i=1}^{\bar{I}} q^* \cdot \hat{x}_i > \sum_{i=1}^{\bar{I}} q^* \cdot w_i$$

(by the property of summing inequalities with at least one strictly greater term).

8. Factoring the price vector q^* from the sum in Step 7:

$$q^* \cdot \left(\sum_{i=1}^{\bar{I}} \hat{x}_i \right) > q^* \cdot \left(\sum_{i=1}^{\bar{I}} w_i \right)$$

(by the distributive property of the scalar product of vectors).

9. Rewriting the inequality from Step 8 by subtracting the right member:

$$q^* \cdot \left(\sum_{i=1}^{\bar{I}} \hat{x}_i - \sum_{i=1}^{\bar{I}} w_i \right) > 0$$

(by term transposition operation and associativity).

10. Since $q^* \in \mathbb{R}_{++}^{\bar{L}\bar{S}}$, the inequality from Step 9 requires that there exists at least one contingent good ls such that:

$$\sum_{i=1}^{\bar{I}} \hat{x}_{lis} > \sum_{i=1}^{\bar{I}} w_{lis}$$

(by the definition of scalar product with strictly positive components).

11. The conclusion of Step 10 directly contradicts the feasibility hypothesis of allocation $\hat{x}$ established in Step 4(c) ($\sum_{i=1}^{\bar{I}} \hat{x}_i \leq \sum_{i=1}^{\bar{I}} w_i$).

12. It is concluded that the Arrow–Debreu equilibrium allocation x^* is ex-ante Pareto efficient. ■

Systematic Resolution of the Formal Question

Diagnosis

- **Area:** Microeconomics / General Equilibrium under Uncertainty.
- **Type of Optimization:** Non-Linear Non-Restricted Static Optimization in the Contingent Subspace (Lagrange Multipliers).
- **Main Tool:** Lagrange Multipliers, First-Order Conditions (FOCs), Walras–Arrow–Debreu Welfare Theorem, and Optimal Risk Sharing Condition (Borch Rule).

Step-by-Step Resolution 1. Formulation of the Consumer's Lagrangian

Consumer i 's optimization problem is:

$$\max_{x_i \in \mathbb{R}_+^{\bar{L}\bar{S}}} \sum_{s=1}^{\bar{S}} \pi_{is} v_i(x_{is})$$

subject to:

$$\sum_{s=1}^{\bar{S}} \sum_{l=1}^{\bar{L}} q_{ls} x_{lis} \leq \sum_{s=1}^{\bar{S}} \sum_{l=1}^{\bar{L}} q_{ls} w_{lis} \quad (\text{with multiplier } \lambda_i \geq 0)$$

Writing the Lagrangian function $\mathcal{L}_i$:

$$\mathcal{L}_i(x_i, \lambda_i) = \sum_{s=1}^{\bar{S}} \pi_{is} v_i(x_{1is}, \dots, x_{\bar{L}is}) + \lambda_i \left(\sum_{s=1}^{\bar{S}} \sum_{l=1}^{\bar{L}} q_{ls} (w_{lis} - x_{lis}) \right)$$

2. First-Order Conditions (FOCs)

Differentiating $\mathcal{L}_i$ with respect to the consumption choice of good l in state s (x_{lis}), assuming interior solution ($x_{lis}^* > 0$):

$$\frac{\partial \mathcal{L}_i}{\partial x_{lis}} = \pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}} - \lambda_i^* q_{ls} = 0$$

Isolating the probability-weighted marginal utility term:

$$\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}} = \lambda_i^* q_{ls}$$

3. Ratio of FOCs and Marginal Rate of Substitution (MRS)

For two physical goods l and k in the **same state** s :

$$\frac{\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}}}{\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{kis}}} = \frac{\lambda_i^* q_{ls}}{\lambda_i^* q_{ks}} \implies \frac{\frac{\partial v_i(x_{is}^*)}{\partial x_{lis}}}{\frac{\partial v_i(x_{is}^*)}{\partial x_{kis}}} = \frac{q_{ls}}{q_{ks}}$$

For the **same physical good** l between **two different states** s and t :

$$\frac{\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}}}{\pi_{it} \frac{\partial v_i(x_{it}^*)}{\partial x_{lit}}} = \frac{\lambda_i^* q_{ls}}{\lambda_i^* q_{lt}} \implies \frac{\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}}}{\pi_{it} \frac{\partial v_i(x_{it}^*)}{\partial x_{lit}}} = \frac{q_{ls}}{q_{lt}}$$

4. System of Solution of Contingent Demands and Equilibrium Prices

The FOCs from Step 3 determine each consumer's contingent demand functions $x_{is}^*(q, q \cdot w_i)$.

Substituting the demand functions into the market clearing condition for each contingent market:

$$\sum_{i=1}^{\bar{I}} x_{lis}^*(q^*, q^* \cdot w_i) = \sum_{i=1}^{\bar{I}} w_{lis}, \quad \forall l \in \{1, \dots, \bar{L}\}, \forall s \in \{1, \dots, \bar{S}\}$$

Normalizing a contingent numeraire good, for example, $q_{11}^* = 1$, the system uniquely determines the equilibrium contingent price vector $q^* \in \mathbb{R}_{++}^{\bar{L}\bar{S}}$.

Interpretation

1. **Optimal Risk Sharing (Borch Rule):** The FOC relationship between two states s and t shows that for any two consumers i and j :

$$\frac{\pi_{is} \frac{\partial v_i(x_{is}^*)}{\partial x_{lis}}}{\pi_{it} \frac{\partial v_i(x_{it}^*)}{\partial x_{lit}}} = \frac{q_{ls}}{q_{lt}} = \frac{\pi_{js} \frac{\partial v_j(x_{js}^*)}{\partial x_{ijs}}}{\pi_{jt} \frac{\partial v_j(x_{jt}^*)}{\partial x_{ijt}}}$$

This implies that the **ratio of marginal utilities between states is equalized among all economic agents**, ensuring that no diversifiable risk is wasted in equilibrium.

2. **Arrow–Debreu Price Decomposition:** If agents have homogeneous probabilities ($\pi_{is} = \pi_s$), the contingent price can be decomposed as $q_{ls} = \pi_s \cdot m_s \cdot p_{ls}^{spot}$, where π_s is the probability of the state, m_s is the stochastic risk discount factor, and p_{ls}^{spot} is the relative price of the physical good in state s .

Synthesis The Arrow–Debreu model extends General Equilibrium theory to uncertain environments by defining contingent goods traded ex ante at date 0, ensuring that decentralization via market prices achieves an ex-ante Pareto efficient consumption allocation and eliminates all diversifiable risk from the economy. However, the validity of this result critically depends on the idealized hypothesis that there exist $L \times S$ markets opened simultaneously and that all contingent contracts can be perfectly monitored and executed ex post without cost.

2.6.3 Risk Sharing, Insurance, and Contingent Pricing

Economic Intuition

The Essence In the static General Equilibrium model without uncertainty, commodities are defined exclusively by their physical characteristics and spatial location. However, the real world is permeated by uncertainties: the same physical good (e.g., a ton of wheat or a liter of water) has radically different marginal utilities and economic values depending on the future scenario that materializes (e.g., drought vs. flood).

The real economic problem: How to allow risk-averse economic agents to protect themselves against idiosyncratic shocks and reallocate wealth between uncertain future states of nature *before* uncertainty is resolved?

The theory of **Arrow–Debreu Equilibrium with Contingent Commodities** resolves this issue by expanding the concept of commodity. A contingent good ls is the right to receive (or the obligation to deliver) one unit of physical commodity l if and only if the specific state of nature $s \in S$ is realized. All markets for all contingent goods open and close at a single initial date ($t = 0$). Ex ante negotiation allows agents to mutualize risks and buy perfect insurance before nature determines which state will prevail.

Example 2.6.4 (Concrete World Analogy — Two Neighboring Farmers). Imagine two neighboring farmers: Farmer 1 produces 100 bags of corn if it’s sunny (s_1), but nothing if it rains (s_2); Farmer 2 produces 100 bags of corn if it rains (s_2), but nothing if it’s sunny (s_1). The aggregate production of the region is always 100 bags of corn, regardless of the weather (absence of aggregate risk). If there were no ex ante market, each farmer would face the risk of extreme hunger in one of the scenarios. In the Arrow–Debreu model, before the start of the season ($t = 0$), both sign a contingent contract in which they agree to exchange 50% of their future crops with each other. When the weather is revealed at date 1, both consume exactly 50 bags, completely eliminating individual uncertainty.

Mechanisms

- **Agent Behavior and Budget Set:** At date 0, each consumer i solves a Von Neumann–Morgenstern expected utility maximization problem $U_i(x_i) = \sum_{s \in S} \pi_s^i u_i(x_{si})$ subject to a **single unified ex ante budget constraint:** $\sum_{s \in S} p_s x_{si} \leq \sum_{s \in S} p_s \omega_{si}$, where p_s is the price quoted at date 0 for the delivery of the numeraire good in state s .
- **Equalization of Marginal Rates of Substitution (MRS):** In any interior Pareto-efficient allocation, the Marginal Rates of Substitution between contingent states must be equalized for all consumers: $MRS_{s,s'}^i = MRS_{s,s'}^k$ for every pair of consumers i, k .

- **Homogeneous Beliefs and the Full Insurance Theorem:** If all agents have **homogeneous beliefs** ($\pi_s^i = \pi_s, \forall i$), strictly concave Bernoulli utilities ($u_i'' < 0$, risk aversion), and the economy has no **aggregate risk** ($\sum_i \omega_{si} = \bar{\omega}, \forall s$), the only efficient point requires constant consumption across all states ($x_{s_1i} = x_{s_2i} = \dots = x_{Si}$). Agents use contingent markets to completely eliminate idiosyncratic risk.
- **Heterogeneous Beliefs and Efficient Bets:** If agents disagree about the probabilities of states ($\pi_s^i \neq \pi_s^k$), the Pareto-efficient allocation **does not** generate full insurance ($x_{s_1i} \neq x_{s_2i}$). Agents who assign higher probability to a given state s will buy more consumption rights for that specific state. Consumption diversity will reflect a combination of risk aversion and mutually agreed bets on the probability of events.

Implications

1. **Risk Decentralization via Market Prices:** Demonstrates that financial and derivatives markets are not mere places of speculation, but essential instruments for allocative efficiency in the Pareto sense.
2. **Interpretation of the Relative Price of Contingent Assets:** The equilibrium price of the Arrow security p_s decomposes into the product between the probability of the state and the Stochastic Discount Factor (the marginal utility of wealth of the economy in that state).
3. **Identification of Market Failures:** When contingent commodity markets are incomplete (for example, absence of insurance contracts against bankruptcy or unemployment due to moral hazard or adverse selection), the equalization of MRS fails, opening space for public social security policies.

Example 2.6.5 (Catastrophe Bonds — Cat Bonds). The market of Catastrophe Bonds (*Cat Bonds*) or Climate Options traded on the Commodity Exchange. Insurers issue securities that pay returns to investors if there is no category 5 hurricane in a given year, but lose the principal in favor of the insurer if the hurricane occurs, transferring the risk of rare shocks to the global capital market.

Formal Definition

Consider a pure exchange economy under uncertainty in a two-period environment $t \in \{0, 1\}$.

- **State Space of Nature:** $S = \{s_1, s_2\}$, with exactly one state realizing at date 1.
- **Consumers:** $i \in I = \{1, 2\}$.
- **Contingent Physical Goods:** $x_i = (x_{s_1i}, x_{s_2i}) \in \mathbb{R}_+^2$, where x_{si} is the quantity of the physical good consumed by agent i if state s occurs.
- **Initial Contingent Endowments:** $\omega_1 = (1, 0)$ and $\omega_2 = (0, 1)$.
- **Aggregate Resource:** $\omega = \omega_1 + \omega_2 = (1, 1)$, characterizing an economy without aggregate risk.
- **Subjective Probabilities:** $\pi^i = (\pi_{s_1}^i, \pi_{s_2}^i) \in \mathbb{R}_{++}^2$ such that $\pi_{s_1}^i + \pi_{s_2}^i = 1$ for each $i \in \{1, 2\}$.

- **Preferences:** $U_i(x_i) = \pi_{s_1}^i u_i(x_{s_1 i}) + \pi_{s_2}^i u_i(x_{s_2 i})$, where $u_i : \mathbb{R}_+ \rightarrow \mathbb{R}$ is by hypothesis continuously differentiable, strictly increasing ($u_i' > 0$) and strictly concave ($u_i'' < 0$) in $\mathbb{R}_{++}$.
- **Arrow–Debreu Contingent Price Vector:** $p = (p_{s_1}, p_{s_2}) \in \mathbb{R}_{++}^2$, quoted at date 0.

Mandatory Structural Hypotheses

1. **Market Completeness:** There are $S = 2$ independent Arrow contingent securities opened ex ante at date 0.
2. **Strict Risk Aversion:** $u_i'' < 0$ for all i , guaranteeing strictly convex indifference curves in the contingent consumption space.
3. **Local Nonsatiability:** $u_i' > 0$, guaranteeing saturation of the budget constraint with equality.

Step-by-Step Proofs

Proposition 1: Equalization of the Marginal Rate of Substitution at the Pareto Optimum *Statement:* In any interior Pareto-efficient allocation $x^* = (x_1^*, x_2^*)$, the Marginal Rates of Substitution between contingent states must be identically equal among consumers.

Step-by-Step Proof:

1. An interior allocation $x^* = (x_1^*, x_2^*)$ is Pareto efficient if and only if it solves the central planner's problem of maximizing Consumer 1's utility subject to a minimum utility level $\bar{U}_2$ for Consumer 2 and the aggregate physical feasibility constraint:

$$\max_{x_1, x_2 \geq 0} \pi_{s_1}^1 u_1(x_{s_1 1}) + \pi_{s_2}^1 u_1(x_{s_2 1})$$

subject to:

$$\begin{aligned} \pi_{s_1}^2 u_2(x_{s_1 2}) + \pi_{s_2}^2 u_2(x_{s_2 2}) &\geq \bar{U}_2 \quad (\text{multiplier } \lambda \geq 0) \\ x_{s_1 1} + x_{s_1 2} &\leq 1 \quad (\text{multiplier } \mu_{s_1} \geq 0) \\ x_{s_2 1} + x_{s_2 2} &\leq 1 \quad (\text{multiplier } \mu_{s_2} \geq 0) \end{aligned}$$

2. Formulating the Lagrangian function of the problem:

$$\begin{aligned} \mathcal{L}(x_1, x_2, \lambda, \mu) &= \pi_{s_1}^1 u_1(x_{s_1 1}) + \pi_{s_2}^1 u_1(x_{s_2 1}) \\ &\quad + \lambda [\pi_{s_1}^2 u_2(x_{s_1 2}) + \pi_{s_2}^2 u_2(x_{s_2 2}) - \bar{U}_2] \\ &\quad + \mu_{s_1} (1 - x_{s_1 1} - x_{s_1 2}) + \mu_{s_2} (1 - x_{s_2 1} - x_{s_2 2}) \end{aligned}$$

3. Differentiating $\mathcal{L}$ with respect to $x_{s_1 1}$ for an interior solution ($x_{s_1 1}^* > 0$):

$$\frac{\partial \mathcal{L}}{\partial x_{s_1 1}} = \pi_{s_1}^1 u_1'(x_{s_1 1}^*) - \mu_{s_1} = 0 \implies \mu_{s_1} = \pi_{s_1}^1 u_1'(x_{s_1 1}^*)$$

4. Differentiating $\mathcal{L}$ with respect to x_{s_21} for an interior solution ($x_{s_21}^* > 0$):

$$\frac{\partial \mathcal{L}}{\partial x_{s_21}} = \pi_{s_2}^1 u_1'(x_{s_21}^*) - \mu_{s_2} = 0 \implies \mu_{s_2} = \pi_{s_2}^1 u_1'(x_{s_21}^*)$$

5. Differentiating $\mathcal{L}$ with respect to x_{s_12} for an interior solution ($x_{s_12}^* > 0$):

$$\frac{\partial \mathcal{L}}{\partial x_{s_12}} = \lambda \pi_{s_1}^2 u_2'(x_{s_12}^*) - \mu_{s_1} = 0 \implies \mu_{s_1} = \lambda \pi_{s_1}^2 u_2'(x_{s_12}^*)$$

6. Differentiating $\mathcal{L}$ with respect to x_{s_22} for an interior solution ($x_{s_22}^* > 0$):

$$\frac{\partial \mathcal{L}}{\partial x_{s_22}} = \lambda \pi_{s_2}^2 u_2'(x_{s_22}^*) - \mu_{s_2} = 0 \implies \mu_{s_2} = \lambda \pi_{s_2}^2 u_2'(x_{s_22}^*)$$

7. Equating the two expressions obtained for μ_{s_1} (Steps 3 and 5) and for μ_{s_2} (Steps 4 and 6):

$$\begin{aligned} \pi_{s_1}^1 u_1'(x_{s_11}^*) &= \lambda \pi_{s_1}^2 u_2'(x_{s_12}^*) \\ \pi_{s_2}^1 u_1'(x_{s_21}^*) &= \lambda \pi_{s_2}^2 u_2'(x_{s_22}^*) \end{aligned}$$

8. Dividing the first equation from step 7 by the second equation from the same step:

$$\frac{\pi_{s_1}^1 u_1'(x_{s_11}^*)}{\pi_{s_2}^1 u_1'(x_{s_21}^*)} = \frac{\lambda \pi_{s_1}^2 u_2'(x_{s_12}^*)}{\lambda \pi_{s_2}^2 u_2'(x_{s_22}^*)}$$

9. Canceling the strictly positive multiplier $\lambda > 0$ on the right side:

$$\frac{\pi_{s_1}^1 u_1'(x_{s_11}^*)}{\pi_{s_2}^1 u_1'(x_{s_21}^*)} = \frac{\pi_{s_1}^2 u_2'(x_{s_12}^*)}{\pi_{s_2}^2 u_2'(x_{s_22}^*)}$$

10. Identifying the expressions as the contingent Marginal Rates of Substitution MRS_{s_1,s_2}^1 and MRS_{s_1,s_2}^2 :

$$MRS_{s_1,s_2}^1(x_1^*) = MRS_{s_1,s_2}^2(x_2^*)$$

(by definition of contingent Marginal Rate of Substitution). ■

Proposition 2: Full Insurance Theorem under Homogeneous Beliefs *Statement:*

If both consumers have homogeneous beliefs about the states of nature ($\pi_{s_1}^1 = \pi_{s_1}^2 = \pi_{s_1}$ and $\pi_{s_2}^1 = \pi_{s_2}^2 = \pi_{s_2}$), then any interior Pareto-efficient allocation satisfies $x_{s_1i}^* = x_{s_2i}^*$ for all $i \in \{1, 2\}$, implying full insurance against nature's risk.

Step-by-Step Proof:

1. By Proposition 1, any interior Pareto-efficient allocation satisfies the MRS equality:

$$\frac{\pi_{s_1}^1 u_1'(x_{s_11}^*)}{\pi_{s_2}^1 u_1'(x_{s_21}^*)} = \frac{\pi_{s_1}^2 u_2'(x_{s_12}^*)}{\pi_{s_2}^2 u_2'(x_{s_22}^*)}$$

2. Substitute the hypothesis of homogeneous beliefs ($\pi_s^1 = \pi_s^2 = \pi_s > 0$ for $s \in \{s_1, s_2\}$) into the equation from step 1:

$$\frac{\pi_{s_1} u'_1(x_{s_1 1}^*)}{\pi_{s_2} u'_1(x_{s_2 1}^*)} = \frac{\pi_{s_1} u'_2(x_{s_1 2}^*)}{\pi_{s_2} u'_2(x_{s_2 2}^*)}$$

3. Cancel the market probability ratio $\frac{\pi_{s_1}}{\pi_{s_2}} > 0$ on both sides of the equation:

$$\frac{u'_1(x_{s_1 1}^*)}{u'_1(x_{s_2 1}^*)} = \frac{u'_2(x_{s_1 2}^*)}{u'_2(x_{s_2 2}^*)}$$

4. Consider the aggregate physical feasibility constraints of the pure exchange economy:

$$\begin{aligned} x_{s_1 1}^* + x_{s_1 2}^* &= 1 \implies x_{s_1 2}^* = 1 - x_{s_1 1}^* \\ x_{s_2 1}^* + x_{s_2 2}^* &= 1 \implies x_{s_2 2}^* = 1 - x_{s_2 1}^* \end{aligned}$$

5. Substitute $x_{s_1 2}^*$ and $x_{s_2 2}^*$ obtained in step 4 into the equation from step 3:

$$\frac{u'_1(x_{s_1 1}^*)}{u'_1(x_{s_2 1}^*)} = \frac{u'_2(1 - x_{s_1 1}^*)}{u'_2(1 - x_{s_2 1}^*)}$$

6. Suppose, for contradiction, that consumer 1 is not fully insured, for example, that $x_{s_1 1}^* > x_{s_2 1}^*$.
7. Since the Bernoulli function u_1 is strictly concave ($u_1'' < 0$), the marginal utility function u'_1 is strictly decreasing. Therefore:

$$x_{s_1 1}^* > x_{s_2 1}^* \implies u'_1(x_{s_1 1}^*) < u'_1(x_{s_2 1}^*) \implies \frac{u'_1(x_{s_1 1}^*)}{u'_1(x_{s_2 1}^*)} < 1$$

8. On the other hand, if $x_{s_1 1}^* > x_{s_2 1}^*$, then by subtraction:

$$1 - x_{s_1 1}^* < 1 - x_{s_2 1}^* \implies x_{s_1 2}^* < x_{s_2 2}^*$$

9. Applying the strict concavity of u_2 to the result from step 8:

$$x_{s_1 2}^* < x_{s_2 2}^* \implies u'_2(x_{s_1 2}^*) > u'_2(x_{s_2 2}^*) \implies \frac{u'_2(1 - x_{s_1 1}^*)}{u'_2(1 - x_{s_2 1}^*)} > 1$$

10. Steps 7 and 9 generate a direct contradiction with the equality proven in step 5.

11. Analogously, assuming $x_{s_1 1}^* < x_{s_2 1}^*$ results in a symmetric contradiction.

12. It is concluded that $x_{s_1 1}^* = x_{s_2 1}^*$, which by aggregate feasibility implies $x_{s_1 2}^* = x_{s_2 2}^*$.

Hence, the Pareto-efficient allocation under homogeneous beliefs and absence of aggregate risk guarantees **full insurance** to all agents ($x_{s_1 i}^* = x_{s_2 i}^*, \forall i$). ■

Systematic Resolution: Logarithmic Solution

Setup Consider a pure exchange economy with $I = 2$ consumers and $S = \{s_1, s_2\}$ states of nature.

- Endowments: $\omega_1 = (1, 0)$ and $\omega_2 = (0, 1)$. The total resource is constant: $\omega = (1, 1)$.
- Consumers have preferences represented by expected utility with logarithmic Bernoulli function: $u_i(x_{si}) = \ln(x_{si})$.
- **Central Trade-off:** Consumer 1 has endowment concentrated in state s_1 , while Consumer 2 has endowment concentrated in state s_2 . Both seek to sell rights over their abundant state to acquire protection against the scarcity state ex ante.

Step-by-Step Resolution 1. First-Order Conditions (FOCs)

Consumer $i \in \{1, 2\}$'s problem is:

$$\max_{x_{s_1i}, x_{s_2i} > 0} \pi_{s_1}^i \ln(x_{s_1i}) + \pi_{s_2}^i \ln(x_{s_2i})$$

subject to:

$$p_{s_1}x_{s_1i} + p_{s_2}x_{s_2i} \leq p_{s_1}\omega_{s_1i} + p_{s_2}\omega_{s_2i} \quad (\text{with multiplier } \lambda_i \geq 0)$$

Differentiating $\mathcal{L}_i$ with respect to x_{s_1i} and x_{s_2i} :

$$x_{s_1i}^* = \frac{\pi_{s_1}^i}{\lambda_i^* p_{s_1}}, \quad x_{s_2i}^* = \frac{\pi_{s_2}^i}{\lambda_i^* p_{s_2}}$$

2. Derivation of Contingent Demand Functions

Substituting into the budget constraint with equality:

$$\frac{\pi_{s_1}^i + \pi_{s_2}^i}{\lambda_i^*} = W_i(p) \implies \lambda_i^* = \frac{1}{W_i(p)}$$

where $W_i(p) \equiv p_{s_1}\omega_{s_1i} + p_{s_2}\omega_{s_2i}$ is agent i 's ex ante wealth.

The contingent Cobb–Douglas demands are:

$$\begin{aligned} x_{s_1i}^*(p, W_i) &= \pi_{s_1}^i \frac{W_i(p)}{p_{s_1}} \\ x_{s_2i}^*(p, W_i) &= \pi_{s_2}^i \frac{W_i(p)}{p_{s_2}} \end{aligned}$$

With endowments $\omega_1 = (1, 0)$ and $\omega_2 = (0, 1)$:

- Consumer 1: $W_1(p) = p_{s_1}$
- Consumer 2: $W_2(p) = p_{s_2}$

Consumer 1:

$$\begin{aligned}x_{s_1 1}^*(p) &= \pi_{s_1}^1 \\x_{s_2 1}^*(p) &= \pi_{s_2}^1 \frac{p_{s_1}}{p_{s_2}}\end{aligned}$$

Consumer 2:

$$\begin{aligned}x_{s_1 2}^*(p) &= \pi_{s_1}^2 \frac{p_{s_2}}{p_{s_1}} \\x_{s_2 2}^*(p) &= \pi_{s_2}^2\end{aligned}$$

3. Market Clearing and Equilibrium Prices

The market clearing condition in state s_1 requires:

$$\pi_{s_1}^1 + \pi_{s_1}^2 \frac{p_{s_2}}{p_{s_1}} = 1$$

Isolating the equilibrium price ratio:

$$\frac{p_{s_1}^*}{p_{s_2}^*} = \frac{\pi_{s_1}^2}{\pi_{s_2}^1}$$

Normalizing $p_{s_2}^* = 1$ as numeraire:

$$p_{s_1}^* = \frac{\pi_{s_1}^2}{\pi_{s_2}^1}$$

4. Analysis of Special Cases

CASE A: Homogeneous Beliefs ($\pi_s^1 = \pi_s^2 = \pi_s$)

The equilibrium price ratio is:

$$\frac{p_{s_1}^*}{p_{s_2}^*} = \frac{\pi_{s_1}}{\pi_{s_2}}$$

Substituting into demands:

- **Consumer 1:** $x_{s_1 1}^* = \pi_{s_1}$ and $x_{s_2 1}^* = \pi_{s_1}$ — full insurance
- **Consumer 2:** $x_{s_1 2}^* = \pi_{s_2}$ and $x_{s_2 2}^* = \pi_{s_2}$ — full insurance

If probabilities are perfectly symmetric ($\pi_{s_1} = \pi_{s_2} = 1/2$):

$$x_{s_1 1}^* = x_{s_2 1}^* = \frac{1}{2}, \quad x_{s_1 2}^* = x_{s_2 2}^* = \frac{1}{2}, \quad p_{s_1}^* = p_{s_2}^* = 1$$

CASE B: Heterogeneous Beliefs ($\pi_{s_1}^1 = 3/4, \pi_{s_1}^2 = 1/4$)

The equilibrium price ratio is:

$$\frac{p_{s_1}^*}{p_{s_2}^*} = \frac{1/4}{1/4} = 1 \implies p_{s_1}^* = p_{s_2}^* = 1$$

Equilibrium consumption allocations:

- **Consumer 1:** $x_{s_1 1}^* = 3/4$ and $x_{s_2 1}^* = 1/4$
- **Consumer 2:** $x_{s_1 2}^* = 1/4$ and $x_{s_2 2}^* = 3/4$

Equilibrium consumption is **no longer constant between states** ($x_{s_1 i}^* \neq x_{s_2 i}^*$). Agent 1 consumes more in state s_1 because they are optimistic about its occurrence, and agent 2 consumes more in state s_2 for the same reason.

Interpretation

1. **Price Ratio as Reconciliation of Evaluations:** The pricing equation $\frac{p_{s_1}^*}{p_{s_2}^*} = \frac{\pi_{s_1}^2}{\pi_{s_2}^1}$ demonstrates that the relative price of the Arrow security in equilibrium reflects agents' optimization and clears markets by unifying divergent beliefs.
2. **Disconnect between Efficiency and Full Insurance under Heterogeneous Beliefs:** Heterogeneous belief transforms the contingent commodity market into a mechanism for efficient betting. Risk sharing in the Pareto sense remains valid ex ante with respect to individual expected utilities, but the resulting physical allocation no longer presents strict consumption stability between states.
3. **Stochastic Discount Factor:** If we redefine risk-neutral probabilities as $q_s^* = p_s^*/(p_{s_1}^* + p_{s_2}^*)$, Arrow asset prices behave as probabilities adjusted by the economy's risk.

2.6.4 Information Structure, Temporal Constraints, and Event Trees

Economic Intuition

The Essence In the static General Equilibrium model without uncertainty, goods are strictly characterized by their physical properties and spatial location. When we introduce uncertainty about future states of nature, the description of a good must necessarily include the **environmental contingency under which physical delivery occurs**.

The real economic problem: Uncertainty does not resolve instantaneously; it unfolds over time. Moreover, economic decisions (such as planting a crop or investing in capital) require the commitment of resources **before** information about the future is revealed.

To formalize this problem, Arrow–Debreu theory models an agent's consumption and endowment as **state-contingent random variables**. Additionally, it formalizes the evolution of agents' knowledge through **partitions of the state space (information structures)**. The central idea is simple: an agent cannot make a decision in the present that depends on information that will only be available in the future.

Example 2.6.6 (Concrete World Analogy — The Farmer and Planting). Think of a corn farmer. At date $t = 0$ (spring), they need to decide how many bags of seeds to bury in the soil. They do not know whether in summer ($t = 1$) there will be drought (s_1) or ideal rain (s_2). The farmer cannot sign a production plan that says “I will only plant the seed if it rains in summer”, because in spring the summer climate is still unknown. The input choice is **non-anticipative** (measurable with respect to present information). Decisions must respect the information structure available at the date they are taken.

Mechanisms

- **Goods and Endowments as Random Variables:** Each physical good $\ell \in \{1, \dots, L\}$ in an economy with finite state space $S = \{s_1, \dots, s_S\}$ becomes a vector $x_\ell = (x_{\ell 1}, \dots, x_{\ell S}) \in \mathbb{R}^S$, which is mathematically equivalent to a **scalar random variable** defined over the model's probability space. Agent i 's initial endowment, $\omega_i : S \rightarrow \mathbb{R}^L$, assigns to each state $s \in S$ a vector of physical goods $\omega_{si} = (\omega_{1si}, \dots, \omega_{Lsi}) \in \mathbb{R}^L$.
- **Production Technology and Temporal Constraint (*Timing*):** A production plan $y_j = (y_{s_1j}, \dots, y_{s_Sj}) \in \mathbb{R}^{LS}$ specifies inputs and products for each state. If an input (such as seeds) needs to be applied at date 0 before the revelation of state $s \in \{s_1, s_2\}$, the consumption of the input at date 0 **must be identical in all states that share the same information at date 0**. That is, $y_{1s_1j} = y_{1s_2j}$. Any production plan that requires different inputs for states indistinguishable at the application date is physically unfeasible.
- **Information Structure and Partitions:** The evolution of knowledge is represented by a sequence of **partitions of the state space** $\mathcal{S}_0, \mathcal{S}_1, \dots, \mathcal{S}_T$.
 - In the initial period ($t = 0$), agents have a trivial partition $\mathcal{S}_0 = \{S\}$ (they do not distinguish any state).
 - In the intermediate period ($t = 1$), information is partially revealed by a partition $\mathcal{S}_1 = \{E_1, \dots, E_K\}$, where each event $E_k \subseteq S$ represents a set of states that the agent cannot distinguish from each other.
 - In the final period ($t = T$), uncertainty is fully resolved and the partition becomes the fine partition $\mathcal{S}_T = \{\{s_1\}, \{s_2\}, \dots, \{s_S\}\}$.
 - The sequence of partitions forms a **filtration**: partition $\mathcal{S}_{t+1}$ is a **refinement** of $\mathcal{S}_t$ ($\mathcal{S}_t \preceq \mathcal{S}_{t+1}$), meaning that information accumulates and agents never forget what they have already learned (perfect memory / gradual learning).
- **Bayesian Updating of Beliefs:** As time advances and event $E \in \mathcal{S}_t$ is observed, agents update their initial subjective probabilities μ_0 through **Bayes' Rule**, eliminating the probability mass of impossible states ($s \notin E$) and proportionally rescaling the probability of compatible states ($s \in E$).

Implications

1. **Measurability Constraint of Contingent Plans:** In stochastic dynamic models, the consumption, investment, and production plans of any public or private agent must be **adapted to the informational filtration**. Regulatory decisions or private contracts that require actions conditioned on unobservable events create executability and information asymmetry problems.
2. **Structuring of Dynamic Financial Markets:** The gradual resolution of uncertainty along an event tree allows substituting the ex ante negotiation of $L \times S$ contingent goods for a sequence of financial asset portfolio rebalancing as the economy moves through the nodes of the informational tree.
3. **Pricing the Value of Information:** The structure of the event tree allows pricing

the economic value of information. An asset whose return concentrates at nodes of the tree where aggregate wealth is low (recession states) will have a higher unit price ex ante (risk premium).

Example 2.6.7 (Climate Insurance Market). Insurance Contracts and Climate Derivatives for Cereal Growers. The insurer establishes that the indemnity trigger (*pluviometric index*) be measured at date $t = 1$ by certified state pluviometers. The producer makes the fertilizer purchase decision at $t = 0$. If the contract required that fertilizer application at $t = 0$ varied with the rain at $t = 1$, the contract would be unfeasible. The partitioned information structure guarantees that the *hedge* and input decisions respect the information available at date 0.

Formal Definition

Consider a stochastic dynamic economic environment with temporal discretization $t \in \{0, 1, \dots, T\}$.

- **State Space of Nature:** $S = \{s_1, s_2, \dots, s_S\}$, finite, exhaustive, and mutually exclusive elements.
- **Event Algebra:** $\Sigma = 2^S$, the power set of S .
- **Dynamic Information Structure (Discrete Stochastic Filtration):** For each date $t \in \{0, 1, \dots, T\}$, agents' common information is represented by a partition $\mathcal{S}_t$ of S :

$$\mathcal{S}_t = \{E_{t,1}, E_{t,2}, \dots, E_{t,K_t}\}$$

satisfying:

1. $\bigcup_{k=1}^{K_t} E_{t,k} = S$.
 2. $E_{t,k} \cap E_{t,k'} = \emptyset, \quad \forall k \neq k'$.
 3. **Refinement Property (Gradual Learning):** For all $t \in \{0, \dots, T-1\}$, if $E_{t+1} \in \mathcal{S}_{t+1}$, there exists a unique event $E_t \in \mathcal{S}_t$ such that $E_{t+1} \subseteq E_t$.
- **Event Tree (Nodes):** $\mathcal{D} = \{(t, E) : t \in \{0, \dots, T\}, E \in \mathcal{S}_t\}$.
 - **Contingent Consumption Vector as Random Variable:** For each good $\ell \in L$, $x_\ell : S \rightarrow \mathbb{R}_+$ is a function from the state space to the non-negative reals, that is, a discrete random variable $x_\ell = (x_\ell(s_1), \dots, x_\ell(s_S)) \in \mathbb{R}_+^S$. Consumer i 's total consumption plan is given by $x_i = (x_{1i}, \dots, x_{\bar{L}i}) \in \mathbb{R}_+^{\bar{L}S}$, which is equivalent to choosing $\bar{L}$ random variables over S .
 - **Contingent Endowment:** A function $\omega_i : S \rightarrow \mathbb{R}_+^{\bar{L}}$, where $\omega_i(s) = (\omega_{1i}(s), \dots, \omega_{\bar{L}i}(s)) \in \mathbb{R}_+^{\bar{L}}$ represents the endowment vector received by consumer i if state s occurs.

Mathematical Definition: Adaptation/Measurability Condition of Contingent Processes A decision plan (consumption x_t or production y_t) at date t is **adapted to the information structure $\mathcal{S}_t$** (or **$\mathcal{S}_t$ -measurable**) if and only if:

$$\forall E \in \mathcal{S}_t, \quad \forall s, s' \in E \implies y_t(s) = y_t(s')$$

(In words: if two states s and s' belong to the same information event E at date t , the decision taken at date t must be strictly identical for both states).

Step-by-Step Proofs

Auxiliary Theorem: Bayes' Rule and Consistency of Belief Updating *Statement:* Let $\mu_0 \in \Delta(S)$ be a strictly positive a priori probability distribution over S . If event $E \in \mathcal{S}_t$ occurs with probability $\mu_0(E) = \sum_{s \in E} \mu_0(s) > 0$, the a posteriori probability distribution $\mu_t^E(s)$ updated via Bayes' Rule satisfies the properties of a legitimate probability measure strictly concentrated on E .

Step-by-Step Proof:

1. By definition of Bayes' Rule for conditioned events:

$$\mu_t^E(s) = \frac{\mu_0(\{s\} \cap E)}{\mu_0(E)}$$

2. Evaluating the numerator for the two mutually exclusive cases:

- Case A: If $s \in E$, then $\mu_0(\{s\} \cap E) = \mu_0(s)$.
- Case B: If $s \notin E$, then $\mu_0(\{s\} \cap E) = \mu_0(\emptyset) = 0$.

3. Substituting, we obtain the two-branch specification:

$$\mu_t^E(s) = \begin{cases} \frac{\mu_0(s)}{\mu_0(E)}, & \text{if } s \in E \\ 0, & \text{if } s \notin E \end{cases}$$

4. Summing the a posteriori probabilities over all states $s \in S$:

$$\sum_{s \in S} \mu_t^E(s) = \sum_{s \in E} \frac{\mu_0(s)}{\mu_0(E)} = \frac{1}{\mu_0(E)} \sum_{s \in E} \mu_0(s) = \frac{\mu_0(E)}{\mu_0(E)} = 1$$

This proves that the a posteriori distribution in the event tree is a well-defined probability measure. ■

Proposition 1: Unfeasibility of Non-Measurable Production Plans *Statement:* Consider an economy with $S = \{s_1, s_2\}$, $L = 2$ (good 1: seeds applied at $t = 0$; good 2: harvest obtained at $t = 1$). Suppose that at date $t = 0$ the information partition is $\mathcal{S}_0 = \{S\} = \{\{s_1, s_2\}\}$. Then, any production plan of technology $y_j = (y_{1s_1j}, y_{2s_1j}, y_{1s_2j}, y_{2s_2j})$ that requires $y_{1s_1j} \neq y_{1s_2j}$ is informationally unfeasible ex ante.

Step-by-Step Proof:

1. At date $t = 0$, the agent's information structure is given by the trivial partition $\mathcal{S}_0 = \{E_{0,1}\}$ where $E_{0,1} = \{s_1, s_2\}$.
2. By definition, two states s_1 and s_2 belong to the same information event $E_{0,1} \in \mathcal{S}_0$ (that is, the agent is not able to distinguish s_1 from s_2 at date $t = 0$).
3. A decision executed at date $t = 0$ for good 1 (seeds) must be an $\mathcal{S}_0$ -measurable function.

4. By the Definition of Measurability, for the vector $y_{1j} = (y_{1s_{1j}}, y_{1s_{2j}})$ to be $\mathcal{S}_0$ -measurable:

$$s_1, s_2 \in E_{0,1} \implies y_{1s_{1j}} = y_{1s_{2j}}$$

5. Consider the alternative plan $y'_j = ((-1, 1), (0, 0))$, where $y'_{1s_{1j}} = -1$ and $y'_{1s_{2j}} = 0$.
6. Since $s_1, s_2 \in E_{0,1}$ at time $t = 0$, but $y'_{1s_{1j}} = -1 \neq 0 = y'_{1s_{2j}}$, plan y'_j violates the equality condition.
7. Therefore, plan y'_j requires that the choice of seed level at date $t = 0$ depends on prior knowledge about whether the future state will be s_1 or s_2 .
8. Since s_1 and s_2 are indistinguishable at $t = 0$, the decision function under y'_j does not exist in the set of adapted strategies at $t = 0$.

Hence, plan y'_j is informationally unfeasible ex ante. ■

Proposition 2: Property of the Information Algebra and Event Trees *Statement:*

If the sequence of information partitions $\{\mathcal{S}_t\}_{t=0}^T$ satisfies the gradual refinement property ($\mathcal{S}_0 \preceq \mathcal{S}_1 \preceq \dots \preceq \mathcal{S}_T$), then the number of distinguishable events $K_t = |\mathcal{S}_t|$ is monotonically non-decreasing in t , that is, $K_0 \leq K_1 \leq \dots \leq K_T \leq S$.

Step-by-Step Proof:

1. Let $\mathcal{S}_t = \{E_{t,1}, E_{t,2}, \dots, E_{t,K_t}\}$ be the partition of space S at date t .
2. By the refinement property, each event $E_{t+1,m} \in \mathcal{S}_{t+1}$ is a subset of exactly one ancestral event $E_{t,k} \in \mathcal{S}_t$.
3. Each cell $E_{t,k} \in \mathcal{S}_t$ is partitioned into non-empty and disjoint subsets:

$$E_{t,k} = \bigcup_{m \in J(t,k)} E_{t+1,m}$$

where $J(t,k) = \{m \in \{1, \dots, K_{t+1}\} \mid E_{t+1,m} \subseteq E_{t,k}\}$.

4. Since each $E_{t,k} \neq \emptyset$, we have $|J(t,k)| \geq 1$ for all $k \in \{1, \dots, K_t\}$.
5. Summing the cardinality:

$$K_{t+1} = \sum_{k=1}^{K_t} |J(t,k)| \geq \sum_{k=1}^{K_t} 1 = K_t$$

6. Inductively:

$$1 = K_0 \leq K_1 \leq K_2 \leq \dots \leq K_T \leq S$$

This proves that the information discrimination capacity accumulates monotonically over time. ■

Example: Dynamic Information Structure with $T = 2$ dates and $S = 6$ states

Consider the process of gradual information revelation in $T + 1 = 3$ dates ($t \in \{0, 1, 2\}$) for the state space $S = \{s_1, s_2, s_3, s_4, s_5, s_6\}$:

- **Date $t = 0$ (Total absence of information):**

$$\mathcal{S}_0 = \{E_{0,1}\} = \{\{s_1, s_2, s_3, s_4, s_5, s_6\}\} \implies K_0 = 1 \text{ event}$$

- **Date $t = 1$ (Partial information revelation):**

$$\mathcal{S}_1 = \{E_{1,1}, E_{1,2}, E_{1,3}\} = \{\{s_1, s_2\}, \{s_3\}, \{s_4, s_5, s_6\}\} \implies K_1 = 3 \text{ events}$$

- **Date $t = 2$ (Total resolution of uncertainty):**

$$\mathcal{S}_2 = \{E_{2,1}, \dots, E_{2,6}\} = \{\{s_1\}, \{s_2\}, \{s_3\}, \{s_4\}, \{s_5\}, \{s_6\}\} \implies K_2 = 6 \text{ events}$$

Verification of hierarchical inclusions:

- For $E_{1,1} = \{s_1, s_2\} \in \mathcal{S}_1$, we have $E_{2,1} = \{s_1\} \subset E_{1,1}$ and $E_{2,2} = \{s_2\} \subset E_{1,1}$.
- For $E_{1,2} = \{s_3\} \in \mathcal{S}_1$, we have $E_{2,3} = \{s_3\} = E_{1,2}$.
- For $E_{1,3} = \{s_4, s_5, s_6\} \in \mathcal{S}_1$, we have $E_{2,4} = \{s_4\} \subset E_{1,3}$, $E_{2,5} = \{s_5\} \subset E_{1,3}$, and $E_{2,6} = \{s_6\} \subset E_{1,3}$.

This algebraically validates the **event tree** structure (*event-tree structure*) necessary to decentralize Dynamic General Equilibrium under uncertainty.

Formal Connection Intuition $\rightarrow$ Proof

Table 2.11: Correspondence between economic intuition and formal elements of the information structure.

Mechanism described in Part 1	Corresponding variable/condition in the proof
Impossibility of predicting the future at $t = 0$	Measurability condition: $s, s' \in E_{0,k} \implies y_0(s) = y_0(s')$
Identical input plan for good and bad weather	Constraint $y_{1s_1j} = y_{1s_2j} = -1$ in the feasible plan
Gradual accumulation of information (Filtration)	Refinement of partitions $\mathcal{S}_t \preceq \mathcal{S}_{t+1}$ and $E_{t+1,m} \subseteq E_{t,m}$
Total resolution of uncertainty in final period T	Singleton partition $\mathcal{S}_T = \{\{s_1\}, \dots, \{s_S\}\}$ and cardinality $K_T = S$
Bayesian updating of beliefs	Conditional probability: $\mu_t^E(s) = \frac{\mu_0(s)}{\mu_0(E)}$ for $s \in E$

2.6.5 Arrow–Debreu Equilibrium with Production under Uncertainty

Economic Intuition

The Essence The introduction of time and uncertainty into General Equilibrium requires understanding how information about states of nature is gradually revealed to economic agents.

The real economic problem: In a stochastic dynamic economy, agents do not learn the final state of nature all at once. Information arrives gradually (e.g., crop reports throughout the year, inflation indicators month by month, technological discoveries in

phases). Investment, production, and consumption decisions must respect the information available at each temporal node. Since agents cannot “guess” the future nor “forget” the past, resource allocation must be structured on an **event tree** (*event tree*), where today’s choices are conditioned to today’s observable events, and contingent contracts price wealth in each future branch.

Example 2.6.8 (Concrete World Analogy — Soccer Tournament). Think of an elimination soccer tournament with 8 teams, where the final champion represents the “state of nature” $s \in \{s_1, \dots, s_8\}$. At date $t = 0$ (start of the tournament), no one knows who will win (null information: trivial partition). At date $t = 1$ (after the quarterfinals), we only know which 4 teams advanced (uncertainty was partially reduced to groups of states). At date $t = 2$ (after the semifinal), only 2 teams remain. At date $t = 3$ (final), the champion is revealed. A financial contract that pays \$100 if Team A is champion cannot require you to deliver the trophy in the middle of the tournament ($t = 1$), because the information has not yet been fully refined; the contract must be quoted at date 0 and settled only at the nodes where the event becomes distinguishable.

Mechanisms

- **Firms and Property Rights** (θ_{ij}): Firms carry out contingent production plans $y_j \in Y_j \subset \mathbb{R}^{LS}$. Each firm’s profits in each state of nature are distributed to consumers according to their shareholdings $\theta_{ij}(s) \geq 0$, with $\sum_{i=1}^I \theta_{ij}(s) = 1$ for every firm j and state s .
- **Economic Logic of Arrow–Debreu Equilibrium:** All $L \times S$ contingent markets for physical delivery of goods in each state open and close at $t = 0$. Consumer i ’s budget constraint is **unified ex ante**: total expenditure on contingent goods quoted at Arrow–Debreu prices $p = (p_{11}, \dots, p_{LS})$ cannot exceed the value of their initial endowment plus their fraction of firms’ contingent profits.

Implications

1. **Feasibility of Corporate Governance under Uncertainty:** Since all trades and shareholdings are settled ex ante at date 0, shareholders unanimously agree that the firm’s objective is to maximize the present market value of their contingent profits $p \cdot y_j$ under Arrow–Debreu prices.

Example 2.6.9 (Project Finance in Oil). Structuring *Project Finance* in the oil industry. A concessionaire raises resources by selling securities whose coupons depend on the success of drilling in phase 1 (node $t = 1$) and the price of the barrel in the international market in phase 2 (node $t = 2$). The concessionaire’s shares (θ_{ij}) and contingent debts are distributed ex ante in the financial market at date 0 before the first drilling.

Formal Definition

Consider a production economy under uncertainty with discrete temporal horizon $t \in \{0, 1, \dots, T\}$.

- **Consumers and Endowments:** $i \in I = \{1, 2, \dots, \bar{I}\}$. Bundle $x_i \in \mathbb{R}_+^{\bar{L}\bar{S}}$, endowment $\omega_i \in \mathbb{R}_+^{\bar{L}\bar{S}}$.

- **Firms and Technologies:** $j \in J = \{1, 2, \dots, \bar{J}\}$. Production set $Y_j \subset \mathbb{R}^{\bar{L}\bar{S}}$ closed and non-empty. Production plan $y_j = (y_{11j}, \dots, y_{\bar{L}\bar{S}j}) \in Y_j$.
- **Property Rights:** $\theta_{ij}(s) \geq 0$ is the fraction of firm j owned by consumer i in state s , with $\sum_{i=1}^{\bar{I}} \theta_{ij}(s) = 1, \forall j, s$. When non-contingent, $\theta_{ij} \geq 0$ with $\sum_{i=1}^{\bar{I}} \theta_{ij} = 1$.
- **Arrow–Debreu Price Vector:** $p = (p_{11}, \dots, p_{\bar{L}\bar{S}}) \in \mathbb{R}_{++}^{\bar{L}\bar{S}}$.

Mathematical Definition: Arrow–Debreu Equilibrium with Production An **Arrow–Debreu Equilibrium** is a tuple (p^*, x^*, y^*) , where $p^* \in \mathbb{R}_{++}^{\bar{L}\bar{S}} \setminus \{0\}$, $x^* = (x_1^*, \dots, x_{\bar{I}}^*) \in \prod_{i=1}^{\bar{I}} X_i$, and $y^* = (y_1^*, \dots, y_{\bar{J}}^*) \in \prod_{j=1}^{\bar{J}} Y_j$, such that:

1. **Firms' Profit Maximization:** For each firm $j \in J$, plan y_j^* solves:

$$y_j^* \in \arg \max_{y_j \in Y_j} p^* \cdot y_j$$

2. **Consumers' Utility Maximization:** For each consumer $i \in I$, bundle x_i^* solves:

$$x_i^* \in \arg \max_{x_i \in B_i(p^*)} U_i(x_i)$$

where the budget set is given by:

$$B_i(p^*) = \left\{ x_i \in \mathbb{R}_+^{\bar{L}\bar{S}} \left| p^* \cdot x_i \leq p^* \cdot \omega_i + \sum_{j=1}^{\bar{J}} \theta_{ij}(p^* \cdot y_j^*) \right. \right\}$$

3. **General Clearing of Contingent Goods Markets (*Market Clearing*):**

$$\sum_{i=1}^{\bar{I}} x_i^* = \sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} y_j^*$$

Step-by-Step Proofs

Proposition 1: Validity of Walras' Law in Contingent Economies with Production *Statement:* Under locally nonsatiable preferences, for any price vector $p \in \mathbb{R}_{++}^{\bar{L}\bar{S}}$, the value of aggregate excess demand is identically zero.

Step-by-Step Proof:

1. Since each consumer $i \in I$'s preferences are locally nonsatiated, the optimal consumption plan x_i^* saturates the budget constraint with strict equality:

$$p \cdot x_i^* = p \cdot \omega_i + \sum_{j=1}^{\bar{J}} \theta_{ij}(p \cdot y_j^*), \quad \forall i \in \{1, \dots, \bar{I}\}$$

2. Sum the equation from Step 1 over all $\bar{I}$ consumers:

$$\sum_{i=1}^{\bar{I}} p \cdot x_i^* = \sum_{i=1}^{\bar{I}} p \cdot \omega_i + \sum_{i=1}^{\bar{I}} \left(\sum_{j=1}^{\bar{J}} \theta_{ij}(p \cdot y_j^*) \right)$$

3. Factor p and reverse the order of summations:

$$p \cdot \left(\sum_{i=1}^{\bar{I}} x_i^* \right) = p \cdot \left(\sum_{i=1}^{\bar{I}} \omega_i \right) + \sum_{j=1}^{\bar{J}} \left(\sum_{i=1}^{\bar{I}} \theta_{ij} \right) (p \cdot y_j^*)$$

4. Since $\sum_{i=1}^{\bar{I}} \theta_{ij} = 1$:

$$p \cdot \left(\sum_{i=1}^{\bar{I}} x_i^* \right) = p \cdot \left(\sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} y_j^* \right)$$

5. Rearranging:

$$p \cdot \left(\sum_{i=1}^{\bar{I}} x_i^* - \sum_{i=1}^{\bar{I}} \omega_i - \sum_{j=1}^{\bar{J}} y_j^* \right) = 0$$

This formally demonstrates **Walras' Law** in an Arrow–Debreu economy with firms. ■

Auxiliary Theorem: First Welfare Theorem with Contingent Production

Statement: Every Arrow–Debreu Equilibrium (p^*, x^*, y^*) in an economy with locally nonsatiable preferences generates a contingent allocation (x^*, y^*) that is ex-ante Pareto efficient.

Step-by-Step Proof:

1. Let (p^*, x^*, y^*) be an Arrow–Debreu Equilibrium.
2. Suppose, for contradiction, that there exists an alternative feasible contingent allocation $(\hat{x}, \hat{y})$ such that:
 - (a) $U_i(\hat{x}_i) \geq U_i(x_i^*)$ for all $i \in \{1, \dots, \bar{I}\}$ (weak dominance).
 - (b) $U_k(\hat{x}_k) > U_k(x_k^*)$ for at least one consumer $k \in I$ (strict dominance for k).
 - (c) $\sum_{i=1}^{\bar{I}} \hat{x}_i \leq \sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} \hat{y}_j$ with $\hat{y}_j \in Y_j, \forall j$ (aggregate physical feasibility).
3. By optimality of consumer choice:
 - For $i \neq k$: $p^* \cdot \hat{x}_i \geq p^* \cdot \omega_i + \sum_{j=1}^{\bar{J}} \theta_{ij}(p^* \cdot y_j^*)$
 - For $i = k$: $p^* \cdot \hat{x}_k > p^* \cdot \omega_k + \sum_{j=1}^{\bar{J}} \theta_{kj}(p^* \cdot y_j^*)$

4. Summing the inequalities:

$$\sum_{i=1}^{\bar{I}} p^* \cdot \hat{x}_i > \sum_{i=1}^{\bar{I}} p^* \cdot \omega_i + \sum_{j=1}^{\bar{J}} \left(\sum_{i=1}^{\bar{I}} \theta_{ij} \right) (p^* \cdot y_j^*) = p^* \cdot \left(\sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} y_j^* \right)$$

5. On the other hand, multiplying feasibility by $p^* \gg 0$:

$$p^* \cdot \left(\sum_{i=1}^{\bar{I}} \hat{x}_i \right) \leq p^* \cdot \left(\sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} \hat{y}_j \right)$$

6. Since y_j^* maximizes profits and $\hat{y}_j \in Y_j$:

$$p^* \cdot \hat{y}_j \leq p^* \cdot y_j^*, \quad \forall j \in \{1, \dots, \bar{J}\}$$

7. Substituting:

$$p^* \cdot \left(\sum_{i=1}^{\bar{I}} \hat{x}_i \right) \leq p^* \cdot \left(\sum_{i=1}^{\bar{I}} \omega_i + \sum_{j=1}^{\bar{J}} y_j^* \right)$$

8. The inequality from Step 7 directly contradicts that from Step 4.

9. It is concluded that the equilibrium allocation $(\hat{x}, \hat{y})$ cannot be Pareto-dominated.

Hence, the Arrow–Debreu allocation is ex-ante Pareto efficient. ■

Systematic Resolution: Shareholder Unanimity

Setup Consider a 2-period economy ($t \in \{0, 1\}$), with $S = \{s_1, s_2\}$ states of nature (probabilities $\pi_1, \pi_2 > 0$, $\pi_1 + \pi_2 = 1$), $L = 1$ physical good in each state, $I = 2$ risk-averse consumers, and $J = 1$ productive firm.

- **Firm:** Chooses input $y_0 \leq 0$ at date 0 to produce contingent product $(y_1, y_2) \geq 0$ at date 1 according to the production function $y_s = f_s(-y_0)$, where f_s is strictly concave.
- **Consumers:** Consumer i maximizes expected utility $u_i(x_{0i}) + \pi_1 v_i(x_{1i}) + \pi_2 v_i(x_{2i})$, subject to holding share fraction θ_i of the firm ($\theta_1 + \theta_2 = 1$).

Step-by-Step Resolution 1. Firm's Problem

The firm chooses (y_0, y_1, y_2) to solve:

$$\max_{y_0 \leq 0, y_1, y_2 \geq 0} p_0^* y_0 + p_1^* y_1 + p_2^* y_2 \quad \text{s.t.} \quad y_1 \leq f_1(-y_0) \quad \text{and} \quad y_2 \leq f_2(-y_0)$$

Defining $K = -y_0 \geq 0$ as the investment level:

$$\frac{\partial \Pi}{\partial K} = -p_0^* + p_1^* f_1'(K^*) + p_2^* f_2'(K^*) = 0 \implies p_0^* = p_1^* f_1'(K^*) + p_2^* f_2'(K^*)$$

2. Consumer i 's Problem

$$\max_{x_{0i}, x_{1i}, x_{2i} \geq 0} u_i(x_{0i}) + \pi_1 v_i(x_{1i}) + \pi_2 v_i(x_{2i})$$

subject to:

$$p_0^* x_{0i} + p_1^* x_{1i} + p_2^* x_{2i} \leq p_0^* \omega_{0i} + p_1^* \omega_{1i} + p_2^* \omega_{2i} + \theta_i \Pi^*$$

3. Consumer's FOCs

$$\begin{aligned} u_i'(x_{0i}^*) &= \lambda_i^* p_0^* \\ \pi_1 v_i'(x_{1i}^*) &= \lambda_i^* p_1^* \\ \pi_2 v_i'(x_{2i}^*) &= \lambda_i^* p_2^* \end{aligned}$$

4. Relative Prices and Production System

Dividing the state FOCs by the period 0 FOC (normalizing $p_0^* = 1$):

$$p_1^* = \frac{\pi_1 v'_i(x_{1i}^*)}{u'_i(x_{0i}^*)} \quad \text{and} \quad p_2^* = \frac{\pi_2 v'_i(x_{2i}^*)}{u'_i(x_{0i}^*)}$$

Substituting into the production FOC:

$$u'_i(x_{0i}^*) = \pi_1 v'_i(x_{1i}^*) f'_1(K^*) + \pi_2 v'_i(x_{2i}^*) f'_2(K^*)$$

This fundamental equation establishes that **the marginal utility of renouncing 1 unit of consumption today must be exactly equal to the expected marginal utility of the additional physical product generated in future states.**

Interpretation

1. **Shareholder Unanimity:** Even if consumers 1 and 2 have different degrees of risk aversion, both fully agree that the firm should choose the investment level K^* that maximizes the present market value $p \cdot y_j^*$. This occurs because contingent prices p_1^* and p_2^* already equalize all agents' marginal rates of substitution in equilibrium.
2. **Shadow Price of Capital:** The capital input K^* is invested until the point where the marginal rate of transformation of the physical product weighted by Arrow–Debreu prices equals the unit price of the present input.

2.6.6 Radner Equilibrium: Sequential Trading and Equivalence with Arrow–Debreu

Economic Intuition

The Essence In the classic Arrow–Debreu model, it is assumed that there exist markets for all contingent commodities for all future states of nature and possible dates, all operating and closing at a single initial instant ($t = 0$). If there are L physical goods and S future states of nature, the Arrow–Debreu arrangement requires the simultaneous opening of $L \times S$ ex ante contingent physical delivery markets.

The real economic problem: In practice, it is institutionally unfeasible and operationally prohibitive for agents to trade physical delivery contracts for each good in each plausible future scenario. The number of contracts required grows combinatorially in an impractical way ($L \times S$).

The theory of **Radner Equilibrium (Sequential Trading with Rational Expectations)** resolves this obstacle by demonstrating that it is not necessary to negotiate all contingent commodities ex ante. If agents can negotiate ex ante a restricted set of financial assets (such as Arrow contingent securities in Good 1) to transfer purchasing power between states, they can wait for the revelation of uncertainty at date 1 and trade physical goods in the local *spot* markets of the realized state.

Example 2.6.10 (Concrete World Analogy — International Trip). Imagine you intend to take an international trip a year from now, but do not know whether you will face snow or extreme heat at the destination. In the Arrow–Debreu arrangement, you would need to sign contracts today in your hometown guaranteeing the purchase and physical delivery

of a specific thermal jacket in the cold scenario and swimwear in the heat scenario. In the Radner arrangement, you simply acquire today a contingent financial security (insurance or Arrow asset) that pays you a certain financial value if it's cold. Upon landing at date 1 and observing the real weather, you settle the asset and walk to the local *spot* market to buy the clothes suitable for the price prevailing on that day.

Mechanisms

- **Agent Behavior:** At date 0, before uncertainty is resolved, each consumer i chooses a portfolio of financial assets $z_i = (z_{1i}, \dots, z_{Si}) \in \mathbb{R}^S$ to reallocate their purchasing power between the possible future states $s \in \{1, \dots, S\}$. At date 1, state s realizes and the financial contract z_{si} is settled at price p_{1s}^* . With their adjusted wealth, the agent participates in the *spot* market of state s , acquiring the bundle of physical goods $x_{si} \in \mathbb{R}_+^L$ that maximizes their utility subject to the prevailing *spot* prices p_s^* .
- **Logic of Self-Fulfilling Expectations:** For the financial choices made at date 0 to result in a viable and optimal consumption plan at date 1, agents need to anticipate with perfect precision the equilibrium *spot* prices p_s^* that will prevail in each future state. In Radner Equilibrium, the price expectations formed ex ante are *self-fulfilling*: the prices p_s^* anticipated by agents at date 0 are exactly the prices that clear all *spot* markets when state s effectively occurs.
- **Absence of Ex Post Pareto-Improving Trades:** Since agents have rational price expectations and asset markets allow unrestricted transfer of wealth between states, sequential equilibrium leads to an allocation efficient in the ex ante Pareto sense. Consequently, **no mutually beneficial trade remains to be performed ex post in *spot* markets**, demonstrating that ex ante efficiency implies ex post efficiency.
- **The Logic of Short Sales:** The theory does not impose $z_{si} \geq 0$. The consumer can take a **short position**, selling more units of Good 1 than they physically own in their endowment in state s (ω_{1si}), that is, $z_{si} < -\omega_{1si}$. Since contract honoring is compulsory (absence of *default*), the agent who is short is obliged to go to the *spot* market at date 1, in state s , and buy the missing units of Good 1 at the prevailing price p_{1s} to deliver them to the counterparty.

Implications

1. **Drastic Reduction in Ex Ante Market Requirement:** The economic system does not need to structure $L \times S$ future physical delivery markets; it suffices to operationalize S financial asset markets at date 0 (or $S - 1$ relative asset prices), letting the L *spot* markets of physical goods open sequentially at date 1 only for the state that effectively occurs.
2. **Structural Role of the Financial System and Derivatives:** The financial sector acts as the infrastructure that completes risk markets, allowing the temporal and contingent reallocation of wealth without the need for anticipated physical delivery contracts.
3. **Fragility Regarding Expectation Errors or Information:** The equivalence between Radner and Arrow–Debreu models depends on the hypothesis of pure

rational expectations. If there is informational rigidity, forecast errors, or information asymmetry, sequential equilibrium breaks down, generating allocative inefficiency or market collapse.

Example 2.6.11 (Energy Derivatives Market). The functioning of commodity and energy derivatives markets. An electricity distributor does not contract ex ante the physical purchase of each megawatt for each possible water reservoir level. It buys futures/options contracts in the financial market and, on a day-to-day basis, settles the energy in the short-term market (*Price of Settlement of Differences - PLD*), adjusting its financial position to the *spot* price of the moment.

Formal Definition

Consider a pure exchange economy under uncertainty with two-period temporal structure $t \in \{0, 1\}$.

Economic Environment and Canonical Notation

- **Agents:** I consumers, indexed by $i \in \{1, \dots, I\}$.
- **States of Nature at Date 1:** $S = \{1, 2, \dots, \bar{S}\}$, with objective probability of state s given by $\pi_s > 0$ and $\sum_{s=1}^{\bar{S}} \pi_s = 1$. At date $t = 1$, exactly one state $s \in S$ realizes. Date $t = 0$ is the initial moment where uncertainty is negotiated.
- **Physical Goods at Date 1:** L physical goods in each state s , indexed by $l \in \{1, \dots, \bar{L}\}$.
- **Consumption Plan:** $x_i = (x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS}$, where $x_{si} \in \mathbb{R}_+^L$ is the consumption bundle in state s .
- **Initial Endowment:** $\omega_i = (\omega_{1i}, \dots, \omega_{Si}) \in \mathbb{R}_+^{LS}$, where $\omega_{si} \in \mathbb{R}_+^L$.
- **Preferences:** Represented by $U_i(x_{1i}, \dots, x_{Si})$, assumed strictly increasing, continuous, and strictly concave (or locally insatiable).

Price Structure

- **Market Prices:**
 - $q = (q_1, \dots, q_{\bar{S}}) \in \mathbb{R}_{++}^{\bar{S}}$: Price vector at date 0 for Arrow contingent securities in **Good 1** (local numeraire). Security s pays 1 unit of good 1 in state s and 0 in the others.
 - $p_s = (p_{1s}, p_{2s}, \dots, p_{\bar{L}s}) \in \mathbb{R}_{++}^{\bar{L}}$: Expected *spot* price vector for physical goods in state s at date 1, with $p_{1s} = 1$ for all $s \in S$ (normalization of the local numeraire).
- **Consumer i 's Choice Variables:**
 - $z_i = (z_{1i}, \dots, z_{\bar{S}i}) \in \mathbb{R}^{\bar{S}}$: Plan of positions in contingent assets negotiated at date 0.
 - $x_i = (x_{1i}, \dots, x_{\bar{S}i}) \in \mathbb{R}_+^{\bar{L}\bar{S}}$: Plan of consumption of physical goods, where $x_{si} = (x_{1si}, \dots, x_{\bar{L}si}) \in \mathbb{R}_+^{\bar{L}}$.

Definition 1: Arrow–Debreu Equilibrium (AD) An **Arrow–Debreu Equilibrium** is a pair (x^*, p^{AD*}) where $x^* = (x_1^*, \dots, x_I^*) \in \mathbb{R}_+^{LSI}$ and $p^{AD*} = (p_1^{AD*}, \dots, p_S^{AD*}) \in \mathbb{R}_{++}^{LS}$ such that:

1. **Optimality:** For each i , x_i^* solves:

$$\max_{x_i \geq 0} U_i(x_i) \quad \text{s.t.} \quad p^{AD*} \cdot (x_i - \omega_i) = \sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq 0$$

2. **Market Clearing (*Market Clearing*):**

$$\sum_{i=1}^I (x_{si}^* - \omega_{si}^*) \leq 0, \quad \forall s \in \{1, \dots, S\}$$

Definition 2: Radner Sequential Equilibrium A **Radner Equilibrium** (or *Equilibrium of Plans, Prices, and Price Expectations*) is a tuple $(x^*, z^*, q^*, (p_1^*, \dots, p_S^*))$ with $q^* \in \mathbb{R}_{++}^S$, $p_s^* \in \mathbb{R}_{++}^L$, and $z^* \in \mathbb{R}^{SI}$ such that:

1. **Optimality:** For each i , (x_i^*, z_i^*) solves:

$$\max_{\substack{x_i \geq 0 \\ z_i \in \mathbb{R}^S}} U_i(x_i)$$

subject to:

$$(i) \quad \sum_{s=1}^S q_s^* z_{si} \leq 0 \quad (\text{Asset constraint at } t = 0)$$

$$(ii) \quad p_s^* \cdot (x_{si} - \omega_{si}) \leq p_{1s}^* z_{si}, \quad \forall s \in \{1, \dots, S\} \quad (\text{Spot constraints at } t = 1)$$

2. **Clearing of the Asset Market at $t = 0$:**

$$\sum_{i=1}^I z_{si}^* \leq 0, \quad \forall s \in \{1, \dots, S\}$$

3. **Clearing of Spot Markets at $t = 1$:**

$$\sum_{i=1}^I (x_{si}^* - \omega_{si}^*) \leq 0, \quad \forall s \in \{1, \dots, S\}$$

Budget Sets Define the set of consumption bundles financeable in each model:

Arrow–Debreu Budget Set:

$$B_i^{AD} = \left\{ (x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} : \sum_s p_s \cdot (x_{si} - \omega_{si}) \leq 0 \right\}$$

Radner Budget Set:

$$B_i^R = \left\{ (x_{1i}, \dots, x_{Si}) \in \mathbb{R}_+^{LS} : \begin{array}{l} \text{there exist } (z_{1i}, \dots, z_{Si}) \text{ such that } \sum_s q_s z_{si} \leq 0 \text{ and} \\ p_s \cdot (x_{si} - \omega_{si}) \leq p_{1s} z_{si} \text{ for all } s \end{array} \right\}$$

Mandatory Structural Hypotheses

1. **Preferences:** $v_i : \mathbb{R}_+^{\bar{L}} \rightarrow \mathbb{R}$ continuous, strictly increasing (local nonsatiability) and strictly quasi-concave.
2. **Asset Market Completeness:** There exist $\bar{S}$ independent Arrow securities, guaranteeing a full-rank return matrix $\bar{S}$.
3. **Self-Confirming Rational Expectations:** All agents anticipate at date 0 exactly the prices p_s^* that zero the *spot* excess demand at date 1.
4. **Absence of Default (No-Default):** All contractual delivery commitments $z_{si} < 0$ are strictly honored at date 1.

What Formally “Breaks” if Any Hypothesis is Removed

- **If price expectations fail** ($p_s^{\text{expected}} \neq p_s^*$): Positions z_{si}^* acquired at date 0 will generate ex post financial balances incompatible with the income necessary to buy the optimal bundle x_{si}^* , breaking market clearing and generating involuntary rationing.
- **If the asset market is incomplete** ($\text{rank}(R) < \bar{S}$): Choices z_i are restricted to a proper subspace of $\mathbb{R}^{\bar{S}}$, preventing the equalization of inter-state marginal rates of substitution and breaking the equivalence with Arrow–Debreu.
- **If $p_{1s} = 0$ for some state s :** The contingent security in Good 1 loses real economic value in state s , making it impossible to transfer purchasing power to the *spot* market of that state.
- **If there is default risk:** If an agent short in $z_{si} < -\omega_{1si}$ cannot honor delivery, the constraint $p_s \cdot x_{si} \leq p_s \cdot \omega_{si} + p_{1s} z_{si}$ fails ex post, breaking equilibrium and Pareto efficiency.

Step-by-Step Proofs

Proposition 1: Radner–Arrow–Debreu Equivalence Theorem *Statement (Proposition 19.D.1 of MWG):* Under strictly increasing and concave preferences, complete asset market, and rational price expectations, an allocation $x^* \in \mathbb{R}_+^{\bar{L}\bar{S}I}$ is an Arrow–Debreu Equilibrium allocation if and only if it is a Radner Sequential Equilibrium allocation.

Proof — Part (i): Direction ($AD \implies \text{Radner}$)

Starting Hypothesis: Let (x^*, p^{AD*}) be an Arrow–Debreu Equilibrium.

Step 1: Construction of the Radner Price System (q^*, p_s^*)

1. **Spot Prices at $t = 1$:** In each state s , we normalize the spot price vector by fixing good 1 as local numeraire ($p_{1s}^* \equiv 1$):

$$p_s^* \equiv \frac{p_s^{AD*}}{p_{1s}^{AD*}} = \left(1, \frac{p_{2s}^{AD*}}{p_{1s}^{AD*}}, \dots, \frac{p_{Ls}^{AD*}}{p_{1s}^{AD*}} \right) \in \mathbb{R}_{++}^L, \quad \forall s \in \{1, \dots, S\}$$

2. **Contingent Security Prices at $t = 0$:** The price of the Arrow security of state s is defined by the Arrow–Debreu contingent price of good 1 in that state:

$$q_s^* \equiv p_{1s}^{AD*} > 0, \quad \forall s \in \{1, \dots, S\}$$

3. Price Factorization:

$$q_s^* p_s^* = p_{1s}^{AD*} \left(\frac{p_s^{AD*}}{p_{1s}^{AD*}} \right) = p_s^{AD*}, \quad \forall s \in \{1, \dots, S\}$$

Step 2: Construction of the Asset Portfolio (z_i^*)

For each consumer i , we define the position in security s as the value of their excess spot expenditure in state s , evaluated at spot prices p_s^* :

$$z_{si}^* \equiv p_s^* \cdot (x_{si}^* - \omega_{si}^*), \quad \forall s \in \{1, \dots, S\}$$

Step 3: Verification of the Feasibility of Radner Constraints

1. **Spot Constraints at $t = 1$ (State s):** Since $p_{1s}^* = 1$, we have:

$$\begin{aligned} p_{1s}^* z_{si}^* &= 1 \cdot z_{si}^* = z_{si}^* = p_s^* \cdot (x_{si}^* - \omega_{si}^*) \\ p_s^* \cdot (x_{si}^* - \omega_{si}^*) &\leq p_{1s}^* z_{si}^* \end{aligned}$$

The spot constraint in state s is satisfied with **exact equality**.

2. **Budget Constraint at $t = 0$:** We calculate the total cost of the portfolio of securities z_i^* :

$$\sum_{s=1}^S q_s^* z_{si}^* = \sum_{s=1}^S q_s^* [p_s^* \cdot (x_{si}^* - \omega_{si}^*)] = \sum_{s=1}^S \underbrace{(q_s^* p_s^*)}_{=p_s^{AD*}} \cdot (x_{si}^* - \omega_{si}^*) = p^{AD*} \cdot (x_i^* - \omega_i^*)$$

By local nonsatiability of preferences, x_i^* reaches the Arrow–Debreu constraint with equality (individual Walras' Law):

$$p^{AD*} \cdot (x_i^* - \omega_i^*) = 0 \implies \sum_{s=1}^S q_s^* z_{si}^* = 0$$

The asset purchase constraint at $t = 0$ is satisfied with exact equality.

Step 4: Equality of Budget Sets and Optimality

Define the set of consumption bundles financeable in Radner as $X_i^{\text{Radner}}(q^*, \{p_s^*\})$. We prove that $X_i^{\text{Radner}} = B_i^{AD}(p^{AD*})$:

- **Inclusion ($\subseteq$) — Feasible in Radner $\implies$ Feasible in AD:** If (x_i, z_i) satisfies Radner constraints, we have $p_s^* \cdot (x_{si} - \omega_{si}) \leq p_{1s}^* z_{si} = z_{si}$ for all s . Multiplying by $q_s^* > 0$ and summing:

$$\sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) = \sum_{s=1}^S q_s^* p_s^* \cdot (x_{si} - \omega_{si}) \leq \sum_{s=1}^S q_s^* z_{si} \leq 0$$

Hence, $p^{AD*} \cdot (x_i - \omega_i) \leq 0 \implies x_i \in B_i^{AD}(p^{AD*})$.

- **Inclusion ($\supseteq$) — Feasible in AD $\implies$ Feasible in Radner:** If $x_i \in B_i^{AD}(p^{AD*})$, choose $z_{si} \equiv p_s^* \cdot (x_{si} - \omega_{si})$. Spot constraints hold by construction with equality and the portfolio cost is:

$$\sum_{s=1}^S q_s^* z_{si} = p^{AD*} \cdot (x_i - \omega_i) \leq 0$$

Hence, $x_i \in X_i^{\text{Radner}}(q^*, \{p_s^*\})$.

- **Optimality:** Since $X_i^{\text{Radner}} = B_i^{AD}$ and the utility function $U_i(x_i)$ is identical in both problems, the allocation x_i^* that maximizes U_i in B_i^{AD} is **necessarily the optimal maximizer of the Radner problem.**

Step 5: Clearing of All Markets

1. **Spot Goods Markets at $t = 1$:** They clear directly by the Arrow–Debreu equilibrium condition:

$$\sum_{i=1}^I (x_{si}^* - \omega_{si}^*) \leq 0, \quad \forall s \in \{1, \dots, S\}$$

2. **Asset Market at $t = 0$:** Summing position z_{si}^* over all I agents:

$$\sum_{i=1}^I z_{si}^* = \sum_{i=1}^I p_s^* \cdot (x_{si}^* - \omega_{si}^*) = p_s^* \cdot \underbrace{\left(\sum_{i=1}^I (x_{si}^* - \omega_{si}^*) \right)}_{\leq 0 \text{ (Market Clearing AD)}} \leq 0, \quad \forall s \in \{1, \dots, S\}$$

Since $q_s^* > 0$, by the aggregate Walras' Law equality holds $\sum_{i=1}^I z_{si}^* = 0$.

Conclusion: The tuple $(x^*, z^*, q^*, (p_1^*, \dots, p_S^*))$ constitutes a **Radner Sequential Equilibrium**. ■

Proof — Part (ii): Direction (Radner $\implies$ AD)

Starting Hypothesis: Let $(x^*, z^*, q^*, (p_1^*, \dots, p_S^*))$ be a Radner Equilibrium.

Step 1: Construction of State Multipliers and Arrow–Debreu Prices (p^{AD*})

1. For each state $s \in \{1, \dots, S\}$, we define the **state multiplier** $\mu_s > 0$ as the ratio between the price of the contingent security and the spot price of good 1 in that state:

$$\mu_s \equiv \frac{q_s^*}{p_{1s}^*} > 0, \quad \forall s \in \{1, \dots, S\}$$

2. We define the Arrow–Debreu contingent price vector for state s by scaling the spot price vector by the multiplier μ_s :

$$p_s^{AD*} \equiv \mu_s p_s^* = \left(\frac{q_s^*}{p_{1s}^*} \right) p_s^* \in \mathbb{R}_{++}^L, \quad \forall s \in \{1, \dots, S\}$$

3. The global vector of Arrow–Debreu contingent prices is:

$$p^{AD*} = (p_1^{AD*}, \dots, p_S^{AD*}) \in \mathbb{R}_{++}^{LS}$$

Step 2: Inclusion 1 — Every consumption feasible in Radner is feasible in Arrow–Debreu ($X_i^{\text{Radner}} \subseteq B_i^{AD}$)

Let x_i be a consumption plan for which there exists a portfolio $z_i = (z_{1i}, \dots, z_{Si})$ that satisfies Radner constraints.

1. In each state $s \in \{1, \dots, S\}$, the spot constraint at $t = 1$ guarantees:

$$p_s^* \cdot (x_{si} - \omega_{si}) \leq p_{1s}^* z_{si}$$

2. Multiply both sides by $\mu_s = \frac{q_s^*}{p_{1s}^*} > 0$:

$$\mu_s p_s^* \cdot (x_{si} - \omega_{si}) \leq \mu_s p_{1s}^* z_{si}$$

3. Substituting $p_s^{AD*} = \mu_s p_s^*$ and $\mu_s p_{1s}^* = q_s^*$:

$$p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq q_s^* z_{si}$$

4. Summing this inequality over all states $s = 1, \dots, S$:

$$\sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq \sum_{s=1}^S q_s^* z_{si}$$

5. Since portfolio z_i satisfies the asset constraint at $t = 0$ ($\sum_{s=1}^S q_s^* z_{si} \leq 0$), we conclude:

$$\sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq 0 \implies p^{AD*} \cdot (x_i - \omega_i) \leq 0$$

This proves that $x_i \in B_i^{AD}(p^{AD*})$.

Step 3: Inclusion 2 — Every consumption feasible in Arrow–Debreu is feasible in Radner ($B_i^{AD} \subseteq X_i^{\text{Radner}}$)

Let $x_i \in B_i^{AD}(p^{AD*})$ be a consumption plan that satisfies the Arrow–Debreu constraint:

$$p^{AD*} \cdot (x_i - \omega_i) = \sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq 0$$

We want to construct an asset portfolio $z_i = (z_{1i}, \dots, z_{Si})$ that finances x_i in the Radner sequential model.

1. For each state $s \in \{1, \dots, S\}$, define the position in security s by:

$$z_{si} \equiv \frac{p_s^* \cdot (x_{si} - \omega_{si})}{p_{1s}^*}$$

2. Testing the spot constraint at $t = 1$ for state s :

$$p_{1s}^* z_{si} = p_{1s}^* \left(\frac{p_s^* \cdot (x_{si} - \omega_{si})}{p_{1s}^*} \right) = p_s^* \cdot (x_{si} - \omega_{si})$$

Hence, $p_s^* \cdot (x_{si} - \omega_{si}) \leq p_{1s}^* z_{si}$ is satisfied with **exact equality**.

3. Testing the portfolio cost z_i at $t = 0$:

$$\sum_{s=1}^S q_s^* z_{si} = \sum_{s=1}^S q_s^* \left[\frac{p_s^* \cdot (x_{si} - \omega_{si})}{p_{1s}^*} \right] = \sum_{s=1}^S \underbrace{\left(\frac{q_s^*}{p_{1s}^*} \right) p_s^*}_{= p_s^{AD*}} \cdot (x_{si} - \omega_{si}) = \sum_{s=1}^S p_s^{AD*} \cdot (x_{si} - \omega_{si}) \leq 0$$

Thus, (x_i, z_i) meets all Radner constraints, which proves that $x_i \in X_i^{\text{Radner}}$.

Step 4: Equality of Budget Sets and Optimality of Bundle x_i^*

From steps 2 and 3, we established the **strict equality of feasible consumption sets**:

$$X_i^{\text{Radner}} = B_i^{AD}(p^{AD*})$$

Since:

1. Preferences $U_i(x_i)$ are rigorously the same in both models.
2. Budget sets coincide perfectly.

The consumption allocation x_i^* that maximizes utility in the Radner problem **is necessarily the optimal choice that solves the Arrow–Debreu problem**.

Step 5: Clearing of All Markets

By the Radner equilibrium hypothesis, spot goods markets clear in all states of nature:

$$\sum_{i=1}^I (x_{si}^* - \omega_{si}^*) \leq 0, \quad \forall s \in \{1, \dots, S\}$$

These are exactly the S market clearing conditions required for the Arrow–Debreu equilibrium.

Conclusion: The pair (x^*, p^{AD*}) constitutes an **Arrow–Debreu Equilibrium**. ■

Proposition 2: Ex Ante Efficiency Implies Ex Post Efficiency in Radner Equilibrium *Statement:* In a Radner Equilibrium under complete asset markets and rational expectations, no feasible ex post physical reallocation of goods in spot markets at date 1 can generate an ex post Pareto improvement among agents.

Step-by-Step Proof:

1. Let $(q^*, \{p_s^*\}_{s \in S}, \{z_i^*, x_i^*\}_{i=1}^{\bar{I}})$ be a Radner Equilibrium.
2. By Proposition 1, x^* coincides with the equilibrium allocation of the Arrow–Debreu model.
3. By the First Welfare Theorem in Arrow–Debreu, x^* is ex ante Pareto efficient.
4. Suppose, for contradiction, that at date 1, after the revelation of state $\hat{s} \in S$, there exists an ex post reallocation $\hat{x}_{\hat{s}}$ such that:
 - (a) $\sum_{i=1}^{\bar{I}} \hat{x}_{i\hat{s}} \leq \sum_{i=1}^{\bar{I}} \omega_{i\hat{s}}$ (physical feasibility in state $\hat{s}$).
 - (b) $v_i(\hat{x}_{i\hat{s}}) \geq v_i(x_{i\hat{s}}^*)$ for all $i \in I$ (weak ex post improvement).
 - (c) $v_k(\hat{x}_{k\hat{s}}) > v_k(x_{k\hat{s}}^*)$ for at least one consumer $k \in I$ (strict ex post improvement).
5. Construct a synthetic ex ante consumption plan $\tilde{x}_i$:

$$\tilde{x}_{is} = x_{is}^*, \quad \forall s \neq \hat{s}; \quad \tilde{x}_{i\hat{s}} = \hat{x}_{i\hat{s}}$$

6. For $i \neq k$:

$$U_i(\tilde{x}_i) = \sum_{s \neq \hat{s}} \pi_s v_i(x_{is}^*) + \pi_{\hat{s}} v_i(\hat{x}_{i\hat{s}}) \geq U_i(x_i^*)$$

7. For k :

$$U_k(\tilde{x}_k) = \sum_{s \neq \hat{s}} \pi_s v_k(x_{ks}^*) + \pi_{\hat{s}} v_k(\hat{x}_{k\hat{s}}) > U_k(x_k^*)$$

8. $\tilde{x}$ is feasible ex ante (by construction).

9. Steps 6, 7, and 8 establish that $\tilde{x}$ Pareto-dominates x^* ex ante.

10. This contradicts the ex ante efficiency of x^* proven in Step 3.

11. It is concluded that the alternative ex post allocation $\hat{x}_{\hat{s}}$ cannot exist.

Thus, **ex ante efficiency guarantees ex post efficiency in spot markets.** ■

Formal Connection Intuition $\rightarrow$ Proof

Table 2.12: Correspondence between economic intuition and formal elements of the Equivalence Theorem.

Mechanism described in Part 1	Corresponding variable/condition
Ex ante wealth transfer via financial assets	Date 0 constraint: $\sum_{s=1}^{\bar{S}} q_s^* z_{si}^* \leq 0$
Physical consumption in the local <i>spot</i> market	<i>Spot</i> constraint: $p_s^* \cdot x_{si}^* \leq p_s^* \cdot \omega_{si}$
State multipliers (μ_s) as numeraire value	$\mu_s \equiv q_s^*/p_{1s}^* = q_s^*$, where $p_s^{AD*} = \mu_s$
Rational and self-fulfilling <i>spot</i> price expectation	Price vector p_s^* anticipated at date 0
Unification of the budget sequence in the Arrow–Debreu constraint	Equality of budget sets: $X_i^{\text{Radner}} = X_i^{\text{Arrow-Debreu}}$

Interpretation of the Canonical MWG Proof The Mas-Colell, Whinston, and Green style is elegant precisely because it uses the equivalence of budget sets ($B_i^{AD} = B_i^R$) as the main “engine” of the proof. The reasoning divides into three logical stages:

1. **Proof of Set Equality ($B_i^{AD} = B_i^R$):** MWG carefully demonstrates both inclusions formally, explicitly constructing the portfolios z_{si} that connect the two problems.
2. **The Implicit “Leap” of Optimality:** Once it is proven that the two budget sets are rigorously the same geometric set in $\mathbb{R}_+^{LS}$, the optimality of the choice becomes an immediate result of set theory:

$$\text{If } B_i^{AD} = B_i^R \implies \max_{x_i \in B_i^{AD}} U_i(x_i) \text{ is identical to } \max_{x_i \in B_i^R} U_i(x_i)$$

For the authors, once $B_i^{AD} = B_i^R$ and the utility function $U_i(\cdot)$ is exactly the same, it is mathematically trivial that the point maximizing utility in one set is the same that maximizes in the other.

3. **The Only Additional Check (Asset Market):** After concluding that the choice x_i^* is the same, the only new thing MWG needs to check is whether the variable z_i^* satisfies the equilibrium conditions of the Radner model:

$$\sum_i z_{si}^* = \frac{1}{p_{1s}} p_s \cdot \underbrace{\left[\sum_i (x_{si}^* - \omega_{si}) \right]}_{\leq 0 \text{ (since } x^* \text{ clears markets in AD)}} \leq 0$$

Since the goods market already cleared in AD and z^* clears the asset market at $t = 0$, the proof concludes with absolute rigor.

Synthesis Under complete asset markets and rational price expectations, the Arrow–Debreu and Radner market structures generate exactly the same consumption and welfare allocations. Sequential trading reduces the number of markets opened ex ante from $L \times S$ to S , but this contractual efficiency depends fundamentally on agents’ ability to anticipate future *spot* prices with perfect precision. The canonical MWG proof demonstrates this equivalence through the strict equality of budget sets ($B_i^{AD} = B_i^R$), guaranteeing that the optimality of consumption choices is preserved in both models.

The Radner Equilibrium replaces the infeasible need for ex ante trading of all contingent physical goods with an efficient sequential trading system, where financial assets transfer wealth between states and *spot* markets allocate physical goods ex post. This perfect equivalence with the Arrow–Debreu model guarantees ex ante and ex post Pareto efficiency, but depends fundamentally on the completeness of financial markets and the self-fulfilling character of future price expectations.

2.6.7 Radner Financial Market: Assets, Arbitrage, and Completeness

Economic Intuition

The Essence The Arrow–Debreu model requires the opening of $L \times S$ contingent markets at date 0, which is operationally prohibitive when L and S are large. The theory of the **Radner Financial Market** resolves this obstacle by introducing a reduced set of **financial assets** (bonds, stocks, derivatives) traded at date 0, whose returns depend on the state of nature realized at date 1. After the revelation of the state, agents use *spot* markets to acquire the desired physical goods.

The real economic problem: How to allow agents to transfer purchasing power between future states without requiring the opening of a market for each contingent good? The answer lies in the capacity of financial assets to **synthesize** contingent income profiles through diversified portfolios.

Example 2.6.12 (Concrete World Analogy — Synthetic Agricultural Insurance). Consider a farmer facing climate risk: drought (s_1) or rain (s_2). Instead of trading contingent physical delivery contracts for inputs (Arrow–Debreu arrangement), they can buy a **financial asset** — an insurance bond — that pays \$100 if drought occurs and \$0 if rain occurs. By combining this bond with other assets (irrigation company stocks, government bonds), the farmer builds a portfolio that guarantees adequate income in both states, allowing them to buy the necessary inputs in the *spot* market after the climate is revealed.

Mechanisms

- **Financial Assets and Return Vector:** An asset k is completely characterized by its **return vector** $r_k = (r_{1k}, \dots, r_{Sk})^T \in \mathbb{R}^S$, where r_{sk} is the asset's payment in state s . The asset pays in units of the **numeraire good** (good 1), avoiding ambiguities about the value of the basket of goods received.
- **Return Matrix R :** When the economy has K assets, we group the return vectors in the columns of an $S \times K$ matrix:

$$R = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1K} \\ r_{21} & r_{22} & \dots & r_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ r_{S1} & r_{S2} & \dots & r_{SK} \end{pmatrix} \in \mathbb{R}^{S \times K}$$

If a consumer chooses a portfolio $z = (z_1, \dots, z_K)^T \in \mathbb{R}^K$ at date 0, the total financial return in state s is $(Rz)_s = \sum_{k=1}^K r_{sk} z_k$.

- **Two Budget Constraints:** The consumer faces a constraint at date 0 (financial market) and S constraints at date 1 (*spot* markets):
 - (a) Date 0: $q \cdot z_i \leq 0$ (the portfolio must be self-financing; purchases require short sales).
 - (b) Date 1 in state s : $p_s \cdot x_{si} \leq p_s \cdot \omega_{si} + (Rz)_s$ (consumption is financed by the physical endowment plus the portfolio return).
- **Absence of Arbitrage:** If agents can sell assets short in unlimited quantities, any price distortion between assets would be exploited on an infinite scale. In equilibrium, the asset price vector q must be **arbitrage-free**, which implies the existence of **implicit state prices** $\mu = (\mu_1, \dots, \mu_S) \geq 0$ such that $q^T = \mu \cdot R$ (Law of One Price).
- **Market Completeness:** If $\text{rank}(R) = S$, the image of R covers all of $\mathbb{R}^S$, allowing any contingent income profile to be synthesized by some portfolio. In this case, the Radner equilibrium is **equivalent** to the Arrow–Debreu equilibrium.

Implications

1. **Drastic Reduction of Markets:** Instead of $L \times S$ contingent markets, only K financial assets at date 0 and L *spot* markets in each state at date 1 are needed.
2. **Role of Synthetic Assets:** It is not necessary to have markets for each physical good. Financial assets such as options, futures, and swaps can complete the market, allowing the decentralization of the Arrow–Debreu efficient allocation.
3. **Allocation Invariance:** What matters for agents' welfare is not the specific assets available, but the **transfer subspace** they can generate ($\text{Range}(R)$). Asset structures with the same image produce the same equilibrium consumption allocation.

Example 2.6.13 (Arrow Securities vs. Derivatives). If there are S purely contingent securities (Arrow securities), the return matrix is the identity matrix $R = I_S$, whose rank is trivially S . Even if there is only 1 primary asset (e.g., a stock whose return varies across states $r_1 > r_2 > \dots > r_S > 0$), if $S - 1$ call options on this asset with appropriate

strike prices are introduced, the resulting return matrix will have an upper triangular form and full rank equal to S . This shows that synthetic financial assets such as options are sufficient to complete the market.

Formal Definition

Definition 19.E.1: Financial Asset and Return Matrix A **financial asset** (or security) k is characterized by its **return vector** $r_k = (r_{1k}, \dots, r_{Sk})^T \in \mathbb{R}^S$, where r_{sk} is the quantity of good 1 (numeraire) paid in state s at date 1.

When the economy has K assets, the **return matrix** $R \in \mathbb{R}^{S \times K}$ is defined by:

$$R = \begin{pmatrix} r_{11} & r_{12} & \dots & r_{1K} \\ r_{21} & r_{22} & \dots & r_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ r_{S1} & r_{S2} & \dots & r_{SK} \end{pmatrix}$$

Special Cases:

- **Arrow security for state s^* :** $r = (0, \dots, 1, \dots, 0)^T$ (canonical basis e_{s^*}).
- **Risk-free asset:** $r = (1, 1, \dots, 1)^T$.

Definition 19.E.2: Radner Equilibrium A **Radner Equilibrium** is a tuple $(q^*, \{p_s^*\}_{s \in S}, \{z_i^*, x_i^*\}_{i=1}^I)$ such that:

1. For each consumer i , the pair (z_i^*, x_i^*) solves:

$$\max_{x_i \in \mathbb{R}_+^{LS}, z_i \in \mathbb{R}^K} U_i(x_{1i}, \dots, x_{Si})$$

subject to:

- (a) $\sum_{k=1}^K q_k^* z_{ki} \leq 0$ (budget constraint at date 0)
- (b) $p_s^* \cdot x_{si} \leq p_s^* \cdot \omega_{si} + \sum_{k=1}^K p_{1s}^* z_{ki} r_{sk}^*$ for all s (budget constraints at date 1)

2. Market clearing:

$$\sum_{i=1}^I z_{ki}^* \leq 0 \quad \forall k, \quad \sum_{i=1}^I x_{si}^* \leq \sum_{i=1}^I \omega_{si} \quad \forall s$$

Normalization: Without loss of generality, $p_{1s}^* = 1$ for all s (good 1 is the local numeraire in each state).

Vector Budget Set:

$$B_i(p, q, R) = \left\{ x_i \in \mathbb{R}_+^{LS} : \exists z_i \in \mathbb{R}^K \text{ such that } q \cdot z_i \leq 0 \text{ and } \begin{pmatrix} p_1 \cdot (x_{1i} - \omega_{1i}) \\ \vdots \\ p_S \cdot (x_{Si} - \omega_{Si}) \end{pmatrix} \leq R z_i \right\}$$

Definition 19.E.3: Market Completeness An asset structure with return matrix $R \in \mathbb{R}^{S \times K}$ is **complete** if $\text{rank}(R) = S$, that is, if there exists a subset of S assets with linearly independent return vectors.

Algebraic Implication: If $\text{rank}(R) = S$, then $\text{Range}(R) = \mathbb{R}^S$, which means that any income transfer vector $w \in \mathbb{R}^S$ can be synthesized by some portfolio $z \in \mathbb{R}^K$ solving $Rz = w$.

Step-by-Step Proofs

Lemma 19.E.1: Absence of Arbitrage and Existence of State Prices *Statement:* If the asset price vector $q \in \mathbb{R}^K$ is arbitrage-free, then there exists a multiplier vector $\mu = (\mu_1, \dots, \mu_S) \geq 0$ such that $q^T = \mu \cdot R$.

Step-by-Step Proof:

1. **Formal Definition of No Arbitrage:** We say that q allows arbitrage if there exists $z \in \mathbb{R}^K$ such that:
 - (1) $q \cdot z \leq 0$ and $Rz > 0$ (non-positive cost today and strictly positive return in at least one state with no losses in the others); OR
 - (2) $q \cdot z < 0$ and $Rz \geq 0$ (strictly positive net gain today and no losses in any future state).

Saying that q is arbitrage-free means that no portfolio z satisfies (1) or (2).

2. **Geometric Construction:** Define the set $V \subset \mathbb{R}^{S+1}$ of all pairs (inverted net cost, future return):

$$V = \{(-q \cdot z, Rz) \in \mathbb{R} \times \mathbb{R}^S : z \in \mathbb{R}^K\}$$

Since V is the image of a linear transformation, V is a vector subspace of $\mathbb{R}^{S+1}$ (convex and closed).

Define the non-zero positive arbitrage cone $K^+ \subset \mathbb{R}^{S+1}$:

$$K^+ = \{(c, y) \in \mathbb{R} \times \mathbb{R}^S : c \geq 0, y \geq 0 \text{ and } (c, y) \neq 0\}$$

The No Arbitrage condition is equivalent to $V \cap K^+ = \emptyset$.

3. **Application of the Separating Hyperplane Theorem:** Since V and K^+ are disjoint convex sets, there exists a non-zero vector $(\mu_0, \mu_1, \dots, \mu_S) \in \mathbb{R}^{S+1}$ such that:
 - $(\mu_0, \mu) \geq 0$ (that is, $\mu_0 \geq 0$ and $\mu_s \geq 0$ for all s);
 - $(\mu_0, \mu) \cdot v = 0$ for all $v \in V$ (since V is a subspace containing the origin).

Substituting $v = (-q \cdot z, Rz) \in V$:

$$\begin{aligned} \mu_0(-q \cdot z) + \mu \cdot (Rz) &= 0 \quad \text{for all } z \in \mathbb{R}^K \\ \implies z \cdot (\mu R - \mu_0 q^T) &= 0 \quad \text{for all } z \in \mathbb{R}^K \end{aligned}$$

For this equality to hold for any z , the vector in parentheses must be zero:

$$\mu_0 q^T = \mu R \tag{1}$$

4. **Normalization** ($\mu_0 > 0$): Suppose, for contradiction, that $\mu_0 = 0$. Then (1) would imply $\mu R = 0$, that is, $\sum_{s=1}^S \mu_s r_{sk} = 0$ for all k . But by hypothesis, each $r_k \geq 0$ and $r_k \neq 0$. Since $(\mu_0, \mu) \neq 0$ and $\mu_0 = 0$, we must have $\mu \geq 0$ with $\mu \neq 0$. Multiplying $\mu \neq 0$ by non-negative non-zero vectors r_k generates a strictly positive product for some k , contradicting $\mu R = 0$. Hence, $\mu_0 > 0$.
5. Dividing (1) by $\mu_0 > 0$ (and redefining $\hat{\mu}_s = \mu_s / \mu_0 \geq 0$):

$$q^T = \mu \cdot R \quad \text{with } \mu = (\mu_1, \dots, \mu_S) \geq 0 \quad \blacksquare$$

Proposition 19.E.1: Equilibrium Prices Are Arbitrage-Free *Statement:* Assume that each return vector is non-negative and non-zero ($r_k \geq 0$ and $r_k \neq 0$ for all k). Then, for every asset price vector $q \in \mathbb{R}^K$ arising from a Radner Equilibrium, it is possible to find multipliers $\mu = (\mu_1, \dots, \mu_S) \geq 0$ such that $q^T = \mu \cdot R$.

Proof: The proof reduces to showing that any price system q^* resulting from a Radner Equilibrium is necessarily arbitrage-free.

1. **Contradiction Hypothesis:** Suppose (q^*, p^*, x^*, z^*) is a Radner Equilibrium, but q^* allows arbitrage.
2. **Explosion of Individual Demand:** If arbitrage exists, there exists $z \in \mathbb{R}^K$ such that $q^* \cdot z \leq 0$ and $Rz > 0$ (or $q^* \cdot z < 0$ and $Rz \geq 0$). Let z_i^* be consumer i 's optimal portfolio. Consider $z_i' = z_i^* + \alpha z$ with $\alpha > 0$.

At date 0: $q^* \cdot z_i' = q^* \cdot z_i^* + \alpha(q^* \cdot z) \leq q^* \cdot z_i^* \leq 0$. The budget constraint remains satisfied.

At date 1 in state s : $p_s^* \cdot x_{si}^* \leq p_s^* \cdot \omega_{si} + (Rz_i^*)_s + \alpha(Rz)_s$. Since $(Rz)_s \geq 0$ for all s (and strictly positive in at least one state), the consumer obtains strictly higher net income in at least one state, without losing income in any other.

3. **Contradiction with Optimality:** Since preferences are strictly increasing / locally nonsatiated, the consumer can let $\alpha \rightarrow +\infty$, increasing their consumption and utility indefinitely without violating any budget constraint. This implies that z_i^* could not be a maximization solution, contradicting the definition of Radner Equilibrium.
4. **Conclusion:** Therefore, in any Radner Equilibrium, q^* is arbitrage-free. By Lemma 19.E.1, the absence of arbitrage guarantees the existence of $\mu \geq 0$ such that $q^T = \mu \cdot R$. $\blacksquare$

Proposition 19.E.2: Equivalence between Arrow–Debreu and Radner under Completeness *Statement:* Suppose the asset structure is complete ($\text{rank}(R) = S$). Then:

- (i) If (x^*, p^{AD*}) is an Arrow–Debreu Equilibrium, then there exist asset prices $q \in \mathbb{R}_{++}^K$ and portfolios $z^* \in \mathbb{R}^{KI}$ such that $(x^*, z^*, q, \{p_s^*\})$ constitutes a Radner Equilibrium.
- (ii) If $(x^*, z^*, q, \{p_s^*\})$ is a Radner Equilibrium, then there exist multipliers $\mu = (\mu_1, \dots, \mu_S) \in \mathbb{R}_{++}^S$ such that $(x^*, \{\mu_s p_s^*\})$ constitutes an Arrow–Debreu Equilibrium.

Proof — Direction (i): AD $\implies$ Radner

1. **Construction of Radner Prices:** Let (p^{AD*}, x^*) be an Arrow–Debreu Equilibrium. For each state s , define:

$$p_s^* \equiv \frac{p_s^{AD*}}{p_{1s}^{AD*}} \in \mathbb{R}_{++}^L \implies p_{1s}^* = 1$$

$$\mu_s \equiv p_{1s}^{AD*} > 0 \implies \mu = (\mu_1, \dots, \mu_S) \in \mathbb{R}_{++}^S$$

Note that $p_s^{AD*} = \mu_s p_s^*$ for all s .

2. **Asset Pricing:** By the Law of One Price:

$$q^T = \mu \cdot R \iff q_k = \sum_{s=1}^S \mu_s r_{sk} \quad \forall k$$

3. **Portfolio Construction via Completeness:** For each consumer i , define the transfer needs vector $w_i \in \mathbb{R}^S$ whose s -th element is:

$$w_{si} \equiv p_s^* \cdot (x_{si}^* - \omega_{si})$$

We want to find $z_i^* \in \mathbb{R}^K$ such that $Rz_i^* = w_i$. Since $\text{rank}(R) = S$, the linear system $Rz_i^* = w_i$ has at least one solution for any w_i .

4. **Verification of the Date 0 Budget Constraint:**

$$q \cdot z_i^* = (\mu \cdot R)z_i^* = \mu \cdot (Rz_i^*) = \mu \cdot w_i = \sum_{s=1}^S \mu_s [p_s^* \cdot (x_{si}^* - \omega_{si})]$$

Substituting $\mu_s p_s^* = p_s^{AD*}$:

$$q \cdot z_i^* = \sum_{s=1}^S p_s^{AD*} \cdot (x_{si}^* - \omega_{si}) = p^{AD*} \cdot (x_i^* - \omega_i) = 0$$

where the last equality follows from Walras' Law in the Arrow–Debreu equilibrium.

5. **Optimality and Market Clearing:** Since the Radner budget set under z_i^* allows replicating exactly any plan feasible in Arrow–Debreu, and x_i^* was the maximizer in Arrow–Debreu, x_i^* remains the optimal choice in Radner. Asset market clearing follows from:

$$R \left(\sum_{i=1}^I z_i^* \right) = \sum_{i=1}^I Rz_i^* = \begin{pmatrix} p_1^* \cdot \sum_i (x_{1i}^* - \omega_{1i}) \\ \vdots \\ p_S^* \cdot \sum_i (x_{Si}^* - \omega_{Si}) \end{pmatrix} = 0$$

Since the columns of R span the space, $\sum_i z_i^* = 0$. ■

Proof — Direction (ii): Radner $\implies$ AD

1. **Uniqueness of Multipliers:** Since (x^*, z^*, q, p) is a Radner Equilibrium, q is arbitrage-free (Proposition 19.E.1). By Lemma 19.E.1, there exist $\mu \in \mathbb{R}_{++}^S$ such that $R^T \mu = q$. Since $\text{rank}(R) = S$, the solution μ is **unique**.
2. **Construction of Arrow–Debreu Prices:** Define:

$$p_s^{AD*} \equiv \mu_s p_s^* \in \mathbb{R}_{++}^L \quad \forall s \implies p^{AD*} = (\mu_1 p_1^*, \dots, \mu_S p_S^*) \in \mathbb{R}_{++}^{LS}$$

3. Equivalence of Budget Sets:

- If x_i is feasible in Radner: $\exists z_i$ such that $q \cdot z_i \leq 0$ and $p_s^* \cdot (x_{si} - \omega_{si}) \leq (Rz_i)_s$ for all s . Multiplying each constraint by $\mu_s > 0$ and summing:

$$\sum_{s=1}^S \mu_s p_s^* \cdot (x_{si} - \omega_{si}) \leq \sum_{s=1}^S \mu_s (Rz_i)_s = \mu \cdot (Rz_i) = (\mu R)z_i = q \cdot z_i \leq 0$$

Since $p_s^{AD*} = \mu_s p_s^*$, this implies $p^{AD*} \cdot (x_i - \omega_i) \leq 0$. Hence, $B_i^{\text{Radner}} \subseteq B_i^{\text{AD}}$.

- If x_i is feasible in Arrow–Debreu ($p^{AD*} \cdot (x_i - \omega_i) \leq 0$): Define $v_{si} = p_s^* \cdot (x_{si} - \omega_{si})$. By completeness ($\text{rank}(R) = S$), $\exists z_i$ such that $Rz_i = v_i$. The cost is:

$$q \cdot z_i = (\mu R)z_i = \mu \cdot (Rz_i) = \mu \cdot v_i = \sum_{s=1}^S \mu_s p_s^* \cdot (x_{si} - \omega_{si}) = p^{AD*} \cdot (x_i - \omega_i) \leq 0$$

Hence, $B_i^{\text{AD}} \subseteq B_i^{\text{Radner}}$.

Therefore, $B_i^{\text{Radner}} = B_i^{\text{AD}}$.

4. **Optimality and Market Clearing:** Since x_i^* maximizes U_i in B_i^{Radner} and the sets coincide, x_i^* is the optimal choice in B_i^{AD} . Spot market clearing in Radner guarantees *market clearing* in Arrow–Debreu. ■

Proposition 19.E.3: Asset Redundancy and Allocation Invariance *Statement:* Suppose (q, p, x^*, z^*) constitutes a Radner Equilibrium for an asset structure with return matrix R . Let R' be the return matrix of a second asset structure. If $\text{Range}(R') = \text{Range}(R)$, then x^* is still the consumption allocation of a Radner Equilibrium in the economy with the second structure.

Step-by-Step Proof:

1. **State Prices:** Since (q, p, x^*, z^*) is a Radner Equilibrium under R , q is arbitrage-free. By Lemma 19.E.1, there exist $\mu \geq 0$ such that $q^T = \mu \cdot R$.
2. **Construction of New Prices:** Define prices for the K' assets of matrix R' using the same μ :

$$(q')^T \equiv \mu \cdot R' \in \mathbb{R}^{K'} \quad \Longleftrightarrow \quad q'_m = \sum_{s=1}^S \mu_s r'_{sm} \quad \forall m$$

By construction, q' is arbitrage-free in the new structure R' .

3. **Portfolio Reconstruction:** For each consumer i , the original portfolio z_i^* generated $v_i \equiv Rz_i^* \in \text{Range}(R)$. Since $\text{Range}(R') = \text{Range}(R)$, there exists $z_i'^* \in \mathbb{R}^{K'}$ such that $R'z_i'^* = Rz_i^* = v_i$.
4. **Cost Verification:**

$$q' \cdot z_i'^* = (\mu \cdot R')z_i'^* = \mu \cdot (R'z_i'^*) = \mu \cdot (Rz_i^*) = (\mu \cdot R)z_i^* = q \cdot z_i^* \leq 0$$

The date 0 budget constraint remains satisfied.

5. **Equivalence of Budget Sets:** Since $\text{Range}(R') = \text{Range}(R)$ and $(q')^T = \mu \cdot R'$, any plan feasible under R is feasible under R' , and vice versa:

$$B_i(p, q', R') = B_i(p, q, R)$$

Since the budget set has not changed, x_i^* remains the optimal choice.

6. **Market Clearing:** Spot markets have not changed (x^* is the same). For the asset market:

$$R' \left(\sum_{i=1}^I z_i'^* \right) = \sum_{i=1}^I R' z_i'^* = \sum_{i=1}^I R z_i^* = R \left(\sum_{i=1}^I z_i^* \right) = 0$$

Since $R'(\sum_i z_i'^*) = 0$, we can choose portfolios such that $\sum_i z_i'^* = 0$. ■

Formal Connection Intuition → Proof

Table 2.13: Correspondence between economic intuition and formal elements of the Radner financial market.

Economic mechanism	Formal element
Financial assets synthesize income profiles	Return matrix R and image $\text{Range}(R) \subseteq \mathbb{R}^S$
Unlimited short sales eliminate arbitrage	No arbitrage $\implies$ existence of $\mu \geq 0$ such that $q^T = \mu \cdot R$
Implicit state prices μ_s	Multipliers of Lemma 19.E.1 (price today of \$1 in state s tomorrow)
Market completeness	$\text{rank}(R) = S \implies \text{Range}(R) = \mathbb{R}^S$
AD–Radner equivalence	$B_i^{\text{Radner}} = B_i^{\text{AD}}$ when $\text{rank}(R) = S$
Asset redundancy	$\text{Range}(R') = \text{Range}(R) \implies$ same allocation x^*

Systematic Resolution of the Formal Question

Diagnosis

- **Area:** Microeconomics / Financial Theory in General Equilibrium.
- **Type of Optimization:** Optimization with Intertemporal and Cross-State Budget Constraints (Lagrange Multipliers and Separating Hyperplane Theorem).
- **Main Tools:** Linear Algebra (rank, image, kernel), Farkas' Alternative Theorem, Law of One Price, Market Completeness.

Interpretation of Results

1. **Lemma 19.E.1 (No Arbitrage $\implies$ State Prices):** The Separating Hyperplane Theorem guarantees that, if there is no “free lunch” (arbitrage), there exists a “ruler” (vector μ) that consistently prices all assets. Each μ_s is the implicit price today of receiving \$1 in state s tomorrow.
2. **Proposition 19.E.1 (Equilibrium $\implies$ No Arbitrage):** If there were arbitrage in equilibrium, agents would exploit the opportunity on an infinite scale ($\alpha \rightarrow \infty$), making the existence of finite optimal choices impossible. Hence, equilibrium prices are necessarily arbitrage-free.

3. **Proposition 19.E.2 (Completeness $\implies$ AD–Radner Equivalence):** When $\text{rank}(R) = S$, any contingent income profile can be synthesized. Budget sets coincide ($B_i^{\text{Radner}} = B_i^{\text{AD}}$), and equilibrium allocations are identical. The First Welfare Theorem holds in its fullness.
4. **Proposition 19.E.3 (Allocation Invariance):** What matters for welfare is not the specific assets, but the transfer subspace they generate. Structures with the same image produce the same allocation. This underpins neutrality theorems such as Modigliani–Miller in corporate finance.

Synthesis The Radner financial market drastically reduces the number of markets needed to decentralize the Arrow–Debreu efficient allocation. Market completeness ($\text{rank}(R) = S$) is the key condition guaranteeing equivalence between the two models. The absence of arbitrage, in turn, ensures the existence of implicit state prices μ that consistently price all assets through the Law of One Price ($q^T = \mu \cdot R$). Asset redundancy demonstrates that the specific structure of securities is irrelevant for welfare — only the capacity for income transfer between states ($\text{Range}(R)$) matters.

2.6.8 Incomplete Markets and Restricted Efficiency

Economic Intuition

The Essence In an economy with incomplete markets ($\text{rank}(R) < S$), we already know that the consumption outcome of a Radner equilibrium does not generally achieve the Arrow–Debreu *First-Best*, since agents do not have enough securities to freely transfer income to all S future states.

The real economic problem: Therefore, the correct question about efficiency is not “whether the equilibrium achieves the *First-Best*”, but rather: if a Central Planner were as limited as the financial market — being able to redistribute only portfolio positions in the K existing assets, but without the capacity to create new markets or impose direct *ex-post* consumption transfers — would the market equilibrium make the most of the available financial structure?

Example 2.6.14 (Concrete World Analogy — The Limited Planner). Imagine a fund manager who can only redistribute asset portfolios among investors, using exactly the same K securities available in the market. They cannot create new derivatives, cannot impose direct future consumption transfers, and cannot interfere with the *spot* prices that will form in each state. The question is: does the allocation decentralized by the market already exhaust all possible gains from trade within this limited structure? If the answer is yes, the equilibrium is **constrained Pareto optimal**.

Mechanisms

- **Constrained Pareto Optimality:** A portfolio allocation $(z_1, \dots, z_I)$ is **constrained Pareto optimal** if there is no other feasible asset allocation ($\sum_i z'_i \leq 0$) that increases some individual’s utility without harming any other.
- **The Case $L = 1$ (Single Good per State):** When there is only 1 consumption good in each state s , the local *spot* price is trivially normalized to $p_s = 1$. There

are no relative goods prices in period $t = 1$. Individual i 's consumption in state s is strictly determined by their initial endowment ω_{si} plus the portfolio return z_i :

$$x_{si} = \omega_{si} + (Rz_i)_s = \omega_{si} + \sum_{k=1}^K r_{sk} z_{ki}$$

- **Induced Utility over Portfolios:** Since x_{si} is a direct function of z_i , the original utility function $U_i(x_{1i}, \dots, x_{Si})$ induces a direct utility function over asset portfolios:

$$U_i^*(z_{1i}, \dots, z_{Ki}) \equiv U_i\left(\omega_{1i} + \sum_k r_{1k} z_{ki}, \dots, \omega_{Si} + \sum_k r_{Sk} z_{ki}\right)$$

- **Isomorphism with a Classical K -Good Economy:** Consumer i 's problem at date $t = 0$ becomes:

$$\max_{z_i \in \mathbb{R}^K} U_i^*(z_{1i}, \dots, z_{Ki}) \quad \text{s.t.} \quad q \cdot z_i \leq 0$$

This is an absolutely traditional pure exchange economy with K commodities (the assets), prices q , and zero initial endowment.

Implications

1. **When $L = 1$:** Since the portfolio economy satisfies all the classical hypotheses of an exchange economy (induced preferences U_i^* locally nonsatiated and concave in z_i), we can directly apply the First Welfare Theorem in the asset space $\mathbb{R}^K$. Hence, every Radner equilibrium is constrained Pareto optimal.
2. **When $L > 1$ (Multiple Goods per State):** Future *spot* prices $p_s(z)$ begin to fluctuate. The *spot* price vector p_s in state s is no longer fixed and comes to depend on the portfolios chosen by agents, since income redistribution via assets alters the *spot* demands for each good. **Pecuniary externalities** arise: when a consumer changes their portfolio z_i , they alter the equilibrium *spot* prices p_s in period $t = 1$, affecting the real purchasing power of all other consumers' asset returns.
3. **Geanakoplos–Polemarchakis (1986) Inefficiency:** Since individual portfolio choices affect *spot* prices but agents take prices as given (*price takers*), the financial market generates uninternalized pecuniary externalities. For this reason, if $L > 1$, the Radner equilibrium in incomplete markets **is generally not even constrained Pareto optimal**. A central planner who slightly alters asset portfolios at date $t = 0$ can manipulate the *spot* prices of date $t = 1$ so as to generate a strict Pareto gain for all consumers.

Formal Definition

Definition 19.F.1: Constrained Pareto Optimality An asset allocation $(z_1, \dots, z_I) \in \mathbb{R}^{KI}$ is **constrained Pareto optimal** if it is feasible (that is, $\sum_i z_i \leq 0$) and if there is no other feasible asset allocation $(z'_1, \dots, z'_I) \in \mathbb{R}^{KI}$ such that

$$U_i^*(z'_{1i}, \dots, z'_{Ki}) \geq U_i^*(z_{1i}, \dots, z_{Ki}) \quad \text{for all } i,$$

with at least one strict inequality.

Proposition 19.F.1: Restricted Efficiency with $L = 1$ *Statement:* Suppose there are two periods and only one consumption good in the second period ($L = 1$). Then any Radner equilibrium is constrained Pareto optimal in the sense that there is no possible redistribution of assets in the first period that leaves every consumer equally well off and at least one consumer strictly better off.

Step-by-Step Proofs

Proof of Proposition 19.F.1 *Proof:*

Step 1: Construction of Induced Preferences over Portfolios

Let $L = 1$. Individual i 's consumption in state $s \in \{1, \dots, S\}$ given a portfolio $z_i = (z_{1i}, \dots, z_{Ki})^T \in \mathbb{R}^K$ is:

$$x_{si}(z_i) = \omega_{si} + \sum_{k=1}^K r_{sk} z_{ki} = \omega_{si} + (Rz_i)_s$$

Since x_{si} is a direct affine function of z_i , we can define an induced utility function over portfolio choices $U_i^* : \mathbb{R}^K \rightarrow \mathbb{R}$:

$$U_i^*(z_{1i}, \dots, z_{Ki}) \equiv U_i(\omega_{1i} + (Rz_i)_1, \dots, \omega_{Si} + (Rz_i)_S)$$

Under standard hypotheses on U_i (continuity, concavity, and local nonsatiation in consumption), the induced utility function U_i^* inherits concavity, continuity, and local nonsatiation in z_i .

Step 2: Mapping to a Standard Exchange Economy

In a Radner equilibrium with asset price vector $q^* \in \mathbb{R}^K$, each consumer i solves:

$$\max_{z_i \in \mathbb{R}^K} U_i^*(z_i) \quad \text{s.t.} \quad q^* \cdot z_i \leq 0$$

This optimization problem is formally identical to a standard consumer choice problem in a pure exchange economy where:

- The “goods” traded are the K financial assets z_i .
- The “price vector” is the asset price vector q^* .
- Each individual's initial asset endowment is the zero portfolio $0 \in \mathbb{R}^K$.

An asset allocation $(z_1^*, \dots, z_I^*)$ is a Radner equilibrium allocation if z_i^* solves consumer i 's utility maximization problem for all i , and the asset market clears:

$$\sum_{i=1}^I z_i^* \leq 0$$

Step 3: Proof by Contradiction (Logic of the First Welfare Theorem)

Suppose, for contradiction, that the equilibrium portfolio allocation $(z_1^*, \dots, z_I^*)$ is not constrained Pareto optimal.

Then there exists another feasible asset allocation $(z'_1, \dots, z'_I) \in \mathbb{R}^{K \cdot I}$ satisfying $\sum_{i=1}^I z'_i \leq 0$ such that:

$$U_i^*(z'_i) \geq U_i^*(z_i^*) \quad \text{for all } i = 1, \dots, I$$

with strict inequality for at least one consumer h .

By the properties of utility maximization in a competitive market:

- For any consumer i , if $U_i^*(z'_i) \geq U_i^*(z_i^*)$, local nonsatiation implies $q^* \cdot z'_i \geq 0$.
- For consumer h , since $U_h^*(z'_h) > U_h^*(z_h^*)$, z'_h must be strictly inaccessible at prices q^* :

$$q^* \cdot z'_h > 0$$

Summing these inequalities over all I consumers:

$$\sum_{i=1}^I q^* \cdot z'_i = q^* \cdot \left(\sum_{i=1}^I z'_i \right) > 0$$

Since $q^* \geq 0$ (by no arbitrage under non-negative asset returns), the condition $\sum_{i=1}^I q^* \cdot z'_i > 0$ implies:

$$\sum_{i=1}^I z'_i \not\leq 0$$

This contradicts the hypothesis that $(z'_1, \dots, z'_I)$ is a feasible asset allocation ($\sum_i z'_i \leq 0$).

Therefore, no Pareto-dominating asset allocation exists, and $(z_1^*, \dots, z_I^*)$ is constrained Pareto optimal. ■

Formal Connection Intuition → Proof

Table 2.14: Correspondence between economic intuition and formal elements of restricted efficiency.

Economic mechanism	Formal element
Planner limited to existing asset structure	Feasible allocation: $\sum_i z'_i \leq 0$
Absence of relative prices when $L = 1$	$p_s = 1$ for all s ; consumption is an affine function of z
Induced utility over portfolios	$U_i^*(z_i) \equiv U_i(\omega_i + Rz_i)$
Isomorphism with K -good exchange economy	Problem: $\max U_i^*(z_i)$ s.t. $q \cdot z_i \leq 0$
Application of the First Theorem in asset space	Contradiction: $q^* \cdot \sum_i z'_i > 0$ violates feasibility
Pecuniary externalities when $L > 1$	Spot prices $p_s(z)$ depend on portfolios; generic inefficiency

Systematic Resolution

Diagnosis

- **Area:** Microeconomics / General Equilibrium with Incomplete Markets (GEI).
- **Type of Optimization:** Static Optimization with Budget Constraints in Asset Space (Isomorphism with Pure Exchange).

- **Main Tools:** First Welfare Theorem applied to asset space, Induced Utility Functions, Pecuniary Externalities.

Table 2.15: Comparison between the single-good and multiple-good cases.

Feature	Single Good ($L = 1$)	Multiple Goods ($L \geq 2$)
<i>Spot</i> prices (p_s)	Trivial ($p_s = 1$, fixed)	Endogenous ($p_s(z)$ depends on portfolios z)
Role of assets	Directly determine real consumption	Determine nominal wealth, which alters <i>spot</i> prices
Externalities	None	Pecuniary externalities via changes in p_s
Restricted efficiency	Guaranteed (Proposition 19.F.1)	Generic failure (Geanakoplos–Polemarchakis)

Comparative Table: $L = 1$ vs. $L \geq 2$

Interpretation

1. **The $L = 1$ Frontier:** The restriction $L = 1$ is the fundamental dividing line in general equilibrium under incomplete markets. When there is only one good per state, there are no relative prices to be distorted, and the asset market functions as a standard exchange economy.
2. **Pecuniary Externalities:** When $L > 1$, portfolio choices alter the wealth distribution in period $t = 1$, which alters the relative prices of physical goods. Since agents are price takers, they do not internalize the effect of their choices on future prices. This pecuniary externality generates inefficiency.
3. **Implications for Economic Policy:** The Geanakoplos–Polemarchakis (1986) result shows that, in economies with multiple goods and incomplete markets, there is room for government intervention in the financial market. A regulator who imposes constraints on traded portfolios (e.g., leverage limits, margin requirements) can, in principle, improve the welfare of all agents.

Synthesis Proposition 19.F.1 establishes that, when there is only one consumption good per state ($L = 1$), the Radner equilibrium with incomplete markets is constrained Pareto optimal: the market makes the most of the available financial structure. The proof exploits the isomorphism between the portfolio choice problem and a pure exchange economy with K goods, allowing the direct application of the First Welfare Theorem. However, when $L > 1$, pecuniary externalities arise via endogenous *spot* prices, and the Radner equilibrium generally fails to be constrained Pareto optimal. This fundamental result, due to Geanakoplos and Polemarchakis (1986), demonstrates the limits of decentralized efficiency in incomplete financial economies with multiple goods.

2.6.9 Inefficiency under Incomplete Markets

Economic Intuition

The Essence In the real world, the existence of incomplete markets prevents the economy from achieving Pareto efficiency. Microeconomic theory shows that this incompleteness causes **two levels of inefficiency**:

1. **Failure of the *First-Best* (Arrow–Debreu Absolute Pareto Efficiency):** In a real economy, the number of independent financial assets is much smaller than the number of possible future states of nature ($\text{rank}(R) < S$). Because of these “gaps” in risk coverage, consumers cannot freely transfer wealth to every future scenario. The First Welfare Theorem ceases to hold, and the market equilibrium allocation falls short of the ideal Arrow–Debreu allocation.
2. **Failure of the *Second-Best* (Restricted Pareto Efficiency):** The more surprising question arises when we ask whether the market at least makes the best possible use of the few assets that exist. In the simplified model with 1 good per state ($L = 1$), we saw in Proposition 19.F.1 that the Radner equilibrium is constrained Pareto efficient. But in the real world with multiple goods per state ($L \geq 2$), this restricted efficiency collapses (classical result of Geanakoplos–Polemarchakis, 1986).

Example 2.6.15 (Central Planner Analogy). Imagine a central planner unable to create new financial assets, but who can reallocate existing portfolios among consumers. When $L = 1$, this planner cannot improve anyone without harming someone else — the market equilibrium already exhausts all possible gains from trade within the limited financial structure. But when $L \geq 2$, financial decisions affect future goods prices, generating side effects (pecuniary externalities) that the market alone cannot correct. A planner who intervenes in the asset market at period $t = 0$ can readjust *spot* prices at period $t = 1$ so as to generate a strict Pareto gain.

Mechanisms

- **When $L = 1$:** There are no relative *spot* prices in the future. Financial redistribution alters income but does not alter relative prices. The equilibrium is efficient in the portfolio space.
- **When $L \geq 2$:** When individuals trade assets in the present ($t = 0$), financial redistribution alters their available income in the different future states ($t = 1$). This change in income alters the physical demand for goods in the future, causing relative *spot* prices (p_s) to fluctuate.
- **Pecuniary Externalities:** Since consumers act as price takers, no one considers how their individual portfolio choice affects the future prices of other agents. This lack of coordination generates pecuniary externalities that the market does not internalize.

Implications

1. **The First Theorem is Conditional:** It is not a universal result; it depends critically on a strict set of hypotheses, with market completeness being one of the most fundamental. Whenever one of these conditions fails in the real world — whether due to incomplete markets, externalities, asymmetric information, or monopoly power — the invisible hand fails.
2. **Room for Public Policy:** In economies with incomplete markets and multiple goods, the market equilibrium is generically inefficient, which opens theoretical room for public policies and financial market regulations that seek to coordinate these income transfers.

3. **The Theorems as *Benchmarks*:** Microeconomics treats the Welfare Theorems not as descriptions of the real world, but as reference models. They show exactly which institutional conditions would be necessary for the free market to allocate resources perfectly.

Accessible Proof via the Envelope Theorem

Statement: In an economy with incomplete markets and $L \geq 2$ goods per state, the Radner equilibrium is generically constrained inefficient (Constrained Pareto Inefficient).

Proof (Central Idea):

1. The Simplified Environment

Consider a 2-period economy ($t = 0, 1$), 2 future states ($s = 1, 2$), 2 goods in each state ($l = 1, 2$), and 2 consumers (A and B). There is 1 financial asset z traded at $t = 0$ at price q . In period $t = 1$, in state s , consumers have income $w_{si}(z)$ arising from the value of their physical endowment plus the asset return. In each state s , goods 1 and 2 are traded at relative *spot* price $p_s = (1, p_{2s})$, where good 1 is the numeraire ($p_{1s} = 1$). Consumer $i \in \{A, B\}$'s indirect utility in state s is given by $V_s^i(p_{2s}, w_{si})$, and their total expected utility is:

$$U_i(z_i) = \sum_{s=1}^2 \pi_s V_s^i(p_{2s}(z), w_{si}(z_i))$$

2. The Effect of the *Spot* Price on Welfare (Roy's Identity)

How does the price p_{2s} affect consumer i 's welfare in state s ? By Roy's Identity (or direct application of the Envelope Theorem to the consumer's problem), the derivative of indirect utility with respect to the *spot* price of good 2 is:

$$\frac{\partial V_s^i}{\partial p_{2s}} = -\mu_{si} \cdot (x_{2si} - \omega_{2si})$$

where:

- $\mu_{si} = \frac{\partial V_s^i}{\partial w_{si}}$ is the marginal utility of income in state s .
- $(x_{2si} - \omega_{2si})$ is the consumer's net demand for good 2 in state s .

Intuition: If consumer A is a net buyer of good 2 in state s ($(x_{2sA} - \omega_{2sA}) > 0$), an increase in price p_{2s} reduces their welfare. If they are a net seller, the price increase raises their welfare.

3. The Central Planner's Intervention

Suppose the economy is in a Radner Equilibrium with equilibrium portfolios (z_A^*, z_B^*) . The Central Planner compulsorily reallocates an infinitesimal quantity of the asset at $t = 0$: forces consumer A to buy dz of the asset and consumer B to sell $-dz$. How does this variation dz affect consumer A 's expected utility?

$$dU_A = \underbrace{\left[\sum_{s=1}^2 \pi_s \mu_{sA} \frac{\partial w_{sA}}{\partial z_A} \right] dz}_{\text{Direct Income Effect (1st Order)}} + \underbrace{\sum_{s=1}^2 \pi_s \frac{\partial V_s^A}{\partial p_{2s}} \cdot \frac{\partial p_{2s}}{\partial z_A} dz}_{\text{Indirect Effect via Spot Prices (Externality)}}$$

4. Where the Market Fails

- **The Direct Income Effect is Zero to 1st Order:** Since consumer A was already individually maximizing their utility at $t = 0$, their First-Order Condition with respect to z_A is satisfied ($\sum_s \pi_s \mu_{sA} r_s = \lambda_A q$). Therefore, any direct portfolio redistribution has a zero net first-order effect on individual welfare.
- **The Indirect Effect via Prices is NOT Zero:** Substituting Roy's Identity into the second term:

$$dU_A = \sum_{s=1}^2 \pi_s [-\mu_{sA} \cdot (x_{2sA} - \omega_{2sA})] \cdot \left(\frac{\partial p_{2s}}{\partial z_A} \right) dz$$

5. The Proof of Pareto Gain

Since consumers take *spot* prices as given (*price takers*), they do not consider the term $\frac{\partial p_{2s}}{\partial z_A}$ when choosing their portfolios at $t = 0$. The Central Planner, on the other hand, sees that by transferring income between states via dz :

1. Aggregate demand for goods in period $t = 1$ changes.
2. This forces a change in equilibrium *spot* prices $\frac{\partial p_{2s}}{\partial z_A}$.
3. This change in *spot* prices transfers real purchasing power between net buyers and net sellers of goods in each state.
4. Since the market is incomplete, the Marginal Rates of Substitution of income between states for consumers A and B are not equal ($MRS_{12}^A \neq MRS_{12}^B$).
5. This gives the Central Planner a mathematical degree of freedom: they can choose the direction of dz so that the change in the *spot* price in State 1 benefits Consumer A more than it harms Consumer B , and compensating in State 2, resulting in:

$$dU_A > 0 \quad \text{and} \quad dU_B > 0$$

Proof Summary: The “magic” is simply the presence of a pecuniary externality. With $L = 1$, there are no relative *spot* prices in the future ($\frac{\partial p_{2s}}{\partial z} = 0$), so the indirect effect is zero and the equilibrium is efficient in the portfolio space. With $L \geq 2$, financial trading at $t = 0$ affects *spot* prices at $t = 1$. Since agents ignore this effect when choosing their portfolios, the decentralized equilibrium cannot coordinate the beneficial manipulation of *spot* prices, allowing a planner's intervention to generate a strict Pareto improvement. ■

Formal Connection Intuition → Proof

Systematic Resolution

Diagnosis

- **Area:** Microeconomics / General Equilibrium with Incomplete Markets (GEI).
- **Type of Optimization:** Marginal Perturbation Analysis via the Envelope Theorem and Roy's Identity.
- **Main Tools:** Envelope Theorem, Roy's Identity, Pecuniary Externalities, First-Order Analysis.

Table 2.16: Correspondence between economic intuition and formal elements of inefficiency under incomplete markets.

Economic mechanism	Formal element
Agents are <i>spot</i> price takers	$\frac{\partial p_{2s}}{\partial z_A}$ ignored in individual FOC
Pecuniary externality	Term $\sum_s \pi_s \frac{\partial V_s^A}{\partial p_{2s}} \cdot \frac{\partial p_{2s}}{\partial z_A} \neq 0$
Roy's Identity / Envelope Theorem	$\frac{\partial V_s^i}{\partial p_{2s}} = -\mu_{si} \cdot (x_{2si} - \omega_{2si})$
Individual FOC saturates direct effect	$\sum_s \pi_s \mu_{sA} r_s = \lambda_A q \implies$ zero 1st-order effect
Divergent inter-state MRS	$MRS_{12}^A \neq MRS_{12}^B$ gives planner a degree of freedom
Strict Pareto gain	$dU_A > 0$ and $dU_B > 0$ for some dz

Interpretation

1. **The $L = 1$ vs. $L \geq 2$ Frontier:** The restriction $L = 1$ is the fundamental dividing line. When there is only one good per state, there are no relative prices to be distorted, and the asset market functions as a standard exchange economy.
2. **Pecuniary Externalities:** When $L > 1$, portfolio choices alter the wealth distribution in period $t = 1$, which alters the relative prices of physical goods. Since agents are price takers, they do not internalize the effect of their choices on future prices.
3. **Implications for Economic Policy:** The Geanakoplos–Polemarchakis (1986) result shows that, in economies with multiple goods and incomplete markets, there is room for government intervention in the financial market. A regulator who imposes constraints on traded portfolios can, in principle, improve the welfare of all agents.

Synthesis The Geanakoplos–Polemarchakis (1986) result demonstrates that, in economies with incomplete markets and multiple goods per state, the Radner equilibrium is generically constrained inefficient. The proof, accessible via the Envelope Theorem and Roy's Identity, reveals that the inefficiency stems from pecuniary externalities: portfolio choices at $t = 0$ affect *spot* prices at $t = 1$, but agents, being price takers, ignore this effect. A central planner who manipulates portfolios can alter *spot* prices so as to generate Pareto gains, exploiting the divergence of inter-state MRS across consumers.

2.6.10 The Firm Problem and *Spanning* Theory

Economic Intuition

The Essence Section 19.G of Mas-Colell, Whinston, and Green studies the Theory of the Firm and the Stock Market in general equilibrium under uncertainty. The central question is: **what should the firm maximize when markets are incomplete?**

The real economic problem: Under complete markets, all shareholders unanimously agree that the firm should maximize its market value. Under incomplete markets, this unanimity collapses — different shareholders, with different assessments of state probabilities and different risk aversions, come to disagree about the production plan the firm should adopt.

Example 2.6.16 (Concrete World Analogy — The Shareholders' Meeting). Imagine an oil company that needs to decide between two projects: (A) exploring a field in the Arctic (high return if the climate is cold, low if hot) and (B) exploring a field in the Gulf of Mexico (stable but lower return). Under complete markets, all shareholders agree on which project maximizes the firm's value, since they can adjust their personal portfolios to neutralize any effect. Under incomplete markets, the shareholder who lives in the Arctic and suffers from cold weather may prefer project A, while the shareholder who lives in Texas may prefer project B. The shareholders' meeting cannot reach a consensus — the firm's objective becomes politically determined.

Mechanisms

- **The Concept of *Spanning*:** A random variable $a \in \mathbb{R}^S$ is in the *span* of the existing asset structure if it belongs to the image of the return matrix R . This means that the financial market already has enough assets to perfectly synthesize/replicate the payments of a in all states.
- **Unanimity under *Spanning*:** If the variation in the firm's profits $\Delta\pi_j$ is in the *span* of available assets, any shareholder who disapproves of the production change can perfectly neutralize it by readjusting their personal portfolio. All shareholders once again agree on the market value maximization rule.
- **Disagreement without *Spanning*:** If the variation in profits cannot be synthesized by existing assets, the firm's decision alters the very structure of the financial market. Since the MRS of income between states varies from shareholder to shareholder, shareholders come to disagree about which production plan the company should adopt.

Implications

1. **Generalized Modigliani–Miller Neutrality:** When the firm's production changes remain within the *span* of available assets, the firm's financial structure is irrelevant — all shareholders agree on market value maximization.
2. **Breakdown of Unanimity:** When the firm creates new risk profiles that cannot be replicated by existing assets, it alters shareholders' opportunity sets heterogeneously, and unanimity breaks down.
3. **Political Objective of the Firm:** Under incomplete markets without *spanning*, the firm ceases to have an objective maximization rule, making the corporation's objective dependent on political control or voting rules among shareholders.

Formal Definition

Definition 19.G.1: *Spanning* of Random Variables A set $A \subset \mathbb{R}^S$ of random variables is **spanned** by the existing asset structure if every vector $a \in A$ belongs to the image of the return matrix R of the asset structure. That is, for each $a \in A$, there exists an asset portfolio $z \in \mathbb{R}^K$ such that:

$$Rz = a \quad \Longleftrightarrow \quad \begin{pmatrix} r_{11}z_1 + \cdots + r_{1K}z_K \\ \vdots \\ r_{S1}z_1 + \cdots + r_{SK}z_K \end{pmatrix} = \begin{pmatrix} a_1 \\ \vdots \\ a_S \end{pmatrix}$$

Economic Interpretation: A random variable a is in the *span* of the market if the risk profile it generates is not “new”. If $a \in \text{Range}(R)$, it means the financial market already has enough assets to perfectly synthesize/replicate the payments of a in all states.

Step-by-Step Proofs

Shareholder Unanimity Theorem (Ekern–Stiglitz, 1973) *Statement:* Consider a Radner economy with incomplete markets and an asset structure with return matrix $R_{S \times K}$. Let $\theta_{ij} \geq 0$ be the initial fraction of shares of firm j owned by consumer i (with $\sum_i \theta_{ij} = 1$). Firm j produces a profit vector across the S states given by $\pi_j = (\pi_{1j}, \dots, \pi_{Sj})^T \in \mathbb{R}^S$. Suppose firm j is considering altering its production plan, which changes its future profit vector from π_j to π'_j , generating a profit variation of:

$$\Delta\pi_j = \pi'_j - \pi_j \in \mathbb{R}^S$$

If the change in the profit vector is in the *span* of the existing asset structure — that is, if $\Delta\pi_j \in \text{Range}(R)$ — then:

1. **Unanimity:** All shareholders of firm j unanimously agree on approving or rejecting the change.
2. **Market Value Maximization:** A change $\pi_j \rightarrow \pi'_j$ increases the welfare of all shareholders if and only if it increases the firm’s market value at period $t = 0$:

$$\Delta V_j = q \cdot \Delta z_j = \mu \cdot \Delta\pi_j > 0$$

where q is the asset price vector and $\mu \geq 0$ are the implicit state prices ($q^T = \mu \cdot R$).

Step-by-Step Proof:

1. The *Spanning* Hypothesis

Since $\Delta\pi_j \in \text{Range}(R)$, there exists by definition an asset portfolio $\Delta z_j \in \mathbb{R}^K$ traded in the financial market that perfectly synthesizes this profit variation:

$$R\Delta z_j = \Delta\pi_j$$

2. The Effect on the Shareholder’s Budget Constraint

Consider shareholder i ’s budget constraint at date $t = 1$ in state s . With the initial production plan π_j , the individual’s income is:

$$p_s \cdot (x_{si} - \omega_{si}) \leq (Rz_i)_s + \theta_{ij}\pi_{sj}$$

If the firm changes its plan to π'_j , the new income in state s becomes:

$$p_s \cdot (x_{si} - \omega_{si}) \leq (Rz'_i)_s + \theta_{ij}\pi'_{sj} = (Rz'_i)_s + \theta_{ij}(\pi_{sj} + \Delta\pi_{sj})$$

Substituting $\Delta\pi_j = R\Delta z_j$:

$$p_s \cdot (x_{si} - \omega_{si}) \leq (R[z'_i + \theta_{ij}\Delta z_j])_s + \theta_{ij}\pi_{sj}$$

3. Personal Portfolio Recomposition

Shareholder i can define a new adjusted portfolio $\hat{z}_i$ given by:

$$\hat{z}_i = z'_i + \theta_{ij}\Delta z_j \quad \implies \quad z'_i = \hat{z}_i - \theta_{ij}\Delta z_j$$

Now let us see how much this choice costs at date $t = 0$:

$$q \cdot z'_i = q \cdot \hat{z}_i - \theta_{ij}(q \cdot \Delta z_j)$$

Since $q^T = \mu \cdot R$, the term $q \cdot \Delta z_j$ is exactly the change in the firm's market value ΔV_j :

$$q \cdot \Delta z_j = (\mu \cdot R)\Delta z_j = \mu \cdot (R\Delta z_j) = \mu \cdot \Delta \pi_j = \Delta V_j$$

Thus, individual i 's budget constraint at date $t = 0$ becomes:

$$q \cdot \hat{z}_i \leq \theta_{ij}\Delta V_j$$

4. Conclusion

- If $\Delta V_j > 0$, consumer i 's net income at period $t = 0$ strictly increases by $\theta_{ij}\Delta V_j > 0$. Their budget set expands in all directions, allowing them to reach a strictly higher utility level.
- If $\Delta V_j < 0$, their budget set shrinks, reducing their utility.

Since the sign of $\theta_{ij}\Delta V_j$ depends only on $\Delta V_j > 0$ (since $\theta_{ij} > 0$), all shareholders, regardless of their preferences or risk aversion, agree that the firm should choose the production plan that maximizes $V_j = \mu \cdot \pi_j$. ■

What Happens if *Spanning* Fails ($\Delta \pi_j \notin \text{Range}(R)$)? If the variation in the firm's profits cannot be synthesized by existing assets, the firm's decision comes to alter the very structure of the financial market (the company creates a new *hedge* opportunity that did not previously exist). In this case:

- **Breakdown of Unanimity:** Since the Marginal Rates of Substitution of income between states vary from shareholder to shareholder ($\text{MRS}^A \neq \text{MRS}^B$), shareholder A may prefer the firm to produce more in State 1, while shareholder B prefers more production in State 2.
- **Nonexistence of a Clear Objective:** The firm ceases to have an objective maximization rule, making the corporation's objective dependent on political control or voting rules among shareholders.

Formal Connection Intuition $\rightarrow$ Proof

Systematic Resolution

Diagnosis

- **Area:** Microeconomics / Theory of the Firm under Uncertainty and Incomplete Markets.
- **Type of Optimization:** Marginal Perturbation Analysis via Linear Algebra (Matrix Image) and the Law of One Price.
- **Main Tools:** *Spanning* Concept, Law of One Price ($q^T = \mu \cdot R$), Portfolio Recomposition, Ekern–Stiglitz Unanimity Theorem (1973).

Table 2.17: Correspondence between economic intuition and formal elements of the unanimity theorem.

Economic mechanism	Formal element
Profit variation replicable by existing assets	$\Delta\pi_j \in \text{Range}(R) \implies \exists \Delta z_j : R\Delta z_j = \Delta\pi_j$
Shareholder can neutralize production change	Recomposition: $\hat{z}_i = z'_i + \theta_{ij}\Delta z_j$
Law of One Price	$q^T = \mu \cdot R \implies q \cdot \Delta z_j = \mu \cdot \Delta\pi_j = \Delta V_j$
Uniform expansion of the budget set	$q \cdot \hat{z}_i \leq \theta_{ij}\Delta V_j$ for all i
Unanimity independent of preferences	Sign of $\theta_{ij}\Delta V_j$ depends only on ΔV_j
Breakdown of unanimity without <i>spanning</i>	$\Delta\pi_j \notin \text{Range}(R) \implies \text{MRS}^A \neq \text{MRS}^B$ generates disagreement

Interpretation

1. **Neutrality under *Spanning*:** When the firm's production changes remain within the *span* of available assets, the firm's financial structure is irrelevant — all shareholders agree on market value maximization. This is the analogue, in general equilibrium under uncertainty, of the Modigliani–Miller Theorem.
2. **Disagreement under Incompleteness:** When the firm creates new risk profiles that cannot be replicated by existing assets, it alters shareholders' opportunity sets heterogeneously. Unanimity breaks down, and the firm's objective becomes politically determined.
3. **Implications for Corporate Governance:** In incomplete markets, the market value maximization rule is not universally valid. The firm needs to adopt voting mechanisms or decision rules that reflect the heterogeneous preferences of shareholders.

Synthesis The Shareholder Unanimity Theorem (Ekern–Stiglitz, 1973) establishes that, under incomplete markets, shareholder unanimity about the firm's objective is restored if and only if the changes in the firm's profits are in the *span* of the existing asset structure. The proof exploits the fact that, when $\Delta\pi_j \in \text{Range}(R)$, any shareholder can neutralize the production change by readjusting their personal portfolio, and the Law of One Price guarantees that the cost of this recomposition is exactly the change in the firm's market value. When *spanning* fails, unanimity breaks down, and the firm loses its objective maximization rule — a fundamental result connecting the theory of the firm to the structure of financial markets.

2.7 MWG Exercises

16.C.1^B Show that if a consumption set $X_i \subset \mathbb{R}^L$ is nonempty, closed, and bounded and the preference relation $\succsim_i$ on X_i is continuous, then $\succsim_i$ cannot be locally nonsatiated. [*Hint:* Show that the continuous utility function representing $\succsim_i$ must have a maximum on X_i ; see Section M.F of the Mathematical Appendix on this.]

16.C.2^A Suppose that the preference relation $\succsim_i$ is locally nonsatiated and that x_i^* is

maximal for $\succsim_i$ in the set $\{x_i \in X_i : p \cdot x_i \leq w_i\}$. Prove that the following property holds: “If $x_i \succ_i x_i^*$ then $p \cdot x_i \geq w_i$.”

16.C.3^B In this exercise you are asked to establish the first welfare theorem under a set of assumptions compatible with satiation. Suppose that every X_i is nonempty and convex and that every $\succsim_i$ is strictly convex (i.e., if $x'_i \succsim_i x_i$ and $x'_i \neq x_i$ then $\alpha x'_i + (1 - \alpha)x_i \succ_i x_i$ whenever $0 < \alpha < 1$). Prove the following:

- (a) Every i can have at most one global satiation point and preferences are locally nonsatiated at any consumption bundle different from the single global satiation point.
- (b) Any price equilibrium with transfers is a Pareto optimum.

16.C.4^A Suppose that for each individual there is a “pleasure function” depending on her own consumption only, given by $u_i(x_i)$. Every individual’s utility depends positively on her own and everyone else’s “pleasure” according to the utility function

$$U_i(x_1, \dots, x_I) = U_i(u_1(x_1), u_2(x_2), \dots, u_I(x_I)).$$

Show that if $x = (x_1, \dots, x_I)$ is Pareto optimal relative to the $U_i(\cdot)$ ’s, then $(x_1, \dots, x_I)$ is also a Pareto optimum relative to the u_i ’s. Does this mean that a community of altruists can use competitive markets to attain Pareto optima? Does your argument depend on the concavity of the u_i ’s, or the U_i ’s?

16.D.1^A Prove that if preferences are locally nonsatiated then the condition: “if $x_i \succ_i x_i^*$ then $p \cdot x_i \geq p \cdot x_i^*$ ” is equivalent to the condition: “ x_i^* is expenditure minimizing for the price vector p in the set $\{x_i \in X_i : x_i \succsim_i x_i^*\}$.”

16.D.2^B Exhibit a one-firm, one-consumer economy in which the production set is convex, the preference relation is continuous and convex, and there is nevertheless a Pareto optimal allocation that can be supported neither as a price equilibrium with transfers nor as a price quasiequilibrium with transfers. Which condition of Proposition 16.D.1 fails?

16.D.3^B Suppose that we have an economy with continuous and strongly monotone preferences (consumption sets are $X_i = \mathbb{R}_+^L$). Suppose also that a strictly positive production is possible; that is, there are $y_j \in Y_j$ such that $\sum_j y_j + \bar{\omega} \gg 0$. Prove that any price quasiequilibrium with transfers must be a price equilibrium with transfers. [Hint: Show first that $w_i > 0$ for some i and then argue that $p \gg 0$.]

16.D.4^C Consider a two-good exchange economy with r identical consumers. The consumption set is $\mathbb{R}_+^2$, the individual endowments are $\omega \in \mathbb{R}_{++}^2$, and the preferences are continuous and strongly monotone but not necessarily convex. Argue that the symmetric allocation in which every consumer gets her initial allocation is either a Walrasian equilibrium (for some price vector p) or, if it is not, then for r large enough it is not a Pareto optimum. [Hint: the differentiable case is simpler.]

16.E.1^B Given a utility possibility set U , denote by $U' \subset U$ the subset actually achieved by feasible allocations:

$$U' = \left\{ (u_1(x_1), \dots, u_I(x_I)) : \sum_i x_i = \sum_j y_j + \bar{\omega} \text{ for some } y_j \in Y_j \right\}.$$

(Relative to U' , the set U allows for free disposal of utility).

- (a) Give a two-consumer, two-commodity exchange example showing that it is possible for a point of U' to belong to the boundary of U and *not* be a Pareto optimum.
- (b) Suppose that every Y_j satisfies free disposal and $0 \in Y_j$. Also, assume that for every i , $X_i = \mathbb{R}_+^L$ and $\succsim_i$ is continuous and strongly monotone. Show that any boundary point of U that belongs to U' is then a Pareto optimum. [Hint: Let $u_i(0) = 0$ for all i and show first that $U' = U \cap \mathbb{R}_+^I$. Next argue that if $u \in U$ is a Pareto optimum and $0 \leq u' \leq u$, $u' \neq u$, then we must be able to reach u' with a surplus of goods relative to u .]
- (c) Consider an exchange economy with consumption sets equal to $\mathbb{R}_+^L$, continuous, locally nonsatiated preferences, and a strictly positive total endowment vector. Show that if $u = (u_1, \dots, u_I)$ is the utility vector corresponding to a price quasiequilibrium with transfers then u cannot be in the interior of U ; that is, there is no feasible allocation yielding higher utility to *every* consumer. [Hint: Show that $w_i > 0$ for some i and then apply Proposition 16.D.2.]

16.E.2^B Show that the utility possibility set U of an economy with convex production and consumption sets and with concave utility functions is convex.

16.F.1^B In text.

16.F.2^A Derive the first-order conditions (16.F.2) and (16.F.3) of the maximization problem (16.F.1).

16.F.3^A Derive conditions (16.F.4), (16.F.5), and (16.F.6) from the first-order conditions (16.F.2) and (16.F.3).

16.F.4^A Derive the first-order conditions (16.F.4), (16.F.5), and (16.F.6) from problems (16.F.7), (16.F.8), and (16.F.9), respectively.

17.B.1^A Show that for a pure exchange economy with $J = 1$ and $Y_1 = -\mathbb{R}_+^L$, “ $y_1^* \leq 0$, $p \cdot y_1^* = 0$, and $p \geq 0$ ” if and only if “ $y_1^* \in Y_1$ and $p \cdot y_1^* \geq p \cdot y_1$ for all $y_1 \in Y_1$.”

17.B.2^B Prove property (v) of Proposition 17.B.2. The proof of Proposition 17.B.2 in the text contains a hint. Recall also the following technical fact: any bounded sequence in $\mathbb{R}^L$ has a convergent subsequence.

17.B.3^B Suppose that $z(\cdot)$ is an aggregate excess demand function satisfying conditions (i) to (v) of Proposition 17.B.2. Let $p^n \rightarrow p$ with some, but not all, of the components of p being zero.

- (a) Show that as n becomes large, the maximal excess demand is always obtained for some commodity whose price goes to zero.
- (b) Argue (if possible by example) that a commodity whose price goes to zero may actually remain in excess supply for all n . [Hint: Relative prices matter.]

17.B.4^B Suppose that there are J firms whose production sets $Y_1, \dots, Y_J \subset \mathbb{R}^L$ are closed, strictly convex, and bounded above. Suppose also that a strictly positive consumption

bundle is producible using the initial endowments and the economy's aggregate production set $Y = \sum_j Y_j$ (i.e., there is an $\bar{x} \gg 0$ such that $\bar{x} \in \{\sum_i \omega_i\} + Y$). Show that the production inclusive aggregate excess demand function $\hat{z}(p)$ in (17.B.3) satisfies properties (i) to (v) of Proposition 17.B.2.

17.B.5^A Suppose that there are J firms. Each firm produces a single output under conditions of constant returns. The unit cost function of firm j is $c_j(p)$, which we assume to be differentiable. The consumption side of the economy is expressed by an aggregate excess demand function $z(p)$. Write down an equation system similar to (17.B.4)–(17.B.5) for the equilibria of this economy.

17.B.6^C [Rader (1972)] Suppose that there is a single production set Y and that Y is a closed, convex cone satisfying free disposal. Consider the following exchange equilibrium problem. Given prices $p = (p_1, \dots, p_L)$, every consumer i chooses a vector $v_i \in \mathbb{R}^L$ so as to maximize $\succsim_i$ on the set

$$\{x_i \in X_i : p \cdot v_i \leq p \cdot \omega_i \text{ and } x_i = v_i + y \text{ for some } y \in Y\}.$$

The price vector p and the choices $v^* = (v_1^*, \dots, v_I^*)$ are in equilibrium if $\sum_i v_i^* = \sum_i \omega_i$. Show that, under the standard assumptions on preferences and consumption sets, the price vector and the individual consumptions constitute a Walrasian equilibrium for the economy with production. Interpret.

17.C.1^A Verify that the correspondence $f(\cdot)$ introduced in the proof of Proposition 17.C.1 is convex-valued.

17.C.2^C Show that a convex-valued correspondence $z(\cdot)$ defined on $\mathbb{R}_+^L$ and satisfying the conditions (i) to (v) listed below (parallel to the corresponding conditions in Proposition 17.C.1) admits a solution; that is, there is a p with $0 \in z(p)$.

- (i) $z(\cdot)$ is upper-hemicontinuous.
- (ii) $z(\cdot)$ is homogeneous of degree zero.
- (iii) For every p and $z \in z(p)$ we have $p \cdot z = 0$ (Walras' law).
- (iv) There is $s \in \mathbb{R}$ such that $z_\ell > -s$ for any $z \in z(p)$ and p .
- (v) If $p^n \rightarrow p \neq 0$, $z^n \in z(p^n)$ and $p_\ell = 0$ for some ℓ , then $\max\{z_1^n, \dots, z_L^n\} \rightarrow \infty$.

[Hint: If you try to replicate exactly the proof of Proposition 17.C.1 you will run into difficulties with the upper-hemicontinuity condition. A possible three-step approach goes as follows: (1) Show that for $\varepsilon > 0$ small enough the solutions must be contained in $\Delta_\varepsilon = \{p \in \Delta : p_\ell \geq \varepsilon \text{ for all } \ell\}$; (2) argue then that for $r > 0$ large enough, one has $z(p) \subset [-r, r]^L$ for every $p \in \Delta_\varepsilon$; finally, (3) carry out a fixed-point argument in the domain $\Delta_\varepsilon \times [-r, r]^L$. For an easier result, you could limit yourself to prove the convex-valued parallel to Proposition 17.C.2. The suggested domain for the fixed-point argument is then $\Delta \times [-r, r]^L$.]

17.C.3^B Consider an exchange economy in which every consumer i has continuous, strongly monotone, strictly convex preferences, and $\omega_i \gg 0$. The peculiarity of the equilibrium problem to be considered is that the consumer will now pay a type of tax on his gross

consumption; moreover, this tax can differ across commodities and consumers. We will also assume that total tax receipts are rebated equally across consumers and in a lump-sum fashion. Specifically, for every i there is a vector of given tax rates $t_i = (t_{i1}, \dots, t_{iL}) \geq 0$ and for every price vector $p \gg 0$ the budget set of consumer i is

$$B_i(p, w_i) = \left\{ x_i \in \mathbb{R}_+^L : \sum_{\ell} (1 + t_{i\ell}) p_{\ell} x_{i\ell} \leq w_i \right\}.$$

An *equilibrium with taxes* is then a price vector $p \gg 0$ and an allocation $(x_1^*, \dots, x_I^*)$ with $\sum_i x_i^* = \sum_i \omega_i$ such that every i maximizes preferences in

$$B\left(p, p \cdot \omega_i + \frac{1}{I} \sum_{\ell, i} t_{i\ell} p_{\ell} x_{i\ell}^*\right).$$

- (a) Illustrate the notion of an equilibrium with taxes in an Edgeworth box. Verify that an equilibrium with taxes need not be a Pareto optimum.
- (b) Apply Proposition 17.C.1 to show that an equilibrium with taxes exists.
- (c) As formulated here, the taxes are on gross consumptions. If they were imposed instead on net consumptions, that is, on amounts purchased or sold, then (assuming the same rate for buying or selling) the budget set would be

$$B_i(p, T_i) = \left\{ x_i \in \mathbb{R}_+^L : p \cdot (x_i - \omega_i) + \sum_{\ell} t_i |p_{\ell}(x_{i\ell} - \omega_{i\ell})| \leq T_i \right\},$$

where the T_i are the lump-sum rebates. In what way does this budget set differ from that described previously for the case of taxes on gross consumptions? Represent graphically. Notice the kinks.

- (d) Write down a budget set for the situation similar to (c) except that the tax rates for amounts bought or sold may be different.
- (e) (More advanced) How would you approach the existence issue for the modification described in (c)?

17.C.4^A Consider a pure exchange economy. The only novelty is that a progressive tax system is instituted according to the following rule: individual wealth is no longer $p \cdot \omega_i$; instead, anyone with wealth above the mean of the population must contribute half of the excess over the mean into a fund, and those below the mean receive a contribution from the fund in proportion to their deficiency below the mean.

- (a) For a two-consumer society with endowments $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$, write the after-tax wealths of the two consumers as a function of prices.
- (b) If the consumer preferences are continuous, strictly convex, and strongly monotone, will the excess demand functions satisfy the conditions required for existence in Proposition 17.C.1 given that wealth is being redistributed in this way?

17.C.5^B Consider a population of I consumers. Every consumer i has consumption set $\mathbb{R}_+^L$ and continuous, strictly convex preferences $\succsim_i$. Suppose, in addition, that every i has a household technology $Y_i \subset \mathbb{R}^L$ satisfying $0 \in Y_i$. We can then define the *induced*

preferences $\succsim_i^*$ on $\mathbb{R}_+^L$ by $x_i \succsim_i^* x'_i$ if and only if for any $y'_i \in Y_i$ with $x'_i + y'_i \geq 0$ there is $y_i \in Y_i$ with $x_i + y_i \geq 0$ and $x_i + y_i \succsim_i x'_i + y'_i$ (i.e., whatever can be done from x'_i , something at least as good can be obtained from x_i).

- (a) Show that induced preferences are rational, that is, complete and transitive.
- (b) Show that if Y_i is convex then induced preferences $\succsim_i^*$ are convex.
- (c) Suppose that goods are of two kinds: marketed goods and nonmarketed household goods. Initial preferences $\succsim_i$ care only about household goods, and initial endowments ω_i have nonzero entries only for marketed goods. Use the concept of induced preferences to set up the equilibrium problem as one that is formally a problem of pure exchange among marketed goods. Discuss.

17.C.6^B Let $L = 2$. Consider conditions (i), (iii), and (v) of Proposition 17.B.2. Exhibit four examples such that in each of the examples only one condition fails and yet the system of equations $z(p) = 0$ has no solution. Why is condition (ii) not included in the list?

17.D.1^B Consider an exchange economy with two commodities and two consumers. Both consumers have homothetic preferences of the constant elasticity variety. Moreover, the elasticity of substitution is the same for both consumers and is small (i.e., goods are close to perfect complements). Specifically,

$$u_1(x_{11}, x_{21}) = (2x_{11}^\rho + x_{21}^\rho)^{1/\rho} \quad \text{and} \quad u_2(x_{12}, x_{22}) = (x_{12}^\rho + 2x_{22}^\rho)^{1/\rho},$$

and $\rho = -4$. The endowments are $\omega_1 = (1, 0)$ and $\omega_2 = (0, 1)$. Compute the excess demand function of this economy and verify that there are multiple equilibria.

17.D.2^A Apply the implicit function theorem to show that if $f(\bar{v}) = 0$ is a system of M equations in N unknowns and if at $\bar{v}$ we have $f(\bar{v}) = 0$ and $\text{rank } Df(\bar{v}) = M$, then in a neighborhood of $\bar{v}$ the solution set of $f(\cdot) = 0$ can be parameterized by means of $N - M$ parameters.

17.D.3^A Carry out explicitly the computations for Proposition 17.D.4.

17.D.4^C Consider a two-commodity, two-consumer exchange economy satisfying the appropriate differentiability conditions on utility and demand functions. There is a total endowment vector $\bar{\omega} \gg 0$. Show that for almost every $\omega_1 \ll \bar{\omega}$ the economy defined by the initial endowments ω_1 and $\omega_2 = \bar{\omega} - \omega_1$ has a finite number of equilibria. This differs from the situation in Proposition 17.D.2 in that total endowments are kept fixed. [*Hint*: You should use the properties of the Slutsky matrix.]

17.D.5^B Consider a two-commodity, two-consumer exchange economy satisfying the appropriate differentiability conditions on utility and demand functions. Set the equilibrium problem as an equation system in the consumption variables $x_1 \in \mathbb{R}_+^2$ and $x_2 \in \mathbb{R}_+^2$, the price variables $p \in \mathbb{R}_+^2$, and the reciprocals of the marginal utilities of wealth $\lambda_1 \in \mathbb{R}$ and $\lambda_2 \in \mathbb{R}$ (neglect the possibility of boundary equilibria). The parameters of the system are the initial endowments $(\omega_1, \omega_2) \in \mathbb{R}_+^4$. Prove without further aggregation that (after deleting one equation and one unknown) the system satisfies the full rank condition of the transversality theorem.

17.D.6^B The setup is identical to Exercise 17.D.5 except that an externality is allowed: The (differentiable) utility function of consumer 1 may depend on the consumption of consumer 2; that is, it has the form $u_1(x_1, x_2)$ where x_i is consumer i 's consumption bundle [but we still have $u_2(x_2)$]. Equilibrium is defined as usual, with the proviso that consumer 1 takes consumer 2's consumption as given. Show that, generically on initial endowments $(\omega_1, \omega_2) \in \mathbb{R}_+^4$, the number of equilibria is finite.

17.D.7^B Suppose the agents of an overall exchange economy are distributed across N islands with no communication among them. Each island economy has three equilibria.

- (a) Argue that the number of equilibria in the overall economy is 3^N .
- (b) Suppose now that the islands' economies are identical and that there is a possibility of communication across the islands: free and costless transportation of commodities. Show that then the number of equilibria is 3.

19.C.1^A There are S states. A consumer has, in every state s , a Bernoulli utility function $u_s(x_s)$, where $x_s \in \mathbb{R}_+^L$. Suppose that, for every s , $u_s(\cdot)$ is concave. Show that the expected utility function $U(x_1, \dots, x_S) = \sum_s \pi_s u_s(x_s)$ defined on $\mathbb{R}_+^{LS}$ is concave.

19.C.2^A For the model described in Example 19.C.2, show that the marginal rates of substitution along the Pareto set are as drawn in Figure 19.C.2; that is, at any point of the Pareto set the marginal rate of substitution is smaller than the ratio of probabilities.

19.C.3^A Consider an Arrow–Debreu equilibrium of the economy described in Section 19.C. Suppose that $L = 1$ and that preferences of every consumer i admit an expected utility representation with continuous, strictly concave, and strictly increasing Bernoulli utility functions (identical across states). For every state s denote by p_s , π_{si} , and x_{si} the equilibrium price for the s -contingent commodity, the subjective probability of consumer i for state s , and the consumption of consumer i in state s , respectively. Denote by $\bar{p} = \sum_s p_s$ the price of uncontingent delivery of one unit of consumption. Show that

$$\sum_s (\pi_{si} \bar{p} - p_s) x_{si} \geq 0$$

for every i . [*Hint:* Use a revealed preference argument.] Interpret.

19.C.4^B There are a single consumption good, two states, and two consumers. Note that this allows the use of Edgeworth boxes. Utility functions are of the expected utility form. Bernoulli utility functions are identical across states. That is,

$$U_1(x_{11}, x_{21}) = \pi_{11} u_1(x_{11}) + \pi_{21} u_1(x_{21})$$

and

$$U_2(x_{12}, x_{22}) = \pi_{12} u_2(x_{12}) + \pi_{22} u_2(x_{22}),$$

where x_{si} is the amount of s -contingent good consumed by consumer i and π_{si} is the subjective probability of consumer i for state s . We assume that every $u_i(\cdot)$ is strictly concave and differentiable. The total initial endowments of the two contingent commodities are $\bar{\omega} = (\bar{\omega}_1, \bar{\omega}_2) \gg 0$. We assume that every consumer gets half of the random variable $\bar{\omega}$, that is, $(\omega_{11}, \omega_{21}) = \frac{1}{2} \bar{\omega}$ and $(\omega_{12}, \omega_{22}) = \frac{1}{2} \bar{\omega}$.

- (a) Suppose that consumer 1 is risk neutral, consumer 2 is not, and both consumers have the same subjective probabilities. Show that at an interior Arrow–Debreu equilibrium consumer 2 insures completely.
- (b) Suppose now that consumer 1 is risk neutral, consumer 2 is not, and the subjective probabilities of the two consumers are not the same. Show then that at an interior (Arrow–Debreu) equilibrium consumer 2 will not insure completely. Which is the direction of the bias in terms of the differences in subjective probabilities? Argue also that consumer 1 (the risk-neutral agent) will not gain from trade.

19.C.5^A Consider an economy such as the one introduced in Section 19.C but with only one commodity in each state. There is a number I of risk-averse consumers. Preferences admit an expected utility representation. Suppose that the Bernoulli utility functions of a consumer for the good are identical across states and that subjective probabilities are the same across individuals. Individual endowments vary from state to state. However, we assume that the total endowment is nonstochastic, that is, uniform across states (if, say, I is large and the realizations of individual endowments are identically and independently distributed, then the total endowment per capita is almost nonstochastic). Set up the Arrow–Debreu trading problem. Show that the allocation in which every individual's consumption in every state is the average across states of his endowments is an equilibrium allocation.

19.D.1^A Consider the model of sequential trade of Section 19.D. The only difference is that we now assume that, for every s , the s -contingent commodity pays 1 dollar (rather than one unit of physical good 1) if state s occurs (and nothing otherwise). Write down the budget constraints corresponding to this model and discuss which price normalizations are possible.

19.D.2^A Show that in Example 19.D.1 the contingent trades of the two consumers are as claimed in the discussion of the example.

19.D.3^A Formulate a model similar to the two-period model of Section 19.D with the difference that consumption also takes place at period $t = 0$. Show that the result of Proposition 19.D.1 remains valid.

19.D.4^B Consider a three-period economy, $t = 0, 1, 2$, in which at $t = 0$ the economy splits into two branches and at $t = 1$ every branch splits again into two. There are H physical commodities and consumption can take place at the three dates.

- (a) Describe the Arrow–Debreu equilibrium problem for this economy.
- (b) Describe the Radner equilibrium problem. Suppose that at $t = 0$ and $t = 1$ there are contingent markets for the delivery of one unit of the first physical good at the following date.
- (c) Argue that the conclusion of Proposition 19.D.1 remains valid.

19.E.1^B Consider an asset trading model such as that considered in Section 19.E. The only difference is that consumption is also possible at date $t = 0$. Assume for simplicity that the Bernoulli utility functions on consumption are state independent and additively separable across time; that is, $u_i(x_{0i}, x_{1i}) = u_{0i}(x_{0i}) + u_{1i}(x_{1i})$, where $x_{0i}, x_{1i} \in \mathbb{R}^L$.

- (a) Argue along the lines of the second proof of Proposition 19.E.1 that the conclusion of Proposition 19.E.1 remains valid.
- (b) Suppose now that there is a single physical good in every period. Express the multipliers μ_s in terms of marginal utilities of consumption.

19.E.2^A Show that if a primary asset with return vector $r \in \mathbb{R}^S$ separates states, that is, if $r_s \neq r_{s'}$ whenever $s \neq s'$, then it is possible to create a complete asset structure by using only options on this primary asset. You can assume that $r_s > 0$ for every s .

19.E.3^A Complete part (i) of the proof of Proposition 19.E.2 in the manner requested in the text.

19.E.4^B In text.

19.E.5^B Complete the verification that at the prices specified in Example 19.E.7 the set of consumptions achievable through sequential trade is the same as the set of consumptions achievable through *ex ante* trade of the four Arrow–Debreu commodities.

19.E.6^A There are two dates; at date 1 there are three states; at date 0 there is trade in assets. There are two basic assets whose return vectors in current dollars are

$$r_1 = (64, 16, 4) \quad \text{and} \quad r_2 = (0, 0, 1).$$

The market prices of these assets are $q_1 = 32$ and $q_2 = 1$, respectively. In the following you are asked to price by arbitrage a variety of derived assets.

- (a) Suppose that one unit of a derived asset is described as “One unit of this asset confers the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs).” Write the return vector of this asset and price it.
- (b) The situation is the same as in (a) except that the asset is modified to read “One unit of this asset confers the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs) provided the spot value is at least 10.”
- (c) Suppose that the asset is as in (b) except that “at least 10” is replaced by “at least 19.” Write down the return vector and argue that this asset cannot be priced by arbitrage with the available primary assets.
- (d) How would the analysis in (c) differ if we had in addition a riskless asset with a price equal to 1? (You do not need to compute the price explicitly.)
- (e) Suppose that now the asset is further complicated to read “One unit of this asset confers, at the choosing of the holder, either 1 dollar in period 1 or the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs).” Write the return vector of this asset and price it.
- (f) The situation is the same as in (e) except that the asset is modified to read “One unit of this asset confers, at the choosing of the holder, either 1 dollar in period 1 or the right to buy one unit of asset 1 at 75% of its spot value in period 1 (after the state of the world occurs) provided this value is at least 10.”

Chapter 3

Conceptual Map for Assessment

This chapter consolidates the analytical framework of the course into a structured reference document organized as a conceptual map. The organization follows the methodological thread:

VNM Axioms and EU $\longrightarrow$ Risk Attitudes $\longrightarrow$ General Equilibrium $\longrightarrow$
Arrow–Debreu / Radner $\longrightarrow$ No-Arbitrage and Assets

3.1 Part I — Choice under Risk

The first part develops the problem of the individual decision maker under risk. The central chain is:

lotteries $\longrightarrow$ preferences $\longrightarrow$ Expected Utility $\longrightarrow$ risk attitudes $\longrightarrow$ applications

3.1.1 Lotteries and VNM Expected Utility

Fundamental Objects and the Lottery Space

- **Set of Consequences (Z):** Finite (or compact) set of certain prizes/outcomes $Z = \{x_1, x_2, \dots, x_N\}$.
- **Simple Lottery (L):** Direct probability distribution over Z , represented by a vector $L = (p_1, p_2, \dots, p_N)$ such that $p_k \geq 0$ and $\sum_{k=1}^N p_k = 1$. The space of all simple lotteries over Z is denoted by $\mathcal{L}$ or $\Delta(Z)$ (the probability simplex).
- **Compound Lottery:** A lottery whose intermediate prizes are themselves other simple lotteries. Formally, given K lotteries $L_1, \dots, L_K$ and weights $\alpha_k \geq 0$ with $\sum \alpha_k = 1$, the compound lottery is $\bigoplus_{k=1}^K \alpha_k L_k$.
- **Reduction of Compound Lotteries (RCLA / Consequentialism):** Axiom/hypothesis that the agent cares only about the final probability distribution over the primary prizes in Z . The reduced lottery $L = \sum_{k=1}^K \alpha_k L_k \in \mathcal{L}$ assigns probability $p_n = \sum_{k=1}^K \alpha_k p_n^k$ to each prize x_n .

The von Neumann–Morgenstern (VNM) Axioms

Let $\succsim$ be a binary preference relation defined over the space of simple lotteries $\mathcal{L}$.

Axiom 3.1.1 (Rationality: Completeness and Transitivity). • **Completeness:** For any $L, L' \in \mathcal{L}$, $L \succsim L'$ or $L' \succsim L$.

- **Transitivity:** For any $L, L', L'' \in \mathcal{L}$, if $L \succsim L'$ and $L' \succsim L''$, then $L \succsim L''$.

Implication: Guarantees the existence of a well-defined weak order over the lottery space.

Axiom 3.1.2 (Continuity). For any $L, L', L'' \in \mathcal{L}$, the sets $\{\alpha \in [0, 1] : \alpha L + (1 - \alpha)L'' \succsim L'\}$ and $\{\alpha \in [0, 1] : L' \succsim \alpha L + (1 - \alpha)L''\}$ are closed in $[0, 1]$.

Intermediate Point Version (C^):* If $L \succ L' \succ L''$, there exists a unique $\alpha \in (0, 1)$ such that:

$$L' \sim \alpha L + (1 - \alpha)L''$$

Economic Role: Rules out lexicographic preferences (e.g., “safety first”) and ensures that small variations in probabilities do not generate discontinuous jumps in preferences.

Axiom 3.1.3 (Independence (Substitution Axiom)). The relation $\succsim$ satisfies independence if, for any $L, L', L'' \in \mathcal{L}$ and any $\alpha \in (0, 1)$:

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''$$

Procedural Meaning: If we mix two lotteries L and L' with a common third lottery L'' in the same proportion $(1 - \alpha)$, the final ranking depends exclusively on the preference between L and L' , being entirely independent of L'' .

The VNM Expected Utility Theorem

Theorem 3.1.1 (vNM / Herstein–Milnor). A preference relation $\succsim$ over $\mathcal{L}$ satisfies Rationality, Continuity, and Independence if and only if there exists a function $u : Z \rightarrow \mathbb{R}$ such that $\succsim$ is represented by the linear expected utility functional $U : \mathcal{L} \rightarrow \mathbb{R}$:

$$L \succsim L' \iff U(L) \geq U(L')$$

where:

$$U(L) = \sum_{k=1}^N p_k u(x_k)$$

Remark 3.1.1 (Critical Distinction of Concepts). • $u(x_k)$ — **Bernoulli Utility (VNM Index):** Assigns a number/value to each deterministic consequence $x_k \in Z$. Reflects the agent’s attitude toward risk when x is monetary.

- $U(L)$ — **VNM Utility Functional:** Assigns a value to the entire lottery L by computing the mathematical expectation of the Bernoulli utilities under the probabilities p_k .

The Role of the Independence Axiom in Linearity

The Independence Axiom is the cornerstone of Expected Utility Theory. It is what imposes linearity in probabilities on the functional $U(L)$:

- **Separation of Probabilities and Valuations:** Independence requires that the impact of a prize’s probability p_k on total utility be strictly additive and linear.

- **Geometry in the Probability Simplex:** In the probability simplex $\Delta(Z)$ (e.g., the Marschak–Machina triangle for $N = 3$), the indifference curves induced by VNM preferences are parallel hyperplanes (straight and parallel lines).
 - *Why straight?* Independence implies that if $L \sim L'$, then any convex mixture $\alpha L + (1 - \alpha)L' \sim L$ (betweenness property).
 - *Why parallel?* If we shift an indifference curve by adding the same lottery $(1 - \alpha)L''$, the marginal rate of substitution between probabilities remains unchanged.
- **Empirical Violations:** Multiple behavioral paradoxes (such as the Allais Paradox) violate the independence axiom because individuals' actual indifference curves tend to “fan out” rather than being parallel.

Uniqueness up to Positive Affine Transformations

Unlike ordinal utility under certainty (invariant to any increasing monotonic transformation), the VNM utility index has a cardinal scale:

Theorem 3.1.2 (Uniqueness). *The functions $u(\cdot)$ and $v(\cdot)$ represent the same VNM preferences over $\mathcal{L}$ if and only if $v(\cdot)$ is a positive affine (monotonic linear) transformation of $u(\cdot)$:*

$$v(x) = \alpha u(x) + \beta \quad \text{with } \alpha > 0 \text{ and } \beta \in \mathbb{R}$$

Intuition of Invariance:

$$\sum_k p_k v(x_k) = \sum_k p_k [\alpha u(x_k) + \beta] = \alpha \left(\sum_k p_k u(x_k) \right) + \beta = \alpha U(L) + \beta$$

Since $\alpha > 0$, the ordering of any pair of lotteries $U(L) \geq U(L')$ is strictly preserved.

Cardinal Property: Affine transformations preserve the ratios of utility differences:

$$\frac{v(x_1) - v(x_2)}{v(x_3) - v(x_4)} = \frac{u(x_1) - u(x_2)}{u(x_3) - u(x_4)}$$

This is indispensable for measuring the intensity of trade-offs between risk and return.

3.1.2 Expected Utility as an Analytical Tool

Translating Preferences into Utility Inequalities

The von Neumann–Morgenstern (VNM) Expected Utility representation establishes a direct bridge between the preference ordering over lotteries and algebraic inequalities.

The Fundamental Translation Rule For two simple lotteries $L = (p_1, \dots, p_N)$ and $L' = (p'_1, \dots, p'_N)$ over the same set of consequences $\{x_1, \dots, x_N\}$:

$$L \succsim L' \iff U(L) \geq U(L') \iff \sum_{k=1}^N p_k u(x_k) \geq \sum_{k=1}^N p'_k u(x_k)$$

Normalization and Extraction of Bernoulli Indices

- **Anchor Fixing:** In a finite set of prizes, we identify the best consequence $\bar{x}$ and the worst consequence $\underline{x}$. By positive affine transformation, we conveniently set $u(\bar{x}) = 1$ and $u(\underline{x}) = 0$.
- **Calibration by Indifference:** For any intermediate prize x_k , by Continuity there exists a unique probability $\alpha_k \in [0, 1]$ such that:

$$x_k \sim \alpha_k \bar{x} + (1 - \alpha_k) \underline{x}$$

- **Valuation:** Substituting into the expected utility function, we have $u(x_k) = \alpha_k \cdot 1 + (1 - \alpha_k) \cdot 0 = \alpha_k$. The Bernoulli utility of a prize is numerically equal to the probability of the best prize that makes the individual indifferent.

Comparing Lotteries Using the VNM Representation

Linearity in Probabilities Property The utility of a convex mixture of two lotteries L and L'' is strictly linear in probabilities:

$$U(\alpha L + (1 - \alpha)L'') = \alpha U(L) + (1 - \alpha)U(L'')$$

This allows simplifying complex lotteries by decomposing them into linear combinations and canceling common components.

Monotonicity in Probabilities Given two lotteries that vary only the probabilities of two fixed prizes $x_1 \succ x_2$:

$$\alpha x_1 + (1 - \alpha)x_2 \succsim \beta x_1 + (1 - \beta)x_2 \iff \alpha \geq \beta$$

The preference ordering strictly follows the higher probability allocated to the best prize.

Identifying Patterns Incompatible with the Independence Axiom

The Independence Axiom requires that the preference between two lotteries L and L' be maintained when both are mixed in the same proportion α with a third lottery L'' :

$$L \succsim L' \iff \alpha L + (1 - \alpha)L'' \succsim \alpha L' + (1 - \alpha)L''$$

The Allais Paradox (Common Consequence Effect) Let the set of prizes be $\{\$5M, \$1M, \$0\}$.

Choice Pair 1:

- $L_1 = (0, 1.0, 0) = \$1M$ for certain.
- $L'_1 = (0.10, 0.89, 0.01) \rightarrow 10\% \text{ of } \$5M, 89\% \text{ of } \$1M, 1\% \text{ of } \0 .
- Typical choice: $L_1 \succ L'_1$.
- Implication in VNM: $u(\$1M) > 0.10u(\$5M) + 0.89u(\$1M) + 0.01u(\$0) \implies 0.11u(\$1M) > 0.10u(\$5M) + 0.01u(\$0)$.

Choice Pair 2:

- $L_2 = (0, 0.11, 0.89) \rightarrow 11\% \text{ of } \$1M, 89\% \text{ of } \$0$.
- $L'_2 = (0.10, 0, 0.90) \rightarrow 10\% \text{ of } \$5M, 90\% \text{ of } \$0$.
- Typical choice: $L'_2 \succ L_2$.
- Implication in VNM: $0.10u(\$5M) + 0.90u(\$0) > 0.11u(\$1M) + 0.89u(\$0) \implies 0.10u(\$5M) + 0.01u(\$0) > 0.11u(\$1M)$.

The Contradiction: The choice $L_1 \succ L'_1$ requires $0.11u(\$1M) > 0.10u(\$5M) + 0.01u(\$0)$, while $L'_2 \succ L_2$ requires exactly the opposite.

Why Does It Violate Independence? Writing as compound lotteries:

- $L_1 = 0.11(\$1M) + 0.89(\$1M)$ and $L'_1 = 0.11(0.10/0.11\$5M + 0.01/0.11\$0) + 0.89(\$1M)$.
- $L_2 = 0.11(\$1M) + 0.89(\$0)$ and $L'_2 = 0.11(0.10/0.11\$5M + 0.01/0.11\$0) + 0.89(\$0)$.

Both pairs differ only by substituting the common event of 89% of \$1M for 89% of \$0. By independence, the ordering should be identical in both pairs.

The Common Ratio Effect

- **Choice 1:** \$3000 for certain $\succ 0.8[\$4000] \oplus 0.2[\$0] \implies u(\$3000) > 0.8u(\$4000)$.
- **Choice 2:** $0.25[\$3000] \oplus 0.75[\$0] \prec 0.20[\$4000] \oplus 0.80[\$0] \implies 0.25u(\$3000) < 0.20u(\$4000) \implies u(\$3000) < 0.8u(\$4000)$.
- **The Contradiction:** Choice 2 is obtained by proportionally reducing (dividing by 4) the winning probabilities of Choice 1 (mixing 1/4 of the choice with 3/4 of winning \$0).

Geometric Signature in the Marschak–Machina Simplex Under VNM theory, indifference curves in the probability simplex are straight and strictly parallel lines. Violation Pattern: Choices incompatible with the Independence Axiom generate non-parallel indifference curves (e.g., the “fanning out” phenomenon, where curves fan out toward the corners of the simplex).

3.1.3 Attitudes Toward Risk

Relationship Between Risk Attitudes and the Curvature of $u(x)$

Given a lottery X with probability distribution over wealth levels and differentiable Bernoulli utility $u(x)$:

Strict Aversion: The individual strictly prefers to receive the expected value of the lottery with certainty rather than face the risk of the lottery.

Jensen’s Inequality and the Preference for Certainty

Theorem 3.1.3 (Jensen’s Inequality). *If $u(\cdot)$ is a strictly concave function and X is a non-degenerate random variable, then:*

$$\mathbb{E}[u(X)] < u(\mathbb{E}[X])$$

Table 3.1: Classification of attitudes toward risk.

Risk Attitude	Preference Condition	Property of $u(x)$	Second Derivative
Risk Aversion	$u(\mathbb{E}[X]) \geq \mathbb{E}[u(X)]$	Strictly Concave	$u''(x) < 0$
Risk Neutrality	$u(\mathbb{E}[X]) = \mathbb{E}[u(X)]$	Linear / Affine	$u''(x) = 0$
Risk Seeking	$u(\mathbb{E}[X]) \leq \mathbb{E}[u(X)]$	Strictly Convex	$u''(x) > 0$

Analytical Role: Jensen's Inequality is the mathematical link between the geometric property of concavity of $u(x)$ and the preference-for-certainty behavior in the Expected Utility model.

Certainty Equivalent (CE) and Risk Premium (P)

Given a lottery X over final wealth:

Certainty Equivalent (CE or $c(F, u)$) *Definition:* The amount of deterministic wealth that yields the same expected utility as the lottery X :

$$u(CE(X)) = \mathbb{E}[u(X)] \implies CE(X) = u^{-1}(\mathbb{E}[u(X)])$$

Characterization by Attitude:

- Risk averse: $CE(X) < \mathbb{E}[X]$ (the agent accepts less than the expected mean to eliminate risk).
- Risk neutral: $CE(X) = \mathbb{E}[X]$.
- Risk seeking: $CE(X) > \mathbb{E}[X]$.

Risk Premium (P or $R(X)$) *Definition:* The maximum amount of wealth the individual is willing to give up to exchange the lottery X for its safe expected value $\mathbb{E}[X]$:

$$P(X) = \mathbb{E}[X] - CE(X)$$

Equivalently: $u(\mathbb{E}[X] - P(X)) = \mathbb{E}[u(X)]$.

Sign: $P(X) > 0$ if and only if the individual is risk averse ($u'' < 0$).

Pratt Approximation for Small Risks (Taylor Expansion): For a local risk $X = w + \epsilon$ where $\mathbb{E}[\epsilon] = 0$ and $\text{Var}(\epsilon) = \sigma^2$:

$$P(w, \epsilon) \approx \frac{1}{2} \sigma^2 A(w)$$

Arrow–Pratt Measures of Risk Aversion

Coefficient of Absolute Risk Aversion ($A(x)$ / $R_a(x)$) *Formal Definition:*

$$A(x) = -\frac{u''(x)}{u'(x)}$$

Why Not Just Use $u''(x)$? The absolute value of $u''(x)$ varies under positive affine transformations $\alpha u(x) + \beta$ (multiplies by $\alpha > 0$). Division by $u'(x)$ normalizes the scale and makes the measure invariant to affine transformations.

Behavior with Respect to Wealth:

- **DARA (Decreasing Absolute Risk Aversion):** $A'(x) < 0 \implies$ as wealth increases, the absolute amount invested in risky assets increases.
- **CARA (Constant Absolute Risk Aversion):** $A(x) = a > 0 \implies u(x) = -e^{-ax}$. The risky monetary amount does not depend on initial wealth.

Coefficient of Relative Risk Aversion ($R(x) / R_r(x)$) *Formal Definition:*

$$R(x) = x \cdot A(x) = -\frac{xu''(x)}{u'(x)}$$

Meaning: Measures aversion to proportional/percentage risks of wealth (elasticity of marginal utility with respect to wealth).

CRRA (Constant Relative Risk Aversion): $R(x) = \rho > 0$:

- If $\rho \neq 1$: $u(x) = \frac{x^{1-\rho}}{1-\rho}$.
- If $\rho = 1$: $u(x) = \ln(x)$.

Portfolio Implication: Under CRRA, the proportion of wealth held in risky assets is constant regardless of the individual's initial wealth.

3.1.4 Stochastic Dominance and Applications

First-Order Stochastic Dominance (FOSD)

FOSD (*First-Order Stochastic Dominance*) formalizes the idea that one lottery offers strictly higher returns/outcomes than another.

Definition by Preferences Lottery F dominates G by FOSD ($F \succsim_{FOSD} G$) if and only if every agent with increasing preferences (who prefers more wealth to less) prefers F to G :

$$\mathbb{E}_F[u(x)] \geq \mathbb{E}_G[u(x)] \quad \forall u \text{ such that } u'(x) \geq 0$$

Equivalent Characterization by CDF

$$F \succsim_{FOSD} G \iff F(x) \leq G(x) \quad \forall x \in \mathbb{R}$$

(with strict inequality on some interval of x).

Geometry: The cumulative distribution function (CDF) of F is always below or to the right of the CDF of G .

Intuition: For any wealth level x , the probability of obtaining an outcome less than or equal to x is always lower under F than under G .

Characterization by Stochastic Construction

$F \succsim_{FOSD} G \iff$ a random variable $Y \sim G$ can be represented as $Y = X + \varepsilon$ where $X \sim F$ and $\varepsilon \leq 0$ almost surely (a non-positive shock).

Implication on Means

$$F \succsim_{FOSD} G \implies \mathbb{E}_F[X] \geq \mathbb{E}_G[X]$$

(*Proof Note:* The converse is not true! Having a higher mean does not guarantee FOSD).

Second-Order Stochastic Dominance (SOSD)

SOSD (*Second-Order Stochastic Dominance*) formalizes the idea that one lottery is less risky / less dispersed than another, maintaining the same expected return.

Definition by Preferences Lottery F dominates G by SOSD ($F \succsim_{SOSD} G$) if and only if every risk-averse agent prefers F to G :

$$\mathbb{E}_F[u(x)] \geq \mathbb{E}_G[u(x)] \quad \forall u \text{ such that } u'(x) \geq 0 \text{ and } u''(x) \leq 0$$

Characterization by Area Under the CDF (Assuming Equal Means $\mathbb{E}_F[X] = \mathbb{E}_G[X]$)

$$F \succsim_{SOSD} G \iff \int_{-\infty}^x F(t)dt \leq \int_{-\infty}^x G(t)dt \quad \forall x \in \mathbb{R}$$

Characterization by Mean-Preserving Spread (MPS)

$$F \succsim_{SOSD} G \iff G \text{ is a Mean-Preserving Spread of } F$$

Stochastic Representation: $Y \sim G$ can be written as $Y = X + \varepsilon$, where $X \sim F$ and $\mathbb{E}[\varepsilon|X] = 0$ (ε is white noise / zero conditional mean shock).

Operation: G is created by taking probability mass from the center of F 's distribution and spreading it to the tails.

Hierarchy

$$F \succsim_{FOSD} G \implies F \succsim_{SOSD} G$$

(If F dominates G by FOSD, it automatically dominates by SOSD; the converse does not hold).

Application 1: Optimal Insurance Demand

Problem Structure

- Initial wealth: w_0 .
- Probability of accident/loss: $\pi \in (0, 1)$.
- Value of loss/damage: $L > 0$.
- Units of insurance purchased: $q \geq 0$, with price/premium per unit of coverage $\gamma > 0$.

Consumer's Maximization Problem

$$\max_{q \geq 0} V(q) = \pi u(w_0 - L + q - \gamma q) + (1 - \pi)u(w_0 - \gamma q)$$

First-Order Condition (FOC)

$$V'(q) = \pi(1 - \gamma)u'(w_0 - L + q(1 - \gamma)) - (1 - \pi)\gamma u'(w_0 - \gamma q) = 0$$

Rearranging:

$$\frac{u'(w_0 - L + q^*(1 - \gamma))}{u'(w_0 - \gamma q^*)} = \frac{\gamma(1 - \pi)}{\pi(1 - \gamma)}$$

Equilibrium Results and Economic Interpretation Case 1: Actuarially Fair Insurance ($\gamma = \pi$)

The ratio on the right becomes $\frac{\pi(1-\pi)}{\pi(1-\pi)} = 1$.

$$\implies u'(w_0 - L + q^*(1 - \pi)) = u'(w_0 - \pi q^*)$$

Since $u'' < 0$ (strictly decreasing marginal utility):

$$w_0 - L + q^*(1 - \pi) = w_0 - \pi q^* \implies q^* = L$$

Conclusion: Faced with actuarially fair insurance, the risk-averse consumer fully insures (*Full Insurance*), equalizing their wealth across all states of nature.

Case 2: Actuarially Unfair Insurance ($\gamma > \pi$)

The ratio $\frac{\gamma(1-\pi)}{\pi(1-\gamma)} > 1 \implies u'(w_{loss}) > u'(w_{no_loss}) \implies q^* < L$.

Conclusion: The consumer purchases only partial insurance (*Partial Insurance*).

Application 2: Portfolio Choice (Demand for Risky Asset)

Problem Structure

- Initial wealth: w_0 .
- Risk-free asset with net return 0 (or risk-free rate r).
- Risky asset with random net return R , with distribution $F(R)$ and mean $\mathbb{E}[R]$.
- Amount of wealth invested in the risky asset: $a \geq 0$.

Investor's Maximization Problem

$$\max_{a \geq 0} v(a) = \mathbb{E}[u(w_0 + aR)] = \int u(w_0 + aR)dF(R)$$

FOC and Corner Decision ($a = 0$) Differentiating with respect to a :

$$v'(a) = \mathbb{E}[u'(w_0 + aR)R] = 0$$

Evaluating the derivative at the zero-investment point ($a = 0$):

$$v'(0) = \mathbb{E}[u'(w_0)R] = u'(w_0)\mathbb{E}[R]$$

Positive Investment Theorem (Arrow–Pratt) Since $u'(w_0) > 0$:

$$v'(0) > 0 \iff \mathbb{E}[R] > 0$$

Result: If the expected return of the risky asset is strictly positive ($\mathbb{E}[R] > 0$), any risk-averse individual will allocate a strictly positive amount of their wealth to the risky asset ($a^* > 0$).

Comparative Statics with Wealth ($a^*(w_0)$) Differentiating the FOC $v'(a^*(w_0), w_0) \equiv 0$ with respect to w_0 :

$$\frac{da^*}{dw_0} = -\frac{\mathbb{E}[u''(w_0 + a^*R)R]}{\mathbb{E}[u''(w_0 + a^*R)R^2]}$$

Since the denominator is strictly negative by concavity ($u'' < 0$):

$$\text{sgn}\left(\frac{da^*}{dw_0}\right) = \text{sgn}(\mathbb{E}[u''(w_0 + a^*R)R])$$

Result:

- If $u(\cdot)$ exhibits DARA ($A'(w) < 0$), then $\mathbb{E}[u''(w_0 + a^*R)R] > 0 \implies \frac{da^*}{dw_0} > 0$ (absolute investment in the risky asset grows with wealth).
- If $u(\cdot)$ is CARA, $\frac{da^*}{dw_0} = 0$ (investment in the risky asset is independent of wealth).

3.2 Part II — General Equilibrium

The central chain is:

$$\text{Feasible Allocations} \longrightarrow \text{Pareto Efficiency} \longrightarrow \text{Core and Blocking} \longrightarrow \\ \text{Walrasian Equilibrium} \longrightarrow \text{Welfare Theorems}$$

3.2.1 Pure Exchange Economies and the Edgeworth Box

Formal Definitions and Objects of the Economy

- **Pure Exchange Economy ($\mathcal{E}$):** An economy without a productive sector composed of I consumers (indexed by $i = 1, \dots, I$) and L consumption goods (indexed by $l = 1, \dots, L$). Each consumer i is characterized by:
 - A consumption set $X_i \subseteq \mathbb{R}_+^L$ (typically $X_i = \mathbb{R}_+^L$).
 - A preference relation $\succsim_i$ over X_i , represented by a utility function $u_i : X_i \rightarrow \mathbb{R}$ that is continuous and concave/quasi-concave.
 - An initial endowment vector $e_i \in \mathbb{R}_+^L$.

Formally, the economy is the set $\mathcal{E} = \{(u_i, e_i)\}_{i=1}^I$.

- **Individual Endowment and Aggregate Resources ($\bar{e}$ or ω):**

- *Individual Endowment:* Vector $e_i = (e_{i1}, e_{i2}, \dots, e_{iL}) \in \mathbb{R}_+^L$, specifying the goods individual i brings to the market.

- *Aggregate Resources (Aggregate Endowment)*: The total supply of goods available in the economy:

$$\bar{e} = \sum_{i=1}^I e_i \in \mathbb{R}_+^L$$

- **Feasible Allocation**: An allocation $x = (x_1, x_2, \dots, x_I) \in (\mathbb{R}_+^L)^I$, where $x_i \in \mathbb{R}_+^L$ is the bundle consumed by individual i , is said to be feasible if aggregate consumption does not exceed aggregate resources:

$$\sum_{i=1}^I x_i \leq \sum_{i=1}^I e_i = \bar{e}$$

Welfare Concepts and Efficiency

- **Pareto Improvement**: Given a feasible allocation x , an alternative feasible allocation y is a Pareto Improvement over x if it does not worsen any individual's situation and strictly improves at least one's:

$$y_i \succsim_i x_i \quad \forall i \in \{1, \dots, I\} \quad \text{and} \quad y_k \succ_k x_k \quad \text{for some } k$$

- **Pareto-Efficient Allocation / Pareto Optimum**: A feasible allocation x^* is Pareto-efficient if there is no other feasible allocation y that is a Pareto improvement over x^* .
 - *Weak Efficiency*: There is no feasible y with $y_i \succ_i x_i^*$ for all i .
 - *Strong Efficiency*: There is no feasible y with $y_i \succsim_i x_i^*$ for all i and $y_k \succ_k x_k^*$ for some k .
 - (*Proof Note*: Under strictly increasing and continuous preferences, Weak and Strong Efficiency coincide).
- **Contract Curve (CC)**: The locus of all feasible Pareto-efficient allocations in the economy.
 - *Interior Algebraic Condition*: For differentiable preferences, x^* belongs to the interior Contract Curve if and only if the Marginal Rates of Substitution (MRS) between any two goods are equalized for all consumers:

$$MRS_{1,2}^1(x_1^*) = MRS_{1,2}^2(x_2^*) = \dots = MRS_{1,2}^I(x_I^*)$$

$$\text{where } MRS_{1,2}^i(x_i) = \frac{\partial u_i / \partial x_{i1}}{\partial u_i / \partial x_{i2}}.$$

The Edgeworth Box: Transition Between Algebraic and Geometric Representation ($I = 2, L = 2$)

Let there be two consumers (A and B) and two goods (1 and 2).

Construction of the Box Geometry

- **Aggregate Endowments (Box Dimensions):**
 - Width = $\bar{e}_1 = e_{A1} + e_{B1}$.
 - Height = $\bar{e}_2 = e_{A2} + e_{B2}$.
- **Dual Coordinate Systems:**
 - *Consumer A*: Origin O_A in the lower left corner (Southwest). Consumption (x_{A1}, x_{A2}) grows to the right and upward.
 - *Consumer B*: Origin O_B in the upper right corner (Northeast — 180° rotation). Consumption (x_{B1}, x_{B2}) grows to the left and downward.
- **Feasibility Conservation (No Waste):** Each interior point of the box (x_{A1}, x_{A2}) automatically defines B's consumption:

$$x_{B1} = \bar{e}_1 - x_{A1} \quad \text{and} \quad x_{B2} = \bar{e}_2 - x_{A2}$$

The Lens of Mutually Beneficial Trades Given the initial endowment point $e = (e_A, e_B)$ in the box:

- Draw the indifference curves $u_A(x_A) = u_A(e_A)$ and $u_B(x_B) = u_B(e_B)$.
- The intersection of A's upper contour set with B's upper contour set forms a lens of trades.
- Any allocation interior to this lens is a Pareto Improvement over e .

Characterization of the Contract Curve in the Box

- *Geometry*: It is the line connecting the points where A's and B's indifference curves are tangent to each other (same slope/same tangent line).
- *Algebra (Planner's Maximization Problem)*: To find the Contract Curve, fix a utility level $\bar{u}_B$ for B and maximize A's utility subject to the feasibility constraint:

$$\begin{aligned} \max_{x_A, x_B} \quad & u_A(x_{A1}, x_{A2}) \\ \text{s.t.} \quad & u_B(x_{B1}, x_{B2}) \geq \bar{u}_B \\ & x_{A1} + x_{B1} = \bar{e}_1, \quad x_{A2} + x_{B2} = \bar{e}_2 \end{aligned}$$

- *First-Order Condition (FOC)*:

$$\frac{\partial u_A / \partial x_{A1}}{\partial u_A / \partial x_{A2}} = \frac{\partial u_B / \partial x_{B1}}{\partial u_B / \partial x_{B2}} \iff MRS^A(x_A) = MRS^B(x_B)$$

3.2.2 The Core and the Edgeworth Conjecture

Fundamental Concepts and the Blocking Mechanism

In a pure exchange economy $\mathcal{E} = \{(u_i, e_i)\}_{i=1}^I$:

- **Coalition (S)**: Any non-empty subset of consumers $S \subseteq \{1, 2, \dots, I\}$.

- **Blocking of an Allocation (Objection):** A coalition S blocks a feasible allocation $x = (x_1, \dots, x_I)$ if there exists an alternative distribution of goods $\{y_i\}_{i \in S}$ exclusively among the members of S such that:

- *Internal Coalition Feasibility:* The allocation is self-sufficient/feasible using only the aggregate endowments of the group itself:

$$\sum_{i \in S} y_i \leq \sum_{i \in S} e_i$$

- *Strict Improvement for the Coalition:* All members of S are at least as well off as in x , and at least one is strictly better off (or, under strictly monotonic preferences, all are strictly better off):

$$y_i \succeq_i x_i \quad \forall i \in S \quad \text{and} \quad y_k \succ_k x_k \quad \text{for some } k \in S$$

- *Economic Intuition of Blocking:* A coalition blocks x if it has the capacity and incentive to “refuse to trade with the rest of the economy”, separating itself and securing a superior outcome for all its members using only its own resources.

Formal Definition of the Core ($\mathcal{C}(\mathcal{E})$)

Definition 3.2.1 (Core). The **Core** of an economy $\mathcal{E}$ is the set of all feasible allocations x that are not blocked by any coalition $S \subseteq \{1, \dots, I\}$.

Two Essential Limit Coalitions:

- **Single-Person Coalitions** ($S = \{i\}$) $\implies$ **Individual Rationality:** If a coalition of a single individual $S = \{i\}$ tries to block x , it uses its own endowment e_i . For x not to be blocked by any $S = \{i\}$, the allocation must satisfy Individual Rationality:

$$u_i(x_i) \geq u_i(e_i) \quad \forall i \in \{1, \dots, I\}$$

- **The Grand Coalition** ($S = \{1, \dots, I\}$) $\implies$ **Pareto Efficiency:** The grand coalition containing all agents has access to $\sum_{i=1}^I e_i = \bar{e}$. For x not to be blocked by $S = \{1, \dots, I\}$, there cannot exist any other feasible allocation y that is a Pareto Improvement over x . Hence, the allocation must be Pareto-efficient.

Geometric Representation in the Edgeworth Box ($I = 2$)

In an economy with 2 consumers (A and B), the only possible coalitions are $S = \{A\}$, $S = \{B\}$, and $S = \{A, B\}$.

- **Non-Blocking Condition by $\{A\}$ and $\{B\}$:** Requires that the allocation be in the **region of mutually beneficial trades** (inside the “lens” delimited by the indifference curves $u_A(e_A)$ and $u_B(e_B)$).
- **Non-Blocking Condition by $\{A, B\}$:** Requires that the allocation belong to the Contract Curve (tangency of indifference curves).

Characterization of the Core in the Edgeworth Box:

$$\text{Core} = \{x \in \text{Contract Curve} : u_A(x_A) \geq u_A(e_A) \text{ and } u_B(x_B) \geq u_B(e_B)\}$$

Geometry: It is the segment of the Contract Curve that lies strictly inside the lens of trades originating from the initial endowment e .

The Edgeworth Conjecture and the Debreu–Scarf Theorem

There is a deep relationship between allocation by bargaining/coalitions (Core) and allocation by market prices (Walrasian Equilibrium):

Basic Inclusion Every Walrasian Equilibrium belongs to the Core (since if it were blockable, it would violate utility maximization or feasibility at equilibrium prices). However, in small economies (with few agents), the Core is “large” and contains many efficient allocations that are not competitive equilibria.

The Economy Replication Process ($\mathcal{E}_r$) Let $\mathcal{E}_1$ be the original economy with I types of agents. The r -replicated economy ($\mathcal{E}_r$) has r identical copies of each of the I types (total of $r \cdot I$ agents).

Properties and the Debreu–Scarf Theorem

- **Equal Treatment Property:** In any allocation belonging to the Core of a replicated economy $\mathcal{E}_r$, all agents of the same type receive exactly the same consumption bundle.
- **Monotonic Shrinking of the Core:** As r increases, the set of possible coalitions grows exponentially. New mixed coalitions (e.g., 2 type-A agents and 1 type-B agent) can block allocations that were previously unblockable. Therefore:

$$\mathcal{C}(\mathcal{E}_1) \supseteq \mathcal{C}(\mathcal{E}_2) \supseteq \cdots \supseteq \mathcal{C}(\mathcal{E}_r)$$

- **Edgeworth / Debreu–Scarf Convergence:** In the limit, when the number of replicas tends to infinity ($r \rightarrow \infty$), the Core shrinks to coincide exactly with the set of Walrasian Equilibria:

$$\lim_{r \rightarrow \infty} \mathcal{C}(\mathcal{E}_r) = \{\text{Walrasian Equilibria}\}$$

Economic Intuition: In small markets, agents possess strategic bargaining power to sustain collusive/local agreements outside the price system. As the market becomes atomized and perfectly competitive (multiple perfect substitutes for each type of agent), individual or small-group bargaining power evaporates. The only way to decentralize sustainable trades immune to collusive blocking is through the vector of competitive prices.

3.2.3 Walrasian Equilibrium: Objects, Properties, and Existence

Fundamental Objects of the Market

- **Price Vector and Normalization:** The price vector is $p = (p_1, p_2, \dots, p_L) \in \mathbb{R}_+^L \setminus \{0\}$. Since demand depends only on relative prices, the price vector can be

normalized without loss of generality to belong to the Price Simplex:

$$\Delta = \left\{ p \in \mathbb{R}_+^L : \sum_{l=1}^L p_l = 1 \right\}$$

Alternatively, one chooses a good l as numeraire by fixing $p_l = 1$.

- **Individual Budget Set ($B_i(p)$):** For consumer i , given their initial endowment $e_i \in \mathbb{R}_+^L$, wealth is endogenous $w_i(p) = p \cdot e_i$. The budget set is:

$$B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot e_i\}$$

- **Individual Walrasian Demand ($x_i(p)$):** It is the solution to the Utility Maximization Problem (UMP):

$$x_i(p) = \arg \max_{x_i \in B_i(p)} u_i(x_i)$$

- **Aggregate Excess Demand Function ($z(p)$):**

- *Individual Excess Demand:* $z_i(p) = x_i(p) - e_i$.
- *Aggregate Excess Demand:* The difference between aggregate demanded consumption and available aggregate resources:

$$z(p) = \sum_{i=1}^I z_i(p) = \sum_{i=1}^I x_i(p) - \sum_{i=1}^I e_i = \sum_{i=1}^I x_i(p) - \bar{e}$$

where $z(p) = (z_1(p), z_2(p), \dots, z_L(p)) \in \mathbb{R}^L$.

- **Market Clearing and Definition of Walrasian Equilibrium:** A Walrasian Equilibrium is a pair (p^*, x^*) composed of a price vector $p^* \in \Delta$ and a feasible allocation $x^* = (x_1^*, \dots, x_I^*)$ such that:

- *Individual Optimization:* For each i , $x_i^* = x_i(p^*)$ maximizes $u_i(x_i)$ in $B_i(p^*)$.
- *Market Compatibility:* $z(p^*) \leq 0$ (with $z_l(p^*) = 0$ for every good with strictly positive price $p_l^* > 0$). If preferences are strictly monotonic, all prices are positive and $z(p^*) = 0$ (supply equals demand in all markets).

Critical Properties of $z(p)$

Zero-Degree Homogeneity (HDG 0)

$$z(\lambda p) = z(p) \quad \forall \lambda > 0$$

Intuition: Multiplying all prices by a constant does not alter the slope of the budget constraint nor the real value of the endowment.

Walras' Law For any price vector p (whether equilibrium or not):

$$p \cdot z(p) = \sum_{l=1}^L p_l z_l(p) = 0$$

Origin: It follows from Local Non-Satiation (LNS). If consumers exhaust their budget ($p \cdot x_i(p) = p \cdot e_i$), summing over all i yields $p \cdot \sum (x_i(p) - e_i) = p \cdot z(p) = 0$.

Practical Consequence: If $L - 1$ markets are in equilibrium ($z_l(p^*) = 0$ for $l = 1, \dots, L - 1$) and $p_L^* > 0$, the L -th market necessarily clears ($z_L(p^*) = 0$).

Continuity If preferences are continuous, strictly convex, and $p \cdot e_i > 0$ for all i , the excess demand function $z(p)$ is continuous on all $p \in \mathbb{R}_{++}^L$.

The Walrasian Equilibrium Existence Argument

The classical proof (Arrow–Debreu–McKenzie / Varian) uses Brouwer's Fixed Point Theorem.

Construction of the Price Mapping Function ($f : \Delta \rightarrow \Delta$) For each good l , a Tâtonnement-style price adjustment rule is defined:

$$f_l(p) = \frac{p_l + \max(0, z_l(p))}{1 + \sum_{k=1}^L \max(0, z_k(p))}$$

If a market has excess demand ($z_l(p) > 0$), its relative price increases in the next period. The denominator ensures that $\sum_{l=1}^L f_l(p) = 1$, keeping the transformed vector inside the simplex Δ .

Application of Brouwer's Fixed Point Theorem Since Δ is a compact and convex set, and $f(\cdot)$ is a continuous function from Δ to Δ , by Brouwer's Theorem there exists at least one fixed point $p^* \in \Delta$ such that:

$$f(p^*) = p^*$$

Demonstration that the Fixed Point Clears Markets Equating $f_l(p^*) = p_l^*$:

$$p_l^* \left[1 + \sum_{k=1}^L \max(0, z_k(p^*)) \right] = p_l^* + \max(0, z_l(p^*))$$

$$p_l^* \sum_{k=1}^L \max(0, z_k(p^*)) = \max(0, z_l(p^*))$$

Multiplying both sides by $z_l(p^*)$ and summing over all l :

$$\left(\sum_{l=1}^L p_l^* z_l(p^*) \right) \cdot \sum_{k=1}^L \max(0, z_k(p^*)) = \sum_{l=1}^L z_l(p^*) \max(0, z_l(p^*))$$

By Walras' Law, the term on the left equals 0 ($p^* \cdot z(p^*) = 0$). Hence:

$$\sum_{l=1}^L z_l(p^*) \max(0, z_l(p^*)) = 0$$

Since each term in the sum is non-negative, this requires that $\max(0, z_l(p^*)) = 0$ for all l .

$$\implies z_l(p^*) \leq 0 \quad \forall l = 1, \dots, L$$

Proving that p^* is a Walrasian equilibrium price vector!

Logical Role of Each Hypothesis in the Existence Proof

Table 3.2: Hypotheses and their roles in the existence theorem.

Hypothesis on Preferences/Economy	Mathematical Consequence	Logical Role in Existence Theorem
Continuity of Preferences	Extreme Value Theorem in individual choice	Guarantees that demand $x_i(p)$ and excess demand $z(p)$ are continuous functions. Without continuity, $f(p)$ is not continuous and Brouwer fails.
Strict Convexity of Preferences	Unique solution in UMP (Demand is a function, not a correspondence)	Allows using Brouwer's Theorem for functions. (If only convex, $z(p)$ is a correspondence and requires Kakutani's Fixed Point Theorem).
Local Non-Satiation (LNS)	Budget constraint holds with equality ($p \cdot x_i = p \cdot e_i$)	Guarantees Walras' Law ($p \cdot z(p) = 0$), essential for zeroing the left side of the equation and proving the fixed point is an equilibrium.
Monotonicity / Boundary Behavior	If $p_l \rightarrow 0$, $z_l(p) \rightarrow \infty$	Guarantees equilibrium occurs in the interior of the simplex ($p^* \gg 0$), preventing zero prices for scarce goods and ensuring exact $z(p^*) = 0$.
Positive Endowment ($e_i \gg 0$)	$p \cdot e_i > 0$ for all $p \in \Delta$	Ensures the budget set does not become empty or discontinuous at the boundary of the price simplex.

3.2.4 Welfare Economics: The Fundamental Theorems

First Fundamental Theorem of Welfare Economics (1st FTWE)

The First Theorem formalizes Adam Smith's "invisible hand" hypothesis, establishing that competitive markets without failures allocate resources in a non-wasteful manner.

Precise Statement

Theorem 3.2.1 (First Welfare Theorem). *Let (p^*, x^*, z^*) be a Walrasian Equilibrium in a pure exchange or production economy with non-negative price vector $p^* \geq 0$. If all consumers have locally non-satiated (LNS) preferences, then the equilibrium allocation x^* is Pareto-efficient.*

Fundamental Hypotheses and Their Roles

- **Local Non-Satiation (LNS):** Guarantees that consumers spend all their income in equilibrium, making Walras' Law hold with strict equality for each individual. Without LNS, an agent at a local satiation point could waste resources that would bring utility to other consumers without harming them.
- **Complete Competitive Markets and Price-Taking:** All goods, services, and future contingencies have publicly defined prices, and no agent possesses market power to alter them.
- **Absence of Externalities and Public Goods:** Each consumer's utility depends only on their own bundle, and each firm's production depends only on its own production plan.

Critical Note on Convexity The 1st FTWE **does not require convexity** of preferences or production sets. The efficiency of a competitive equilibrium holds even in the presence of local indivisibilities or technological non-convexities, as long as the equilibrium exists.

Second Fundamental Theorem of Welfare Economics (2nd FTWE)

The Second Theorem addresses the opposite relationship: whether any desirable allocation can be obtained through the market mechanism.

Precise Statement

Theorem 3.2.2 (Second Welfare Theorem). *Let x^* be a feasible Pareto-efficient allocation in an economy where the production sets Y_j and consumers' preferences $\succsim_i$ are convex. If preferences are continuous and locally non-satiated, there exists a non-zero price vector $p^* \neq 0$ and a zero-sum redistribution of income/endowments (lump-sum transfers) that decentralize x^* as a Walrasian Equilibrium (or Quasi-Equilibrium).*

Where Does Convexity Enter?

- *Mathematical Necessity:* The proof of the 2nd FTWE is based on applying the Separating Hyperplane Theorem between the set of technologically feasible allocations and the set of allocations strictly preferred by consumers.
- *Economic Role:* For there to exist a supporting price vector p^* that separates feasible production from preferred consumption, the set of utility imputations must be convex. If there are non-convexities (e.g., increasing returns to scale in production or non-convex preferences), a Pareto-efficient allocation may lie in a “re-entrance”

or non-convex boundary where no price plane can simultaneously tangent the indifference curves for all agents.

Where Does Redistribution (*Lump-Sum Transfers*) Enter?

- *Budgetary Necessity*: The Pareto-efficient allocation chosen by society x^* generally does not coincide with the budgets generated by agents' gross initial endowments e_i at supporting prices p^* .
- *The Solution*: The social planner applies zero-sum wealth transfers T_i (where $\sum_i T_i = 0$) such that the consumer's new wealth becomes $w_i = p^* \cdot x_i^*$. The transfers need to be *lump-sum* (levied on fixed/exogenous attributes) so as not to distort relative prices and not alter agents' marginal choice incentives.

Critical Distinction: Efficiency vs. Decentralization via Prices

One of the most important conceptual focuses is clearly separating these two concepts:

Table 3.3: Efficiency vs. decentralization by prices.

Dimension	Pareto Efficiency	Decentralization by Prices
Nature of Concept	Normative and Physical/Quantity: Concerns purely with feasibility and non-waste of resources in the economy.	Institutional Mechanism of In-formation: Concerns with autonomy and coordination through market signals.
Information Required	To be calculated by a central planner, requires complete and centralized knowledge of p^* and their own local preferences/technologies.	Each agent needs to know only the public price vector and their own local preferences/technologies, ignoring the rest of the economy.
Decision Mechanism	Evaluates whether some physical reorganization (y, x) can increase one agent's utility without reducing another's.	Agents act in a decentralized and selfish manner, maximizing utility or profit subject to the price vector p^* .

The Connection Established by the Theorems:

- **1st FTWE (Market $\implies$ Efficiency)**: Shows that the decentralized price system spontaneously leads to a physically efficient outcome without need for central coordination.
- **2nd FTWE (Efficiency + Redistribution $\implies$ Market)**: Shows that society can separate the discussion into two independent stages:
 1. The *Distribution / Equity Branch* decides the socially desirable allocation and effects the necessary *lump-sum* wealth transfers.
 2. The *Allocation / Efficiency Branch* lets the decentralized price mechanism operate freely to reach that allocation without market distortions.

3.3 Part III — Arrow–Debreu, Radner, and Financial Markets

The key progression is:

Contingent Trade (Arrow–Debreu) $\longrightarrow$ Sequential Trade (Radner) $\longrightarrow$
 Assets, Payoffs, and Portfolios $\longrightarrow$ No-Arbitrage and State Prices $\longrightarrow$
 Risk-Neutral Valuation

3.3.1 Arrow–Debreu Equilibrium under Uncertainty

Temporal Structure and State Space

- **Two-Period Structure:**

- *Date 0 (Ex-ante / Moment of Trade):* The state of nature is uncertain. All contingent contracts are negotiated and financially settled on this date.
- *Date 1 (Ex-post / Moment of Consumption):* Uncertainty is resolved; nature reveals the realized state $s \in S$. The contracts negotiated on date 0 are physically executed (delivery of goods).

- **State Space ($\mathcal{S}$ or Ω):** Finite set of mutually exclusive and exhaustive states of nature:

$$\mathcal{S} = \{1, 2, \dots, S\}$$

Only one state $s \in \mathcal{S}$ will realize on date 1.

The Crucial Distinction: Physical Good $\neq$ State-Contingent Good

This is the central ontological definition of Arrow–Debreu theory:

- **Physical Good ($l \in \{1, \dots, L\}$):** The description of the physical, spatial, and temporal characteristics of a commodity (e.g., 1 kilogram of wheat on date 1).
- **State-Contingent Good ((l, s)):** The contractual promise to deliver 1 unit of physical good l on date 1 if and only if state $s \in \mathcal{S}$ realizes (and nothing otherwise).
- **Dimension of the Goods Space:** If there are L physical goods and S states of nature, the total space of commodities in the economy has dimension:

$$N = L \times S$$

Objects of the Arrow–Debreu Model

- **Contingent Consumption (x_i):** For consumer i , the consumption plan is a vector of dimension $L \times S$:

$$x_i = (x_{i1}, x_{i2}, \dots, x_{iS}) \in \mathbb{R}_+^{L \times S}$$

where $x_{is} = (x_{1s}, x_{2s}, \dots, x_{Ls}) \in \mathbb{R}_+^L$ is individual i 's physical consumption bundle if state s occurs.

- **Contingent Endowment (e_i):** Consumer i 's vector of initial contingent endowments:

$$e_i = (e_{i1}, e_{i2}, \dots, e_{iS}) \in \mathbb{R}_+^{L \times S}$$

- **Arrow–Debreu Contingent Prices (p):** The price vector fixed and paid on date 0 for contingent delivery on date 1:

$$p = (p_{11}, \dots, p_{LS}) = (p_1, p_2, \dots, p_S) \in \mathbb{R}_+^{L \times S} \setminus \{0\}$$

where p_{ls} is the monetary price paid on date 0 for one unit of physical good l to be delivered on date 1 if state s occurs.

Consumer Optimization and the Single Budget Constraint

The Single Budget Constraint on Date 0 Unlike sequential models, in the Arrow–Debreu model all markets for all $L \times S$ contingent goods open simultaneously and only once on date 0. There is no renegotiation, budgetary uncertainty, or use of money on date 1. Consumer i faces a single global ex-ante budget constraint:

$$\sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot e_{is} \iff \sum_{s=1}^S \sum_{l=1}^L p_{ls} x_{ils} \leq \sum_{s=1}^S \sum_{l=1}^L p_{ls} e_{ils}$$

The Utility Maximization Problem (UMP) Under the VNM Expected Utility structure, with subjective beliefs $\pi_{is} > 0$ (where $\sum_{s=1}^S \pi_{is} = 1$):

$$\begin{aligned} \max_{x_i \in \mathbb{R}_+^{L \times S}} \quad & U_i(x_i) = \sum_{s=1}^S \pi_{is} u_i(x_{is}) \\ \text{s.t.} \quad & \sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot e_{is} \end{aligned}$$

First-Order Condition (FOC) For interior consumption of good l between states s and s' :

$$\frac{\pi_{is} \frac{\partial u_i(x_{is})}{\partial x_{ils}}}{\pi_{is'} \frac{\partial u_i(x_{is'})}{\partial x_{ils'}}} = \frac{p_{ls}}{p_{ls'}}$$

The Marginal Rate of Substitution between consumption in different states of nature equals the ratio of Arrow–Debreu contingent prices.

Feasibility, Market Clearing, and Equilibrium Definition

Feasibility / Market Clearing The consumption allocation is feasible if, for each physical good l and in each state of nature s , aggregate demand does not exceed aggregate contingent resources:

$$\sum_{i=1}^I x_{ils} \leq \sum_{i=1}^I e_{ils} \quad \forall l = 1, \dots, L \quad \text{and} \quad \forall s = 1, \dots, S$$

Definition of Arrow–Debreu Equilibrium (ADE) An **Arrow–Debreu Equilibrium** is a pair (p^*, x^*) composed of a contingent price vector $p^* \in \mathbb{R}_+^{L \times S} \setminus \{0\}$ and a feasible contingent allocation $x^* = (x_1^*, \dots, x_I^*)$ such that:

1. For each consumer i , x_i^* solves the UMP subject to the single Arrow–Debreu budget constraint at prices p^* .
2. All $L \times S$ contingent goods markets clear: $\sum_{i=1}^I x_{ils}^* = \sum_{i=1}^I e_{ils}^*$ for all l and all s .

The Arrow Isomorphism and Validity of Welfare

The Arrow Isomorphism By redefining a “good” as a pair (l, s) , the choice problem under uncertainty becomes mathematically and formally identical to a traditional static Walrasian General Equilibrium with $N = L \times S$ deterministic goods.

Validity of the Fundamental Welfare Theorems Since the ADE is a special case of Walrasian Equilibrium in a space of dimension $L \times S$:

- **1st Welfare Theorem:** Every Arrow–Debreu Equilibrium is ex-ante Pareto-efficient (no feasible contingent allocation can increase one agent’s expected utility without reducing another’s).
- **2nd Welfare Theorem:** Any Pareto-efficient contingent allocation can be decentralized as an ADE through *lump-sum* wealth transfers effected on date 0.

3.3.2 Radner Equilibrium and Sequential Markets

The Temporal Architecture of Sequential Trades

While the Arrow–Debreu model requires all contingent markets to open simultaneously ex-ante, Radner Equilibrium introduces the realistic structure of sequential markets with financial assets:

Date 0 (Finance) $\longrightarrow$ Interval (Realization of State s) $\longrightarrow$ Date 1 (Spot Markets)

- **Date 0 (Asset Market):** There is no physical consumption (in the basic model). Agents trade portfolios of financial assets $z_i \in \mathbb{R}^K$ at the asset price vector $q \in \mathbb{R}^K$ to transfer wealth between future states.
- **Interval (*Interim*):** Nature reveals the realized state $s \in \mathcal{S} = \{1, \dots, S\}$.
- **Date 1 (Spot Market):**
 - Financial assets pay their returns (*payoffs*) in the realized state s .
 - Spot markets open for the L physical goods at spot prices $p_s \in \mathbb{R}_+^L$.
 - Agents use wealth from endowments and assets to consume $x_{is} \in \mathbb{R}_+^L$.

Fundamental Objects of the Radner Model

- **Asset Prices (q):** Vector $q = (q_1, q_2, \dots, q_K) \in \mathbb{R}^K$ indicating the monetary price on date 0 of each of the K assets.
- **Asset Payoff Matrix (R):** Matrix of dimension $S \times K$ specifying each asset’s return in each state:

$$R = \begin{bmatrix} R_{11} & R_{12} & \dots & R_{1K} \\ R_{21} & R_{22} & \dots & R_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ R_{S1} & R_{S2} & \dots & R_{SK} \end{bmatrix}$$

where R_{sk} is the payoff (in units of account) of asset k if state s realizes. Row $R_s = (R_{s1}, \dots, R_{sK})$: Vector of payoffs of all assets in state s .

- **Portfolio Position (z_i):** Vector $z_i = (z_{i1}, \dots, z_{iK}) \in \mathbb{R}^K$ held by consumer i .
 - $z_{ik} > 0$: Long position.
 - $z_{ik} < 0$: Short position.
- **State-Contingent Spot Prices (p_s):** Vector $p_s = (p_{1s}, \dots, p_{Ls}) \in \mathbb{R}_+^L$ of physical goods prices in the spot market of state s on date 1.
- **Interim Wealth (w_{is}):** The total monetary income available to consumer i in state s on date 1, composed of the value of their endowment plus the portfolio return:

$$w_{is}(q, z_i, p_s) = p_s \cdot e_{is} + R_s \cdot z_i = p_s \cdot e_{is} + \sum_{k=1}^K R_{sk} z_{ik}$$

Radner's Budget Constraints

Unlike Arrow–Debreu (which has 1 single ex-ante budget constraint), the consumer in the Radner model faces $S + 1$ distinct budget constraints:

Date 0 Budget Constraint (Asset Purchase)

$$q \cdot z_i \leq 0 \iff \sum_{k=1}^K q_k z_{ik} \leq 0$$

(Asset purchases must be fully financed by sales of other assets).

Date 1 Budget Constraints (One for each state $s \in \{1, \dots, S\}$)

$$\begin{aligned} p_s \cdot x_{is} &\leq p_s \cdot e_{is} + R_s \cdot z_i \quad \forall s = 1, \dots, S \\ \iff \sum_{l=1}^L p_{ls} x_{ils} &\leq \sum_{l=1}^L p_{ls} e_{ils} + \sum_{k=1}^K R_{sk} z_{ik} \quad \forall s = 1, \dots, S \end{aligned}$$

Consumer Optimization and Market Clearing

Radner's Maximization Problem (UMP) Given the asset price vector q^* and the family of spot prices $(p_s^*)_{s=1}^S$, consumer i chooses their portfolio z_i and their spot consumption plan $x_i = (x_{i1}, \dots, x_{iS})$:

$$\begin{aligned} \max_{z_i \in \mathbb{R}^K, x_i \in \mathbb{R}_+^{L \times S}} \quad & \sum_{s=1}^S \pi_{is} u_i(x_{is}) \\ \text{s.t.} \quad & q^* \cdot z_i \leq 0 \\ & p_s^* \cdot x_{is} \leq p_s^* \cdot e_{is} + R_s \cdot z_i \quad \forall s = 1, \dots, S \end{aligned}$$

Market Clearing Conditions

- **Asset Market Clearing (Date 0):**

$$\sum_{i=1}^I z_i^* = 0 \quad (\in \mathbb{R}^K)$$

- **Spot Goods Markets Clearing (Date 1 — for each state s):**

$$\sum_{i=1}^I x_{is}^* = \sum_{i=1}^I e_{is}^* \quad \forall s = 1, \dots, S \quad (\in \mathbb{R}^L)$$

Definition of Radner Equilibrium (RE) A **Radner Equilibrium with Rational Expectations** is a profile of prices and choices $(q^*, (p_s^*)_{s=1}^S, (z_i^*)_{i=1}^I, (x_i^*)_{i=1}^I)$ such that:

1. For each consumer i , the pair (z_i^*, x_i^*) solves the Radner UMP at prices $(q^*, p_1^*, \dots, p_S^*)$.
2. All agents possess correct/common expectations about the future spot prices p_s^* that will prevail in state s .
3. The asset market clears on date 0 and all S spot markets clear on date 1.

Rational Expectations Hypothesis (*Perfect Foresight*)

An indispensable conceptual element in Radner Equilibrium is that agents on date 0 need to perfectly predict the spot price vector p_s^* that will arise on date 1 for each state s . Without this perfect prediction, agents could not calculate the real income generated by portfolio $R_s \cdot z_i$ nor optimize their future consumption.

3.3.3 Arrow–Debreu / Radner Equivalence and Financial Completeness

The Central Message: Financial Spanning $\iff$ Contingent Trade

The fundamental result of this block establishes the bridge between finance and general equilibrium:

The Spanning Principle: The ability to effect direct contingent trades for all future states (Arrow–Debreu) is substantially and allocatively equivalent to the ability to freely transfer wealth between all states through the asset market (Radner), if and only if the financial market is complete (spanning completeness).

The Canonical Case: Pure Arrow Assets ($R = I_S$)

- **Arrow Security:** An elementary financial title that pays exactly 1 unit of account in state s and 0 in all other states $s' \neq s$.

- **Canonical Payoff Structure ($R = I_S$):** If there are $K = S$ Arrow assets (one for each state), the payoff matrix $R_{S \times S}$ is the identity matrix of order S :

$$R = I_S = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

- **Price Vectors and Direct Equivalence:**

- Let $q = (q_1, q_2, \dots, q_S) \in \mathbb{R}_{++}^S$ be the vector of Arrow asset prices on date 0.
- Let $p_s = (p_{1s}, \dots, p_{Ls}) \in \mathbb{R}_+^L$ be the vector of spot prices in state s on date 1.
- *Arrow–Debreu Price Mapping:* The Arrow–Debreu contingent price for physical good l in state s is given by multiplying the state asset price by the spot price:

$$p_{ls}^{AD} = q_s \cdot p_{ls}^{spot} \quad \forall l, \forall s$$

- **Budget Constraint in the Canonical Case:** The state s budget constraint in Radner becomes $p_s \cdot x_{is} \leq p_s \cdot e_{is} + z_{is}$. Isolating $z_{is} = p_s \cdot (x_{is} - e_{is})$ and substituting into the date 0 constraint ($\sum_s q_s z_{is} \leq 0$):

$$\sum_{s=1}^S q_s [p_s \cdot (x_{is} - e_{is})] \leq 0 \iff \sum_{s=1}^S \sum_{l=1}^L (q_s p_{ls}) x_{ils} \leq \sum_{s=1}^S \sum_{l=1}^L (q_s p_{ls}) e_{ils}$$

This is exactly the single Arrow–Debreu budget constraint with prices $p_{ls}^{AD} = q_s p_{ls}$.

The General Condition of Complete Markets

When the available assets are not pure Arrow assets, but rather complex assets (stocks, bonds with mixed returns, options) described by a generic payoff matrix $R \in \mathbb{R}^{S \times K}$:

The Full Rank Condition The financial market is strictly complete if and only if:

$$\text{rank}(R) = S \iff \text{Range}(R) = \mathbb{R}^S$$

Geometric-Algebraic Interpretation:

- $\text{Range}(R) = \{Rz : z \in \mathbb{R}^K\}$ is the subspace of all wealth transfers between states that can be synthesized by combining the available assets.
- If $\text{Range}(R) = \mathbb{R}^S$, the image of the payoff-generating matrix covers the entire S -dimensional space. That is, any wealth transfer profile $w \in \mathbb{R}^S$ can be exactly achieved by some portfolio z .
- *Necessary Quantity Condition of Assets:* There must be at least as many assets as states of nature ($K \geq S$), containing at least S linearly independent assets.

The General Case as a Change of Basis from the Canonical Case

Why does the case of a generic complete market ($\text{rank}(R) = S$) reduce to the canonical Arrow case ($R = I_S$)? By a simple Change of Basis in linear algebra!

- **Synthesizing Arrow Assets (Replicating Pure Assets):** If $\text{rank}(R) = S$, there exist S linearly independent columns in R that form an invertible matrix $\hat{R}_{S \times S}$. To synthetically create the s -th Arrow asset (whose payoff is the canonical vector $e_s = (0, \dots, 1, \dots, 0)^T$), simply choose the portfolio $z^{(s)}$ that solves:

$$\hat{R}z^{(s)} = e_s \implies z^{(s)} = \hat{R}^{-1}e_s$$

The columns of $\hat{R}^{-1}$ contain the exact portfolio recipes needed to construct each Arrow asset from the market assets.

- **Change of Coordinates in the Transfer Space:**
 - In the canonical (Arrow) space, the agent chooses direct wealth transfers $w \in \mathbb{R}^S$ with matrix I_S .
 - In the general market, the agent chooses a portfolio $z \in \mathbb{R}^S$ that produces the same wealth vector $w = \hat{R}z$.
 - The passage from real asset portfolio to wealth transfers is a bijective linear transformation (change of basis of $\mathbb{R}^S$).
- **Prices by No-Arbitrage:** If $q^{Arrow} \in \mathbb{R}_{++}^S$ is the vector of implicit state prices, the vector of real asset prices $q \in \mathbb{R}^S$ satisfies the consistency condition:

$$q = \hat{R}^T q^{Arrow} \iff q^{Arrow} = (\hat{R}^T)^{-1}q$$

Formal Statement of the Arrow–Debreu / Radner Equivalence Theorem

Theorem 3.3.1 (Equivalence (Arrow–Radner)). *Suppose the asset market is complete ($\text{rank}(R) = S$).*

1. **(Arrow–Debreu $\implies$ Radner):** *If (p^*, x^*) is an Arrow–Debreu Equilibrium, there exist asset prices q^* , spot prices $(p_s^*)_{s=1}^S$, and portfolios $(z_i^*)_{i=1}^I$ such that $(q^*, (p_s^*)_{s=1}^S, (z_i^*)_{i=1}^I, x^*)$ is a Radner Equilibrium.*
2. **(Radner $\implies$ Arrow–Debreu):** *Reciprocally, if $(q^*, (p_s^*)_{s=1}^S, (z_i^*)_{i=1}^I, x^*)$ is a Radner Equilibrium under complete markets, the consumption allocation x^* is exactly the allocation of an Arrow–Debreu Equilibrium.*

Welfare Consequence: When $\text{rank}(R) = S$, Radner Equilibrium achieves complete ex-ante Pareto efficiency (the 1st and 2nd Welfare Theorems hold). If the market is incomplete ($\text{rank}(R) < S$), the transfer space is restricted, generating allocative inefficiency in Radner Equilibrium.

3.3.4 Asset Markets and No-Arbitrage

Canonical Notation and Financial Objects

In a two-period financial economy with K assets and S states of nature:

- **Payoff Vector of Asset k (r_k):** Column vector containing asset k 's returns in each of the S states of nature:

$$r_k = \begin{bmatrix} r_{1k} \\ r_{2k} \\ \vdots \\ r_{Sk} \end{bmatrix} \in \mathbb{R}^S$$

- **Global Payoff Matrix (R):** Matrix of dimension $S \times K$ whose columns are the payoff vectors of each of the K assets:

$$R = \begin{bmatrix} r_1 & r_2 & \cdots & r_K \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1K} \\ r_{21} & r_{22} & \cdots & r_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ r_{S1} & r_{S2} & \cdots & r_{SK} \end{bmatrix} \in \mathbb{R}^{S \times K}$$

- **Portfolio Vector (z):** Allocation of units bought or sold short of each asset:

$$z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_K \end{bmatrix} \in \mathbb{R}^K$$

- **Portfolio Cost ($q \cdot z$):** Given the vector of asset prices on date 0, $q = (q_1, \dots, q_K) \in \mathbb{R}^K$, the total monetary cost of portfolio z is:

$$q \cdot z = \sum_{k=1}^K q_k z_k \in \mathbb{R}$$

- **Portfolio Payoff (Rz):** The vector of wealth generated by portfolio z on date 1 in each state s :

$$Rz = \sum_{k=1}^K z_k r_k = \begin{bmatrix} \sum_{k=1}^K r_{1k} z_k \\ \vdots \\ \sum_{k=1}^K r_{Sk} z_k \end{bmatrix} \in \mathbb{R}^S$$

Formal Definition of Arbitrage Opportunity

An arbitrage opportunity is the possibility of obtaining a strictly positive financial return without assuming any risk and without net initial investment.

Definition 3.3.1 (Arbitrage). A portfolio $z \in \mathbb{R}^K$ is an arbitrage if it satisfies one of the two equivalent conditions:

1. Type I Arbitrage (“Free Lunch” on Date 1):

- $q \cdot z \leq 0$ (zero cost or the investor receives money to set up the portfolio on date 0)
- $Rz \geq 0$ (non-negative payoff in all states of nature on date 1)

- $Rz \neq 0$ (strictly positive payoff in at least one state s)

2. Type II Arbitrage (“Free Lunch” on Date 0):

- $q \cdot z < 0$ (receives strictly positive money on date 0)
- $Rz \geq 0$ (non-negative payoff in all states of nature on date 1)

Equilibrium Implication: Strict Monotonicity $\implies$ No-Arbitrage

Theorem 3.3.2. *If the preferences of at least one consumer are strictly monotonic (LNS in all states), then in any Radner Equilibrium $(q^*, (p_s^*)_{s=1}^S, z^*, x^*)$, the asset price vector q^* does not admit arbitrage opportunities.*

Formal Demonstration (By Contradiction). **Absurd Hypothesis:** Suppose q^* is an equilibrium price vector, but there exists an arbitrage portfolio $z_{arb} \in \mathbb{R}^K$ such that $q^* \cdot z_{arb} \leq 0$, $Rz_{arb} \geq 0$, and $Rz_{arb} \neq 0$.

Deviation Construction: Let z_i^* be consumer i ’s equilibrium portfolio. Consider the new perturbed portfolio $z'_i = z_i^* + z_{arb}$.

Budget Feasibility Verification:

- On date 0: $q^* \cdot z'_i = q^* \cdot z_i^* + q^* \cdot z_{arb} \leq 0 + 0 = 0$. The new position is perfectly financeable.
- On date 1: For each state s , the new wealth is $w'_{is} = p_s^* \cdot e_{is} + R_s z_i^* + R_s z_{arb} \geq w_{is}^*$, with strict inequality in at least one state.

Optimality Violation: Since the consumer possesses strictly monotonic preferences, consuming the strictly greater additional wealth generated by z_{arb} in at least one state strictly increases their expected utility:

$$U_i(x(z'_i)) > U_i(x(z_i^*))$$

This contradicts the hypothesis that z_i^* was the consumer’s optimal choice.

Conclusion: No price that allows arbitrage can sustain an equilibrium. Hence, Equilibrium $\implies$ No-Arbitrage. $\square$

3.3.5 State Prices and Risk-Neutral Probability Measures

The State-Price Representation

In a financial market without arbitrage opportunities, the market prices of all assets can be expressed as linear combinations of Arrow prices (state prices):

State Price Vector (μ) There exists a strictly positive vector $\mu = (\mu_1, \mu_2, \dots, \mu_S) \in \mathbb{R}_{++}^S$, where $\mu_s > 0$ represents the monetary value on date 0 of receiving 1 unit of account in state s on date 1.

Individual Pricing Equation The price q_k of any asset $k \in \{0, 1, \dots, K\}$ with payoff vector $r_k = (r_{1k}, \dots, r_{Sk})^T$ is the weighted sum of its payoffs by state prices:

$$q_k = \sum_{s=1}^S \mu_s r_{sk} \quad \forall k = 0, \dots, K$$

Global Matrix Representation Given the payoff matrix $R \in \mathbb{R}^{S \times K}$ and the asset price vector $q \in \mathbb{R}^K$:

$$q^\top = \mu^\top R \iff \begin{bmatrix} q_1 & \cdots & q_K \end{bmatrix} = \begin{bmatrix} \mu_1 & \cdots & \mu_S \end{bmatrix} \begin{bmatrix} r_{11} & \cdots & r_{1K} \\ \vdots & \ddots & \vdots \\ r_{S1} & \cdots & r_{SK} \end{bmatrix}$$

The Risk-Free Asset as Normalization Factor

The inclusion of a risk-free asset establishes the pure interest rate of the economy and provides the mathematical anchor to convert state prices into probabilities:

Risk-Free Asset Structure (Asset 0)

- *Payoff vector:* $r_0 = (1 + r)\mathbf{1}_S = \begin{bmatrix} 1 + r \\ 1 + r \\ \vdots \\ 1 + r \end{bmatrix}$, where $r > 0$ is the risk-free interest rate and $\mathbf{1}_S$ is the unit vector of dimension S .
- *Price on date 0:* Normalized to $q_0 = 1$.
- *Pricing Condition of the Safe Asset:* Applying the state-price representation to asset 0:

$$q_0 = \sum_{s=1}^S \mu_s r_{s0} \implies 1 = \sum_{s=1}^S \mu_s (1 + r) = (1 + r) \sum_{s=1}^S \mu_s$$

- *The Sum of State Prices:*

$$\sum_{s=1}^S \mu_s = \frac{1}{1 + r}$$

Interpretation: The sum of all state prices equals the economy's risk-free discount factor (the present value of 1 unit of account with certainty on date 1).

Derivation and Definition of the Risk-Neutral Measure (π)

Since state prices $\mu_s > 0$ are strictly positive and their sum is $\frac{1}{1+r}$, we can normalize them by multiplying by the factor $(1 + r)$ to construct a legitimate probability distribution:

Definition of Risk-Neutral Probability (π_s)

$$\pi_s \equiv (1 + r)\mu_s \quad \forall s = 1, \dots, S$$

Verification of Probability Axioms

- *Positivity:* Since $\mu_s > 0$ and $1 + r > 0$, we have $\pi_s > 0$ for all s .
- *Normalization (Sum 1):*

$$\sum_{s=1}^S \pi_s = \sum_{s=1}^S (1+r)\mu_s = (1+r) \sum_{s=1}^S \mu_s = (1+r) \cdot \frac{1}{1+r} = 1$$

Hence, the vector $\pi = (\pi_1, \dots, \pi_S)$ belongs to the simplex of strictly positive probabilities: $\pi \in \Delta_{++}^{S-1}$.

The Set of Risk-Neutral Measures ($\mathcal{Q}_M$) and Pricing by Expectation

For a financial market $\mathcal{M} = \{(r_k, q_k)\}_{k=0}^K$:

The Set of Risk-Neutral Measures ($\mathcal{Q}_M$)

$$\mathcal{Q}_M = \left\{ \pi \in \Delta_{++}^{S-1} : q_k = \frac{1}{1+r} \sum_{s=1}^S \pi_s r_{sk}, \quad k = 0, \dots, K \right\}$$

Pricing in Discounted Expectation Form Substituting $\mu_s = \frac{\pi_s}{1+r}$ into the pricing equation, we obtain the fundamental asset pricing formula:

$$q_k = \frac{1}{1+r} \sum_{s=1}^S \pi_s r_{sk} \iff q_k = \frac{\mathbb{E}_\pi[r_k]}{1+r} \quad \forall k = 0, \dots, K$$

Central Message: A risk-neutral probability measure (or equivalent martingale measure) is a fictitious probability distribution π that simultaneously prices all assets traded in the market through the expected value of their payoffs under π , discounted by the risk-free rate $1 + r$.

Crucial Distinction: State Prices (μ) vs. Risk-Neutral Probabilities (π)

This conceptual differentiation is a frequent verification point in proofs:

The First Fundamental Theorem of Asset Pricing (1st FTAP)

Synthesizing all of Part III into a single logical theorem:

Theorem 3.3.3 (1st Fundamental Theorem of Asset Pricing). *The financial market $\mathcal{M}$ does not admit arbitrage opportunities if and only if there exists at least one strictly positive risk-neutral probability measure ($\mathcal{Q}_M \neq \emptyset$).*

Completeness and Uniqueness:

- If the market is complete ($\text{rank}(R) = S$), the risk-neutral measure is **UNIQUE** ($|\mathcal{Q}_M| = 1$). The price of any derivative or new asset is determined univocally.
- If the market is incomplete ($\text{rank}(R) < S$), there exist infinitely many valid risk-neutral measures ($\mathcal{Q}_M$ is a convex set of dimension $S - \text{rank}(R)$), defining an arbitrage-free price interval.

Table 3.4: State prices vs. risk-neutral probabilities.

Property	State Price (μ_s)	Risk-Neutral Probability (π_s)
Definition	Market monetary price on date 0 for 1 unit of account in state s .	Synthetic/normalized probability factor for risk valuation.
Sum of Elements	$\sum_{s=1}^S \mu_s = \frac{1}{1+r} < 1$ (is a discount factor).	$\sum_{s=1}^S \pi_s = 1$ (is a valid probability distribution).
Incorporated Elements	Contains three things simultaneously: risk aversion, real subjective probabilities (π_s^{real}), and time preference.	Adjusts payoffs for risk without altering the discount rate (simulates that all agents are risk-neutral).
Connection Relation	$\mu_s = \frac{\pi_s}{1+r}$	$\pi_s = (1+r)\mu_s$ (the risk-free asset connects both).

3.3.6 Completeness, Spanning, and Payoff Replication

The Generated Payoff Space ($\text{Range}(R)$)

The central geometric object in theoretical finance is not the individual list of traded assets, but rather the space of wealth transfer opportunities they collectively generate:

Payoff Space / Image of R Given the payoff matrix $R \in \mathbb{R}^{S \times (K+1)}$ whose columns are the payoff vectors of assets $\{r_0, r_1, \dots, r_K\}$, the space of attainable payoffs is the vector subspace of $\mathbb{R}^S$ generated by these columns:

$$\text{Range}(R) = \text{Span}\{r_0, r_1, \dots, r_K\} = \{w \in \mathbb{R}^S : w = Rz \text{ for some portfolio } z \in \mathbb{R}^{K+1}\}$$

Dimension of the Payoff Space The dimension of $\text{Range}(R)$ is given by the rank of the payoff matrix:

$$\dim(\text{Range}(R)) = \text{rank}(R) \leq \min(S, K+1)$$

Differentiation of Key Concepts

Complete vs. Incomplete Financial Markets

- **Complete Market:** The space generated by assets covers all possible future contingency dimensions in the economy:

$$\text{Range}(R) = \mathbb{R}^S \iff \text{rank}(R) = S$$

- **Incomplete Market:** The asset space cannot cover all contingencies; there exist wealth transfers that financial technology cannot reach:

$$\text{Range}(R) \subsetneq \mathbb{R}^S \iff \text{rank}(R) < S$$

Linearly Independent vs. Redundant Assets

- **Redundant Asset:** Asset k is redundant if its payoff vector r_k already belongs to the space generated by the other assets in the economy:

$$r_k \in \text{Span}\{r_j\}_{j \neq k}$$

Implication: Removing or adding a redundant asset does not alter the financial opportunity space $\text{Range}(R)$ nor affect the set of equilibrium allocations.

- **Linearly Independent Assets:** No asset can be written as a linear combination of the others. If all $K + 1$ assets are independent, $\text{rank}(R) = K + 1$.

Replicable / Attainable Payoffs and Cost of Replication

- **Replicable Payoff (*Attainable Payoff*):** Any future contingent flow $x \in \mathbb{R}^S$ is said to be replicable if it belongs to the space generated by the economy's assets:

$$x \text{ is replicable} \iff x \in \text{Range}(R) \iff \exists z \in \mathbb{R}^{K+1} \text{ such that } Rz = x$$

- **Replicating Portfolio:** The vector z that satisfies $Rz = x$.
- **Cost of Replication:** The monetary value necessary on date 0 to set up the replicating portfolio:

$$\text{Cost}(x) = q \cdot z = \sum_{k=0}^K q_k z_k$$

- **Law of One Price:** Under no-arbitrage, if two different portfolios $z^{(1)}$ and $z^{(2)}$ replicate the same payoff x (i.e., $Rz^{(1)} = Rz^{(2)} = x$), they necessarily have the same market cost: $q \cdot z^{(1)} = q \cdot z^{(2)}$.

Completeness and Uniqueness of the Risk-Neutral Measure

The relationship between the payoff space structure $\text{Range}(R)$ and the set of risk-neutral measures $\mathcal{Q}_M$ under no-arbitrage is the highest point of the theorem:

Complete Market Benchmark ($\text{Range}(R) = \mathbb{R}^S$)

- The system of state price equations $q^\top = \mu^\top R$ has a unique strictly positive solution $\mu^* \in \mathbb{R}_{++}^S$.
- Since $\pi^* = (1 + r)\mu^*$, the risk-neutral probability measure is **UNIQUE**:

$$\mathcal{Q}_M = \{\pi^*\}$$

- *Unique Valuation:* Any asset or derivative with payoff $x \in \mathbb{R}^S$ has its price determined univocally by:

$$q_x = \frac{\mathbb{E}_{\pi^*}[x]}{1 + r}$$

Incomplete Market Case ($\text{Range}(R) \subsetneq \mathbb{R}^S$)

- There exist contingent directions in $\mathbb{R}^S$ that are not reached by combinations of assets.
- The system $q^\top = \mu^\top R$ is underdetermined: there exist multiple state price vectors μ compatible with the prices of traded assets.
- The set $\mathcal{Q}_M$ is not a singleton (contains infinitely many valid risk-neutral measures). Non-replicable payoffs do not possess a unique price determined purely by arbitrage.

The Central Structural Idea: Span Invariance

Spanning Invariance Principle: What determines economic possibilities and pricing in the financial economy is exclusively the generated vector subspace $\text{Range}(R)$.

Two completely different collections of assets (with distinct payoff matrices R_1 and R_2 and different numbers of assets) offer exactly the same investment opportunities and generate the same equilibrium if and only if they generate the same subspace:

$$\text{Range}(R_1) = \text{Range}(R_2)$$

3.4 An Integrated View

The concepts above are connected by a common methodological thread:

$$\begin{aligned} \text{VNM Axioms and EU} &\longrightarrow \text{Risk Attitudes} \longrightarrow \text{General Equilibrium} \longrightarrow \\ &\text{Arrow-Debreu and Radner} \longrightarrow \text{No-Arbitrage and Assets} \end{aligned}$$

3.4.1 Decision Matrix: From Statement to Solution

This section functions as a Quick Diagnostic Index: when you read a problem statement, this flow immediately indicates which object to construct, which theorem to cite, and which calculation to perform.

If the Problem Involves Individual Choice under Risk

- **Trigger in Statement:** Lotteries, choice paradoxes, insurance contracting, simple portfolio allocation, comparison of risk distributions.
- **Object to Construct:** $U(L) = \sum p_k u(x_k)$ or $\mathbb{E}[u(W)]$.
- **Hypothesis / Axiom Tested:** VNM Independence (if violated $\rightarrow$ Allais / Common Ratio), concavity $u'' < 0$ (Risk Aversion).
- **Critical Formula / Calculation:**
 - *Certainty Equivalent / Risk Premium:* $u(CE) = \mathbb{E}[u(X)] \implies P = \mathbb{E}[X] - CE \approx \frac{1}{2}\sigma^2 A(w)$.
 - *Optimal Insurance:* FOC $\frac{u'(w_{loss})}{u'(w_{no_loss})} = \frac{\gamma(1-\pi)}{\pi(1-\gamma)}$. Fair insurance ($\gamma = \pi$) $\implies q^* = L$.

- *Stochastic Dominance*: FSD ($F(x) \leq G(x)$ for all x) vs. SSD ($\int F \leq \int G$ + equal means / MPS).

If the Problem Involves Static General Equilibrium (Pure Exchange)

- **Trigger in Statement:** Two or more people trading goods, Edgeworth Box, efficiency in consumption, income transfers, equilibrium price p^* .
- **Object to Construct:** Feasibility constraint $\sum x_i \leq \bar{e}$, excess demand vector $z(p) = \sum x_i(p) - \bar{e}$.
- **Theorem / Property:**
 - *Contract Curve / Pareto*: $MRS_A(x_A) = MRS_B(x_B)$.
 - *1st FTWE*: LNS $\implies$ Walrasian Equilibrium is Pareto-efficient (without needing convexity).
 - *2nd FTWE*: Convexity + LNS + Lump-Sum Transfers $\implies$ Any Pareto-efficient is decentralizable by prices.
 - *Existence*: Walras' Law ($p \cdot z(p) = 0$), Zero-Degree Homogeneity, and Brouwer's Theorem on Simplex Δ .

If the Problem Involves Uncertainty in General Equilibrium (Arrow–Debreu vs. Radner)

- **Trigger in Statement:** State-contingent goods $s \in \mathcal{S}$, asset markets at time $t = 0, 1$, payoff matrix $R_{S \times K}$, portfolio positions z_i .
- **Object to Construct:**
 - *Arrow–Debreu*: 1 single ex-ante budget constraint on date 0 with prices p_{is}^{AD} .
 - *Radner*: $S+1$ budget constraints ($q \cdot z_i \leq 0$ on date 0 and $p_s \cdot x_{is} \leq p_s \cdot e_{is} + R_s z_i$ on date 1).
- **Theorem / Equivalence Relation:**
 - *Arrow Isomorphism*: A-D is a standard Walrasian Equilibrium in dimension $L \times S$.
 - *Complete Spanning*: $\text{rank}(R) = S \iff \text{Range}(R) = \mathbb{R}^S \iff \text{A-D} \iff \text{Radner Equivalence}$.
 - *Arrow Prices*: $p_{is}^{AD} = q_s p_{is}^{\text{spot}}$.

If the Problem Involves Theoretical Finance, Arbitrage, and Pricing

- **Trigger in Statement:** Asset prices q , matrix R , risk-neutral probability π , state prices μ , derivative replication.
- **Object to Construct:**
 - *Arbitrage Condition*: $q \cdot z \leq 0, Rz \geq 0, Rz \neq 0$.
 - *State Price System*: $q^\top = \mu^\top R$.

- **Theorem / Critical Formula:**

- *1st FTAP: No-Arbitrage* $\iff$ There exists at least one risk-neutral measure $\pi \in \Delta_{++}^{S-1}$.
- *Fundamental Connection:* $\mu_s = \frac{\pi_s}{1+r} \implies \sum \mu_s = \frac{1}{1+r}$.
- *Risk-Neutral Pricing:* $q_k = \frac{\mathbb{E}_\pi[r_k]}{1+r}$.
- *Uniqueness of π :* $\mathcal{Q}_M = \{\pi^*\} \iff \text{rank}(R) = S$ (Complete Market).

3.4.2 Summary of Quick Triggers ($X \rightarrow Y$)

3.4.3 Final Formatting Tips for the Exam Document

- Keep one page per block (Block I: Risk; Block II: General Equilibrium; Block III: Arrow–Debreu/Radner/Finance).
- Add this Decision Matrix on the initial page as an analytical index.
- Use colored boxes or bold highlights on the rank conditions ($\text{rank}(R) = S$) and pricing equations ($q_k = \frac{\mathbb{E}_\pi[r_k]}{1+r}$).

Table 3.5: Diagnostic triggers for the exam.

Question	Answer
Do compound lotteries give the same value as reduced ones?	$\rightarrow$ Consequentialism / RCLA
Are indifference curves in the simplex straight and parallel?	$\rightarrow$ Independence Axiom
Which transformation preserves the Expected Utility structure?	$\rightarrow$ Only positive affine transformations $v = \alpha u + \beta$ ($\alpha > 0$)
Can I apply $f(u(x)) = u(x)^3$ to Expected Utility?	$\rightarrow$ No! Non-linear transformations destroy the mathematical expectation property
$F \succsim_{FOSD} G$?	$\rightarrow$ Every $u' \geq 0$ prefers F . Graphically $F(x) \leq G(x)$ for all x
$F \succsim_{SOSD} G$?	$\rightarrow$ Every $u' \geq 0, u'' \leq 0$ prefers F . Equal mean + G is an MPS of F
Fair premium in insurance ($\gamma = \pi$)?	$\rightarrow$ Full Insurance ($q^* = L$)
Risky asset with $\mathbb{E}[R] > 0$?	$\rightarrow$ Every risk-averse person buys $a^* > 0$
Is allocation Pareto-efficient?	$\rightarrow$ Impossible to improve i without worsening j . Tangency in the Box ($MRS^A = MRS^B$)
Where is the Core in the Edgeworth Box?	$\rightarrow$ Intersection of the Contract Curve with the trade lens from e
What happens when the economy is replicated ($r \rightarrow \infty$)?	$\rightarrow$ Debreu–Scarf Theorem: Core converges to Walrasian Equilibria
What is Walras' Law for in the existence proof?	$\rightarrow$ Transform fixed point $f(p^*) = p^*$ into $z(p^*) \leq 0$
If $L - 1$ markets clear, does the L -th clear?	$\rightarrow$ Yes, directly by Walras' Law ($p \cdot z(p) = 0$)
Does the 1 st FTWE require convexity?	$\rightarrow$ No! It only requires LNS and absence of market failures
Where is convexity indispensable?	$\rightarrow$ In the 2 nd FTWE, to guarantee separating hyperplane
How many budget constraints in Radner vs. A-D?	$\rightarrow$ Radner: $S + 1$; A-D: only 1 global ex-ante
What defines a financially complete market?	$\rightarrow \text{rank}(R) = S \iff \text{Range}(R) = \mathbb{R}^S$
What is the matrix formula for state prices?	$\rightarrow q^\top = \mu^\top R$
How to go from state price to risk-neutral probability?	$\rightarrow \pi_s = (1 + r)\mu_s$
How to price any asset under π ?	$\rightarrow q_k = \frac{\mathbb{E}_\pi[r_k]}{1+r}$
What guarantees $\mathcal{Q}_M \neq \emptyset$?	$\rightarrow$ Absence of Arbitrage (1 st FTAP)
What guarantees uniqueness of π ?	$\xrightarrow{181}$ Completeness of the Financial Market ($\text{rank}(R) = S$)